

PRODUCTION AND OPERATIONS MANAGEMENT

DEPARTMENT OF
BUSINESS MANAGEMENT



Prof L P Krüger

Updates by
Jonathan de Villiers

Only study guide for
MNO2601

UNIVERSITY OF SOUTH AFRICA,
PRETORIA

© 2011 University of South Africa

First edition 1998

Revised editions 2001, 2004, 2007 and 2011

All rights reserved

Printed and published by the
University of South Africa
Muckleneuk, Pretoria

MNO2601/1/2012–2018

98780646

InDesign

CONTENTS

	<i>Page</i>
TOPIC	
PREFACE	iv
STEPS IN THE STUDY PROCESS	v
PRESCRIBED BOOK	vi
ICONS USED IN THIS STUDY GUIDE AND THEIR MEANING	vii
MODULE FRAMEWORK: MNO2601	viii
MODULE AIM	ix
MODULE OUTCOMES	ix
FOUNDATIONS OF PRODUCTION/OPERATIONS AND PRODUCTIVITY MANAGEMENT – PART I	1
1. INTRODUCTION TO PRODUCTION/OPERATIONS MANAGEMENT	2
2. PERFORMANCE OBJECTIVES OF PRODUCTION/OPERATIONS MANAGEMENT	16
3. PRODUCTIVITY MANAGEMENT	22
DESIGNING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM – PART II	83
4. DESIGN OF PROCESSES FOR PRODUCTS AND SERVICES	84
5. DESIGN AND LOCATION OF PRODUCTION/OPERATIONS CAPACITY	104
6. DESIGN OF LAYOUT AND WORKFLOW	113
7. JOB DESIGN AND WORK ORGANISATION	119
PLANNING AND CONTROLLING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM – PART III	131
8. CAPACITY PLANNING AND CONTROL	132
9. AGGREGATE PLANNING AND MASTER SCHEDULING	156
10. INVENTORY AND SUPPLY CHAIN PLANNING AND CONTROL	182
REFLECTION AND A LOOK AHEAD	201

PREFACE

Welcome to the MNO2601 module! This is the fifth rewrite of this study guide and I wish to thank all those, including students and colleagues, who in one way or another contributed to its initial and ongoing design, production and continued improvement. You will shortly read much more about the terms “design”, “production” and “improvement”, as in a way they personify what production and operations management is all about – so enjoy!

This module specifically examines production and operations management. We argue (and hopefully you will come to agree with us) that this is one of the more interesting and exciting specialist fields within business management. When we think of a business, whether it be a large motor vehicle manufacturer/assembler such as BMW (South Africa) or a communications provider such as TELKOM, we usually recognise and associate the company name with the products it manufactures and/or the services it provides. In effect, therefore, the core of any business (or, as we like to refer to it, the heart of any business) is what it manufactures or produces (goods or products) and/or what services it provides or renders. Without these products and/or services to offer customers/clients, we cannot really call the enterprise a “business” as we know the term in countries that have western-style, free-market economies.

While the value to businesses of specialised fields like marketing, human resources and financial management is widely acknowledged, this value is not always as clear in the case of production and operations management. This may be because some people consider it to be a “technical” discipline or function reserved for industrial engineers. However, we do not believe this is true. All managers in all businesses can benefit from having a fundamental knowledge and some practical experience (however rudimentary the concepts and methods, tools and techniques of application) of production and operations management.

If the manufacturing/service processes that manufacture the products or render the services of the business are “better” managed (by this we mean more efficiently and effectively, which is the ultimate goal of production and operations management in any case), this would make it possible to create, maintain and improve the competitive advantages of the production/operations-based performance objectives of higher quality, greater speed, dependability, flexibility and lower costs.

Now that you have a general idea of what production and operations management entails and why it is considered an important specialised discipline within the field of business management, you should realise the potential applications of knowledge of the concepts and methods involved. You will also appreciate the importance of mastering the skills required to apply the available tools and techniques, the possible value of production and operations management and the contribution it can make to successful business enterprises.

Good luck with your studies!

STEPS IN THE STUDY PROCESS

We suggest that you approach your study of the module MNO2601, Production and Operations Management, by following the steps indicated below.

- Step 1 Get an overview of the entire syllabus for MNO2601. You can do this by reading the contents page of this study guide. You will notice that the syllabus is divided into three main parts (parts I, II and III). Each part contains a number of topics (part I – topics 1, 2 and 3; part II – topics 4, 5, 6 and 7; part III – topics 8, 9 and 10).
- Step 2 Get an overview of the contents of each of the three parts of the syllabus. You can do this by reading the title page of each part (these can also be regarded as a “mindmap” for each part) of this study guide (part I, part II and part III). You should note the various topics covered in each part and the reference to the applicable chapter in the prescribed book (except in the case of topic 3, where the study material is contained in the study guide itself). Each title page also contains a further breakdown of the topics into one or more study units (topic 1 – study unit 1; topic 2 – study unit 2; topic 3 – study units 3 and 4; topic 4 – study units 5 and 6; topic 5 – study unit 7; topic 6 – study unit 8; topic 7 – study unit 9; topic 8 – study units 10 and 11; topic 9 – study unit 12; topic 10 – study units 13 and 14). See the diagrammatic representation of the MNO2601 syllabus, which consists of three parts, ten topics, 14 study units and the references to the relevant chapters of the prescribed book in the “MODULE FRAMEWORK: MNO2601” on page xi.
- Step 3 Begin with the first topic in part I by reading its title page, the topic aim and the learning outcomes for this topic. The learning outcomes indicate “what you should be able to do” when you have finished studying the topic. The outcomes will be either to know or understand some portion of the theory or to demonstrate that you are able to apply your knowledge by solving practical problems.
- Step 4 You are now ready to begin your in-depth study of study unit 1. Read the contents of the first study unit under the heading “GETTING AN OVERVIEW”. This will indicate which learning outcomes within a particular topic will be addressed in the specific study unit and what they entail.
- Step 5 Read the contents under the next heading “INTRODUCTION IN THE PRESCRIBED BOOK”, which will refer you to the chapter summary at the end of the applicable chapter in the prescribed book. Having gained a good idea of what you will be expected to know, understand or achieve (ie the learning outcomes), and by referring to the applicable chapter in the prescribed book, you should now be able to continue with your intensive study of study unit 1. The headings, which follow next, incorporate the various portions of the study material that you will need to study and answer questions on. Begin with the first portion of the study material and complete activity 1.1. The first activity of a study unit will always place the theme to be studied in the context of the business environment (except in the case of topic 3, where the “stand-alone” method rather than the “wrap-around” method has been used).

- Step 6 Continue with the first study unit by completing the remainder of the activities (ie 1.2, 1.3, etc) that follow the first activity referred to in step 5. These will focus specifically on one or more sections of your prescribed book. Note that feedback is given at the end of each activity and may contain either detailed explanations or general comments only.
- Step 7 Once you have completed all the activities of the study unit, the next heading you will encounter is "CHECKLIST". This checklist lists what you should be able to do after you finish studying the study unit.
- Step 8 Read the "SUMMARY" at the end of the first study unit and answer all the questions under the heading "ASSESSMENT". After you have completed this step you should have acquired a good knowledge of the contents and mastered the skills necessary for this part of the study material. The "new" knowledge you have gained and the skills you have mastered, coupled with your appreciation of the relative importance of the study material, should add value to your career development. Ask yourself whether this has happened.
- Step 9 Repeat steps 3 to 8 for all topics (and the study units) of part I. Note that the study material for topic 3 is incorporated in this study guide and its layout is therefore somewhat different from that of the other topics.
- Step 10 Repeat step 9 for all topics and study units of part II and part III.

GOOD LUCK WITH YOUR STUDIES!

PRESCRIBED BOOK

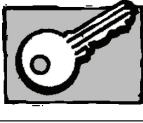
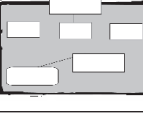
The prescribed book for the second-level module MNO2601 is:

Pycraft, M, Singh, K, Phihlela, K, Slack, N, Chambers, S & Johnston, R. 2010. *Operations management*. 2nd edition. Cape Town: Pearson Education South Africa. ISBN: 978 177025 5630

IMPORTANT PRACTICAL ELEMENTS/ CALCULATIONS WITHIN MNO2601 IN PREPARATION FOR THE EXAM:

- The master production schedule – study unit 12
- Materials requirement planning – study unit 2
- Little law – study unit 5
- Johnson's rule – study unit 10
- Sequencing rules – study unit 10
- Utilisation and efficiency ratios – study unit 11
- EOQ and EBQ formulae – study unit 11

ICONS USED IN THIS STUDY GUIDE AND THEIR MEANING

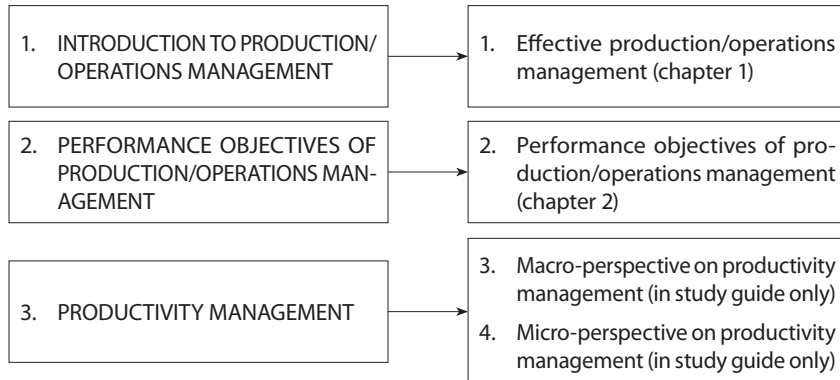
	Aim. This icon broadly indicates the objectives of each of the respective topics.
	Activity. This icon shows the activities that you must do in order to develop a deeper understanding of the learning material.
	Assessment. This icon indicates that you are required to test your knowledge and understanding of the material, and demonstrate the ability to apply your knowledge in a work situation.
	Checklist. After completion of a particular study unit, you should confirm that all learning outcomes were in fact achieved and that you comply with the assessment criteria.
	Feedback. This icon indicates that you will receive feedback on your answers to the self-assessment activities.
	Key concepts. This icon draws your attention to certain keywords or concepts that you will encounter in a topic or study unit.
	Learning outcomes. This icon indicates which aspects of the particular topic or study units you have to master.
	Mind-map. Mind-maps illustrate relationships between various parts of the learning material.
	Read. This icon directs you to read certain sections of the prescribed books for background information.
	Study. This icon indicates which sections of the prescribed book or the study guide you need to study and internalise.



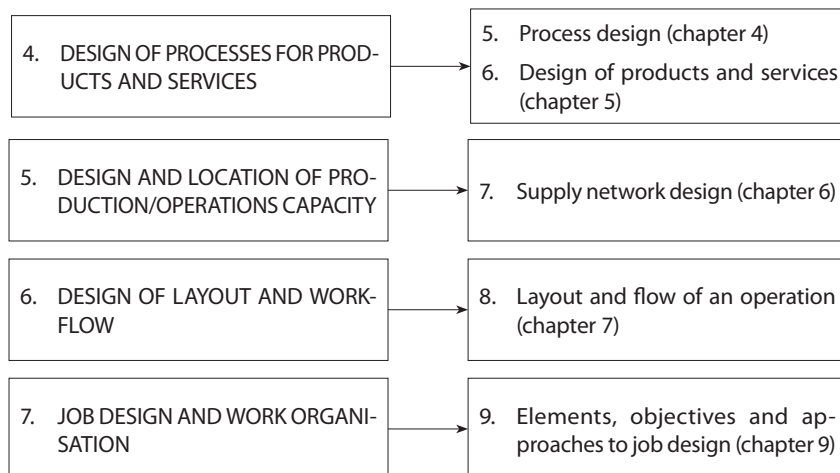
Summary. This section provides a brief summary of what was covered in a particular study unit and what can be expected in the following study unit(s).

MODULE FRAMEWORK: MNO2601

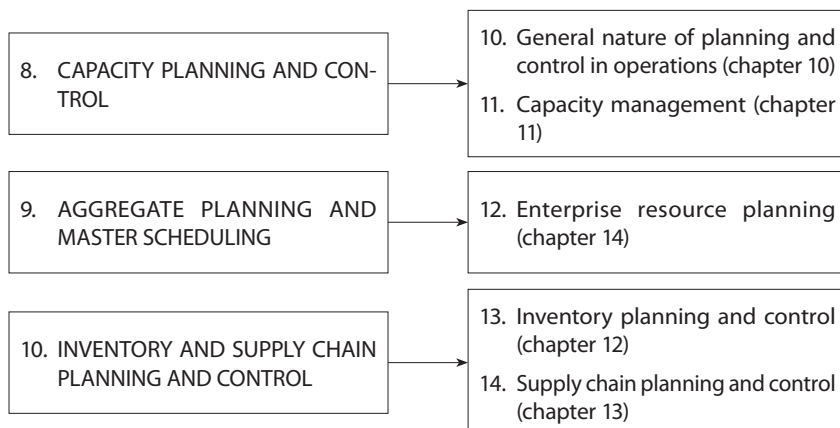
FOUNDATIONS OF PRODUCTION/OPERATIONS AND PRODUCTIVITY MANAGEMENT



DESIGNING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM



PLANNING AND CONTROLLING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM



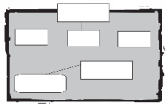
MODULE AIM

To provide learners with a basic knowledge of the role and functioning of production and operations and productivity management, including some of the theoretical concepts and practical skills with regard to selected aspects of the design, planning and control of the production/operations management system.

MODULE OUTCOMES

After making a thorough study of this module, you should:

- understand the basic role of production and operations management (POM) in businesses, organisations or other institutions
- be conversant with the use of selected productivity terminology and be able to explain the basic components of a productivity management model
- understand certain selected aspects involved in the design of the POM system and demonstrate their practical application
- understand certain selected aspects involved in planning and the control of the POM system and demonstrate their practical application



PART I

Foundations of production/operations and productivity management

TOPIC 1

INTRODUCTION TO PRODUCTION/OPERATIONS MANAGEMENT

Topic 1 outlines the overall nature of production/operations management and explains the importance of the interface between the production/operations function and the other business functions.

— Study unit 1: Effective production/operations management (chapter 1)

TOPIC 2

PERFORMANCE OBJECTIVES OF PRODUCTION/OPERATIONS MANAGEMENT

Topic 2 identifies and explains what the performance objectives of production/operations management entail.

— Study unit 2: Performance objectives of production/operations management (chapter 2)

TOPIC 3

PRODUCTIVITY MANAGEMENT

Topic 3 presents both a macro-perspective and a micro-perspective of productivity management and emphasises the importance of productivity improvement.

— Study unit 3: Macro-perspective on productivity management (in study guide only)
— Study unit 4: Micro-perspective on productivity management (in study guide only)

Topic 1

Introduction to production/operations management

AIM



To provide learners with an understanding of what production/operations management entails and why it is important for the business/organisation/other institution.



Learning outcomes

When you have worked through this study unit, you should be able to do the following:

- explain what production/operations management entails
- give reasons why production/operations management focuses on managing processes
- distinguish between different operations processes based on the key dimensions of volume, variety, variation and visibility
- identify and describe the various activities of production/operations management
- illustrate and explain the constituent parts of a general production/operations management model
- use the knowledge, insight and skills you have acquired regarding
 - production and operations management
 - the focus of production/operations management on managing processes
 - distinguishable operations processes based on the key dimensions of volume, variety, variation and visibility
 - the activities of production/operations management
 - the general model of production/operations management
- to justify the relevance of production/operations management and demonstrate this by identifying both correct and incorrect statements with regard to the study material and also practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the introduction to production and operations management by endorsing the theoretical concepts and demonstrating the practical application of the relevant production/operations skills

Study unit 1

Effective production/operations management



Key concepts

production and operations management (POM)
production/operations function
production/operations managers
effective POM
effective operation
core functions of an organisation
support functions of an organisation
broad definition of operations
not-for-profit (nonprofit) organisations
commercial (for profit) organisations
transformation process model
input resources
outputs of goods and services
transformed resources
materials
information
customers/clients
transforming resources
facilities
staff
dominant materials processing
dominant information processing
dominant customer/client processing
tangibility
“pure” product producers
“pure” service producers
facilitating products
facilitating services
operations as service providers
managing processes
supply network
internal supplier
internal customer
hierarchy of operations
macro/micro-operations
external customers/clients and suppliers
operations as a function

operations as an activity
“end-to-end” business processes
business process re-engineering – BPR
operations processes characteristics
volume of output
repeatability and systemisation
variety of output
standardisation
variation in demand for output
predictability in output
visibility of customers/clients
portion of operation exposed to customers/clients
short waiting tolerance
high received variety
customer contact skills
mixed high and low visibility operations
front-office
back-office
four V’s of operations
a typology of operations
activities of operations
management
the operation’s strategic objectives
operations strategy
designing the operation’s products, services and processes
planning and controlling the operation
improving the performance of the operation
broad responsibilities of operations managers
impact on strategic success
reduces costs
increases revenue
reduces investment
basis for future innovation
the “new operations agenda”
input-transformation-output model
operations management activity areas
operations management model

GETTING AN OVERVIEW

In this study unit we address the learning outcomes of topic 1. We explain what production/operations management entails; give reasons why production/operations management focuses on managing processes; distinguish between different operations processes based on the key dimensions of volume, variety, variation and visibility; identify and describe the various activities of production/operations management; and illustrate and explain the constituent parts of a general production/operations management model.

1.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 1 in your prescribed book. Also note the basic composition of the figure illustrated on page 1 of the prescribed book (Part one – Introduction title page). This figure depicts an overall model of production and operations management and should be remembered as it can serve as a mind map for the whole module. Then read the points listed under the following questions at the end of chapter 1:

- (1) "What is operations management?" (2 marks)
- (2) "What are the similarities between all operations?" (5 marks)
- (3) "How are operations different from each other?" (2 marks)
- (4) "What do operations managers do and why is this so important?" (4 marks)

1.2 OPERATIONS IN PRACTICE



Activity 1.1

Read the case example of IKEA in chapter 1 of your prescribed book and then answer the following questions:

- (1) What are the outstanding performance areas in which IKEA excels?
- (2) With specific reference to the definitions of the operations function, operations managers and operations management in chapter 1 of your prescribed book, which outstanding performance do you think originates from the production/operations function?



Feedback on Activity 1.1

- (1) *You could have identified many areas in which IKEA excels, including value for money, affordability and knowledge of its market.*
 - (2) *As you are now studying production/operations management (POM), the second question focuses on this field. Production/operations-based areas that you might have identified include location of the stores (outside the city centres); the design, layout and organisation of IKEA'S stores; and its genuine concern for its shoppers' "peace of mind" as indicated by the provision of a supervised children's play area and other facilities. Also see the discussion at the end of the case study entitled "Operations management is a vital part of IKEA's success".*
-

1.3 EFFECTIVE PRODUCTION/OPERATIONS MANAGEMENT



Activity 1.2

Study the first section of chapter 1 of your prescribed book entitled "What is operations management?" and then answer and/or complete the following:

- (1) The production/operations function is devoted to ...
- (2) Production/operations managers have a particular responsibility for ...
- (3) Production/operations management is the term which is used for ...
- (4) What does production and operations management in the business entail?
- (5) Why do all businesses need production and operations management (POM)?
- (6) What does the term "effective" mean in the context of production/operations management?
- (7) Broadly speaking, what are the various activities that the different functions of organisations carry out in conjunction with the production/operations function?
- (8) Why is it said that the production/operations function occupies a central position in the organisation (or business)? Distinguish between the so-called "core" functions and "support" functions of a business/organisation.
- (9) What does the "broad scope of operations management's responsibilities" imply in terms of the other business/organisation functions? Is production/operations management relevant in smaller businesses/organisations and nonprofit (not-for-profit) organisations?
- (10) The three main components of the transformation model are inputs that represent the ..., the transformation process, by which is meant ... and outputs that are either or or a combination of the two.
- (11) What is the difference between transformed and transforming resources that serve as the inputs to an operation and how would each present itself in various practical examples of operations?
- (12) How do the outputs from operations differ in terms of tangibility and storability? Why is it said that "most operations produce both products and services"? What is the distinction between facilitating services and facilitating products? What does the argument that "services and products are merging" imply?



Feedback on Activity 1.2

- (1) *With reference to questions 1, 2 and 3, please consult the sections of your prescribed book that deal with basic definitions of the following concepts: operations function, operations managers and operations management.*
- (2) *With reference to question 4, production and operations management deals with the specialised field in a business organisation that is primarily responsible for managing the conversion processes (also referred to as the transformation processes) whereby goods (or products) are manufactured and/or services are rendered. Because of the interrelationships between its different functional components, a business must have a reasonable knowledge of the internal workings (activities and tasks) of this discipline. Without this knowledge, it is conceivable that the business will not be able to manufacture products and/or render services that meet the specific operations-based requirements (in terms of low cost, high quality, dependability, flexibility and speed) of its customers and/or clients.*
- (3) *With reference to question 5, all organisations need production and operations management. This is because they all have a function that is specifically responsible for manufacturing products and/or rendering services, which the organisation (be it a profit-making or nonprofit-making organisation) hopes to deliver in its endeavour to satisfy the needs of its customers or clients.*
- (4) *With reference to question 6, being “effective” means that a particular goal (or multiple goals) must be achieved. Then, we may ask, “What is the goal of production and operations management?” Simply stated, the goal is to manufacture goods (products) and/or render services that will satisfy the needs of the customers or clients of the business in such a way that it is able to make money (or more directly stated: show a profit or realise an acceptable return on investment for all its interested stakeholders). Let us focus on the three key issues of this goal. (Note: they are not necessarily presented in the exact order of their perceived importance.)*
 - (a) *It deals with the primary function of a business, that is, it is responsible for managing all the activities and different steps needed to manufacture the product (a chair, a motor vehicle, a fridge, etc) or to actually provide the service (legal advice and representation, postal delivery, a haircut, etc). The essence of this issue is therefore the management of the production/operations function. If it is not managed well (both efficiently and effectively) the result is usually either a partial or total business failure. Failures that typically stem from the production/operations function include*
 - *unreliable operation (many breakages occur or deliveries are unpredictable)*
 - *inflexibility (the business cannot respond to demand changes in product features, volumes or delivery processes)*
 - *poor quality (noncompliance with product specifications or perceived discourteous, inefficient service)*
 - *slow response time (nonavailability of products or long queues)*
 - *high production/operation costs (unrealistically high product prices, apathetic customers/clients and loss of market share) – as affected by all failures previously mentioned*
 - (b) *It implies that the products manufactured or the services rendered must satisfy the specific and/or actual needs of the customers or clients of the business. The essence of this issue is therefore the needs of the customers or clients of the business. If products are made or services are provided that do not meet the requirements of the market, the business will suffer consequences similar to those stated above. From a production/operations management viewpoint, the customers’ or clients’ needs may be expressed in “performance objectives”. These were previously referred to when possible reasons*

for partial or total business failure were mentioned, namely the need for high dependability, flexibility, quality, high speed and low costs.

- (c) This is the essence of a business as we know it in western-style free economies. The goal is to conduct all production/operations activities in such a way that they help the business make money or realise a profit. Obviously there are many stakeholders in a business who have an interest in its long-term growth and survival. A few groups that spring to mind are the owners of the business (or the shareholders, if it is a company), the employees or personnel who offer their labour input, the customers or clients who buy and use its products or services and, in a broader context, the community at large. Note that while each stakeholder group may have somewhat different interests (the owners envisage an acceptable return on their investment, the employees earn a salary and hope to have a stimulating career, the customers and clients want their needs to be satisfied, and the community at large demands that the business is responsible in the way it conducts its activities in terms of the environment, community involvement and employee empowerment, etc), none of these interests could be realised if the business was continually losing money and had to close its doors .*
- (5) With reference to question 7, it is often said that nothing and nobody exists in isolation. This is also true of the production/operations function in a business organisation. Production/operations is responsible for and primarily involved with the production of products and services. Each of the other business functions has a similar responsibility and primary involvement in the business organisation. Stated in very simple terms, marketing researches the needs of the business's customers/clients for products or services, determines pricing policies, decides on the physical distribution or place where it will market its products/services, and is responsible for the promotion (including advertising) of products/services. The financial function is responsible for all financial and accounting activities, including the preparation of annual financial statements and budgets, paying suppliers, creditors and employees, collecting income from sales and debtors, managing cash flow, considering and looking after fixed investments, and generally establishing and maintaining the sound financial position of the business. The personnel or the human resources function is responsible for all aspects directly related to the recruitment, training, appraisal etc of staff or employees. The purchasing function is responsible for the procurement of all materials, consumables and equipment, and the development of reliable supplier relationships. Please note that the list of responsibilities and areas of involvement of the other business functions mentioned here is by no means complete. Not only are there many other areas of involvement, but the exact breakdown of the different functional responsibilities could also differ significantly from business to business. Our concern in this module, however, is the production/operations function.*
- (6) With reference to question 8, while the point of view that the production/operations function occupies a central position in an organisation is rather persuasive (see the discussion in your prescribed book), it is may be more important to acknowledge the interactive nature of the different functions in an organisation. Certain functions with which the production/operations function must interact are seen as fundamental or "core" to the business organisation (eg marketing, new product/service development and operations), while others are seen as supportive (eg accounting and finance, human resources, and IT). It is also not really very important to reach consensus on how to divide the functions of the business organisation exactly in terms of these "core" and "supportive" roles. We should accept that interactions do and must take place. See table 1.1, which shows the activities of the three "core" functions, namely "selling" (marketing), "making" (operations) and "creating for future" (new product/service development) for the operations of an ISP (internet service provider) like Yahoo or Global, a fast chain food outlet like Steers or Nando's, an International aid charity like the Red Cross, and a furniture manufacturer like Montani Leather or Joshua Doore.*

- (7) With reference to question 9, study figure 1.1 in your prescribed book and note the graphic illustration of the “broad scope of operations management’s responsibilities” as well as the indication of the “core” and “support” functions of the business/organisation (see the answer to question 8 above). With regard to operations management in the smaller organisation, note that – theoretically at least – it should be the same. However, as pointed out by the authors of your prescribed book, company size does matter in the end. In the case of smaller companies, resource restrictions may limit the number of dedicated members of staff who can be appointed, and it may be more difficult to separate operational issues from the mass of other issues affecting the company. In larger companies, the more formal structures make it difficult to respond quickly as opportunities or problems present themselves. However, there is greater clarity on the decision-making responsibility, and individuals’ roles are better defined. Take note that operations management is also relevant to organisations whose aim is not necessarily to earn profits while pursuing the primary goal for which they were created. Here we refer to state (or government) departments (like those responsible for public works, transport, defence and policing), institutions (hospitals, schools, fire stations, etc) or other semi-state owned entities (telecommunications [Telkom], postal service [SAPO], national air carrier [SAA], etc). As in any commercial enterprise, similar operational decisions (eg design of products and services, investment in technology, location, managing capacity and delivery) need to be taken, but may even be more difficult to make because of the wide-ranging and conflicting strategic objectives of such not-for-profit organisations, which affect many more people and need to satisfy the needs of an even more diverse group of stakeholders. Read the short case study on Oxfam in this regard and answer the question at the end.
- (8) With reference to question 10, also look at figure 1.2 in your prescribed book, which illustrates the “transformation process model”, and shows how all the various components fit together.
- (9) With reference to question 11, note that an operation’s input resources are usually a mixture of materials, information and customers/clients, although one of these may be more dominant than the others. See table 1.2 in your prescribed book where examples of various types of operations and their dominant type of input- transformed materials are listed. Input resources are further divided into “transformed resources” and “transforming resources”. Transformed resources are resources that are **to be** treated, transformed or converted in the process, as mentioned above, and transforming resources are resources that **actually do** the treating, transforming or converting, such as facilities (buildings, equipment, plant, machines, etc) and staff (people who operate, plan and control the actions).
- (10) With reference to question 12, look at figure 1.3 in your prescribed book to see how various types of operations produce/deliver outputs that differ in tangibility. Products are tangible – you can physically touch them – while services are not, although the results of the service can usually be seen or felt. Products also usually have a longer shelf life than services. Services are offered on a “now only” basis and those not taken or used when on offer cannot be “kept” in an inventory because they expire and fall away. A “pure” goods producer (like crude oil producers) may also deliver a facilitating service like technical advice, while a “pure” service provider (like a management consultancy firm), may actually produce goods (such as a report). One possible implication of the argument that “services and products are merging” is the acceptance of the name “operations management” (OM) instead of “production and operations management” (POM) by more and more academics and business people. However, we agree with the authors of your prescribed book, among others, that too much time and effort have been and are still being wasted trying to select appropriate names, to define boundaries too narrowly, and to list the exact responsibilities of the various functional areas in a business/organisation.
-

1.4 DOMINANT FOCUS OF PRODUCTION/OPERATIONS MANAGEMENT



Activity 1.3

Study the second section of chapter 1 in your prescribed book, entitled “Operations management is about managing processes” and then answer the following questions:

- (1) Why is it said that “operations management is about managing processes”? How can the operations of a wide range of businesses, organisations or institutions such as an airline, department store, police service or frozen food manufacturer be described in terms of their processes?
- (2) What does the “hierarchy of operations” entail and how can the idea of the input-transformation-output model in operations management be used to describe the three levels of operations analysis?
- (3) The authors of your prescribed book argue that “operations management is relevant to all parts of the business”. What does this imply for operations viewed as a function and/or viewed as an activity? How can our understanding of “end-to-end” business processes be used in combination with BPR to improve processes including those in nonoperations functions like marketing and sales, finance and accounting, human resource management and information technology?



Feedback on Activity 1.3

- (1) *With reference to question 1, note that the “mechanisms” needed to treat, to convert, or to transform inputs to outputs are processes. Operations management is therefore all about managing these processes to manufacture, produce and make products, or to render, deliver or provide services in the most effective and efficient manner. An operation is usually made up of a collection of such interconnected processes and subprocesses. Table 1.3 of your prescribed book shows how certain types of operations may be described in terms of their processes.*
- (2) *With reference to question 2, note that the concept of a “hierarchy of operations” refers to the notion that any particular operation is made up of a number of smaller operations or suboperations, which have their own set of inputs that they treat, transform or convert to produce their output in the form of finished goods or services. Look at the example of the business that makes television programmes and videos, as illustrated in figure 1.4 in your prescribed book. This operation can be broken down into the three levels of operations analysis, namely (a) the business level (the whole supply network with the flow of people and resources between operations), (b) the operational level itself (the flow between processes) and finally (c) the process level (the flow between the resources – people and facilities). Also note that the concept of “internal customer/client and supplier” versus “external customer/client and supplier” can be better understood against the backdrop of the breakdown of an operation into its supply network, the operation itself, and the process. However, note the critical comments by the authors of your prescribed book on the internal customer/client concept.*
- (3) *With reference to question 3, one possible implication of the argument that “operations management is relevant to all parts of the business” is that all members of staff or employees of a business/organisation could benefit from some basic knowledge and experience of production/operations management. As the authors of your prescribed book very aptly put it in an older edition of the prescribed textbook (Slack et al 2004:18): “Every manager in*

all parts of an organization is to some extent an operations manager.” Their current view (2010:15) expands on this: “So, operations management is relevant for all functions within the organization and all managers should have something to learn from the principles, concepts, approaches and techniques of operations management.” Given that operations management is relevant to all functions, we can distinguish between operations “as a function” (the part of a business that produces the products and services for external customers/clients) or “as an activity” (the management of any process within any of the businesses functions). Table 1.4 in your prescribed book illustrates some examples of the processes in nonoperations functions. Reorganising seemingly loose or disjointed processes to reflect a single end-to-end business process that contains all the necessary elements of a particular dedicated process from start to finish provides an advantage in terms of focus in meeting the needs of customers/clients. Reorganising or re-engineering of this nature is the philosophy that drives business process re-engineering (BPR). See figure 1.5 for the example of the television and video business that decided to reorganise itself into two end-to-end business units, one for producing television programmes and the other dedicated to making music videos. Compare this with figure 1.4. However, note the greater duplication of functions needed to ensure dedicated end-to-end business units, even though it comes with the advantage of focused customer/client service.

1.5 DIFFERENT CHARACTERISTICS OF PRODUCTION/ OPERATIONS PROCESSES



Activity 1.4

Study the third section of chapter 1 of your prescribed book, entitled “Operations processes have different characteristics”, and then answer the following questions:

- (1) The four important measures that can be used to distinguish between the different types of operation are as follows: the volume of their output, which refers to, the variety of their output, which refers to, the variation in the demand for their output, which refers to, and the degree of visibility to customers/clients of the production/rendering of the output, which refers to
- (2) What are the distinguishing characteristics of the following types of operations: high or low volume, high or low variety, high or low variation in demand, and mixed, high, or low visibility operations for customers/clients?



Feedback on Activity 1.4

- (1) *With reference to question 1, we are of the opinion that this very important aspect of operations processes is probably the least understood by people not schooled in production/ operations management. When matching the product or service to be offered with the most appropriate or natural process for its manufacture or rendering, these people fail to see the catastrophic implications of a possible mismatch; they try to fix something with the wrong tools. Consequently they forget that operations processes can and do differ from one another in respect of these measures of volume, variety, variation and degree of visibility. Finally, note that in the earlier editions of the prescribed book (1st [Pitman 1995] and 2nd [Pitman 1998]), the authors referred to the last measure (how visible the production of the product/service is to customers/clients) as “the degree of customer/*

client contact which is involved in producing the output". The authors used the front-office (high customer/client contact with primarily customer/client processing activities) versus back-office (low customer/client contact with higher materials or information processing activities) analogy to illustrate a difference in types of operation.

- (2) *With reference to question 2, we suggest that you read the short case study on "Two very different hotels" in your prescribed book, which describes two types of hotel namely, Formula 1 and Mwagusi Safari Lodge, and answer the two questions at the end. These are two real-life examples of operations that differ with regard to variety and degree of visibility. Please make sure that you understand the differences and implications between various types of operations processes resulting from changes in the volume, variety, variation and degree of visibility measures. These four dimensions/measures are collectively referred to as the four V's of operations. Make a careful study of figure 1.6 ("A typology of operations") and note both operational implications of the two extreme poles of each dimension. Lastly, study the worked example with figure 1.7, which illustrates the above two types of hotel with their differences in terms of the four V's of operations. The net effect of such operational differences is evident in the level of service (highly individualised with many options, versus standard with little choice), and cost (high prices versus budget prices).*
-

1.6 ACTIVITIES OF PRODUCTION/OPERATIONS MANAGEMENT



Activity 1.5

Study the fourth section of chapter 1 of your prescribed book, entitled "The activities of operations management" and then answer the following questions:

- (1) Production/operations managers generally take responsibility for all the activities in the organisation that contribute to the effective production of goods and/or rendering of services. However, while the exact nature of the responsibilities may vary, depending on the way the business is organised and the boundaries of its various functions are defined, certain general classes of activity apply to all types of operations. These are (a) understanding ..., (b) developing ..., (c) designing ..., (d) planning ..., and (e) improving
- (2) What does the broad operations responsibility entail? Which five responsibilities are currently of particular relevance to production/operations managers, and how should they deal with them?
- (3) How can the operations function contribute to the success of the business/organisation, and what advantages could result from having an effective and efficient operation?
- (4) What does the "new operations agenda" entail? How does the advent of increased globalisation of production/operations networks, new internet-based technologies, the supply chain perspective and the notion of "lean" operations add to the new pressures on the production/operations function?
- (5) Given the four types of advantage that an operation may contribute to a business, it should be obvious that the operation could affect profits. This is illustrated in the fictitious example of two companies described in table 1.6 of your prescribed book. Note the impact that operations management can have on the "general workings" of the companies. In this case, company A is said to be "managing its operations" better than company B. Do you agree? Also consider the example of Kandy Kitchens in your prescribed book, and the three options it is considering to boost its earnings. Which option do you think would be the most beneficial?



Feedback on Activity 1.5

- (1) With reference to question 1, note that production/operations managers are generally responsible for (a) understanding how the operation should help the business to achieve its long-term goals, (b) developing its own operations strategy or set of general principles and guidelines to conduct its activities, (c) designing the products, services and processes of the operation, (d) planning and controlling the activation of the operation itself, and (e) improving the performance of what the operation is producing or delivering.
 - (2) With reference to question 2, first note that while a production/operations manager may be directly responsible for most of the above activities, being indirectly responsible for certain other organisational activities does not mean that he/she has no responsibility for them. Even if it only means that production/operations management must provide information, render assistance, or offer suggestions to improve and benefit the business organisation as a whole, this role (or the various activities involved) should never be neglected. Remember, the organisation is not likely to achieve the third goal of effective production/operations management – to make money (see topic 1, study unit 1) – if individual functions try to pursue their own (or local) optimisation goals to the detriment of overall organisational performance. The broad responsibilities of production/operations managers are increasingly seen as important for the long-term survival and growth of the business and include a concern for the effects of globalisation, the pressure for environmental protection, the relevance of social responsibility, the need for technological awareness, and knowledge management.
 - (3) With reference to question 3, note the discussion in your prescribed book under the subheading “Why is operations management so important?” Fundamental to its importance is the critical role that operations management plays in using the resources for efficient production of goods and services that will effectively satisfy the needs of its customers/clients. Effective and efficient operations management results in the following advantages:
 - (a) reduced costs of production through less waste, reworking, etc
 - (b) increased revenue from increased customer satisfaction with better quality
 - (c) reduced investment in facilities and equipment through better utilisation
 - (d) new knowledge and skills development promote future innovation
 - (4) With reference to question 4, it should be noted that the list of items in the “new operations agenda” in table 1.5 is not necessarily complete or relevant to every type of business/organisation. Different companies in different industries may need to respond to different pressures brought about through changes in the various business/organisational environments. What does apply to all companies is that the business/organisational environments are now more complex and are changing more rapidly than ever before!
 - (5) With reference to question 5, note the discussion in table 1.6 of the two companies, A and B, in your prescribed book under the subheading “How operations can affect profits”. Company A appears to be “better” at following operations management principles in the way it produces and delivers its services to its customers for long-term advantage. With regard to the example of Kandy Kitchens, options 2 and 3, as illustrated in table 1.7, both focus on reducing operating costs and improving customer service, which may more than offset the benefits from merely improving sales volume.
-

1.7 GENERAL MODEL OF PRODUCTION/OPERATIONS MANAGEMENT



Activity 1.6

Study the fifth section of chapter 1 in your prescribed book, entitled “The model of operations management” and then answer the following questions:

- (1) The general model of production/operations management comprises two ideas. They are and
- (2) What are the main components of the general production/operations management model?
- (3) What do the linking arrows and interconnected loops between the various components of the general production/operations model signify?



Feedback on Activity 1.6

- (1) *With reference to question 1, it should be noted that when we combine the input-transformation-output model (fig 1.2 in your prescribed book) and the activities of production/operations management (see the previous section, 1.5 in this study guide), we are in a position to suggest a general production/operations management model that can be broadly applied to any type of organisation. Such a general model is illustrated in figure 1.8 of your prescribed book, and the chapter layout for the remainder of the book is indicated. Note, however, that not all of these chapters in your prescribed book will be covered in this second-year module, MNO2601. However, those not covered (part of ch 2, chs 3, 8, 14, 15, 16, 17, 18, 19, 20 and 21) will be discussed in the third-level follow-on module in production and operations management, namely, MNO3701. When selecting your third-year modules, consider MNO3701, in which we continue our discussion of this most exciting subject field!*
- (2) *With reference to question 2, note that when we view the proposed general model of production/operations management by focusing on its two constituent parts, we would first identify the same components that we saw in the transformation model, namely, inputs-transformation process(es)-outputs, and then, secondly, look at three of the five direct responsibility activities of production/operations managers, namely, design, planning and control, and improvement. These activities are included in the transformation process. In addition, because we accept that the business organisation as a whole does not exist in isolation from its external environment, we need to make provision for the other two of the five direct responsibility activities in the model, namely, developing the operation's strategic objectives and the production/operations strategy. Finally, although it is not actually identified as a specific direct responsibility of production/operations managers, but rather as a result of the improvement activities undertaken and the market's reaction to the operation's products and/or services, we include an additional component, which encapsulates the effect of the two influences mentioned, namely, the operation's competitive role and position.*
- (3) *With reference to question 3, it should be noted that the various linking arrows and interconnected loops in the model of production/operations management signify the flow of information between the constituent parts. For example, the arrows that link the direct responsibility activities of design, planning and control, and improvement, more or less represent the chronological sequence in which these activities would be carried out. The arrows and interconnected loops that link the production/operations strategy and*

the design and improvement activities signify the need for interaction and information flow between these components. Similar observations could be made for all other linking arrows and interconnected loops. Remember, we are visually portraying the components of a system (and the interaction among them), in this case, a production/operations system. You should also see that the two interconnected loops between the top and the bottom of the model represent the subject fields of production/operations management (bottom loop) and production/operations strategy (top loop). Finally, take note of the critical comments made by the authors of your prescribed book on the suggestion that all businesses/organisations have operations processes and produce products and services that are essentially similar. Some scholars may object to this suggestion because as the authors of your prescribed book note, the nature of processes, products and services is lost, distorted or depersonalised and the human element is disregarded if businesses or organisations are seen from this process perspective. We could counter this argument by saying that the management effectiveness and efficiency of the organisation would probably be suboptimal if an organisation were not viewed like this. If managers do not understand the various processes in an organisation or the interaction and interrelationships between processes, they cannot manage (improve, streamline, etc) them. Do you agree?

CHECKLIST



Now that you have completed study unit 1 on effective production/operations management, you should be able to do the following:

- explain what production/operations management entails
- give reasons why production/operations management focuses on managing processes
- distinguish between different operations processes based on the key dimensions of volume, variety, variation, and visibility
- identify and describe the various activities of production/operations management
- illustrate and explain the constituent parts of a general production/operations management model
- use the knowledge, insight and skills you have acquired regarding
 - production and operations management
 - the focus of production/operations management on managing processes
 - distinguishable operational processes based on the key dimensions of volume, variety, variation and visibility
 - the activities of production/operations management
 - the general model of production/operations management
- justify the relevance of production/operations management and demonstrate this by identifying both correct and incorrect alternative statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for effective production and operations management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



You should now have a better understanding of the importance of effective production/operations management in all types of businesses/organisations. If you are still not quite convinced, please re-read the short case studies entitled “Oxfam” (an international aid and emergency relief organisation) and “Two very different hotels” (Formula 1 and Mwagusi Safari Lodge) to get a further glimpse of some operational issues faced by three completely different businesses/organisations/institutions.

ASSESSMENT



- (1) Consider the case study entitled “Greensite High School – Operational leadership and accountability” at the end of chapter 1 of your prescribed book and answer the five questions posed at the end.
- (2) Do problems 1 to 8 at the end of chapter 1 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 1 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to an equivalent local business/organisation or other institution to do your own investigation.

Topic 2

Performance objectives of production/operations management

AIM



To provide learners with an understanding of what the performance objectives of production/operations management entail and why they are important for the business/organisation/other institution.



Learning outcomes

When you have worked through this study unit, you should be able to do the following:

- identify and describe the five performance objectives of production/operations management
- explain how the five performance objectives of production/operations management can contribute to achieving production/operations-based advantages
- use the knowledge, insight and skills you have attained regarding
 - the performance objectives of production/operations management
 - the contribution of the performance objectives to achieving “production/operations-based advantages”
- justify the importance of production/operations management and demonstrate this by identifying both correct and incorrect alternative statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the performance objectives of production and operations management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 2

Performance objectives of production/operations management



Key concepts

production/operations
performance objectives
internal benefits
external benefits
stakeholders
quality objective
speed objective
dependability objective
flexibility objective
product/service flexibility
mix flexibility
volume flexibility

delivery flexibility
mass customisation
cost objective
agility measure of performance
productivity measure of performance
polar representation of performance objectives
productivity improvement
single factor measure of productivity
multifactor productivity
labour productivity
output
input

GETTING AN OVERVIEW

In this study unit we address the learning outcomes of topic 2. We identify and describe the five performance objectives of production/operations management, and explain how the five performance objectives of production/operations management can contribute to achieving “production/operations-based advantages”.

2.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 2 of your prescribed book. Then study figure 2.1 in your prescribed book and specifically note which part(s) of the model of production and operations management is (are) discussed in this particular topic. Finally read the six points listed under the question “What are the performance objectives of operations and what are the internal and external benefits which derive from excelling in each of them?” at the end of chapter 2 of your prescribed book.

2.2 OPERATIONS IN PRACTICE



Activity 2.1

Read the case study on XPS – delivery service in chapter 2 of your prescribed book and answer the following questions:

- (1) In which areas of its business does XPS try to excel?
- (2) How does XPS hope to provide outstanding levels of customer satisfaction?



Feedback on Activity 2.1

- (1) *XPS primarily tries to excel in the following area: it specialises in “on time, every time, door to door” deliveries of documents and packages throughout southern Africa and the rest of the world.*
 - (2) *To provide outstanding levels of customer satisfaction, XPS must be able to fulfil local and international customers’ needs. It achieves this by specialising in “on time, every time” deliveries (speed, dependability and quality), a “range of services to suit every time profile for the delivery of urgent and important packages” (speed and flexibility), “sophisticated computer systems to track the movement of each and every parcel” (dependability and quality). The operational philosophy at XPS is based on a “clear understanding that the cost to the customer of a delay is higher than the cost of movement, irrespective of the premium attached to it”. It would seem that while XPS understands the need to manage costs, it prioritises quality over cost. Were they to think the other way, that is prioritise cost over quality, they could potentially lose clients and longer-term profitability*
-

2.3 PERFORMANCE OBJECTIVES FOR ACHIEVING PRODUCTION/ OPERATIONS-BASED ADVANTAGES



Activity 2.2

Study the second section of chapter 2 of your prescribed book entitled “Operations performance objectives” and then answer and/or complete the following:

- (1) What or who are the stakeholders of the operation? Why are they important?
- (2) The five performance objectives for production/operations management are the following:
 - (a) quality, which means
 - (b) speed, which means
 - (c) dependability, which means
 - (d) flexibility, which means
 - (e) cost, which means
- (3) What effect could the achievement of the five performance objectives have on gaining a production/operations-based advantage for each of the following types of operation: a hospital, an automobile plant, a bus company and a supermarket?

- (4) What do each of the following performance objectives of production/operations management entail and how could their achievement benefit the external and internal customers/clients of the operation: quality, speed, dependability, flexibility and cost?
- (5) Why is a distinction drawn between product/service, mix, volume and delivery flexibility? What does mass customisation entail?
- (6) What are the external and internal effects of the five performance objectives and why would cost affect all other performance objectives?
- (7) What do the “agility” and “productivity” measures of judging the performance of an operation entail? How can single-factor productivity and multifactor productivity be calculated for an operation? Finally, how can one go about improving the productivity of an operation?
- (8) How can a polar representation of the performance objectives help to illustrate the relative importance of each objective (required performance) and to assess the actual performance of the operation?



Feedback on activity 2.2

- (1) *With reference to question 1, stakeholders refer to the people and groups who could be influenced by, or who could influence the activities of the operation. Stakeholders may be either internal (employees) or external (customers, suppliers, shareholders, and community groups or society at large). Each group has a different concern, but collectively these concerns will affect the long-term prospects or outlook for the company. For instance, some may have a direct commercial relationship with the business (suppliers and customers) while others may not (industry regulators). Some stakeholders' interests may actually overlap (eg in not-for-profit charity organisations, stakeholders may be the permanent employees, voluntary workers, aid recipients, community groups, government, etc) and in some cases they may even be in conflict with each other. On a somewhat sobering note we must caution against any one group hijacking the business/organisation by recklessly pushing its own agenda, to the detriment of the business/organisation as a whole. In a way we still believe that “the customer/client is king” – the very reason why the business/organisation exists.*
- (2) *With reference to question 2, note the basic description and explanation of the advantage to be gained by each objective under the heading “The five performance objectives” in chapter 2 of your prescribed book.*
- (3) *With reference to question 3, study the various illustrations for each of the performance objectives with regard to a hospital, a bus company, an automobile plant and a supermarket in chapter 2 of your prescribed book (the quality objective – figure 2.4; the speed objective – figure 2.5; the dependability objective – figure 2.6; the flexibility objective – figure 2.7; the cost objective – figure 2.8).*
- (4) *With reference to question 4, read the discussions under each of the subheadings for the five performance objectives in chapter 2 of your prescribed book. Note specifically that each of the performance objectives inside the operation may lead to the following benefits/advantages: the quality objective – reduces costs and increases dependability; the speed objective – reduces inventories and risks; the dependability objective – saves time and money, and provides stability; the flexibility objective – speeds up response, saves time, and maintains dependability; the cost objective – a universally attractive objective because “every unit of money less to cost is a unit added to profit”. Also study the short case studies for each and answer the questions that follow (quality – “Farm productivity”; speed – “When speed means life or death”; dependability – “Taxi wars”; flexibility – “Flexibility and dependability in the newsroom”; cost – “The Mini–Supermarkets debate” and “Being cheap is our speciality”).*

- (5) With reference to question 5, the performance objective of flexibility for customers/clients means that the operation must be able to change to meet four types of requirement, namely, product/service flexibility (new or modified products/services); mix flexibility (wide range of products/services); volume flexibility (different quantities of products/services over time); and delivery flexibility (timing of delivery of products/services). Mass customisation means both high variety (usually obtained at a higher cost but a lower volume) and high volume of products/services. Products/services are customised for individuals, but mass production brings about high volumes. Is mass customisation possible at Unisa?
- (6) With reference to question 6, it should be noted that while all individual performance objectives will have external and internal beneficial effects if they improve – see figure 2.9 in chapter 2 of your prescribed book – cost is also collectively affected by the improvement of all other performance objectives within the operation. Error-free processes (higher quality), fast throughput (greater speed), reliable operations (higher dependability) and the ability to change quickly (greater flexibility) all have the potential to improve total profitability through lower costs. See the worked example of the internet retailer of speciality cosmetics in chapter 2 of your prescribed book and indicate whether you agree with the analysis.
- (7) With reference to question 7, these are combined measures (measures that are a combination of quality, speed, dependability, flexibility and cost) used to judge the performance of the operation. Agility (under the heading, “The flexibility objective”) is a combination of all five performance objectives, but in particular flexibility and speed. It refers to the ability to respond to market requirements by producing new and existing products fast. Productivity (under the heading, “The cost objective”) is the ratio of output (what is produced) divided by input (what is required to produce it) and reflects the general interest of the operation in keeping the costs as low as possible with the required levels of quality, speed, dependability and flexibility for optimal customer/client satisfaction. A single-factor measure of productivity (eg labour) is a partial measure that may be used to make overall comparisons such as number of cars per employer per year, or number of patients per employee per week, while the multifactor productivity measure is calculated by taking all input factors (labour, materials, capital, etc) into account. See the worked example of the health-check clinic in chapter 2 of your prescribed book and indicate whether you agree with the calculations. Productivity may be improved in many different ways, such as by increasing outputs while maintaining inputs, increasing outputs while reducing inputs, reducing inputs while maintaining outputs, reducing inputs and increasing outputs, increasing outputs more than inputs. In your prescribed book the authors mention the obvious methods of reducing the input costs while maintaining the level of outputs and making better use of inputs, thereby cutting out waste of materials or employee time, or waste in the form of underutilisation of machines.
- (8) With reference to question 8, study figure 2.10 in chapter 2 of your prescribed book. The zero level point for each performance objective is in the centre of the diagram, and the required level of performance is denoted by the distance from the centre. The further a particular performance objective is from the centre, the higher its relative importance.

Note: The red connecting lines denote the required performance level set for the operation, and the blue lines indicate the actual performance at any given time. Both a higher and a lower actual performance should be investigated in an attempt to find out why the performance deviates from the required level. See the worked example of the environmental services department of a city in chapter 2 of your prescribed book and indicate whether you agree with the analysis.

CHECKLIST



Now that you have completed study unit 2 on the performance objectives of production/operations management, you should be able to do the following:

- describe the five performance objectives of production/operations management and explain how they can contribute to achieving production/operations-based advantages
- explain how the five performance objectives of production/operations management can contribute to achieving production/operations-based advantages
- use the knowledge, insight and skills you have gained in the areas of
 - the performance objectives of production/operations management, and
 - the contribution of the performance objectives to achieving "production/operations-based advantages"
- to justify the importance of production/operations management and to demonstrate this by identifying both correct and incorrect alternative statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the performance objectives of production and operations management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



You will have noticed that in this module, topic and study unit we confined our attention to the second section of chapter 2 of the prescribed book.

The other section (section 2.1) will be covered in MNO3701 (our third-year module on production and operations management).

Frequent reference is made throughout the remainder of the study material to the effect that certain aspects of the five performance objectives have on production/ operations management.

ASSESSMENT



- (1) Consider the case study entitled "Operations objectives at the Penang Mutiara" at the end of chapter 2 of your prescribed book and then answer question 3 at the end.
- (2) Do problems 1 to 7 at the end of chapter 2 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 2 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experience of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 3

Productivity management

AIM



To provide learners with an understanding of what the macro-perspective and micro-perspective on productivity entail and why these perspectives are important for South Africa and production/operations management.



Learning outcomes for Study Unit 3:

MACRO-PERSPECTIVE ON PRODUCTIVITY MANAGEMENT

When you have worked through this study unit, you should be able to do the following:

- explain what the macro- productivity perspective entails
- explain the complexities involved in measuring productivity
- describe the various perspectives on productivity held by different parties
- explain why macro-productivity management is important
- describe the various means of measuring macro-productivity
- use the knowledge, insight and skills you have gained regarding
 - South Africa's industrial sectors, their contributions to GDP and year-on-year growth
 - macro-productivity indicators in a reporting context
 - measurement of sectoral productivity
 - trends in sectoral productivity
- describe the world competitiveness report and the factors used to determine world competitive rankings
- describe South Africa's world competitiveness position, including its overall rankings in the four competitive factors
- use the knowledge, insight and skills you have gained in terms of the breakdown of factors, subfactors and variables used by the IMD to determine world competitiveness
- describe South Africa's international position with regard to GDP and overall labour productivity
- describe ways of improving macro-productivity
- demonstrate your understanding of this study unit by identifying both correct and incorrect alternative statements with regard to the study material and also practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your understanding for macro-productivity management by endorsing the theoretical concepts and demonstrating the application of the relevant production/ operations skills in a practical manner.



MICRO-PERSPECTIVE ON PRODUCTIVITY MANAGEMENT

When you have worked through this study unit, you should be able to do the following:

- explain what the micro-productivity perspective entails
- explain and demonstrate with practical measures how productivity should be managed within a business and how it could be improved
- explain why micro-productivity is important for production/operations management
- use the knowledge, insight and skills you have gained regarding
 - the micro-productivity perspective
 - productivity management within a business and its improvement
 - micro-productivity management and the effect on production/operations management and its improvement
- demonstrate your understanding of this study unit by identifying both correct and incorrect alternative statements with regard to the study material and also practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your understanding for micro-productivity management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

Study unit 3

Macro-perspective on productivity management



Key concepts

macro-perspective
macro-productivity
productivity management
input
output
competitiveness

gross domestic product (GDP)
GDP per capita
labour productivity
capital productivity
multifactor productivity

GETTING AN OVERVIEW

In this study unit we will address the learning outcomes of study unit 3. We begin by introducing you to the concept of productivity (section 3.1). We then define macro-productivity and raise some of the complexities it embodies (section 3.2). This is followed by an explanation of the importance of macro-productivity (section 3.3) and a description of five different means of measuring macro-productivity (section 3.4). At this point, you will have sufficient knowledge and understanding of macro-productivity to view its application in a real world context. We achieve this by presenting you with evidence of theory in practice through local reporting on productivity (section 3.5) and international reporting on productivity (section 3.6). We end by describing ways of improving macro-productivity in South Africa (section 3.7) and current challenges and opportunities (section 3.8).

3.1 INTRODUCTION

Productivity is a concept we apply each and every day of our lives. We apply it either consciously or subconsciously. Let's take an example of productivity which applies in the context of a goal that you as a student of MNO2601 have to aspire to. You would like to pass this module, or perhaps perform very well in this module. Suppose your goal is to achieve a final mark of 70%. The achievement of this goal could be classified as an output. What are the inputs you will require to achieve this goal? The inputs would be your time, motivation, study material etc. Could you achieve this goal without the variables of time, motivation and study material? No, you would need a minimum amount of each variable. You would need all the study material (as prescribed in Tutorial Letter 101), time (to read/learn the study material) and motivation (a desire to achieve 70%). By ensuring an adequate level of each of these variables, you will achieve a mark of 70% and therefore achieve your output. But how productive were you in achieving this output? Should you achieve the goal of being awarded 70% with small levels of each input variable, your achievement

could be considered to be highly productive. Should you achieve the goal with high levels of each input variable, your achievement could be considered less productive.

Am I suggesting the former, that you qualify as “highly productive” by using lower levels of each input variable? Productivity differs from person to person. Having had study experience within the tertiary environment, you will have an idea of how you need to manage your input variables and ultimately increase your personal productivity levels.

The aspiration of the author of this particular topic (productivity management) and study unit (macro-perspective on productivity management) is that you grow your perspective on productivity in both your own life and the world around you. Your life and the “world around you” have multitudes of input and output variables. By outlining the principle of productivity and a number of these variables, you will become more acutely aware of the way in which productivity can be managed and improved.

One of the primary responsibilities of a manager is to achieve productive use of an organisation’s resources. The term productivity can be applied here. Stevenson (2009) defines productivity as an index that measures output (goods and services) relative to the input (labour, materials, energy and other resources) used to produce it.

Consider the above definition in the context of our opening example. Given the categories of labour, materials and energy, where would you categorise the input variables of time, study material and motivation? Are there any “other resources” we have not considered?

Stevenson reflects productivity as an equation, saying that it is usually expressed as the ratio of output to input, in which:

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$

A productivity ratio such as the one expressed above can be computed for a number of entities. In terms of our topic (productivity management), productivity in a single operation, department or organisation is of importance. With respect to this specific study unit (macro- perspective on productivity management), the productivity of an entire economy is of importance.

For all of the above entities, productivity has important implications. For non-profit organisations, higher productivity means lower costs; for profit-based organisations, productivity is an important factor in determining how competitive a company is. For a nation, both the rate of productivity and growth in productivity are of great importance (Stevenson 2009).

This study unit focuses on macro-productivity. Productivity from a macro-economic perspective encompasses the whole economy (ie all sectors that make up the economy of a country, such as agriculture, mining, manufacturing, electricity, government, construction, commerce, transport, finance, etc).

Productivity can also be examined from a micro-perspective, where the emphasis falls on the individual firm, business, enterprise etc.

This approach, where productivity is viewed on two different, though interconnected, levels, is the approach advocated by the National Productivity Institute (1996). The National Productivity Institute (NPI) is now known as Productivity SA. Section 3.2 provides more information on the background and mandate of Productivity SA.

The previous paragraphs referred to the interconnection between micro- and macro-productivity. Keep the link between these two concepts in mind. The overall productivity of a country (macro-productivity) is the sum of the productivity of all its individual enterprises (micro-productivity). Micro-productivity will be covered in detail in the next study unit, study unit 4.

3.2 MACRO-PRODUCTIVITY DEFINED



Activity 3.1

Read the discussion below and then answer the following questions:

- (1) Describe the various perspectives on productivity by government, business and labour.
- (2) Describe the dangers of using a purely economic definition of productivity such as the “well researched” definition referenced in the discussion.
- (3) What are the three essential elements we need to consider to fully appreciate productivity and the improvement thereof in a given context?
- (4) Define productivity as advocated by Productivity SA. Do you agree with this definition of macro-productivity? Justify your answer.

DISCUSSION

In a paper by NEDLAC (National Economic Development and Labour Council) and the NPI, NEDLAC (2000) asserts that the word “productivity” is the new buzzword in the workplace, saying it is a word that is used loosely by economists, businesspeople and labour representatives. They state that the aforementioned parties use the word with equal zeal, but simultaneously ascribe different meanings to the word. Consequently, when these parties discuss the issue of productivity (eg productivity and its link to the output of a particular enterprise or sector), they generally move in circles as there is no common agreement on the meaning of productivity or how to achieve it. In this section, we outline:

1. The various perspectives on productivity by different parties
2. The complexities of measuring productivity in view of the many variables one needs to consider

We then go on to offer the formal definition of productivity as advocated by the NPI (2006), providing reasons why we strongly advocate this definition.

NEDLAC (2000) states that government, business and labour have both similar and different interests in relation to the concept of productivity. They have a common interest in increasing productivity in order to promote and generate economic growth. Following

this, however, the authors comment on the way the various parties' interests begin to diverge.

For the state, the debate on unemployment and employment creation is of importance in relation to productivity (NEDLAC 2000). The most unproductive person is an unemployed person. In the context of a developing country with high unemployment figures, the authors state that any accord or agreement on productivity must include mechanisms for job creation.

For business (NEDLAC 2000), profitability is of importance in relation to productivity, in order to interact meaningfully in the global economy. Mechanisms to measure productivity must measure it against global benchmarks. These benchmarks must be linked to the overall economic growth of the country, must accommodate both large and small businesses, and permit the introduction of sector-specific productivity.

For labour, the authors (NEDLAC 2000) state that the word "productivity" provokes a general response that productivity must increase in order to grow the overall economy. At whose expense should this growth in productivity take place? Here labour would cite the statistics on dismissals for operational requirements, and the constant battle with employers to develop a better working environment. A better working environment (in the context of labour) would include higher wages, shorter working hours and more social benefits.

The NEDLAC Paper (2000) goes on to say that there is no single definition of the concept of productivity and that attempts to develop a strict definition of productivity tend to be divisive rather than harmonising. The authors state (2000:5) that the best-researched definition of productivity to have emanated from South Africa, appears to be the one that sees productivity as "the ability to combine and convert inputs of labour, materials and capital into outputs of goods and services". Take note, this is the definition which the author of this study unit referred to in section 3.1.

The above definition is an economic definition, which essentially characterises productivity as something that can be accurately measured statistically by calculating the output and input costs in a particular enterprise or sector. Economically, a higher productivity rate will be achieved by doing more with less, or at least accomplishing more, with constant inputs.

Section 3.5.4 of this study unit reflects productivity increases within selected sectors in South Africa. One cannot afford to look at these statistics from a narrow perspective. An analysis of these statistics could reveal that the increase in productivity is being achieved by a decrease in the use of resources, or a decrease in the cost of inputs, instead of using the same resources to achieve more. In itself, this is not problematic since it indicates more efficient use of resources. However, in a developing country like South Africa with a high rate of unemployment, NEDLAC (2000) cautions us against narrow mindsets. The long-term consequences of further unemployment, but with an increased rate of productivity, would be an increased burden on the state for the provision of minimum social services. High levels of unemployment together with a vastly unequal distribution of the benefits of economic productivity inevitably create an unstable and volatile environment, which is not conducive to further foreign investment. The overall effectiveness of an enterprise,

sector and in fact the country as a whole must be considered when analysing productivity and determining the consequences of increasing it.

Because the purely economic definition was considered inadequate, a multifaceted definition of productivity was framed by NEDLAC (2000). The definition (NEDLAC 2000:5) states that productivity is “creating wealth through generation and application of knowledge to make available products and services that meet the needs of the user and are consistent with social, environmental, and economic goals of the society”.

NEDLAC (2000) cautions us that in a given context, to appreciate productivity fully, one needs to look at three essential elements:

1. The nature of the applicable production system (whether it is international, national, regional, an enterprise or an individual) and how it can be appropriately described. .
2. An understanding of the desired outcome of the process. The fact that productivity needs to be measured as a ratio of output over input strongly suggests that it is equally important to understand the desired outcome of the process in order to measure the right outputs.
3. The different ways in which productivity can be improved. The guidelines outlined below could help to ensure the active participation of all contributors and beneficiaries in the achievement of growth in productivity.

If we accept that productivity is measured as the quotient of output divided by input, it follows (NEDLAC 2000) that productivity may be improved in five different ways:

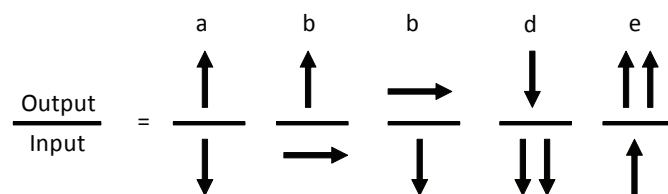


FIGURE 3.1: Five different ways of improving productivity

Productivity SA (2007:10) explains these five ways of increasing the productivity ratio:

- a. **Increase output by using fewer inputs.** This method usually produces the largest productivity gains but may not be the ideal way to realise productivity growth in countries with unemployment problems, as job losses could follow.
- b. **Increase output by using the same quantity of input.** This method could *inter alia* imply no change in employment, and may not be suitable for enterprises with a bloated labour force, or could impede the introduction of new technology.
- c. **Maintain the same output with fewer inputs.** Overall, inputs can be reduced, without reducing employment, by focusing on the other inputs such as raw materials or capital services. This method excludes an increase in output and most likely also employment growth and, therefore does not address unemployment.
- d. **Produce a smaller output with even fewer inputs.** From a macro point of view, this option is the least desirable in developing countries, which need labour absorption capacity. The reduction of products and services, as well as resources needed to

produce these, could likely aggravate unemployment. In mature economies with labour shortages, this option could help to maintain or enhance competitiveness.

- e. **Produce a larger output with more input.** In this scenario, the marginal increase in output is greater than the marginal increase in input, resulting in a larger number, which is, therefore, an increase in productivity. This is the ideal way to improve productivity in developing or transitional economies, because it integrates economic growth, employment creation and higher productivity, on condition that the growth in output can be maintained at a faster rate than the growth in inputs. The proceeds from these productivity gains need to be equitably shared by the employees in the form of increased wages and/or better conditions of employment accompanied by appropriate training and improvements in technology. These conditions imply well-established R&D facilities, sound policies and good governance, if the gains are to be sustainable over the long term.

The purpose of this section was to increase your understanding of how to measure productivity by outlining all the factors one needs to consider when measuring productivity. Improved productivity is influenced by a complex set of variables and must therefore be understood and measured contextually.

We conclude this section by outlining the definition of macro productivity as formally advocated by Productivity SA. However, before, we do that, it is important that you understand the reasons why we have decided to “buy in” to their definition.

1. Productivity SA is the official institution in South African that has been mandated by government, labour and business to enhance productivity in South Africa.
2. We believe that the Productivity SA definition (and the key elements that follow the definition) represent the broader collection of factors that need to be considered when measuring productivity.

Productivity SA (2010:3) “supports the government’s strategic objective of creating more sustainable and quality jobs through an increasingly productive and competitive economy”. Consider this sentence carefully in the light of option e (the five ways to raise the productivity ratio). Take note of Productivity SA’s holistic approach; we need an increasingly productive and competitive economy (increased output) with more sustainable and better jobs (increased inputs).

PRODUCTIVITY SA DEFINITION

Productivity SA defines productivity as “the efficiency with which inputs of capital, materials and labour are used in the conversion of inputs (resources) into outputs (goods and services) as efficiently and effectively as possible, with the optimum use of human capital and physical resources for the benefit of society, the economy and the environment” (NPI 2006:1).

The following are the key elements of productivity (Productivity SA 2011a):

1. The outcome of productivity is continuous improvement of performance.

2. The improvement must be measurable.
3. Key drivers of productivity are:
 - a. Effectiveness (doing the right things)
 - b. Efficiency (doing things the right way)
 - c. Utilisation (optimum use of human capital and physical resources)
 - d. Elimination of all forms of waste.
4. The beneficiaries of productivity must be:
 - a. the environment
 - b. the economy
 - c. society.



Feedback on Activity 3.1

The discussion which followed Activity 3.1 ought to provide enough information to answer questions on the various perspectives on productivity, the dangers of using a purely economic definition of productivity, the three essential elements to be considered in order to understand productivity fully and the definition of macro productivity as advocated by Productivity SA.

3.3 IMPORTANCE OF MACRO PRODUCTIVITY



Activity 3.2

Read the discussion below and then answer the following questions:

- (1) Why is productivity at a macro-level important?
- (2) How do we know that productivity at a macro-level is important?

DISCUSSION

Let's start by considering the perspective gained from sections 3.1 and 3.2 regarding the importance of macro-productivity. We began by outlining the basic principle of productivity, which is the efficiency with which we use inputs to create outputs. An understanding of this basic principle provides us with an intuitive understanding of its value and importance. Extending our understanding into the area of macro-productivity tells us that productivity is not only about the efficiency with which inputs are used to create outputs; there can and should also be an ultimate benefit to the environment, the economy and society.

This broader perspective clearly illustrates why macro-productivity is important. Productivity growth can and should benefit the environment, the economy and society. It is possible, however, that growth in productivity can take place at the expense of the environment, the economy and society. A holistic understanding of productivity can help

us to avoid the latter scenario, focus on the former and ultimately use macro-productivity to benefit the environment, the economy and society.

Productivity SA (2010) explains how empirical studies, which emphasise the importance of productivity in the economy, attempt to highlight the correlation between productivity and certain variables. For example, it would appear that there is a link between productivity growth and economic growth or economic development. Higher productivity growth is associated with higher output growth and higher income per capita. Ultimately, what we require is not only productivity growth but sustained productivity growth. Sustained productivity would boost the value of goods produced, create more jobs, improve the standard of living, and alleviate poverty. How could poverty be alleviated? It could be alleviated through better incomes, which would ultimately lead to a further increase in demand. Note how one factor influences another and how productivity growth ultimately fuels an increase in demand.

Productivity SA (2011b) notes that a comparison of national economies has shown that the more productive the national economy becomes, the higher the personal income of its workers and the lower the rate of inflation in the long run. A productive enterprise or organisation leads to competitiveness, which raises incomes and saving for investment, which lead to growth, job creation and ultimately a decline in the unemployment rate.

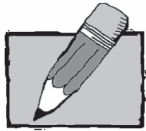
Policymakers, economists, sociologists, business analysts, academics and labour unions have all given considerable attention to macro-productivity growth and with good reason. Arguably, productivity is at the centre of national output growth, bargaining councils' discussions and imperative national objectives (Productivity SA 2009). Consequently, the main challenge for any economy is to create conditions in which companies and employees can function more productively, thereby increasing competitiveness.



Feedback on Activity 3.2

The discussion which followed Activity 3.2 ought to provide enough information to tell us why productivity at a macro-level is important and how we know that it is important. Take note of the link between the two questions. The first paragraph in the discussion tells us that macro-productivity is important because of its ultimate benefit to the environment, the economy and society. The rest of the discussion informs us how empirical studies and national comparisons between economies show that higher productivity growth leads to higher output growth, a higher standard of living and a decline in unemployment. Can you see how the aforementioned aspects relate to both the economy and society?

3.4 MEASURING MACRO-PRODUCTIVITY



Activity 3.3

Read the discussion below and then answer the following questions:

- (1) Why is it important that we measure productivity?
- (2) What is your understanding of GDP and the value-added approach?
- (3) Why can the GDP per capita be regarded as a broad measure of productivity for the country as a whole, and how can it be calculated?
- (4) What is labour productivity, capital productivity and multifactor productivity and how may these performance indicators be used to monitor the productivity performance of the private economy of a country?

DISCUSSION

Productivity SA (2010) emphasises the importance of productivity measurement in cultivating a productive and competitive economy, mentioning that this important aspect cannot be ignored. Some of the important drivers of growth, employment and poverty reduction, as identified by the South African government in its strategic objectives for 2010, include the need for an overall improvement in productivity. This call for improvement goes beyond policy implementation; it also involves monitoring and evaluation of progress made by the relevant sectors on a continuous basis. The ability to measure productivity performance and critically assess the efficiency of the economic sectors of South Africa enables a quantifiable evaluation of progress made and challenges faced, informing the need for specific interventions.

This monitoring and evaluation of progress made by South Africa's sectors is carried out by Productivity SA and released annually in a report entitled *Productivity Statistics*. This report provides detailed assessments of national productivity issues, as reflected in the performance of the various industrial sectors. The 2009 *Productivity Statistics* report contains a descriptive statistical analysis in which the productivity performances of the 11 main economic sectors and 20 selected subsectors of the manufacturing industry in South Africa are examined (Productivity SA 2010).

In section 3.3, we provided a broad definition of macro-productivity as formally advocated by Productivity SA. In the productivity statistics reports, Productivity SA make use of more focused productivity performance indicators. Before we view these performance indicators in an applied real-world context (see section 3.5.4), it is imperative that you understand five basic productivity performance indicators.

3.4.1 Gross domestic product (GDP)

Before we delve into GDP, it is important that you gain an understanding of national output. Moss (2007) expounds on national output and begins by stating it to be the heart of macroeconomics. The total amount of output (goods and services) that a country produces, constitutes its ultimate budget constraint. A country can use more output than it produces only if it borrows the difference from foreigners. Large volumes of output,

not large quantities of money, are what make nations prosperous. Moss defends this statement, saying that national government could print and distribute all the money it wanted, turning all of its residents into millionaires. But collectively, they would be no better off than before unless national output increased as well. Even with all that money, Moss states that they would find themselves worse off if national output declined.

The most widely accepted measure of national output is gross domestic product (GDP). Two important aspects to remember about GDP are, firstly, that GDP represents the market value of final output and, secondly, that GDP represents a flow concept. We expound on these two aspects below:

GDP MEASURES FINAL OUTPUT

Understanding that GDP is the market value of final output helps us understand how GDP is measured. Moss (2007) describes the central challenge in measuring national output (GDP), namely that we need to avoid counting the same output more than once. He makes use of a simple example to illustrate the problem. Imagine that Company A cuts trees in a forest it owns and sell the wood to Company B, a furniture company for R1000. Company B uses the wood to make tables and sells them to Company C, a retailer, for R2500. Company C sells the tables to a consumer for R3000.

In calculating the output, if one added up the sales ($R1000 + R2500 + R3000$), the result (R6500) would overstate the amount of output. A way to avoid the overcounting problem is to focus only on “value added”, which only takes account of “the new output” created at each stage of production. Because company A sold the raw wood for R1000 and had purchased no material input, it added R1000 in value to the economy. Company B added R1500 in value, since it paid R1000 for inputs (from Company A) and sold its output (to Company C) for R2500. Finally, Company C added R500 in value, having purchased R2500 in inputs (from Company B) and sold R3000 worth of final output to the customers. If one sums the value added at each stage ($R1000 + R1500 + R500$), one finds that a total of R3000 worth of output was created.

Another and, far simpler, way to avoid overcounting problems is to focus exclusively on final sales, which implicitly account for the output created in all prior stages of production. Since consumers paid Company C, the retailer, R3000 for the final table, we can conclude that R3000 worth of total output was created. Note that this was precisely the same answer we got by using the value-added approach in the previous paragraph.

Although both methods are correct, the second method, known as the expenditure method, has emerged as the standard approach for calculating GDP in most countries. As a result, the standard definition of GDP is “the market value of all final goods and services produced within a country over a given year” (Moss 2007:9).

GDP IS A FLOW CONCEPT

In economics, it is important to distinguish between flow and stock. GDP is a flow concept: the amount of total final output a country produces “per year”. The “per year” is often left unstated, but it is important to remember the annualised basis for GDP measurement. GDP is usually reported quarterly or every three months, but it is reported on an annualised

basis, meaning the applicable reporting body which compiles the GDP figures uses quarterly figures to estimate total output for the whole year (Colander 1998).

Now that we have an understanding of GDP and how it is derived, we have a sufficient basis of knowledge to understand the other relevant productivity performance indicators.

3.4.2 GDP per capita

One broad measure of “productivity” for the country as a whole (against which we can compare South Africa with other countries in both the developed and developing world) is the gross domestic product (GDP) per capita. The GDP per capita is calculated by taking the value of the total output of a country (goods and services, as outlined in the previous section) and dividing it by that country’s total population (ie the “input” which is used to produce the outputs). The “total population” includes all the people of that country, both employed and unemployed.

3.4.3 Labour productivity

Moss (2007) states that in general countries with high labour productivity enjoy higher wages and living conditions than countries with low labour productivity. Productivity SA (2010:14) reflects a simple and often-used macro-measure of labour productivity for an economy:

$$\text{Real GDP per person-hour worked in the economy} = \frac{\text{Real GDP}}{\text{Number of hours worked by all workers}}$$

Labour productivity may be viewed from two distinct perspectives, namely productivity **levels** or productivity **growth**. (Blandy, Dawkins, Gannicott, Kain, Kasper & Krieglner 1985, cited in Productivity SA 2007).

If a report states that “hourly productivity **grew** by 3 percent last year”, this would mean that real GDP (output) divided by the total number of hours worked nationwide (input) was 3% higher at the end of last year than it was at the end of the previous year.

Examples from the IMD report (2005:521–522) illustrate the point further. South African productivity, measured as GDP per employed person in 2004, stood at a **level** of \$17 782, comparing well with countries like Mexico, Poland, Malaysia and Turkey. These levels were far below the \$84 261 in the USA and \$63 758 in Australia. Productivity **growth** provides a different perspective which is also important. South Africa’s productivity growth of 0.5% in 2004 was comparable to that of countries like Scotland and Spain. These growth levels were low in comparison with countries like Venezuela, which registered a growth of 11.7% and China, which registered a growth of 8.1%.

There are many reasons why labour productivity might be higher in one country than another, or why it might grow in a given country from one year to the next. In particular, greater availability of machinery and capital equipment is associated with higher labour

productivity (Moss 2007). Productivity SA (2007:15) associates higher labour productivity with “advances in technology” and “improvements in education and training”.

Another important matter with regard to labour productivity growth is to increase employment at a rate that will reduce unemployment. This means that the annual growth in employment should exceed the number of people who enter the labour market each year, such as school leavers and higher education graduates (South-Western 2005, cited in Productivity SA 2007)

3.4.4 Capital productivity

Similar to labour productivity, capital productivity measures the real value of goods produced (output), but relates this value to the units of capital or capital services (input) required to produce the products or services. Capital productivity at the macro-level is measured by taking the single or aggregated outputs (ie across several products or services) and dividing these outputs by the capital inputs (also at constant prices and/or machine hours). These capital inputs refer not only to the machines used in actual production but also to the buildings in which the machines and products are kept, the land on which the buildings were erected, and other equipment such as transport equipment to deliver products to the markets (Prinsloo & Smith 1996, cited in Productivity SA 2007).

Productivity SA (2007) takes cognisance of the two different ways in which capital inputs are measured. We could consider capacities such as machine-hours used and/or capital outlays at constant prices. The number of machine hours used is a more accurate measure of input than capital input costs at constant values. Take note that in the context of capital input and machine hours, the potentially largest source of increased productivity is “full utilisation of machine capacity” because machines can run 24-hour shifts, interrupted only for essential maintenance. Multiple shifts can therefore boost productivity as well as employment and export growth.

3.4.5 Multifactor productivity

Multifactor Productivity (MFP), sometimes referred to as Total Factor Productivity (TFP), is a more informative measure of productivity changes than labour or capital productivity, because it relates output to a combined index of capital and labour inputs. This combination of inputs is necessary in the light of the fact that few products or services are produced by a machine only, or labour only. More often than not, labour and capital are complementary in the production process and in measuring productivity (Productivity SA 2007).

Multifactor productivity compares the growth of real output (value of all goods and services) with the growth of all inputs, or as many as possible (usually the value of capital and labour, and land if relevant). It is therefore a more holistic approach than the use of the partial productivity indices like labour productivity or capital productivity, where real output is divided by one set of inputs only.

In order to calculate the multifactor productivity indicator for the private economy as a whole, we would take the value of the total real output of all economic sectors (including public corporations like Eskom but excluding general government – thus, GDP minus

general government contribution), and divide it by the total inputs (labour and capital) used to produce such output. If this indicator is calculated annually and plotted on a graph, it is possible to monitor productivity performance for the private economy as a whole.



Feedback on Activity 3.3

The discussion which followed Activity 3.3 ought to provide enough information to answer the questions on why it is important to measure productivity, GDP and the value-added approach, why GDP per capita may be regarded as a broad measure of productivity and how the productivity measurement indicators of labour productivity, capital productivity and multifactor productivity may be used to monitor the performance of the private economy of a country.

3.5 MACRO-PRODUCTIVITY MEASUREMENT IN SOUTH AFRICA (SECTORAL AND NATIONAL)



Activity 3.4

Read the discussion below and then answer the following questions:

- (1) What are the sectors contributing to South Africa's GDP?
- (2) What are the respective sectors' contributions (in percentages) to South Africa's GDP for 2009? Were these figures significantly different for 2007 and 2008?
- (3) Describe the general trend in the growth of the GDP for the various sectors and for South Africa as a whole over 2008 and 2009. How do you explain this trend?
- (4) Describe the general trends in capital, labour and MFP productivity over the period 2000 to 2009 for the selected sectors.
- (5) In the comparative analysis of sectoral MFP productivity, is there a relationship between the MFP growth rate and average annual output growth for the applicable periods?

DISCUSSION

The Productivity Statistics 2009 report provides a comprehensive reference on and analysis of South Africa's productivity and competitiveness performance at national, international, sector and subsector levels (Productivity SA 2010). In this section, in the context of our understanding of macro-productivity and the five productivity performance measurement indicators (section 3.4), we reflect and discuss selected productivity measures from the Productivity Statistics 2009 report. The purpose of this section is fourfold:

1. To increase your knowledge and understanding of South Africa's industrial sectors, and their contributions to GDP and year-on-year growth.
2. To observe and understand the productivity performance indices, as reflected in the *Productivity Statistics 2009* report, in the light of their real-world application.

3. To increase your knowledge and understanding of South Africa's industrial sectors and their respective status with regard to productivity measurement and productivity trends

Regarding points 1 and 3, it should be noted that any specific annual report provides mainly snapshots or period-specific information. Keep in mind that Productivity SA does release a productivity statistics report on an annual basis. One can therefore keep track of trends by monitoring and evaluating past data. Such data can be sourced from newspapers, business magazines, journals and organisations such as Productivity SA and the Swiss-based Institute of Management Development.

Please do not be overwhelmed by the large number of figures and tables in the forthcoming section. Bear points 1, 2 and 3 above in mind. While knowledge and understanding of South Africa's sectors is important (points 1 and 3), an understanding of the productivity performance indices in an applied real-world context is even more important (point 2).

3.5.1 Sectoral structure in South Africa

Productivity SA (2010) outlines the three sectors under which all economic activities in South Africa are grouped. They include the primary, secondary and tertiary sectors. The primary sector consists of:

- Agriculture, forestry and fishing (dominated by the agricultural sector)
- Mining and quarrying (dominated by the mining sector)

The secondary sector consists of:

- Manufacturing (the most dominant)
- Electricity, gas and water
- Construction

The tertiary sector consists of:

- Wholesale, retail trade, catering and accommodation
- Transport, storage and communication
- Finance, real estate and business
- Community, social and personal
- General government services

All the industrial sectors influence overall economic growth to a varying extent and they are therefore all important.

3.5.2 Sectoral contributions to GDP

Table 3.1 indicates the proportionate gross value added (in percentages) to South Africa's total GDP by each sector. This allows us to gain an idea of the varying contributions of each of the sectors and their respective subsectors to overall economic growth for the years 2007 to 2009.

TABLE 3.1: Gross value added (in percentages) from 2007 to 2009

National/Economic Sectors	Sector contribution in percentage 2007	Sector contribution in percentage 2008	Sector contribution in percentage 2009
Primary sector	8	7	7
Agriculture, forestry and fishing	2	2	2
Mining and quarrying	6	5	5
Secondary sector	20	20	18
Manufacturing	15	15	14
Construction	2	2	2
Electricity, gas and water	3	3	3
Tertiary sector	73	73	75
Wholesale, retail trade, catering & accommodation	12	11	11
Transport, storage & communication	8	8	8
Finance, real estate and business	19	19	20
Community, social and personal	17	17	18
General government services	12	12	12
Other	5	5	5
Total	100	100	100

Source: Productivity SA (2010:12)

The pie chart diagram below outlines the respective sectors and their gross value added to overall SA GDP (in percentages) for 2009.

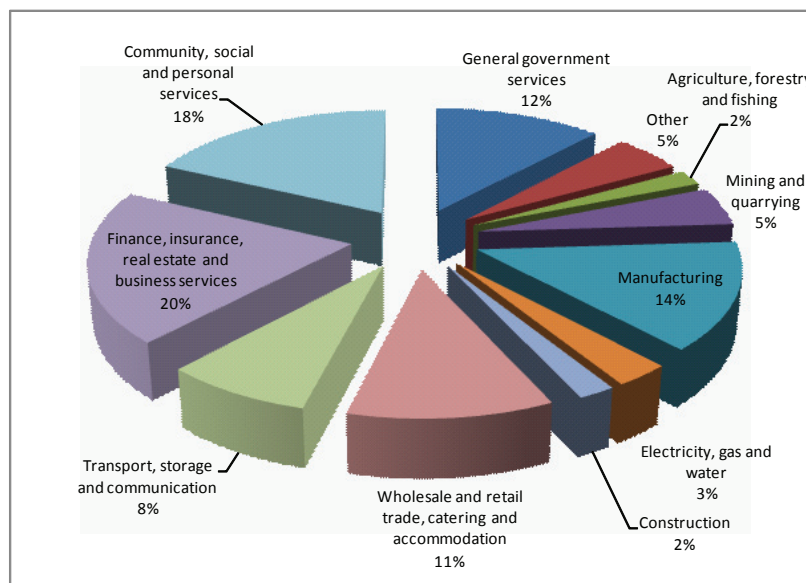


FIGURE 3.2: Contributions per sector (%) to South Africa's total GDP, 2009

Source: Figures extracted from table 3.1 for 2009

A glance at the structure of the economic sectors during the periods 2007, 2008 and 2009 as reflected in table 3.1, reveals only gradual shifts in the percentage gross value added to South Africa's total GDP by the various sectors. For example, the mining and quarrying sector's contribution, which was 6% in 2007, went down to 5% by 2009 as international demand for commodities declined. The manufacturing sector also experienced a drop in

its overall contribution, from 15% in 2007 to 14% in 2009. Overall, however, the percentage contributions during the period 2007 to 2009 remained reasonably constant. Figure 3.3 below reflects how percentage contributions have remained reasonably constant for the three main sectors, with slight changes over the period 2007 to 2009.

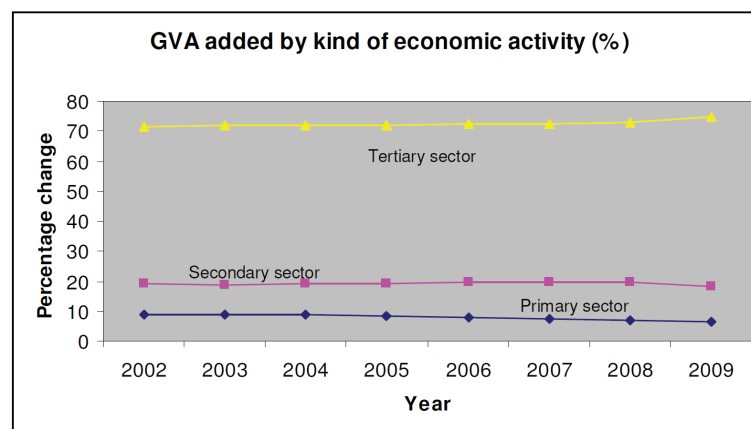


FIGURE 3.3: Gross value added by type of economic activity (%), 2002 to 2009

3.5.3 Sector growth

TABLE 3.2: Evolution of GDP growth over the periods 2008 and 2009

National/Economic Sectors	Production Growth %	
	2008	2009
South Africa	3.7	-1.8
Agriculture, forestry and fishing	10.9	-3.2
Mining and quarrying	-5.4	-7.2
Manufacturing	2.7	-10.7
Construction	9.3	7.8
Electricity, gas and water	1.0	-0.5
Wholesale, retail trade, catering & accommodation	1.1	-2.9
Transport, storage & communication	3.9	0.5
Finance, real estate and business services	7.9	1.3
General government services	4.2	4.2
Personal services	3.6	2.7

Source: Productivity SA (2010:10)

NATIONAL GDP GROWTH

Table 3.2 reflects GDP growth for South Africa and its nine sectors for the periods 2008 and 2009. The annual real GDP growth rate in 2009 declined to -1.8% from the 3.7% recorded in 2008. A quarterly assessment of the performance in 2009 shows that the South African economy bounced back in the third quarter, registering a GDP growth rate of 0.9%, and continued to gather momentum in the fourth quarter, posting a growth rate of 3.2%.

Among comparator countries that made up the “BRIC SM bloc” (Brazil, Russia, India, China, South Africa and Mexico) in 2009, the annual GDP growth of South Africa (-1.8%) was

greater than that of Russia (-7.9%) and Mexico (-6.5%) but less than that of Brazil (-0.2%), India (-2.2%), and China (8.7%). (Productivity Statistics 2010:11)

A comparison with other advanced economies of the G8, reveals that South Africa performed better than the United Kingdom (-4.9%), Germany (-5%), France (-2.2%), Japan (-5%), Italy (-5%), Russia (-7.9%), Canada (-2.6%) and the USA (-2.4%). The trend reveals that the developed economies were more influenced by the recession than the emerging economies in 2009 (Productivity Statistics, 2010:11).

The overall decline in real gross domestic product in South Africa by -1.8% in 2009 was the first contraction since 1992 and though it was “generally disappointing”, Productivity SA (2010:25) states that the economy displayed its resilience during the last two quarters of the year.

SECTORAL GDP GROWTH

As indicated in table 3.2, growth in production declined in the agriculture, forestry and fishing sector from 10.9% in 2008 to -3.2% in 2009. Productivity SA (2010) states that this poor performance can be attributed, among other factors, to field crop production, especially wheat production, which was adversely affected by poor weather conditions. Furthermore, maize production declined.

Other industrial sectors (Productivity SA, 2010) also recorded significant decreases in growth in production. The mining and quarrying sector registered a contraction from -5.4% in 2008 to -7.2% in 2009; the manufacturing sector recorded a decline from 2.7% in 2008 to -10.7% in 2009 and the wholesale, retail trade, catering and accommodation sector registered a decline from 1.1% in 2008 to -2.9% in 2009. These contractions contributed significantly to the annual decline in the overall real GDP growth rate of -1.8% in 2009.

3.5.4 Sectoral trends in productivity, 2000–2009

The section below tracks the evolution of three productivity performance indicators, namely MFP, capital productivity, and labour productivity for selected industrial sectors in South Africa, from 2000 to 2009.

AGRICULTURE, FORESTRY AND FISHING SECTOR

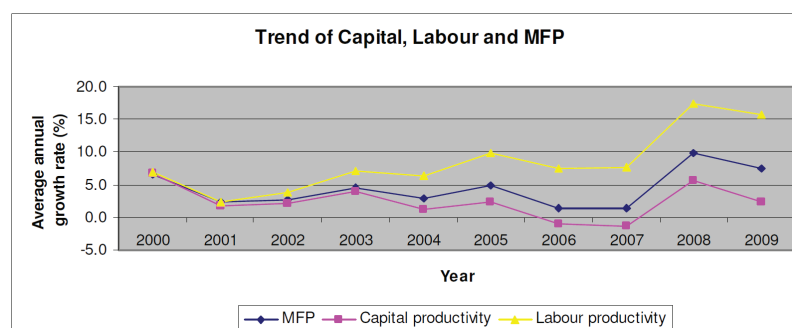


FIGURE 3.4: Trends in capital, labour and MFP in agriculture, forestry and fishing

Source: Productivity SA (2010:40)

It is evident from figure 3.4 that the best performing year for the relevant productivity performance indicators in this sector was 2008, where labour productivity peaked at 17.4%, MFP productivity at 9.9% and capital productivity at 7.5%.

MINING AND QUARRYING SECTOR

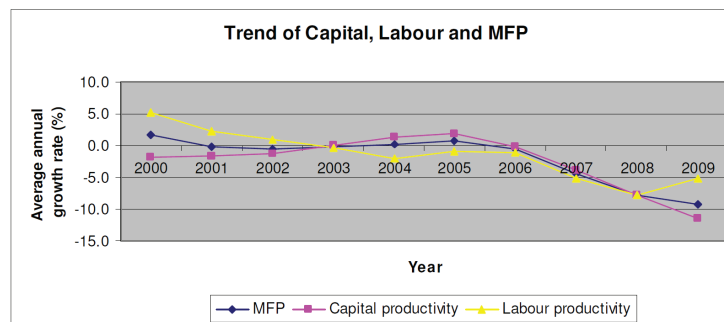


FIGURE 3.5: Trends in capital, labour and MFP in the mining and quarrying sector, 2000–2009

Source: Productivity SA (2010:41)

Figure 3.5 shows a constant decline in all the productivity performance indicators, especially after the year 2005, with labour productivity showing only a slight improvement after 2008 from a growth rate of -7.8% in 2008 to -5.2% in 2009.

MANUFACTURING SECTOR

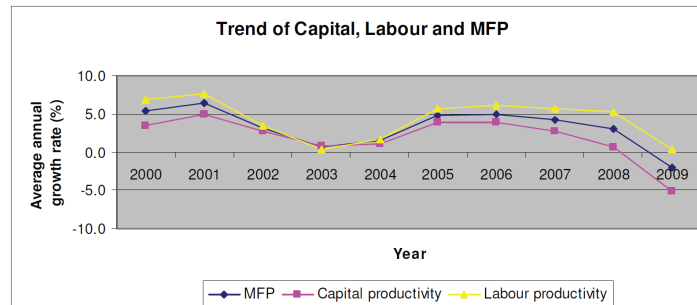


FIGURE 3.6: Trends in capital, labour and MFP in the manufacturing sector, 2000–2009

Source: Productivity SA (2010:42)

An analysis of the productivity performance trends from 2000 to 2009 in figure 3.6 reveals that labour productivity, capital productivity, and MFP seem to co-move, with labour productivity trending higher, followed by MFP and capital productivity. In the years 2002 to 2004, the co-movement appears to be very close, although the growth in labour productivity exceeded that in the other two measures. The best-performing indicator in this sector over this period was definitely labour productivity, followed by multifactor productivity.

ELECTRICITY GAS AND WATER SECTOR

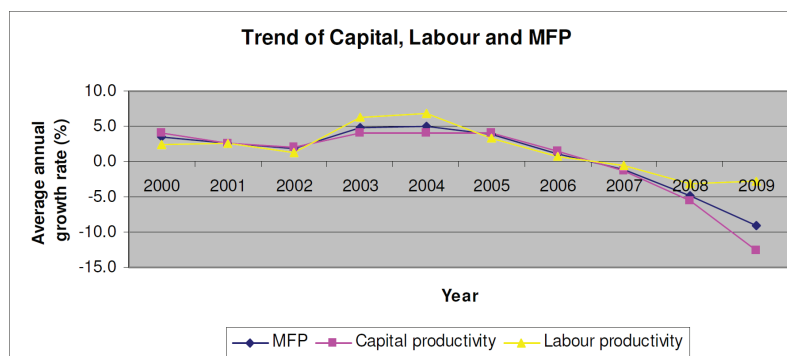


FIGURE 3.7: Trends in capital, labour and MFP – Electricity, Gas and Water, 2000–2009

Source: Productivity SA (2010:43)

The trend in Figure 3.7 illustrates a reasonable co-movement of all the productivity indicators from 2000 to 2007; with labour productivity occasionally trending higher (as seen from 2002 to 2005 and 2007 to 2009). These periods coincided with ‘down turn phases of the South African economy’ (Productivity Statistics 2009:43).

GENERAL GOVERNMENT SERVICES SECTOR

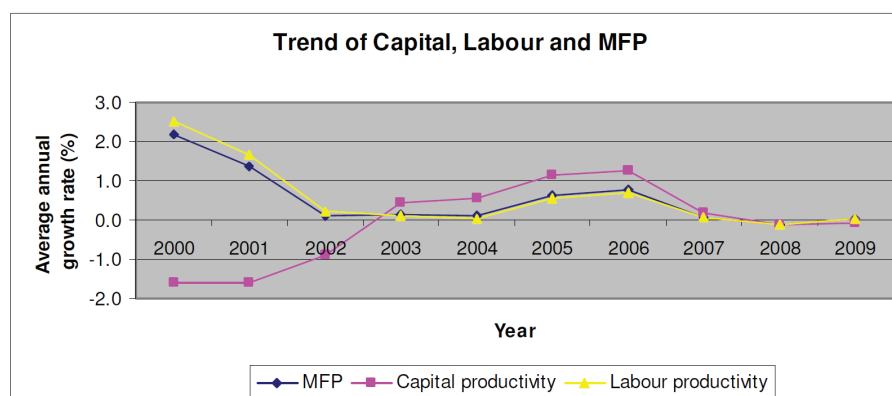


FIGURE 3.8: Trends in labour productivity in the general government services sector, 2000–2009

Source: Productivity SA (2010:46)

Figure 3.8 illustrates how labour productivity and MFP both performed better than capital productivity before 2003. From 2003 to 2007, capital productivity trended higher than the other productivity indicators. From 2007 to 2009, all three productivity indicators co-moved, with MFP and labour productivity trending slightly higher in 2009.

3.5.5 Comparative analysis of sectoral MFP productivity

We conclude this section with table 3.3, which outlines specific performance figures relating to average multifactor productivity growth and average annual output growth for the

periods 1996 to 1999, 2007 to 2009 and 2009. Keep in mind the comments in section 3.4.5, that multifactor productivity is a “more informative measure of productivity changes” than labour or capital productivity because it relates output to a “combined index of capital and labour inputs”. One benefit of seeing the growth in both these measurement indicators in the same table is that it makes it possible to compare the growth of the two separate measurement indicators for the periods under review.

TABLE 3.3: Performance of MFP and output of the main economic sectors

Economic Sectors	Annual average MFP growth rate			Average annual output growth		
	1996-1999	2007-2009	2009	1996-1999	2007-2009	2009
Agriculture, Forestry and Fishing	6.8	7.1	7.6	6.4	3.7	3.8
Mining and Quarrying	4.1	-8.0	-9.2	-0.2	-4.2	-6.3
Manufacturing	1.1	-0.1	-2.0	1.1	-1.0	-4.0
Construction	5.3	4.3	2.1	-0.4	10.5	8.6
Electricity, Gas and Water	2.1	-6.8	-9.1	2.0	1.3	0.3
Services	1.3	1.3	0.8	3.1	3.9	2.9
Wholesale, Retail Trade, Catering & Accommodation	0.2	-1.5	-2.6	3.6	1.2	-0.9
Transport, Storage & Communication	7.4	-0.2	-1.9	6.1	3.6	2.2
Finance, Insurance and Business services	1.2	3.0	3.0	4.7	5.7	4.6
General Government services	0.7	0.1	0.0	0.3	4.2	4.2
Total Private Economy	1.8	0.6	-0.4	2.4	2.6	1.2

Source: Productivity SA (2010:28)

Table 3.3 summarises the growth rates of MFP and output of the private economy for the main economic sectors of South Africa, for the periods 1996 to– 1999, 2007 to 2009 and 2009 respectively. We begin with an analysis of the average annual growth rate of MFP and its relation to the average annual output growth for selected main economic sectors.

As recorded in table 3.3, the private economy (excluding government) registered an annual MFP growth rate of 1.8% (supporting an output level of 2.4%) during the period 1996 to 1999 and a further MFP growth rate of 0.6% (supporting an output level of 2.6%) during the period 2007 to 2009. The decline in the growth of MFP for 2009 (Productivity SA 2010) was attributed to a reduction in the growth rate of labour intake in the private economy as a result of the recession in 2009. The effect of the global recession cannot be underestimated; the private economy was weakened considerably in the first two quarters of 2009, recording the lowest quarterly growth rate in ten years (South African Reserve Bank 2010, cited by Productivity SA 2010). This is evidenced by the deceleration of the MFP growth rate in 2009, with a rate of -0.4% supporting a real output growth of 1.2%.

The agriculture, forestry and fishing sector registered an average annual MFP growth rate of 6.8%, supporting a real output growth of 6.4% during the period 1996 to 1999. For the period 2007 to 2009, MFP registered an average growth rate of 7.1%, contributing to a real output growth of 3.7%. In 2009, the MFP growth rate in this sector was 7.6%,

supporting real output growth of 3.8%. This sector recorded the best MFP growth rates of all the economic sectors in the periods under review.

The construction sector registered an average annual MFP growth rate of 5.3% during the period 1996 to 1999. In the period 2007 to 2009, this sector registered an average MFP growth rate of 4.3%. In 2009, the MFP growth rate in the construction sector dropped to 2.1%, driving a real output growth rate of 8.6%. Productivity SA (2010:29) attributes this “to a drop in the volume of civil construction amid the completion of certain infrastructure projects related to the FIFA World Cup.”

Overall, table 3.3 reflects the negative effect of the global recession in 2009, as most industrial sectors of the local economy registered a deceleration in the growth of productivity. There was a general slowdown in the average annual growth rate of MFP in 2009. Productivity SA (2010) concluded this part of the publication of the 2009 productivity statistics by reiterating the importance of focusing not only on labour or capital productivity but also on how all factors of production have to be managed more efficiently to ensure consistently high MFP growth rates.



Feedback on Activity 3.5

The discussion which followed Activity 3.5 ought to provide enough information to answer the questions relating to South African sectors, their respective contributions to GDP, the consistency of these contributions, GDP, capital, labour and MFP productivity growth trends, and MFP productivity and its relationship to average annual output growth.

3.6 SOUTH AFRICA'S WORLD COMPETITIVENESS POSITION



Activity 3.5

Read the discussion below and then answer the following questions:

- (1) What is the World Competitiveness Report and what are the factors used to determine its rankings?
- (2) How does South Africa fare in the four competitiveness factors and overall in international rankings?
- (3) What is South Africa's international position with regard to GDP and overall labour productivity?

DISCUSSION

3.6.1 The world competitiveness report

We begin section 3.6 by describing the origin of the World Competitiveness Report, after which we provide an outline of the rest of section 3.6. Productivity SA has established itself as the partner institute of the Swiss-based Institute of Management

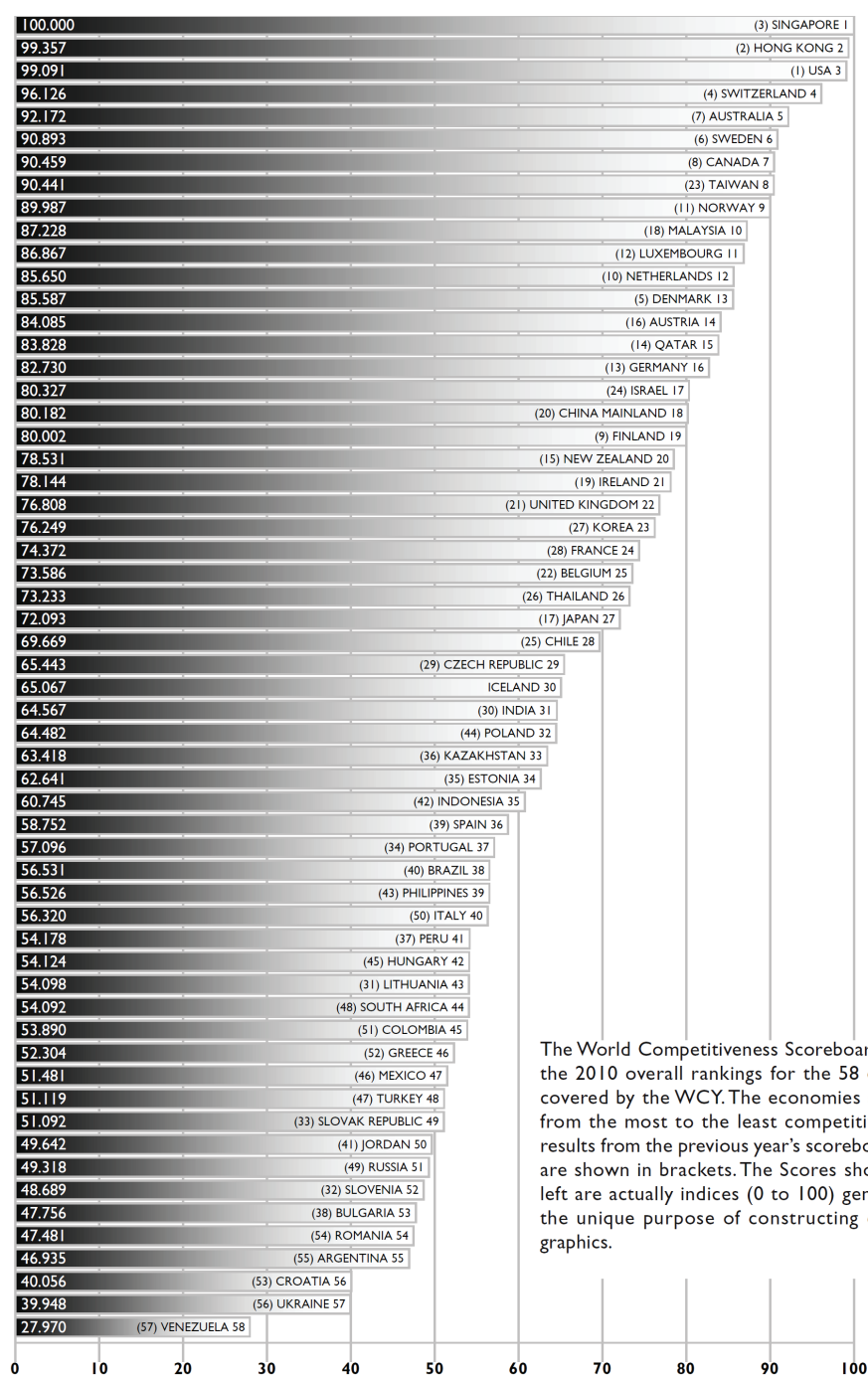
Development (IMD) (Productivity SA 2010). In 1989, the IMD (2009) launched the IMD World Competitiveness Center (WCC), which is dedicated to the advancement of knowledge on world competitiveness. The WCC achieves this by gathering the latest and most relevant data on the subject and by analysing policy consequences. The WCC conducts its mission in cooperation with a network of 54 partner institutes (of which Productivity SA is a partner) worldwide to provide the government, business and academic community with an annual publication called the *IMD World Competitiveness Yearbook (WCY)*. Data reflected in the IMD *WCY* forms the reference point for much of our discussion in this section of the study unit.

The IMD (2009) *WCY* is reputed to be the worldwide reference point on the competitiveness of nations. It analyses and ranks economies on the way they manage the totality of their resources and competencies to increase the prosperity of their populations. It compares the competitiveness of roughly 58 economies on the basis of 327 criteria, grouped into four competitiveness factors: economic performance, government efficiency, business efficiency and infrastructure. Focusing primarily on hard facts taken from international and regional organisations and private institutes, the statistics are complemented with results from an annual executive opinion survey. The collaboration with 54 partner institutes worldwide helps ensure that the data are as reliable and up-to-date as possible.

The following is an outline of what you can expect to find in the remainder of section 3.6. In section 3.6.2 we discuss the IMD World Competitiveness Scoreboard (WCS) for 2010 and give a table outlining the rankings of the roughly 58 economies analysed in respect of the four competitiveness factors for the period 2006 to 2010. The figure in section 3.6.3 provides a breakdown of the four competitive factors and references a listing of each of the 327 criteria in their respective competitive factor categories. Section 3.6.4 contains an in-depth analysis carried out by the IMD on South Africa and its international rankings for 2009. In this section we include an analysis by Productivity SA on South Africa's GDP and labour productivity relative to selected international economies. In section 3.6.5 we outline South Africa's international ranking for 2010 by other reputable organisations measuring international competitiveness.

3.6.2 IMD world competitive scoreboard and competitive factors

The graph below reflects the competitive ranking of each of the 54 economies covered by the IMD



The World Competitiveness Scoreboard presents the 2010 overall rankings for the 58 economies covered by the WCY. The economies are ranked from the most to the least competitive and the results from the previous year's scoreboard (2009) are shown in brackets. The Scores shown to the left are actually indices (0 to 100) generated for the unique purpose of constructing charts and graphics.

FIGURE 3.9: World Competitiveness Scoreboard for 2010

Source: IMD World Competitiveness Yearbook (2010:19)

The competitive ranking, as per figure 3.9, is based on 327 criteria that can be grouped into four competitiveness factors. Table 3.4 below outlines the rankings of the roughly 58 economies (certain economies were not included in certain years) in respect of the four competitiveness factors for the period 2006 to 2010. The table covers two pages.

TABLE 3.4: Overall ranking and competitiveness factors from 2006 to 2010



	OVERALL					Economic Performance				
	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010
Argentina	47	51	52	55	55	30	34	37	29	32
Australia	6	12	7	7	5	13	36	15	15	7
Austria	13	11	14	16	14	27	21	17	18	18
Belgium	26	25	24	22	25	21	20	19	10	13
Brazil	44	49	43	40	38	38	47	41	31	37
Bulgaria	41	41	39	38	53	39	31	38	26	46
Canada	7	10	8	8	7	12	13	11	16	12
Chile	23	26	26	25	28	15	28	43	35	26
China Mainland	18	15	17	20	18	3	2	2	2	3
Colombia	34	38	41	51	45	35	26	46	50	35
Croatia	51	53	49	53	56	48	50	40	43	53
Czech Republic	28	32	28	29	29	23	29	20	25	29
Denmark	5	5	6	5	13	29	18	26	23	30
Estonia	19	22	23	35	34	11	9	23	48	52
Finland	10	17	15	9	19	34	32	36	40	44
France	30	28	25	28	24	16	19	13	17	17
Germany	25	16	16	13	16	20	8	6	6	9
Greece	36	36	42	52	46	43	46	48	52	48
Hong Kong	2	3	3	2	2	5	6	5	3	4
Hungary	35	35	38	45	42	37	38	39	33	40
Iceland					30					51
India	27	27	29	30	31	7	10	18	12	20
Indonesia	52	54	51	42	35	53	55	52	41	27
Ireland	11	14	12	19	21	9	25	24	37	22
Israel	24	21	20	24	17	31	33	27	38	36
Italy	48	42	46	50	40	44	39	45	47	33
Japan	16	24	22	17	27	14	22	29	24	39
Jordan	40	37	34	41	50	52	44	51	53	45
Kazakhstan				36	33				44	43
Korea	32	29	31	27	23	36	49	47	45	21
Lithuania		31	36	31	43		37	28	36	57
Luxembourg	9	4	5	12	11	2	3	4	4	11
Malaysia	22	23	19	18	10	10	12	8	9	8
Mexico	45	47	50	46	47	22	30	33	28	25
Netherlands	15	8	10	10	12	17	5	7	7	14
New Zealand	21	19	18	15	20	28	40	34	30	31
Norway	12	13	11	11	9	18	23	10	19	19
Peru			35	37	41			14	22	28
Philippines	42	45	40	43	39	45	45	42	51	34
Poland	50	52	44	44	32	46	41	31	39	24
Portugal	37	39	37	34	37	42	48	44	42	38
Qatar				14	15				5	2
Romania	49	44	45	54	54	41	35	35	32	47
Russia	46	43	47	49	51	51	52	49	49	49
Singapore	3	2	2	3	1	4	4	3	8	5
Slovak Republic	33	34	30	33	49	47	42	32	34	54
Slovenia	39	40	32	32	52	33	24	25	21	42
South Africa	38	50	53	48	44	40	54	55	56	56
Spain	31	30	33	39	36	32	27	30	46	41
Sweden	14	9	9	6	6	26	17	22	20	15
Switzerland	8	6	4	4	4	24	14	9	13	10
Taiwan	17	18	13	23	8	25	16	21	27	16
Thailand	29	33	27	26	26	19	15	12	14	6
Turkey	43	48	48	47	48	50	53	53	54	50
Ukraine		46	54	56	57		43	50	55	55
United Kingdom	20	20	21	21	22	8	7	16	11	23
USA	1	1	1	1	3	1	1	1	1	1
Venezuela	53	55	55	57	58	49	51	54	57	58

**TABLE 3.4: Overall ranking and competitiveness factors from 2006 to 2010
(continued)**



Government Efficiency					Business Efficiency					Infrastructure					
2006	2007	2008	2009	2010	2006	2007	2008	2009	2010	2006	2007	2008	2009	2010	
49	53	54	57	57	48	51	54	57	52	41	44	47	47	47	Argentina
6	7	5	8	4	8	7	6	7	5	17	16	16	12	18	Australia
15	10	20	24	27	5	11	15	12	12	12	13	13	13	10	Austria
32	28	42	37	43	26	23	23	23	30	13	17	15	15	19	Belgium
51	54	51	52	52	35	40	29	27	24	46	49	50	46	49	Brazil
37	38	29	28	32	42	54	48	47	56	40	41	41	43	48	Bulgaria
9	11	8	9	10	9	12	11	9	10	11	12	8	7	4	Canada
13	16	9	13	14	17	21	22	14	21	38	39	38	36	44	Chile
17	8	12	15	25	27	26	33	37	28	33	28	31	32	31	China Mainland
31	36	35	47	38	29	30	39	40	39	36	46	44	51	53	Colombia
48	50	50	51	55	53	55	51	55	58	43	43	40	41	42	Croatia
29	41	33	31	33	30	36	34	36	40	25	27	24	25	26	Czech Republic
3	4	4	4	11	3	3	5	2	11	3	4	7	6	5	Denmark
11	13	10	22	24	21	24	27	41	36	31	31	26	28	27	Estonia
5	17	13	6	15	10	20	17	5	16	7	11	12	3	6	Finland
42	42	45	46	42	41	42	35	42	35	19	18	11	14	14	France
28	23	26	27	28	28	25	28	19	25	9	7	6	9	8	Germany
40	30	46	53	54	40	35	42	49	45	29	34	35	35	33	Greece
1	2	2	2	1	1	1	1	1	2	14	20	19	19	23	Hong Kong
35	40	47	50	51	33	41	45	52	47	28	25	27	33	35	Hungary
				48					33					9	Iceland
30	33	23	35	30	18	19	20	11	17	47	50	49	57	54	India
45	46	38	33	23	49	45	44	38	34	53	54	53	55	55	Indonesia
7	5	7	12	19	6	8	4	17	18	24	24	23	22	24	Ireland
27	25	21	23	16	24	16	16	20	14	15	14	14	18	16	Israel
52	51	53	54	49	47	47	46	48	48	34	35	33	34	32	Italy
26	34	39	40	37	22	27	24	18	23	2	6	4	5	13	Japan
33	32	25	32	39	39	31	21	44	46	35	36	34	40	52	Jordan
			21	20				34	29				44	39	Kazakhstan
41	31	37	36	26	38	38	36	29	27	22	19	21	20	20	Korea
	26	36	25	34		29	41	35	41		30	32	29	30	Lithuania
16	9	14	16	12	16	5	9	15	6	20	15	18	17	21	Luxembourg
19	21	19	19	9	19	15	14	13	4	27	26	25	26	25	Malaysia
44	44	40	45	46	46	49	55	46	51	51	53	54	50	50	Mexico
18	18	17	14	17	14	10	12	8	15	16	10	9	11	12	Netherlands
12	6	6	7	5	20	18	18	21	22	23	23	22	21	22	New Zealand
10	15	15	11	7	15	14	13	10	8	8	9	10	10	7	Norway
		32	41	35			30	33	42			52	49	57	Peru
39	47	41	42	31	37	39	31	32	32	49	51	48	56	56	Philippines
50	52	49	44	36	52	52	50	50	38	39	40	37	39	36	Poland
36	24	27	29	44	43	44	43	43	50	30	32	28	24	29	Portugal
		5	8					24	19				30	37	Qatar
46	45	48	49	50	50	50	47	56	49	48	42	43	53	43	Romania
38	39	30	39	40	45	48	49	54	53	44	37	45	38	38	Russia
2	1	1	1	2	7	4	2	4	1	5	3	3	8	11	Singapore
22	37	31	34	41	31	28	26	26	43	37	38	36	37	40	Slovak Republic
43	43	43	38	53	44	43	32	39	57	32	33	29	27	34	Slovenia
25	35	28	26	21	32	32	38	30	31	52	55	55	54	51	South Africa
34	29	34	43	45	36	33	40	45	44	26	29	30	31	28	Spain
21	14	11	10	13	12	13	8	6	9	6	5	5	2	2	Sweden
8	3	3	3	3	11	9	7	3	7	4	2	2	4	3	Switzerland
23	20	16	18	6	13	17	10	22	3	18	21	17	23	17	Taiwan
20	27	22	17	18	25	34	25	25	20	42	48	39	42	46	Thailand
47	49	44	48	47	34	37	37	31	37	45	45	42	45	45	Turkey
	48	52	56	56		46	52	53	54	47	46	48	41		Ukraine
24	22	24	30	29	23	22	19	28	26	21	22	20	16	15	United Kingdom
14	19	18	20	22	4	6	3	16	13	1	1	1	1	1	USA
53	55	55	55	58	51	53	53	51	55	50	52	51	52	58	Venezuela

Source: IMD World Competitiveness Yearbook (2010:50–51)

3.6.3 IMD competitive factor breakdown

Figure 3.10 below outlines the four factors, as well as the various subcategories beneath each of the four factors, that contribute to the overall scoreboard as reflected in figure 3.9. Figure 3.10 is in fact a referencing framework for the 327 criteria used by the IMD to

analyse economies. The 327 criteria are listed in Annexure A, which is included at the back of your study guide. Browse through the figure below in conjunction with the 327 criteria listed in Annexure A.

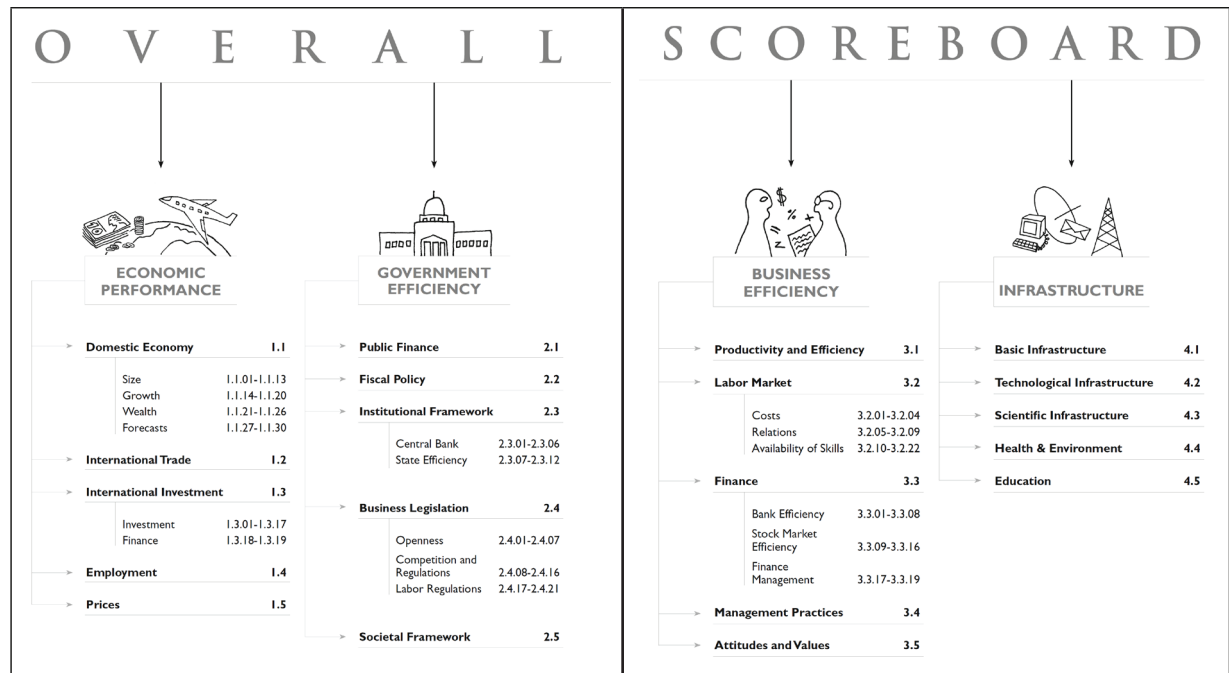


FIGURE 3.10: Factors and subfactors contributing to the overall World Competitiveness Scoreboard

Source: IMD World Competitiveness Yearbook (2009:36–37)

With regard to Annexure A, you are not expected to memorise a list of the 327 factors. It is important, however, that you browse through the variables and consider their significance. We have included the whole list to help you broaden your understanding of the multitude of factors which determine the competitiveness of a country. Reading the 327 factors will give you a more intimate understanding of how the “net result” of figure 3.9 (the WCS for 2010) was achieved. While figure 3.9 and table 3.4 provide snapshots of rankings for particular years, and thus increase our knowledge on international rankings for these periods, what is more important is that we develop an understanding of the theoretical basis that is used to develop these snapshots.

3.6.4 IMD analysis of South Africa

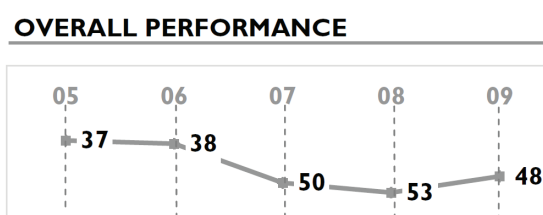
We begin our section on the 2009 IMD analysis of South Africa by reporting on essential information relating to South Africa. You will note that some of the figures in table 3.5 constitute important input variables for the productivity measurement indicators discussed in section 3.4.

TABLE 3.5: Basic facts relating to South Africa for 2009

Basic Facts		
	Value	Year
Population (million)	49.30	2009
Land area (km ² '000)	1,219.1	2008
GDP (\$ billion)	284.1	2009
GDP (PPP) per capita (\$)	9,803	2009
Real GDP growth (%)	-1.8	2009
Consumer price inflation (%)	7.1	2009
Unemployment rate (%)	24.00	2009
Labor force (million)	17.14	2009
Current account balance (%)	-3.98	2009
Direct investment		
Stocks inward (\$ billion)	125.09	2009
Flows inward (% of GDP)	2.00	2009

Source: IMD World Competitiveness Online, 2011

The figures below outline South Africa's overall performance for the period 2005 to 2009.

**FIGURE 3.11: South Africa's WCY competitiveness trends – overall for the period 2005 to 2009**

Source: IMD World Competitiveness Yearbook (2010:244)

Figure 3.11 is supported by table 3.6, which reflects trends in South Africa's competitiveness performance in terms of its overall placing as well as the four competitiveness factors for the period 2005 to 2009.

TABLE 3.6: Trends in South Africa's competitive performance in the four competitive factors and overall – rankings from 2005 to 2009

Categories	2005	2006	2007	2008	2009
Overall performance	38	50	53	48	44
Economic performance	40	54	55	56	56
Government efficiency	25	35	28	26	21
Business efficiency	32	32	38	30	31
Infrastructure	52	55	55	54	51

Source: IMD World Competitiveness Yearbook (2010:245), as cited by Productivity SA (2010:48)

Table 3.6 summarises South Africa's competitive performance in the four competitive factors and overall for the period 2005 to 2009. South Africa's overall competitiveness ranking improved consistently, from 53rd position in 2007 to 48th position in 2008 and 44th position in 2009.

Among comparator countries South Africa performed better than Russia, one of the "BRIC SM bloc" countries and ranked 51st. The results for the other BRIC SM bloc countries in 2009 were as follows: Brazil (37th), India (31st), China (18th) and Mexico (47th) (IMD 2010, cited by Productivity Statistics 2010).

A better way of reflecting the values in table 3.6 is to represent the rankings of each competitive factor graphically. By doing so, we are easily able to monitor the trends in each of the four factors over the period 2005 to 2009. We begin with economic performance. See figure 3.12.

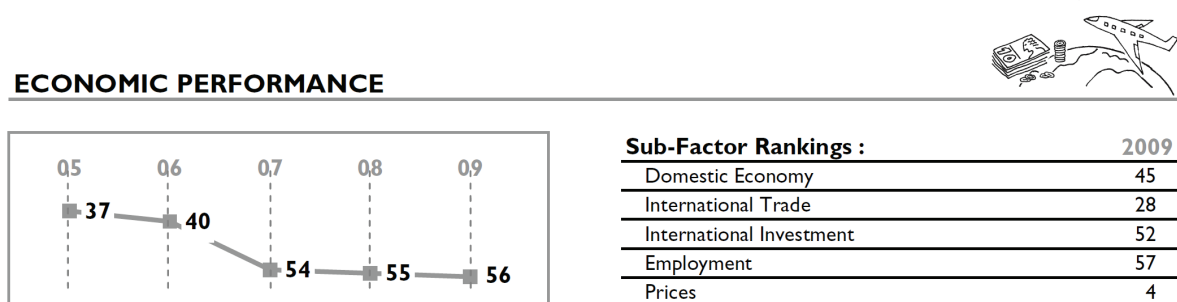


FIGURE 3.12: South Africa's rankings in economic performance for the period 2005 to 2009 and its respective subfactor rankings for 2009

Source: IMD World Competitiveness Yearbook (2010:246)

The table on the right in figure 3.12 outlines the subfactor rankings for economic performance for 2009. Note the overall decline in economic performance over the period 2005 to 2009. Following this, note the subfactor rankings to see which factors pulled South Africa down in 2009. Bear in mind that 57 economies were analysed in 2009. In this context, South Africa fared very poorly on the subfactors of "employment" and "international Investment". To gain a more detailed understanding of these two subfactors, refer to Annexure A, table 1, which outlines the variables that are considered under each of these two subfactors.

Can you see from the above discussion how reflection on the rankings of the factors, subfactors and ultimately the variables under these subfactors not only gives you a "feel" for the multitude of factors affecting an economy, but also an ability to assess these rankings more critically.

To further assist you with your critical assessment of South Africa's overall position regarding economic performance for 2009, table 3.7 identifies the five strongest variables and five weakest variables out of the 76 variables considered under economic performance.

TABLE 3.7: Five strengths and five weaknesses among the 76 variables considered under the competitive factor: economic efficiency

Strengths		
I.2.19	Terms of trade index Unit value of exports over unit value of imports (2000 = 100)	1
I.5.02	Cost-of-living index Index of a basket of goods & services in major cities, including housing (New York City = 100)	1
I.1.20	Resilience of the economy to economic cycles is strong (Survey)	17
I.1.14	Real GDP Growth Percentage change, based on national currency in constant prices	22
I.1.16	Real GDP Growth per capita Percentage change, based on national currency in constant prices	31
Weaknesses		
I.4.06	Unemployment rate Percentage of labor force	57
I.4.02	Employment Percentage of population	56
I.4.08	Youth unemployment Percentage of youth labor force (under the age of 25)	54
I.4.07	Long-term unemployment Percentage of labor force	50
I.1.22	GDP (PPP) per capita Estimates; US\$ per capita at purchasing power parity	48

Source: IMD World Competitiveness Yearbook (2010:246)

Productivity SA (2009) comments on how closely linked an analysis of government and fiscal policy is to an analysis of economic efficiency and how, surprisingly, given the extremely high rating in South Africa's fiscal policy (see fiscal policy in figure 3.13), the economic performance rating of the country saw a continuing downward trend from 37th place in 2005 to 56th place in 2009. Productivity SA makes the further observation that the figures tell a fairly strong story, with South Africa's subfactors "international investment", "domestic economy", "employment" and "international trade" counting heavily against us. It was only the subfactor "price division" which provided relief, and lifted South Africa's ranking. Productivity SA (2009) comments further that for the most part these figures remain unsurprising and are largely indicative of previously identified trends, adding (2009:2) that it is not sufficient for SA to be seen as a "cheap economy" it must also be seen as one that is able to "balance our needs, such as employment, with our offerings such as export manufacture".

We now move to the second of the four factors, namely government efficiency. See figure 3.13 below:

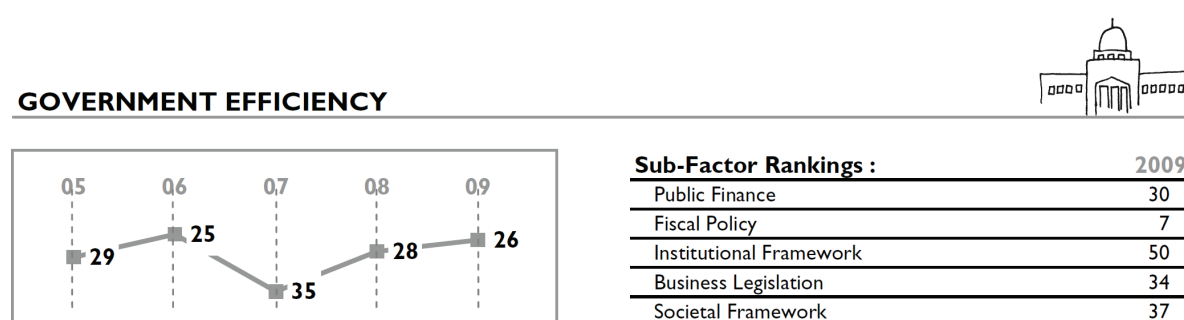


FIGURE 3.13: South Africa's rankings in government efficiency for the period 2005 to 2009 and its respective subfactor rankings for 2009

Source: IMD World Competitiveness Yearbook (2010:246)

Note how the rankings are a little more staggered in this instance. However, for the period 2007 to 2009, there is undoubtedly a positive trend in South Africa's rankings. These rankings are high when one considers that roughly 58 economies were included. One can conclude that government efficiency had a "pull up" effect on South Africa's overall ranking in 2009, when it achieved 48th position. Do you agree with these rankings? Think about this critically and arrive at an answer. First look at the subfactor rankings on the right of figure 3.13. Then, delve deeper by looking at the variables under these subfactors as presented in Annexure A, table 2.

To further assist you in your critical assessment of South Africa's overall position in 2009 in relation to government efficiency, table 3.8 reflects, among the 71 variables considered in relation to government efficiency, the five strongest variables and the five weakest variables.

TABLE 3.8: Five strengths and five weaknesses among the 71 variables considered under the competitive factor: government efficiency

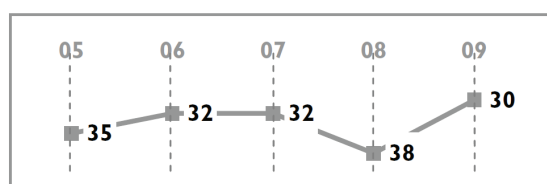
Strengths		
2.2.07	Effective personal income tax rate Percentage of an income equal to GDP per capita	1
2.2.10	Employee's social security contribution rate Compulsory contribution as a percentage of an income equal to GDP per capita	1
2.2.11	Employer's social security contribution rate Compulsory contribution as a percentage of an income equal to GDP per capita	2
2.4.08	Government subsidies To private and public companies as a percentage of GDP	9
2.5.03	Ageing of society is not a burden for economic development. (Survey)	9
Weaknesses		
2.5.08	Equal opportunity legislation in your country encourages economic development. (Survey)	57
2.3.06	Exchange rate stability Parity change from national currency to SDR, 2008 / 2006	53
2.3.02	The cost of capital encourages business development. (Survey)	53
2.5.02	Personal security and private property are adequately protected. (Survey)	52
2.3.04	Country credit rating Rating on a scale of 0-100 assessed by the Institutional Investor Magazine ranking	42

Source: IMD World Competitiveness Yearbook (2010:246)

Productivity SA (2009) states that it is encouraging to note that the SA government was ranked in 26th place in 2009, adding (2009:2) that the IMD ranking system provides a clear indication that, not only is the country in "strong sensible hands from a governmental point of view" but also that the country is being "well managed, with policies in place that are conducive to doing business, and encouraging for outside investors". Productivity SA concludes that although critical issues do need to be addressed, South Africa's "firm fiscal policy has made for a stable emerging economy".

We now move to the third of the four factors, namely business efficiency. See figure 3.14 below:

BUSINESS EFFICIENCY



Sub-Factor Rankings :	2009
Productivity & Efficiency	44
Labor Market	54
Finance	10
Management Practices	18
Attitudes and Values	35

FIGURE 3.14: South Africa's rankings in business efficiency for the period 2005 to 2009 and its respective subfactor rankings for 2009

Source: IMD World Competitiveness Yearbook (2010:247)

The business efficiency rankings appear to be reasonably stable over the five-year period, apart from 2008, when SA dropped to 38th place out of the 55 countries analysed in that year. Business efficiency and government efficiency, can be said to be two competitive factors that helped lift South Africa's overall ranking over the period 2005 to 2009. Take note of the subfactor rankings on the table on the right as well as the business efficiency strengths and weaknesses below.

TABLE 3.9: Five strengths and five weaknesses among the 67 variables considered under the competitive factor: business efficiency

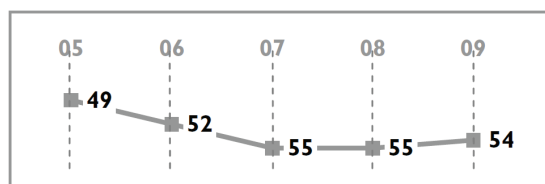
Strengths		
3.3.09	Stock markets provide adequate financing to companies. (Survey)	2
3.3.08	The risk factor in the financial system (new financial instruments, non-performing loans...) is adequately addressed. (Survey)	3
3.3.11	Stock market capitalization Percentage of GDP	4
3.4.08	Social responsibility of business leaders is high (Survey)	4
3.3.14	Stock market index Percentage change on index in national currency	6
Weaknesses		
3.2.19	The brain drain (well-educated and skilled people) does not hinder competitiveness in your economy. (Survey)	57
3.2.16	Skilled labor is readily available. (Survey)	56
3.2.17	Finance skills are readily available. (Survey)	56
3.2.11	Labor force Percentage of population	55
3.2.22	Competent senior managers are readily available. (Survey)	55

Source: IMD World Competitiveness Yearbook (2010:247)

Productivity SA (2009) comments that it is encouraging to note the big upswing in South Africa's business ranking, which saw the country climb from 38th place in 2008 to 30th place in 2009. Regarding the subfactors of "labour market", "productivity & efficiency" and "attitudes & values", Productivity SA states that it is clear where South Africa's key areas for development lie, adding (2009:3) that the "efficient utilization of our labour market, and the close association with financing companies within the country could unlock a vast wealth resource and open new avenues of employment", which "increases the wealth of the country and our ability to draw wealth in from external sources".

We now move to the last competitive factor, infrastructure. See figure 3.15 below:

INFRASTRUCTURE



Sub-Factor Rankings :	2009
Basic Infrastructure	47
Technological Infrastructure	52
Scientific Infrastructure	39
Health and Environment	52
Education	53

FIGURE 3.15: South Africa's rankings in infrastructure for the period 2005 to 2009 and its respective subfactor rankings for 2009

Source: IMD World Competitiveness Yearbook (2010:247)

Infrastructure over the five-year period is definitely a weak point when one considers that in each year (2005 to 2009) roughly 58 countries were analysed. SA appears to fare poorly over all the subfactor rankings of infrastructure too. "Scientific infrastructure" achieved the highest ranking among the subfactor rankings for 2009. Review the variables under "scientific infrastructure" in Annexure A, table 4, and consider why this might be the case. Also take note of the infrastructure strengths and weaknesses below.

TABLE 3.10: Five strengths and five weaknesses among the 67 variables considered under the competitive factor: infrastructure

Strengths		
4.1.25	Electricity costs for industrial clients US\$ per kwh	1
4.5.01	Total public expenditure on education Percentage of GDP	7
4.4.01	Total health expenditure Percentage of GDP	20
4.3.18	Intellectual property rights are adequately enforced. (Survey)	20
4.2.10	Fixed Broadband Tariffs Monthly fee (residential), US\$	22
Weaknesses		
4.4.05	Life expectancy at birth Average estimate	57
4.4.11	Health problems (sicknesses, AIDS, alcohol, drug abuse, etc.) do not have a significant impact on companies. (Survey)	57
4.5.04	Pupil-teacher ratio (secondary education) Ratio of students to teaching staff	56
4.4.10	Human development index Combines economic - social - educational indicators / Source: Human Development Report	55
4.5.03	Pupil-teacher ratio (primary education) Ratio of students to teaching staff	55

Source: IMD World Competitiveness Yearbook (2010:247)

Productivity SA (2009) comments that infrastructure requires urgent attention, arguing (2009:3) that in the context of the "difficult times through which South Africa has recently emerged, as well as damage done to infrastructure through previous neglect", it is encouraging to note how South Africa has "turned a corner", rising from 55th place in 2008, to 54th in 2009. Productivity SA has also stated (2009:3) that the key challenges lie within the "health care sector, with the lack of concise direction in terms of HIV/Aids", as well as "the deficient schooling system". Productivity SA then goes on to provide a possible rationale for these low rankings, stating both of these problems to be "historical problems arising from the integration of two health and education systems at opposite ends of the economic scale".

Against a backdrop of the four competitive factors and their varied rankings, which ultimately determine South Africa's overall ranking, the following graph provides a helpful graphic depiction of all four competitive factors, and their respective subfactor rankings for 2009.

COMPETITIVENESS LANDSCAPE

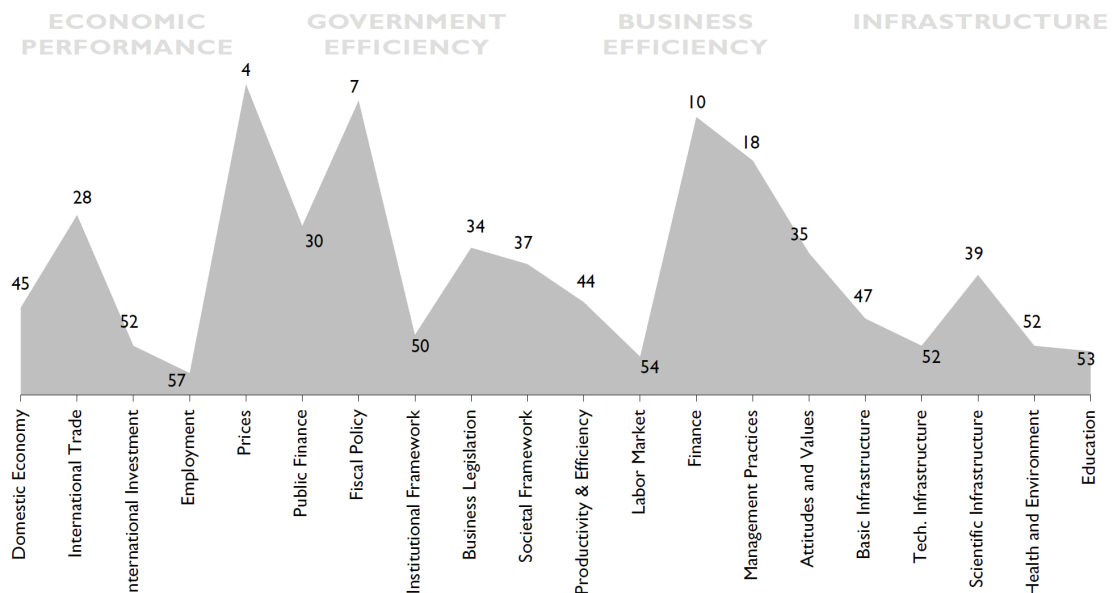


FIGURE 3.16: Competitive landscape of South Africa for the subfactors of the four competitive factors used by the IMD to determine rankings on the WCS

Source: IMD World Competitiveness Yearbook (2010:244)

The numbers in the graph represent the rankings of the various subfactors for 2009. Keep in mind that the figures represent rankings. Therefore, the lower figures represent high rankings, while the higher figures represent low rankings. In the former part of this section, we noted how South Africa's overall ranking was positively influenced by government efficiency and business efficiency and negatively influenced by economic performance and infrastructure. Note how clearly this is evident in this representation of the competitive landscape.

Another valuable report that forms part of the IMD World Competitive Yearbook is a table indicating the 15 areas of greatest improvement in South Africa and the 15 areas where there was the greatest decline in the economy over the period 2008 to 2009 in terms of the 327 variables considered by the IMD. The 15 variables were selected by virtue of their having experienced the highest percentage change from one year to the next (15 inclines and 15 declines).

TABLE 3.11: Fifteen biggest improvements in the SA economy for the period 2008–2009

IMPROVEMENTS		WCY 2008	WCY 2009
4.1.18	Energy infrastructure is adequate and efficient. (Survey)	1.03	2.56
4.1.19	Future energy supply is adequately ensured. (Survey)	1.38	2.68
1.3.07	Direct investment flows inward % of GDP	2.02	3.48
1.2.19	Terms of trade index Unit value of exports over unit value of imports (2000 = 100)	117.1	197.9
1.3.06	Direct investment flows inward US\$ billions	5.74	9.63
4.1.17	Maintenance and development of infrastructure are adequately planned and financed. (Survey)	2.80	4.28
4.5.16	Qualified engineers are available in your labor market. (Survey)	2.12	3.18
3.2.16	Skilled labor is readily available. (Survey)	1.88	2.76
4.3.13	Science in schools is sufficiently emphasized. (Survey)	2.15	3.04
2.4.08	Government subsidies To private and public companies as a % of GDP	0.55	0.33
2.1.07	Central government foreign debt % of GDP	4.13	2.68
2.5.02	Personal security and private property are adequately protected. (Survey)	2.00	2.65
2.3.11	Bureaucracy does not hinder business activity. (Survey)	1.82	2.36
1.1.20	Resilience of the economy to economic cycles is strong (Survey)	3.78	4.84
2.2.11	Employer's social security contribution rate Compulsory contribution	1.37	1.00

Source: IMD World Competitiveness Yearbook (2010:245)

TABLE 3.12: Fifteen largest declines in the SA economy for the period 2008–2009

DECLINES		WCY 2008	WCY 2009
2.1.02	Government budget surplus/deficit % of GDP	0.91	-1.00
1.1.16	Real GDP Growth per capita % change, based on national currency in constant prices	4.09	1.29
1.5.01	Consumer price inflation Average annual rate	7.2	10.6
1.1.14	Real GDP Growth % change, based on national currency in constant prices	5.1	3.1
1.3.01	Direct investment flows abroad US\$ billions	3.73	2.93
1.3.02	Direct investment flows abroad % of GDP	1.32	1.06
2.5.08	Equal opportunity legislation in your country encourages economic development. (Survey)	4.06	3.36
4.5.11	The educational system meets the needs of a competitive economy. (Survey)	2.31	1.96
3.2.20	Foreign high-skilled people are attracted to your country's business environment. (Survey)	4.49	3.96
4.2.20	High-tech exports % of manufactured exports	6.49	5.73
2.1.04	Total general government debt % of GDP	26.18	29.23
3.1.11	Productivity of companies is supported by global strategies (supplies, offshoring, outsourcing). (Survey)	5.26	4.76
1.4.07	Long-term unemployment % of labor force	20.61	22.50
3.3.17	Credit is easily available for businesses. (Survey)	5.60	5.12
2.2.01	Collected total tax revenues % of GDP	25.82	28.00

Source: IMD World Competitiveness Yearbook (2010:245)

What do you think the value is of knowing figures like those presented in tables 3.11 and 3.12 for South Africa (or other countries)? There are a multitude of benefits to tracking the performance of individual variables. An obvious reason for reviewing the fifteen biggest areas of growth and decline respectively would be to view these areas as “raised flags” warranting more serious investigation. We could investigate strong areas of growth to further analyse what factors are stimulating this growth. We could also investigate strong areas of decline, analyse the causes of the decline, deal with the causes and, ultimately, steer the trend towards one of growth. Productivity SA (2009) comments that the strengths

and weaknesses chart gives us good insight into the factors requiring urgent attention and those that should be capitalised upon.

In the conclusion to section 3.6.4, we consider an analysis by Productivity SA (2010) of South Africa's GDP and labour productivity relative to selected international economies. Table 3.13 reflects a number of productivity measures across selected countries. We refer you to the following discussion on table 3.13 in which we consider:

- 1) Overall labour productivity per person employed
- 2) Labour productivity growth (percentage change in GDP, per person employed per hour)

Keep in mind that "overall labour productivity per person employed" (column 4) is calculated by dividing "GDP in US\$ billions" (column 2) by "employment in millions" (column 3).

All the countries below form part of the "BRIC SM bloc" and the G8. Productivity SA (2009) commented on each of these productivity measures and their findings for the respective countries. Their comments have been placed below table 3.13.

TABLE 3.13: Gross domestic product (GDP), employment and overall labour productivity level, 2009

Country	GDP (US\$ billions at purchasing power parity)	Employment (Total employment in millions)	Overall Labour Productivity (Estimates: GDP,(PPP) per person employed, US\$)	Labour Productivity growth (% change of GDP (PPP) per person employed per hour)
Australia	842	10.85	77,634	2.02
Brazil	1,973	84.49	23,396	6.14
Canada	1,267	16.85	75,219	-1.00
China	8,591	779.95	11,015	14.65
Denmark	193	2.77	69,612	2.67
France	2,075	25.71	80,725	-0.01
Germany	2,759	42.02	65,665	-4.43
Hong Kong	298	3.50	85,106	0.05
India	3,598	438.08	8,213	9.05
Indonesia	949	104.87	9,045	-0.24
Ireland	172	1.93	89,008	1.88
Italy	1,777	23.03	77,195	-0.93
Japan	4,139	62.82	65,892	-13.38
Korea	1,347	23.51	57,307	0.20
Malaysia	377	10.68	35,314	2.01
Mexico	1,448	43.68	33,155	-1.74
New Zealand	116	2.18	53,242	-4.65
Norway	276	2.51	109,953	-4.12
Russia	2,080	69.40	29,972	-7.60
Singapore	234	2.99	78,172	-0.45
South Africa	483	12.97	37,251	9.68
Spain	324	4.50	72,032	10.47
Sweden	324	4.50	72,032	-0.55
Taiwan	736	10.28	71,602	9.30
Thailand	532	37.71	14,108	-5.66
United Kingdom	2,071	28.98	71,474	2.86
USA	14,235	139.88	101,770	6.19

Source: IMD World Competitiveness Yearbook (2010:245), as cited by Productivity SA (2010:44)

OVERALL LABOUR PRODUCTIVITY PER PERSON EMPLOYED

South Africa's labour productivity level was compared to that of the block commonly referred to as "BRIC SM" (Brazil, Russia, India, China, South Africa and Mexico). The estimated overall productivity level per person employed in South Africa was 37,251, surpassing that of Brazil (23,396), Russia (29,972), India (8,213), China (11,015) and Mexico (33,155) (Productivity SA 2010).

Comparing South Africa's labour productivity performance to that of member countries of the G8 group reveals that SA performed better than Russia (29,972), but registered lower productivity levels per person than the United Kingdom (71,474), Germany (65,665), France (80,725), Japan (65,892), Italy (77,195), Canada (75,219) and the United States (101,770) (Productivity SA, 2010).

Productivity SA (2009) reminds us that it is important to consider the productivity measure of "overall labour productivity per person employed" in a broader light. Although South Africa surpassed all the BRIC SM countries in their labour productivity per person ratio, the BRIC SM countries all posted higher GDP in billions of US\$ than South Africa. See these figures in the second column of table 3.13. The reason for the lower labour productivity levels per person employed than those of the other BRIC SM countries is that all the BRIC SM countries had significantly and "proportionately" larger labour forces than South Africa did in 2009.

LABOUR PRODUCTIVITY GROWTH (% CHANGE IN GDP, PER PERSON EMPLOYED PER HOUR)

Refer to column 5 of table 3.13. South Africa performed relatively well in this particular measure by posting a growth rate of 9.68%. This figure surpassed that of most of the BRIC SM countries, namely Brazil (6.14%), Russia (7.16%), India (9.05%) and Mexico (-1.74%), but was lower than that of China, which achieved a growth rate of 14.65% (Productivity Statistics 2009).

3.6.5 Other publications on competitiveness

Within section 3.6, in which we cover South Africa's competitive position internationally, we have only reported on research undertaken by the IMD (including Productivity SA's comments on this research). This raises the question why the IMD and their WCR were the selected source for this section. This question is answered in the paragraph that follows.

Another major organisation conducting research into world competitiveness is the World Economic Forum (WEF), which produces an annual publication known as the Global Competitiveness Report (GCR). One of the major differences according to the IMD (2009) between the GCR (WEF) and the WCY (IMD) is the number of criteria that the two reports take into account when determining the rankings. The WCY considers 327 variables, compared with the 116 variables considered in the GCR. The GCR places more emphasis on survey data (66%), whereas the WCY focuses more on hard statistics from international, national and regional organisations. This is one reason why the WEF can cover so many more economies (134) than the IMD (roughly 58). It is nearly impossible to find hard data

for many of the 134 economies covered by the WEF. This raises the issue of rankings and credibility. The IMD (2009:476) prefers “a more objective approach based on hard facts”, rather than an approach “almost entirely based on subjective opinion data that is difficult to manage efficiently and reliably”. Productivity SA (2010) is the official South African partner of the IMD and therefore assists the IMD and endorses its findings.

In table 3.14 below, we present a summary of findings on South Africa’s world competitiveness for 2010, based on data from three different annual publications.

1. The IMD WCR
2. The WEF GCR
3. The Ease of Doing Business report

TABLE 3.14: South Africa’s world competitiveness rankings, strengths and opportunities for improvement for 2010 from three different institutions

International competitiveness reports, 2010	3 main strengths or competitiveness of South Africa	3 main areas of concern or opportunities for improvement in South Africa
IMD WCY (N=58, Ranked 44th)	Fiscal policy-effective personal income tax rate	High unemployment rate - a deterrent to economic development
	Prices-competitive pricing and a low cost of living index in major cities	Brain drain and dwindling of the skilled labour force.
	Finance-low risk factor in the banking system, finance and banking regulation is sufficiently effective.	Business costs of crime and violence impose costs on businesses.
Global Competitiveness report (WEF) (N=133, Ranked 45th)	Accountability of private institutions and goods market efficiency	Labour market efficiency (inflexible hiring and firing practices)
	Strong confidence in South Africa’s financial markets	Low innovative efficiency as university enrolment rate drops
	Quality of institutions and factor allocation such as intellectual property protection	Infrastructure (though good in regional standard, requires upgrading)
The Ease of doing business report (N=183, Ranked 34 th)	Getting credit S.A ranked 2 nd . Measures on credit information sharing and legal rights of borrowers and lenders are evident.	Trading across borders. The costs and procedures involved in importing and exporting a standardised shipment of goods. Starting from the final contractual agreement between the two parties and ending with the delivery of the goods.
	Protecting investors, S.A ranked 10 th . (including transparency of transactions, liability for self dealing, shareholders’ ability to sue officers and directors for misconduct and strength of investors’ protection plan)	Employing workers. The difficulties that employers face in hiring and firing workers are discussed. The difficulty of hiring index and the rigidity of employment index both have high index scores.
	Paying taxes, S.A ranked 23 rd . Frequency of medium companies for paying or withholding taxes within a given year. A good measure of the administrative burden in paying taxes. The number of payments an entrepreneur must make, the number of hours spent preparing, filing and paying.	Registering of property. The ease with which businesses can secure rights to property. All the steps involved, time and cost in registering of property.

Source: Productivity Statistics (2010:49)



Feedback on Activity 3.5

The discussion that followed Activity 3.5 should provide enough information to answer the questions relating to the WCR and the factors used to determine its rankings, South Africa's overall ranking, its ranking in the four competitiveness factors, and its international position with regard to GDP and overall labour productivity.

3.7 IMPROVING MACRO-PRODUCTIVITY IN SOUTH AFRICA



Activity 3.6

Read the discussion below and then answer the following questions:

- (1) What are the suggestions from the NPI (the former Productivity SA) regarding policies for productivity growth?
- (2) What are the twelve pillars used by the WEF in determining its global competitiveness index for measuring a nation's competitiveness?

DISCUSSION

With specific reference to policies for productivity growth, the NPI (1996) provides relevant and timeless information. The NPI (1996:21) notes that "the trendiest issues in South Africa's economic debate" such as inflation, the balance of payments, the budget deficit and gold mine output are mostly short term but from a long term view-point, such issues "pale into insignificance". Therefore, economic policy should focus on the long run, with productivity growth as its central goal. The problem (1996:21) is that productivity growth is linked to long-term trends in society, like savings and investment rates, technological progress, export performance and education levels, which "have their roots in political history, cultures and religions" and these aspects cannot be changed in a hurry. The NPI comments that the government "should consider a range of specific measures to encourage productivity growth". The NPI (1996:21–28) then goes on to offer some valuable suggestions regarding specific policies for productivity growth under the following headings:

1. *A market-friendly social environment.* Such market systems are seen as the most efficient and, generally speaking, the government should only intervene when it is clear that the market mechanisms are failing or the state would be better able to coordinate individual actions. This market-friendly social environment should be combined with stable macro-economic conditions (ie with inflation under control and drastic currency movements minimised), a properly functioning legal and accounting framework (with the widespread crime and violence brought under control), the protection of competition with open and free markets, and "effective social safety nets, to assist the temporary victims of market forces and support people in extreme need".
2. *Investment and savings.* While this is not the only means of stimulating output growth, it is one of the most direct ways of boosting labour productivity by providing the

workforce with more and better machinery and factories for turning out more each hour than was previously produced. Investment and savings can be stimulated in South Africa by the following: government investment in infrastructural programmes, provided these replace and do not merely add to government's current spending and, furthermore, provided the investments themselves are necessary and productive; more competent management of government by significantly cutting back on the current budget deficit and redirecting government spending on basic needs into areas such as education, health and child welfare, water and electricity; an overall reduction in tax levels as these are fairly high when compared to South Africa's developing country rivals. The aim of all these provisos is to encourage foreign investment.

3. *Trade and exports.* Trade stimulates productivity growth and industrial exports enable poorer countries to make full use of their labour resources and encourage the diffusion of technical progress across the country by redirecting the economy towards industrial exports and reducing tariffs, possibly accepting a weaker currency, credit assistance, trade fairs, and so on.
4. *The labour market.* An efficiently functioning labour market boosts growth directly since labour is a major production input. This requires more flexible policies that could raise employment levels and provide incentives to develop "human capital". Wages should be brought into line with the forces of supply and demand; there should be new labour legislation aimed at trying to improve current tense industrial relations; and "millions" of new low-wage formal sector jobs should be created even though this would probably be a politically unpopular move.
5. *Education and skills.* A high proportion of productivity growth stems from advances in knowledge that arise from finding the most appropriate technology and upgrading the quality of labour through education and training so that the education system has the capacity to provide the right quality and quantity of human resources. The need is for vocational training with sufficient depth and breadth to make skills transferable between businesses and to enable workers to learn new tasks as work conditions change. This calls for an integrated, restructured education and training system geared to supplying the human resources necessary to ensure high productivity and competitiveness.
6. *Technology, research and development.* When advances in technological knowledge are applied to processes, they become a vital source of productivity growth. This should be enhanced by developing technologies that are suited to the country's own resource base and focused in directions where South Africa can develop a competitive advantage in export markets through the acquisition of technology from foreign sources. Foreign technology should, however, be adapted to South African conditions.
7. *Small, medium and micro-enterprises (SMMEs).* These will probably be the main source for the creation of new jobs in the economy as neither the public sector nor large firms or businesses in the private sector are expected to contribute significantly to employment growth. SMMEs should be increased and supported by easing their access to finance, favouring physical investment through tax incentives, and promoting partnerships among SMMEs and between SMMEs and large businesses or other institutions.

We mentioned in section 3.6.5 that there are other reputable institutions that produce publications on competitiveness, one of them being the WEF, which publishes the GCR. The WEF (2010) has developed a global competitiveness index (GCI) by means of which a

nation's competitiveness can be measured. The GCI captures this open-ended dimension by providing a weighted average of many different components, each of which reflects one aspect of the complex reality called competitiveness. All these components are grouped together in what the WEF calls the 12 pillars of economic competitiveness. We briefly outline the 12 pillars below, as presented by the WEF (2010:4-8).

Pillar 1: Institutions

The institutional environment forms the framework within which individuals, firms and governments interact to generate income and wealth in the economy. The institutional framework has a strong bearing on competitiveness and growth and plays a central role in the ways in which societies distribute the benefits and bear the costs of development strategies and policies, furthermore, it influences investment decisions and the organisation of production.

Pillar 2: Infrastructure

Extensive and efficient infrastructure is an essential driver of competitiveness. It is critical for ensuring the effective functioning of the economy, as it is an important factor determining the location of economic activity and the kinds of activities or sectors that can develop in a particular economy.

Pillar 3: Macro-economic stability

The stability of the macro-economic environment is important for business, and is therefore important for the overall competitiveness of a country. It is acknowledged that macro-economic disarray harms the economy. For example, firms cannot make informed decisions when inflation is raging out of control.

Pillar 4: Health and primary education

A healthy workforce is vital if a country is to be competitive and productive. Workers who are ill cannot function optimally and will be less productive. Investment in the provision of health services is therefore critical for purely economic reasons, as well as on account of moral considerations.

Pillar 5: Higher education and training

Quality higher education and training is crucial for economies that want to move up the value chain beyond simple production processes and products. In particular, today's globalising economy requires economies to nurture pools of well-educated workers, who are able to adapt rapidly to their changing environment.

Pillar 6: Goods market efficiency

Countries with efficient goods markets are well positioned to produce the right mix of products and services for the applicable supply-and-demand conditions, as well as to ensure that these goods can be most effectively traded in the economy. Healthy market competition, both domestic and foreign, is important in driving market efficiency and

hence business productivity, because it is market competition that ensures that the most efficient firms, producing goods demanded by the market, are those that thrive.

Pillar 7: Labour market efficiency

The efficiency and flexibility of the labour market are critical for ensuring that workers are allocated to jobs in which they can be most efficient in the economy and provided with incentives to give of their best in their jobs. Labour markets must therefore have the flexibility to shift workers from one economic activity to another rapidly, and at low cost, to allow for wage fluctuations without much social disruption.

Pillar 8: Financial market sophistication

The present global financial crisis has highlighted the critical importance of financial markets. For the functioning of national economies, an efficient financial sector is necessary to allocate the resources saved by a nation's citizens, as well as those entering the economy from abroad, to their most productive uses.

Pillar 9: Technological readiness

This pillar measures the agility with which an economy adopts existing technologies to enhance the productivity of its industries. In today's globalised world, technology has become an increasingly important element for firms to compete and prosper. In particular, information and communication technologies (ICT) have evolved as the "general purpose technology" of our time, given the spillovers to the other economic sectors and their role as efficient infrastructure for commercial transactions.

Pillar 10: Market size

The size of the market affects productivity because large markets allow firms to exploit economies of scale. Traditionally, the markets available to firms have been constrained by national borders. In the era of globalisation, international markets have become a substitute for domestic markets, especially in small countries.

Pillar 11: Business sophistication

Business sophistication is conducive to higher efficiency in the production of goods and services. This leads, in turn, to increased productivity, thus enhancing a nation's competitiveness. Business sophistication is concerned with the quality of a country's overall business networks, as well as the quality of individual firms' operations and strategies.

Pillar 12: Innovation

Innovation is particularly important for economies as they approach the frontiers of knowledge, where the possibility of integrating and adapting imported technologies tends to disappear. Although less advanced countries can still improve their productivity by adopting existing technologies or making incremental improvements in other areas, for countries that have reached the innovation stage of development, this is no longer sufficient to increase productivity. Firms in these countries must design and develop cutting edge products and processes to maintain a competitive edge. This means sufficient

investment in research and development (especially by the private sector), the presence of high-quality scientific research institutions, extensive collaboration in research between universities and industry and the protection of intellectual property.

Although the 12 pillars of competitiveness are described separately, this should not obscure the fact that they are not independent: not only are they related to each other, but they tend to reinforce each other. For example, innovation (12th pillar) is not possible in a world without institutions (1st pillar) that guarantee intellectual property rights. Furthermore, innovation cannot be performed in countries with a poorly educated and poorly trained labour force (5th pillar) and will never take place in economies with inefficient markets (6th, 7th and 8th pillars) or without extensive and efficient infrastructure (2nd pillar).

The NPI (1996:28) concluded their section on macro-productivity improvement by stating that “the challenge to government is harsh”, because in many areas “the economy is weak, while fiscal pressures are strong”, and that the “world markets are keenly competitive, punishing perceived policy errors ruthlessly” so that the government has little room to make policy mistakes. They comment further that the government has tough decisions looming, which may harm the short-term interests of many of its constituents and that making the wrong decisions could mean that the people of South Africa will be “no better off in a decade or two than they are at present”.

It is interesting to look back more than a decade after this statement was made and reflect on whether or not policies were managed correctly. What do you think? In the concluding section which follows, we consider South Africa in terms of its present challenges and opportunities and supply some thoughts on the future.



Feedback on Activity 3.6

The discussion which followed activity 3.6 should provide enough information to answer the questions relating to the suggestions offered by the NPI regarding policies for productivity growth and the twelve pillars used by the WEF in its global competitiveness index for measuring the competitiveness of nations.

3.8 MACRO-PRODUCTIVITY IN SOUTH AFRICA – CHALLENGES AND OPPORTUNITIES

The IMD (2009:244) prioritised the following as the five major challenges for South Africa:

- Protecting the poor
- Building capacity for long-term growth by accelerating investment spending
- Sustaining employment growth and expanding training opportunities
- Addressing sectoral barriers to growth and investment
- Maintaining a sustainable debt level

The challenges, as expressed above, are certainly significant. While the author of this study unit acknowledges the weightiness of dealing with such challenges, the author also sees signs of hope in current local and international developments that could be a catalyst to progress in all of the above challenges.

The NPI (1996:14) says that it is easier to “divide a rapidly-expanding cake to every-one’s satisfaction than one which grows by a few crumbs each year”. This study unit is being updated in January 2011. Recent developments show positive signs that South Africa’s “cake” is expanding and could continue to expand. One of these developments (and the focus of this concluding section) is South Africa’s networking on an international scale through bodies such as “BRICS” and the G8’s Outreach 5. Such involvement could promote foreign investment in South Africa and hence help to “grow South Africa’s cake”. Assuming the South African government harnesses these opportunities by means of correct policy decisions, the author believes there are signs of hope.

Brazil, Russia, India and China make up the BRIC group of nations. According to Goldman Sachs (2010), these countries are all deemed to be at a similar stage of newly advanced economic development. Goldman Sachs argues that since they are all developing rapidly, by 2050 the combined economies of the BRICs could eclipse the combined economies of the current richest countries of the world. From 2000 to 2008, the BRIC countries contributed almost 30% to global growth (USD), compared to approximately 16% in the previous decade. The four countries currently account in combination for more than a quarter of the world’s land area and more than 40% of the world’s population.

Goldman Sachs did not argue that the BRICs would organise themselves into an economic bloc, or a formal trading association (Investopedia 2011) However, in May 2008, the leaders of the BRIC countries held their first summit in Yekaterinburg, and issued a declaration calling for the establishment of an equitable, democratic and multipolar world order (Mortished 2008).

Despite lagging behind the other members in terms of economic growth, South Africa was formally invited by China and the other BRIC countries to join the BRICs in 2010, and so the acronym changes from “the BRICs” to “the BRICS”. South Africa thus became the first African country to be admitted to BRICS. South Africa was officially admitted as a BRICS nation on December 24, 2010 (Smith 2011).

Smith (2011:2) states that South Africa is a gateway to Southern Africa and Africa in general, as well as Africa’s largest economy, giving BRICS a “four-continent breadth, influence and trade opportunities”. Furthermore, with China as South Africa’s largest trading partner and India’s desire to “increase commercial ties to Africa”, it is imperative that South Africa aligns itself with the growth potential of the BRIC nations (Smith 2011).

South Africa will need to improve its own competitiveness and importantly maintain strategic alliances with all four BRIC countries to ensure that the lessons learned in those countries can be put to good use in South Africa.

Van der Merwe (2009) states that many South African companies are well placed to provide services to the BRIC nations looking to invest and do business in South Africa. Numerous South African companies have a presence in other African countries and therefore have

inside information about the different business environments. He also says that South African expertise and “sometimes capital” are very important for the facilitation of business in many parts of Africa, and that the BRIC nations will have to tap into this. The banking, telecommunications, retail and construction sectors will be particularly well positioned to continue to facilitate the BRIC nations’ investment in Africa. The BRIC nations are already delivering on investment promises and continuing to seek new opportunities. It is crucial for South Africa to attract part of that investment, to be utilised as a facilitator and lastly to compete with BRIC businesses.

South Africa is also internationally recognised as a rising economic force in the world, and is included in the G8’s Outreach 5, which also includes India, Brazil, China and Mexico (Van der Merwe 2009). In the light of the argument on foreign investment and “growing South Africa’s cake”, the author believes these are positive signs for the future of South Africa.

We end this study unit with two points made by the previous author of this study unit, which the current author believes are relevant and need to be reiterated.

- (1) We emphasise the importance and perceived need to regularly and freely discuss matters related to productivity and competitiveness in South Africa among government, private business, academics, individuals and groups, and other interested stakeholders, including the population at large, in order to generate interest (productivity awareness), improve understanding (productivity management) and foster a climate of nonpoliticised productivity improvement.
- (2) While the “productivity” of participants in the private economy is the focus of attention in terms of productivity statistics etc, the productivity of the government and the state (as well as their functionaries at provincial and local government level), which in combination are a major and in some cases a dominant participant in the economy, also needs to be scrutinised. Government has a similar if not even greater need to demonstrate responsible conduct in all its activities, including the utilisation of resources (taxpayers’ money) to provide services to the population as a whole or individual members of communities.

CHECKLIST



Now that you have completed study unit 3 on the macro-perspective on productivity management, you should be able to do the following:

- explain what the macro-productivity perspective entails
- explain the complexities involved in measuring productivity
- describe the various perspectives on productivity of different parties
- explain why macro-productivity management is important
- describe the various means of measuring macro-productivity
- use the knowledge, insight and skills you have acquired on
 - South Africa’s industrial sectors, their contributions to GDP and year-on-year growth
 - macro-productivity indicators in a reporting context

- measurement of sectoral productivity
- trends in sectoral productivity
- describe the world competitiveness report and the factors used to determine world competitiveness rankings
- describe South Africa's world competitiveness position, including its overall rankings in the four competitive factors
- use the knowledge, insight and skills you have acquired in term of the breakdown of factors, subfactors and variables used to determine world competitiveness by the IMD
- describe South Africa's international position with regard to GDP and overall labour productivity
- describe ways of improving macro-productivity
- demonstrate your understanding of this study unit by identifying both correct and incorrect alternative statements with regard to, specifically, the study material but also to practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your understanding for macro-productivity management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



This study unit provided a macro-perspective on productivity management. It covered various means of measuring productivity and provided evidence of their application in a national and international context. The next study unit, study unit 4, will focus on a micro-perspective (within the individual firm or business) on productivity management.

ASSESSMENT



- (1) *Against the backdrop of the knowledge gained in this study unit and your perspective on South African productivity trends, write an essay in which you argue the merits of and/or criticise government policy decisions (for the past decade) on macro-productivity issues. Your discussion should include a section where you offer suggestions for government on how to proceed in order to promote productivity awareness among everyone in the country and improve future macro-productivity performance.*

SOURCE REFERENCES

Colander, DC. 1998. *Macroeconomics*. 3rd edition. Boston: McGraw-Hill.

Goldman Sachs. 2010. *Goldman Sachs BRIC Fund*. Available at: http://www2.goldmansachs.com/gsam/docs/funds/investor_education/investor_education/wp-133_brics.pdf (accessed on 14 January 2011.)

- Institute for Management Development (IMD). 2005. *IMD World Competitiveness Yearbook*, 2005.
- Institute for Management Development. 2009. *IMD World Competitiveness Yearbook 2009*. Selected information available at: <http://www.npi.co.za/MediaLib/Downloads/Home/SiteSections/AnnualReports/CurrentAnnualReports/IMD%20Report%202009.pdf> (accessed on 13 February 2011).
- <http://www.imd.org/research/publications/wcy/upload/FAQs.pdf> (accessed on 13 February 2011).
- Institute for Management Development. 2010. *IMD World Competitiveness Yearbook 2010*. Selected information available at: http://www.imd.org/research/publications/wcy/upload/Overall_ranking_5_years.pdf (accessed on 14 February 2011).
- <http://www.imd.org/research/publications/wcy/upload/scoreboard.pdf> (accessed on 15 February 2011).
- Institute for Management Development. 2011. *IMD World Competitiveness Online*. Selected information available via Unisa subscription at: <http://www.worldcompetitiveness.com> (accessed on 15 February 2011).
- Investopedia. 2011. *Brazil, Russia, India and China – BRIC*. Available at: <http://www.investopedia.com/terms/b/bric.asp> (accessed on 26 February 2011).
- Mortished, C 2008. Russia shows its political clout by hosting Bric summit. *The Times*. Available at: <http://business.timesonline.co.uk/tol/business/markets/russia/article3941462.ece> (accessed on 26 February 2011).
- Moss, DA. 2007. *A concise guide to macro economics: what managers, executives, and students need to know*. Boston, MA: Harvard Business School Press.
- Nedlac, 2000. *Productivity, economic efficiency and equity: a National Productivity Institute (NPI) and Nedlac document*. Available at: <http://www.nedlac.org.za/research/occasional-papers/productivity.aspx>. (accessed on 25 January 2011).
- National Productivity Institute. 2006. *Productivity Statistics 2005*. Pretoria: Government Printer.
- National Productivity Institute. 1996. *Productivity Focus 1996*. Pretoria: Government Printer.
- Productivity SA. 2007. *Concise handbook on productivity*. Midrand: Productivity SA. ISBN 0947015701
- Productivity SA. 2009. *World Competitiveness Report Executive Summary*. Available at: [http://www.npi.co.za/MediaLib/Downloads/Home/SiteSections/AnnualReports/CurrentAnnualReports/IMD %20Report%202009%20-%20Executive%20Summary.pdf](http://www.npi.co.za/MediaLib/Downloads/Home/SiteSections/AnnualReports/CurrentAnnualReports/IMD%20Report%202009%20-%20Executive%20Summary.pdf) (accessed on 14 December 2010).

- Productivity SA. 2010. *Productivity Statistics 2009*. Pretoria: Government Printer. ISBN 9780620478427
- Productivity SA. 2011a. Available at: <http://www.productivitysa.co.za/pebble.asp?reli-d53> (accessed on 10 January 2011).
- Productivity SA. 2011b. Available at: <http://www.productivitysa.co.za/pebble.asp?reli-d56> (accessed on 13 January 2011).
- Smith, JA. 2011. BRIC becomes BRICS: changes on the geopolitical chessboard. *Foreign Policy Journal*. Available at: <http://www.foreignpolicyjournal.com/2011/01/21/bric-becomes-brics-changes-on-the-geopolitical-chessboard/> (accessed on 22 February 2011).
- Stevenson, WJ. 2009. *Operations management*. 10th edition. New York: McGraw-Hill.
- Van der Merwe, JP. 2009. BRIC and South Africa: Can it become BRICS? *Trade Invest-South Africa*. Available at: http://www.tradeinvestsa.co.za/feature_articles/345544.htm. (accessed on 20 February 2011).
- World Economic Forum. 2010. *Global Competitiveness Report 2010–2011*.

Study unit 4

Micro-perspective on productivity



Key concepts

micro-perspective
effectiveness of production
efficient use of resources
utilisation of resources
productivity (P)
 $P = TE \times RU$
productivity process
productivity formula
technical efficiency (TE)
resource utilisation (RU)
usable outputs
actual inputs
available inputs
value-added productivity
measurement

productivity measurement
cooperative labour management
approaches to productivity
measurement
gain-sharing
profit-sharing
DPA deterministic productivity approach
DPA productivity management process
DPA model
price recovery formula
"hard" and "soft" components productivity
improvement
micro-productivity

GETTING AN OVERVIEW

In this study unit we will address the first, second and third learning outcomes for topic 3 study unit 4. We will look at the micro-perspective of productivity, productivity management within a business and its improvement, and micro-productivity management and the effect on production/operations management and its improvement.

4.1 INTRODUCTION

At the beginning of the previous study unit (study unit 3), we mentioned that we would apply a similar distinction to the one advocated by the former NPI (now known as Productivity SA) in their publication *Productivity Focus 1996*, namely in viewing productivity from both a macro-perspective and a micro-perspective. In study unit 3 we dealt with the macro-perspective on productivity management for the South African economy as a whole (including its sectoral breakdowns). Because ultimately the whole economy (with all its sectors) is made up of individual firms and businesses, and the overall productivity of the country is the sum of the productivity of all its individual enterprises, large and small, this study unit examines productivity management from a micro-perspective (or from the perspective of the individual firm or business).

4.2 MICRO-PRODUCTIVITY PERSPECTIVE DEFINED



Activity 4.1

Read the discussion below and then answer the following questions:

- (1) What does the productivity process within a business entail and how do the concepts “efficiency” and “utilisation” in the traditional definition and the inclusion of “effectiveness” in the recent definition figure in the process?
- (2) Which of the popular misconceptions regarding productivity (and productivity measurement) in business are still heard regularly?

DISCUSSION

The NPI (1996:29) notes that “productivity has traditionally been explained as the ratio of output to input” where “productivity improvement therefore involves the better and more efficient utilisation of all resources to maximise output with the minimum input”. However, according to the NPI, more recently the concepts of “efficiency” and “effectiveness” have been used to explain what productivity (or micro-productivity within the individual business or enterprise, as the term is used in this study guide) entails. The NPI defines “effectiveness” as the degree to which the output of the production process is related, in both quantitative and qualitative terms, to market needs (thus whether the output has value in the market).

Figure 4.1 illustrates how the different factors in the productivity process are related.

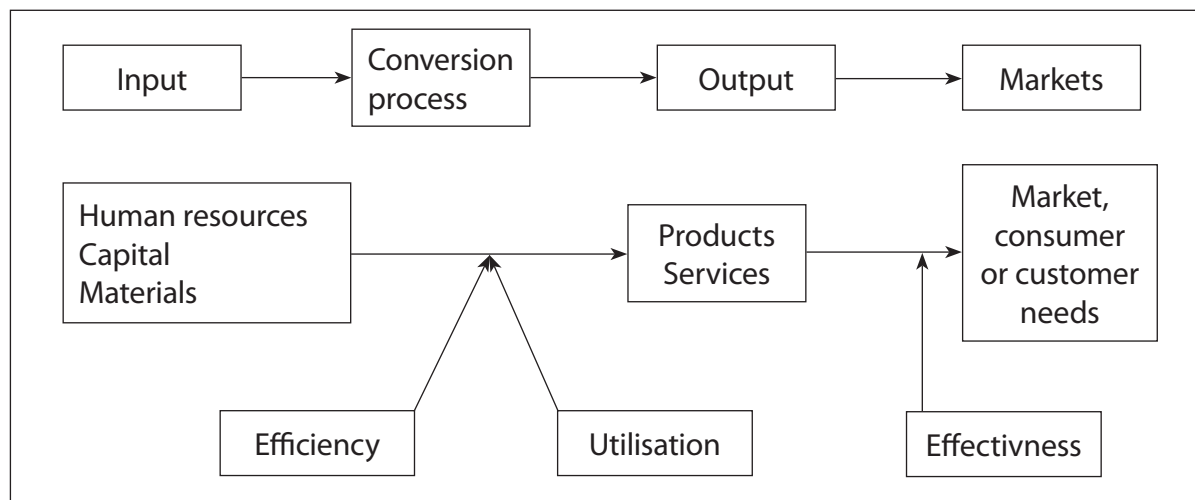


FIGURE 4.1: The productivity process

Source: NPI (1996:29, fig 1)

The traditional definition therefore focused largely on the efficiency and utilisation of resources or was expressed in terms of the following formula:

$$\text{Productivity (P)} = \text{Technical Efficiency (TE)} \times \text{Resource Utilisation (RU)}$$

where

$$TE = \frac{\text{Usable outputs}}{\text{Actual inputs}}$$

$$RU = \frac{\text{Actual inputs}}{\text{Available inputs}}$$

thus

$$P = \frac{\text{Usable outputs}}{\text{Available inputs}}$$

If we were to include the criterion of an output “having value in the market” in our definition of “usable outputs”, we would find ourselves agreeing with the more recent view of what productivity within a business or enterprise entails, namely the efficient and effective use of all resources in the production or conversion process in order to produce outputs that can satisfy market, consumer, customer/client needs.

Historically, “productivity” used to be associated with output per employee or labour productivity only. It is the use of this partial productivity measure (output is compared with only one input, ie labour) that, as the NPI (1996:7) so eloquently puts it, causes some people to experience “distaste: visions of layoffs, exploitation of workers and everybody playing games with simplistic and unfair measures”. Although such partial measures may be relatively easy to calculate, the NPI (1996:29) again cautions that “they may not provide a complete picture of productivity trends because they can increase if the quantities of other inputs are raised” and further “they do not isolate the effects of qualitative improvements in inputs, such as higher levels of education, technological progress and better allocation of resources”. These qualitative improvements in inputs are “actually what productivity is all about, because they enable output to increase without any additional manpower” (or people power) “or capital inputs”.

While many other definitions of productivity appear in the literature, as do various approaches to determining the level of productivity in enterprises, Webber (1996:11) notes that each approach, especially in the area of productivity measurement, provides “its own form of information to management for their use and benefit”. He then goes on to highlight certain misconceptions with regard to the task of measuring productivity (1996:12–13) namely:

- (1) Productivity is not only labour efficiency or labour productivity because the latter is also an input into the greater productivity formula – although not the sole determinant.
- (2) Judging an enterprise simply by its output is not a determinant of productivity. An enterprise’s output may increase, but if the input costs of production are not controlled, an increase in output does not constitute an improvement in productivity.
- (3) Productivity and profitability are often confused. Profit per se in an enterprise can be increased through improved price recovery, irrespective of a simultaneous decline in productivity.
- (4) Productivity is often confused with efficiency. Efficiency means producing high-quality goods in the shortest possible time. But if the goods are not needed and they have been produced for nothing, then the enterprise has been counterproductive.

- (5) Cost-cutting in the enterprise does not always contribute to an improvement in productivity. Indiscriminate cost-cutting can lead to long-term difficulties for an enterprise.
- (6) Productivity can only be applied to a manufacturing enterprise. This is incorrect. Although this study is concerned with manufacturing enterprises, there are service functions within these enterprises, such as a drawing office, that provide a service and require control and assessment. Typical service functions are financial functions, engineering support, procurement divisions, human resource management and public relations.

Finally, Webber (1996:14) identifies and discusses twelve different approaches to productivity measurement as found in the literature. They are the following: productivity measurement and evaluation system (ProMES), deterministic productivity accounting (DPA), the National Productivity Institute (NPI) methodology, theory of constraints (TOC), the total productivity model (TPM), Alan Lawlor's approach, Applied productivity – Gold's approach, operation function analysis (OFA), the approach of the International Labour Organisation (ILO), quick productivity appraisal (QPA), Kurosawa and Goshi – Japan Productivity Center, and finally the multifactor productivity measurement model (MFPMM).



Feedback on Activity 4.1

- (1) *The discussion which followed activity 4.1 should provide enough information to answer the question on what micro-productivity entails.*
 - (2) *It would be a good idea to rephrase the six points highlighted by Webber (1996:12–13) in question format (true/false) and then ask several of your supervisors, colleagues and friends to indicate which ones they believe are true or false.*
-

4.3 MANAGING PRODUCTIVITY WITHIN A BUSINESS



Activity 4.2

Read the discussion which appears below and then answer the following questions:

- (1) What does a successful strategy to improve productivity within a business entail and how could it be enhanced by cooperative labour-management relations, gain-sharing and profit sharing?
- (2) What is value-added productivity measurement and why is it considered a better measurement approach than physical indicators?
- (3) What does the DPA productivity management process entail and how can the DPA model be used in conjunction with the management process to measure and improve productivity within the business?

DISCUSSION

The NPI (1996:29) believes that “a successful strategy to improve productivity requires a systematic approach to measurement at both macro- and micro-levels, since productivity is relevant to individuals, firms, industries and the national economy”. They further note that “on the shop floor and at operations level, physical indicators can be used to compare productivity performance” but that such indicators “can only be used in situations where output is homogeneous” (ie the number of television sets produced per worker on a particular line of production).

The NPI (1996:30) then argues that “at the company level, where goods produced may vary widely in terms of type and quality, a value indicator is needed”, because “another shortcoming of physical indicators is that they measure only efficiency not effectiveness”. According to the NPI, value-added productivity measurement “can be used to supplement physical indicators because they not only reflect the efficiency of production, but are a measure of the additional wealth created by the company” where the value added is calculated in the following two ways:

Value added = total sales minus purchased materials and services

or

Value added = labour costs + profit + depreciation + other distributed costs (ie taxation and interest)

Value-added productivity measurement has the following advantages:

- (1) “Attaching a rand value to each product type, allows output of different products to be aggregated”, which “permits management to assess company-wide productivity”.
- (2) “It shows how the company creates wealth through its production process, and how it distributes this wealth to those who have contributed to its creation”, which can “encourage employees to participate in improving the performance of the enterprise, because they see themselves as the beneficiaries of higher output and productivity”.
- (3) “It allows quick comparison with standards”, where “published statistics on value-added for specific industries can be used by companies to evaluate their own standing” and it “highlights the crucial operational factors that cause differences in performance”, so that “management can then decide how to allocate organisational resources – labour, machines and materials”.
- (4) “Value added can easily be computed from the profit and loss accounts of companies” and “it is not necessary to set up a separate data collection system to monitor value-added productivity”, and therefore “because it is an extension of conventional accounting concepts, using it to measure productivity enables management to relate productivity to various financial analyses, such as those concerning profitability and liquidity”.

Cooperative labour-management relations, according to the NPI (1996:30–31), may be one of the key elements of improving productivity in the business. Since an increasing number of people agree that “high levels of productivity depend on people’s ‘hearts and minds’ even more than on hardware and capital” it follows that “labour-management cooperation requires mutual recognition and accommodation of the divergent interests

of workers and employers, so that differences are dealt with equitably and peacefully, in ways that change opposition to cooperation". Such necessary cooperative worker-management relations are characterised by recognition of the dignity of the individual and acceptance of the assumption that workers want to do high quality work. Managers should be reminded that their task is to "encourage, lead and train employees, and develop attitudes of positive thought and continual improvement", and should assure employees of the commitment that "job security depends on high quality, productivity, flexibility, profitability and acceptance of change – each of which depends on the others".

Another key element of improving productivity in a business, which the NPI (1996:31) also highlights, involves gain sharing and profit sharing. For the individual business the productivity challenge is to "create an environment in which workers and managers trust each other, and where the benefits coming from increased productivity are perceived to be distributed fairly" so that "workers need to consider themselves an integral part of their companies – not only in terms of their work contributions, but also in terms of the distribution of rewards". Pay and performance should therefore be directly linked in order to "create incentives for workers to pursue the interests of the team and the organisation", which will "increase worker effort and align workers more closely with the long-term interests of their firm, resulting in better communication, increased product quality, longer job tenure and greater acceptance of technological change". According to the NPI (1996:32–33), gain sharing, unlike profit sharing which applies to the whole business and does not necessarily involve employees, "is a management system in which an organisation seeks higher performance through the involvement and participation of its people". In gain-sharing, "as performance improves, employees share financially in the gain" through various gain-sharing formulae which favour operational measures rather than profit. Four principles which form the foundation of gain sharing are:

- (1) Identity. It is imperative for employee and company goals to converge so that employees acquire a sense of ownership of the business.
- (2) Equity. Gain-sharing is accepted as a fair practice and employees are rewarded through a bonus system as performance improves.
- (3) Involvement. The people who actually do the work are often the best at solving problems related to it and employee involvement can be enhanced through communication, cooperation and teamwork.
- (4) Commitment. Committed, competent managers who value human resources greatly are the best at recognising the need for change and higher performance levels.

We noted previously that Webber (1996:14) identified twelve approaches in the literature that could be used to measure productivity. The remainder of our discussion in this section focuses on one of these, the DPA (deterministic productivity method) approach. A recognised authority in South Africa on this approach, Basil van Loggerenberg, proposes that productivity improvement involves a process of applying five sequential steps. Webber (1996:27–39) summarised the DPA approach in his MCom dissertation on which we shall now draw heavily in order to present this study material. The contributions of Webber (1996) and Van Loggerenberg (1990) are again gratefully acknowledged.

The DPA productivity management process is illustrated in figure 4.2. The five sequential steps encompass step 1 – productivity measurement, step 2 – productivity diagnosis,

step 3 – productivity plan, step 4 – productivity disclosure, and step 5 – productivity accountability. All steps lead and contribute to productivity improvement.

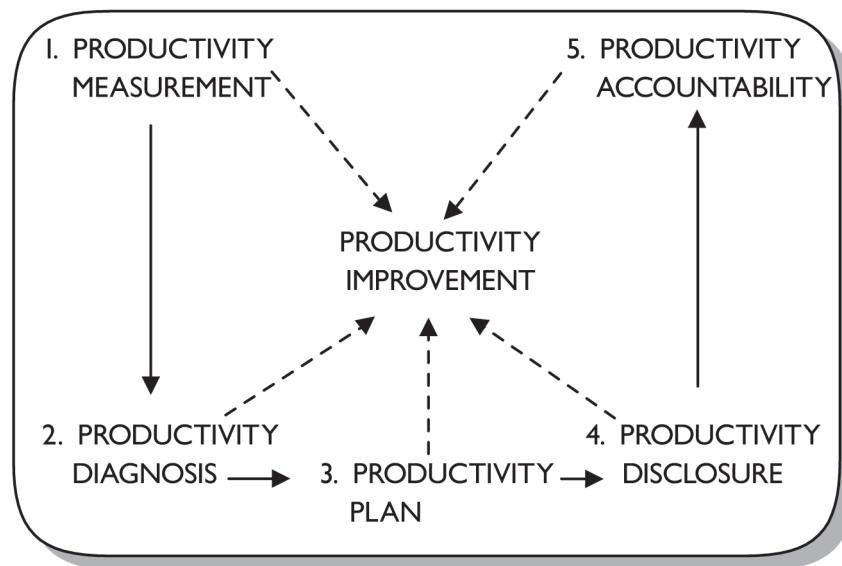


FIGURE 4.2: The DPA productivity management process

Source: Webber (1996:27, fig 2.3)

Productivity measurement is the first step in the process of productivity improvement and its function is to monitor and provide information on the financial standing of the business; the financial standing may vary according to changes in productivity over a period of time, whether from one month to the next or from one financial quarter to the next. The development of the DPA approach starts with the following equations:

$$\text{Value (in rand)} = \text{Quantity (in units)} \times \text{Price (in rand/units)} \quad \text{equation 6.1}$$

$$\text{Productivity} = \frac{\text{Product quantity}}{\text{Resource quantity}} \quad \text{equation 6.2}$$

$$\text{Price recovery} = \frac{\text{Product price}}{\text{Resource price}} \quad \text{equation 6.3}$$

Working from equation 4.1 and replacing equations 4.2 and 4.3 in it, the equation for profitability is developed as follows:

$$\frac{\text{Product value}}{\text{Resource value}} = \frac{\text{Product quantity}}{\text{Resource quantity}} \times \frac{\text{Product price}}{\text{Resource price}}$$

Thus

$$\text{Profitability} = \text{Productivity} \times \text{Price recovery}$$

(We will refer to this as equation 4.4.)

If we first focus on profit (or change in profit) we know it is the difference between income (or product revenue) and cost (or resource value). A change in profit can therefore be brought about by the following means:

- (1) a change in product revenue (which, following equation 4.1, is in turn influenced by a change in product quantity and/or a change in product price), and/or
- (2) a change in resource value (which following equation 4.1, is in turn influenced by a change in resource quantity and/or a change in resource price).

Incorporating equations 4.2 (productivity), 4.3 (price recovery) and finally equation 6.4 (profitability), we may complete the DPA model as illustrated in figure 4.3.

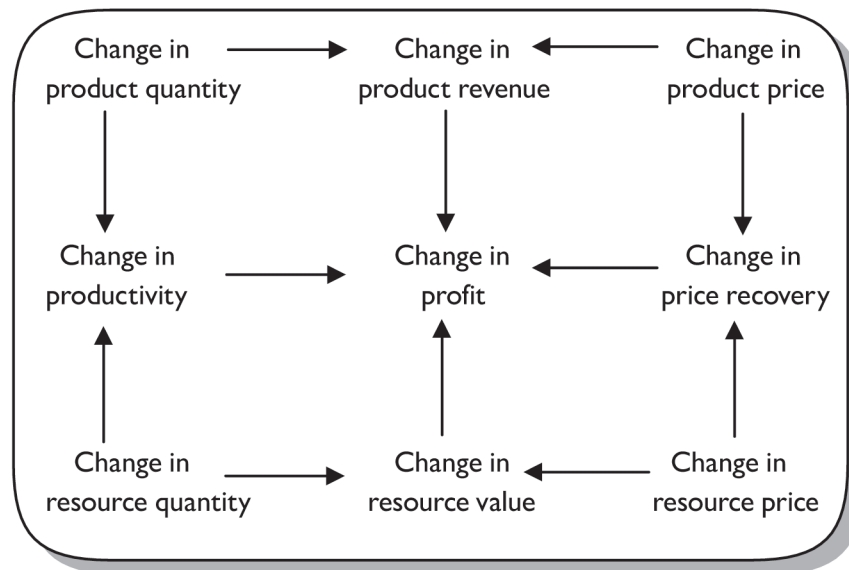


FIGURE 4.3: The DPA model

Source: Webber (1996:35, fig 2.10)

To illustrate how we could use the DPA model to gain a better understanding of how productivity improvement could affect a business and its profitability, let us consider the following scenario:

- (1) Company A has just had an unexpected increase in the price of its input resource (block on the far right at the bottom of fig 4.3). What action does it take? Normally, in order to maintain profitability, it increases its product price just enough to counter the increase (block on the far right at the top of fig 4.3). Company A has just passed the price increase on to its customers/clients.
- (2) Company B, which is company A's main competitor, has had the same unexpected price increase but, through tightening up on its internal production process control, is able to reduce its input resource quantity (block on the far left at the bottom of fig 4.3) by an amount that allows its resource value (middle block at the bottom of fig 4.3) to remain constant, as does its profitability.

In the immediate short term, both companies A and B kept their profitability the same, but company B's product price (or selling price) was not increased as in the case of company

A. If we assume that their products were exactly the same, as was their selling price before the increase in their input prices, and that the product price was a determining factor in deciding whom to purchase from, company B now has a distinct advantage because its price has remained the same. Should any of company A's customers now choose to purchase the product from company B, the change in product quantity (or selling quantity) (block on the far left at the top of fig 4.3) will increase, leading to an increase in profitability.

Let us now also briefly examine the other four steps of the DPA productivity management process illustrated in figure 4.2. Step 2 encompasses productivity diagnosis and is aimed at identifying causes or reasons for changes in productivity as detected by productivity measurement (step 1). Step 3, the productivity plan, focuses on the "hard" and "soft" components of productivity to develop a plan by which productivity can be improved. The soft components include, but are not limited to, paradigm shifts, education, training, culture and reorganisation of the enterprise's structure while the hard components include acquisition or sale of machinery, value engineering, use of material requirements planning (MRP), etc. Step 4 involves productivity disclosure, where index numbers obtained from reports on profit and/or cost performance are divulged to internal parties (disclosure which brings an increased awareness of the productivity concept within the enterprise) and external parties. Although it is unlikely that this information will be disclosed freely, it represents the "acid test" of management's commitment to productivity improvement. Finally, in step 5, which involves productivity accountability, both management and labour assume joint accountability for productivity. The requirement is that both management and labour will share in the benefits of any improvement in productivity.



Feedback on Activity 4.2

The discussion which followed activity 4.2 should provide enough information to answer the questions on what a successful strategy for improving productivity within a business (including possible contributions by cooperative labour-management relations and gain sharing), what value-added productivity measurement is, and what the DPA productivity management process entails (including how the DPA model could be used in conjunction with this management process to measure and improve productivity within a business).

4.4 IMPORTANCE OF MICRO-PRODUCTIVITY FOR PRODUCTION/ OPERATIONS MANAGEMENT



Activity 4.3

Read the discussion below and then answer the following question:

- (1) Why is micro-productivity important for production/operations management?

DISCUSSION

As pointed out earlier by the NPI (1996:29), productivity “is relevant to individuals, firms, industries and the national economy”. In a nutshell, to everybody! While macro-productivity focuses on the issues for the country as a whole (ie GDP per capita, multifactor productivity in the various economic sectors, etc), micro-productivity focuses on the issues at the level of the individual business or enterprise. This is because “ultimately ... an economy or sector is made of individual firms, and its overall productivity is the sum of the productivity of all its enterprises, large and small”.

If we recall the model of production and operations management (see topic 4) which comprised a transformation model to which we added the production/operations management’s activity areas of design, planning and control, and compare it with figure 4.1, it is not difficult to see the similarity. In both cases, the basic building block involves the input-transformation-output (or input-conversion-output) process and is found in all types of operation. The fundamental purpose of production/operations management is the transformation of input resources into goods (or products) and services. Micro-productivity is also primarily concerned with outputs and inputs. Logically, therefore, the responsibility for productivity within a business should fall largely to production and operations management. Improved micro-productivity means better efficiency and utilisation of resources within the production (or transformation) process (a direct responsibility of production/operations managers) and greater effectiveness in satisfying customer/client needs (probably a direct marketing function but an indirect responsibility for production/operations managers).

To summarise, micro-productivity is important for production and operations management because it is another measure or indicator of how well the management of this function is performing, and if the business is hoping to improve its overall financial performance, productivity improvement is one of the obvious means of doing so. Many of the ways of improving productivity lie directly in the responsibility field of production and operations management.



Feedback on Activity 4.3

The discussion which followed activity 4.3 should provide enough information to answer the questions on why micro-productivity is considered important for production and operations management.

CHECKLIST



Now that you have completed study unit 4 on a micro-perspective on productivity management, you should be able to do the following:

- explain what the micro-productivity perspective entails
- explain and demonstrate by means of practical measures how productivity should be managed within a business and how it could be improved
- explain why micro-productivity is important for production/operations management
- use the knowledge, insight and skills you have gained regarding
 - the micro-productivity perspective
 - the management of productivity within a business and its improvement
 - micro-productivity management and the effect on production/operations management and its improvement
- justify its importance and demonstrate this by identifying both correct and incorrect alternative statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the micro-perspective on productivity management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



This study unit provided a micro-perspective on productivity management. It focused on the issues of productivity management within the context of the individual business or enterprise. Together with study unit 3, which dealt with the macro-perspective on productivity management (for the country as a whole), this study unit should have given you a good idea of what productivity management entails and why it is important for production and operations management.

This topic completes part I of this module, which concentrated on the foundations of production/operations and productivity management. In the next part of this module we will examine some of the aspects of the design of production/operations management systems.

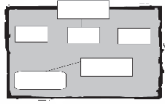
ASSESSMENT



- (1) *Study figure 4.3 again and then write an essay in which you explain how the DPA model was developed and how you would use it to manage productivity within a business.*
- (2) *Assume you have been asked by your immediate supervisor to give a lecture on productivity management and its importance in South Africa. Pay special attention to both the macro-issues and the micro-issues of productivity management and offer concrete suggestions on how productivity for the country as a whole could be improved. Include suggestions for improving productivity within the business or other institution in which you may be an employee.*

SOURCE REFERENCES

- Adendorff, SA & De Wit, PWC. 1997. *Production and operations management: a South African perspective*. 2nd edition. Halfway House: Thomson.
- National Productivity Institute. 1996. *Productivity Focus 1996*. Pretoria.
- Van Loggerenberg, BJ. 1992. *Productivity decoding of financial signals, vol 1: a primer on deterministic productivity accounting for novices*. Pretoria: Productivity Measurement Associates.
- Webber, AE. 1996. *Productivity measurement in small manufacturing enterprises in the steel and engineering industry of South Africa*. MCom dissertation, University of South Africa.



PART II

Designing the production/operations management system

TOPIC 4

DESIGN OF PROCESSES FOR PRODUCTS AND SERVICES

Topic 4 explains the general nature of design in operations and examines the different stages involved in designing products and/or services for competitive advantage.

- Study unit 5: Process design (chapter 4)
- Study unit 6: Design of products and services (chapter 5)

TOPIC 5

DESIGN AND LOCATION OF PRODUCTION/ OPERATIONS CAPACITY

Topic 5 introduces the supply network perspective of operations and examines the location decision of the production/ operations capacity.

- Study unit 7: Supply network design (chapter 6)

TOPIC 6

DESIGN OF LAYOUT AND WORKFLOW

Topic 6 explains the layout procedure and examines the advantages and disadvantages of each basic layout type.

- Study unit 8: Layout and flow of an operation (chapter 7)

TOPIC 7

JOB DESIGN AND WORK ORGANISATION

Topic 7 explains the elements, objectives and various approaches to job design and the organisation of work in operations.

- Study unit 9: Elements, objectives and approaches to job design (chapter 9)

Topic 4

Design of processes for products and services

AIM



To provide learners with an understanding of what is involved in the design of products/services and the processes by which they are manufactured or provided, and why the design activity is important for production/operations management.



Learning outcomes

When you have worked through this study unit, you should be able to do the following:

- explain what the design activity in production/operations management entails
- distinguish between the various process types and explain the volume-variety effect on process design
- explain what the detailed process design in production/operations management entails
- explain the effects of process variability on process design
- explain why good design is important
- distinguish between the sequence of stages in the design of products/services, namely concept generation, concept screening, preliminary design, design evaluation and improvement, prototyping and final design
- justify the use of interactive design
- use the knowledge, insight and skills you have gained regarding
 - the design activity in production/operations management
 - the various process types and the volume-variety effect on process design
 - the detailed process design
 - the effects of process variability on process design
 - good design
 - the sequence of design stages, namely concept generation, concept screening, preliminary design, design evaluation and improvement, prototyping and final design
 - interactive design
- to justify the importance of design and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the design of processes for products and services by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills

Study unit 5

Process design



Key concepts

process design
design activity
products/service design
design overlap/interrelatedness
process design objectives
strategic performance objectives
benefits of good process design
throughput rate
throughput time
work in process
utilisation
environmentally sensitive design
life cycle analysis
volume-variety positions
process types
process types in manufacturing
project processes
jobbing processes
batch processes
mass processes
continuous processes
process types in services

professional services
service shops
mass services
product-process matrix (p-p matrix)
natural diagonal line in p-p matrix
detailed process design
process mapping
process blueprinting (process analysis)
process mapping symbols
high-level process mapping
outline process mapping
process performance
flow process charts
work content
cycle time
Little's Law
 $\text{throughput time} = \text{work in process} \times \text{cycle time}$
 $\text{throughput efficiency} = \frac{\text{work content}}{\text{throughput times} \times 100}$
value-added throughput efficiency
process variability
simulation in design

GETTING AN OVERVIEW

In this study unit we address the first four learning outcomes of topic 4. We explain what the design activity in production/operations management entails, distinguish between the various process types and explain the volume-variety effect on process design, explain what the detailed process design in production/operations management entails, and explain the effects of process variability on process design.

5.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 4 of your prescribed book. Then study figure 4.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is (are) discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 4 of your prescribed book:

- "What is process design?" (2 marks)
- "What objectives should process design have?" (2 marks)
- "How do volume and variety affect process design?" (3 marks)
- "How are processes designed in detail?" (3 marks)

5.2 OPERATIONS IN PRACTICE



Activity 5.1

Read the case study entitled "Process redesign through product redesign – a mobile tele-communications case" in chapter 4 of your prescribed book and then answer the following questions:

- (1) What challenge is the telcos industry facing with regard to contract upgrades?
- (2) What challenge does this pose from a design point of view?
- (3) How is the telcos industry dealing with this challenge from a design point of view?



Feedback on Activity 5.1

- (1) *The challenge faced by the telcos industry is the ongoing support required by a large number of customers at the end of their contract periods of 1-2 years, which amounts to millions of clients in any given year.*
 - (2) *From a design point of view, the telcos industry need to manage client upgrades effectively (in relation to customer satisfaction) and efficiently (achieving customer satisfaction without wasting resources). Processes need to be designed with these two objectives in mind.*
 - (3) *The telcos industry has chosen the hybrid solution of providing clients with the choice of a device or a prepaid gift card through an external supplier called Tutuka. The advantage of the gift card is its flexibility. The expensive process of managing the issuing of cell phones with the associated stock, logistics and security challenges has been totally replaced with a simple process of transferring the data to Tutuka, who produce these cards in a batch run based on the planned production campaigns. In essence the telcos have cut down on process complexity by, firstly, transferring responsibility for what was normally an internal process to an external supplier and, secondly, adapting the product offering (clients receiving a gift card in accordance with their contract selection).*
-

5.3 THE DESIGN ACTIVITY IN PRODUCTION/OPERATIONS MANAGEMENT



Activity 5.2

Study the first section of chapter 4 entitled “The design activity” in your prescribed book and then answer the questions and/or complete the following:

- (1) To “design” is to
- (2) Why should the design of products and services and the design of processes for their manufacture/rendering be interrelated?
- (3) What is the overriding objective of process design?
- (4) The operations performance objectives should translate directly into process design objectives. The impact of strategic performance objectives on (a) process design objectives and (b) benefits in the case of
 - (i) quality, are ; and ;
 - (ii) speed, are ; and ;
 - (iii) dependability, are ; and ;
 - (iv) flexibility, are ; and ;
 - (v) cost, are ; and ;
- (5) Which micro performance flow objectives can be used to describe the process flow performance?
- (6) What do environmentally sensitive designs entail? How could life cycle analyses help to achieve such designs?



Feedback on Activity 5.2

- (1) *With reference to question 1, the authors of your prescribed book note the following about design: “It is to conceive the looks, arrangement and workings of something before it is constructed”; “it is a conceptual exercise”; “[it] must deliver a solution that will work in practice”.*
- (2) *With reference to question 2, the authors of your prescribed book insist that “it would be foolish to commit the organization to the detailed design of any product or service without some consideration of how it is to be produced”. Also see figure 4.2 in chapter 4 of your prescribed book and note the overlapping area of the two circles, where it is stated that “product/service design has an impact on the process design and vice versa”.*
- (3) *With reference to question 3, according to the authors of your prescribed book, the overriding objective of process design is “to make sure that the performance of the process is appropriate for whatever it is trying to achieve”. Thus if the operation primarily wants to compete on, for example, its ability to respond quickly to customer needs, then its processes must be designed to achieve fast throughput times.*
- (4) *With reference to question 4, note the contents of table 4.1 in your prescribed book. For each strategic performance objective (ie quality, speed, dependability, flexibility and cost), typical process design objectives as well as some benefits of good process design are listed. Make sure you understand the connection between the performance objective (eg dependability), what the process design objective should focus on (eg reliable process output in terms of both timing and volume), and the potential benefits of such good process design (eg less disruption, confusion and rescheduling within the operation because it was not able to deliver the quantity of products/services in the time it committed to).*

- (5) With reference to question 5, measures that may be used to describe the process flow performance at a micro level comprise the following:
- (a) the throughput rate (number of units emerging from the process over a unit of time, for example 30 hamburgers per hour @ nonpeak times [6 hrs per day] and double that @ peak times [2 hrs per day])
 - (b) the throughput time (average time from entry to exit as output, for example 2 minutes per hamburger)
 - (c) work in process (average number of units within the process being “worked on” over a period of time, for example 3712 hamburgers per hour)
 - (d) utilisation (available time for a process to perform and deliver usable outputs, for example 8 hours per day).
- (6) With reference to question 6, environmentally sensitive designs deal with the concerns of environmentalists or “greenies” (this is a tongue in cheek reference to the adherents of the Green Peace movement). On a more serious note, environmental protection has become a very important focus point for most businesses/organisations and is well supported by legislation and international standards of conduct – ISO 14 000 (ISO is the abbreviation for International Organisation for Standardisation and the 14 000 standard refers to Environmental Management Systems). The need for environmentally sensitive designs can no longer be ignored – they must deal with fundamental issues related to the sources of inputs, quantities and sources of energy, amounts and types of waste material, and the life as well as the end-of-life of the product itself. Since designers face very complex trade-offs between these issues when they attempt to make the best choices or rational decisions in the design activity, it may be useful to perform a life cycle analysis. Here the total energy used from the point when the first production inputs are made until the final disposal of the product (ie the complete life cycle of the product) is analysed. Study the short case study entitled “Mittal Steel innovates processes with Voest Alpine Concast solution” in chapter 4 of your prescribed book and answer the questions at the end.
-

5.4 PROCESS TYPES IN MANUFACTURING AND SERVICES



Activity 5.3

Study the second section of chapter 4 of your prescribed book entitled “Process types – the volume-variety effect on process design” and then answer the questions and/or complete the following:

- (1) The five process types in manufacturing are
 - (i) project processes, which entail (Add your own three examples.)
 - (ii) jobbing processes, which entail (Add your own three examples.)
 - (iii) batch processes, which entail (Add your own three examples.)
 - (iv) mass processes, which entail (Add your own three examples.)
 - (v) continuous processes, which entail (Add your own three examples.)
- (2) The three process types in services are
 - (i) professional services, which entail (Add your own three examples.)
 - (ii) service shops, which entail (Add your own three examples.)
 - (iii) mass services, which entail (Add your own three examples.)
- (3) Where would you place a project process that is done in high volume (eg multiple-unit project processes such as a housing scheme involving the construction of 50 identical houses in nine different provinces, or a shipyard where 100 identical patrol

boats are built over a period of three years) in terms of figure 4.3 (left-hand figure for manufacturing process types) in your prescribed book?

- (4) Why do you think it is important to categorise the process types in manufacturing and services? What is the basis of these classifications?
- (5) What is the value of the product-process matrix illustrated in figure 4.12 of your prescribed book? Where in the matrix would you prefer to position an operation? What happens if the operation is positioned to the left or to the right of the diagonal line?



Feedback on Activity 5.3

- (1) *With reference to question 1, the process types in manufacturing are illustrated on the left-hand side of figure 4.3 of your prescribed book. Compare your own examples with those suggested in your prescribed book. Also note the photograph and the illustrative process flow chart for each of the process types in manufacturing (project – fig 4.4, jobbing – fig 4.5, batch – fig 4.6, mass – fig 4.7, and continuous – fig 4.8).*
 - (2) *With reference to question 2, the process types in services are illustrated on the right-hand side of figure 4.3 of your prescribed book. Compare your own examples with those suggested in your prescribed book. Also note the photograph and the illustrative process flow chart for each of the process types in services (professional – fig 4.9, service shop – fig 4.10, and mass – fig 4.11).*
 - (3) *With reference to question 3, a multiple-unit project process could cause difficulties when we try to position it in accordance with figure 4.3 in your prescribed book. Clearly we are dealing with a project process, but the volume is greater than one. The characteristics displayed here show lower variety but greater volume (more of a limited selection of units) versus the “pure” project processes, which typically display high variety and low volume. Ideally we could place the multiple-unit project process in a position somewhat similar to the batch process to indicate slightly less variety and greater volume.*
 - (4) *With reference to question 4, note that similar process types can (and should) be managed in the same way. Conversely, different process types must (and should) be managed in a manner that suits their unique operational characteristics. The basis of the classifications referred to stems from their relative position on two continuums, namely, a variety continuum and a volume continuum. The process types in manufacturing and services that have been identified display varying degrees of variety and volume characteristics. However, also note the critical comments of the authors of your prescribed book and their warning that the classifications of the process types for manufacturing and services can be considered somewhat “simplistic”, because in reality there are no such clear boundaries.*
 - (5) *With reference to question 5, the value of the product-process matrix as illustrated in figure 4.12 of your prescribed book lies in identifying the “natural” lowest cost position for an operation, namely the diagonal line. If the operation is either to the left or to the right of this line (resulting in either less or more process flexibility) it would lead to cost increases.*
-

5.5 DETAILED PROCESS DESIGN IN PRODUCTION/OPERATIONS MANAGEMENT



Activity 5.4

Study the third section of chapter 4 of your prescribed book entitled “Detailed process design” and then answer the following questions:

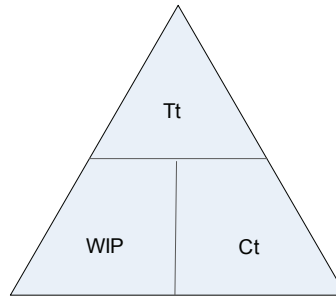
- (1) What does the detailed process design in production/operations management entail, and how can process mapping help to improve processes?
- (2) Why is it important to determine the process performance of an operation?
- (3) How can the measurement of the work content, throughput, cycle time and work-in-process be used to improve the performance of a process?
- (4) What is the implication if the throughput time of a particular process is different (not equal) to the work content time? What does value-added throughput efficiency mean and how can it be calculated?



Feedback on Activity 5.4

- (1) *With reference to question 1, the authors of your prescribed book note that the individual activities must be configured after the overall process design. The detailed design of a process in its simplest form “involves identifying all the individual activities that are needed to fulfil the objectives of the process, and deciding on the sequence in which these activities are to be performed and who is going to do them”. A visual approach to process design involves process mapping (also termed process blueprinting or process analysis), where a process is described in terms of how the activities within the process relate to each other. Note the examples of the symbols that may be used in process mapping (see fig 4.13 in your prescribed book) and the difference between “high level process mapping” (see fig 4.14 in your prescribed book) and an “outline process map” (see fig 4.15 in your prescribed book). The main difference between these is the level of detail involved. When processes are mapped it is possible to focus attention systematically on each and every activity (question its relevance, duration, etc) in an attempt to improve the process as a whole. Look at the example of a process flow chart at Intel Corporation before and after it was critically examined. See figure 4.16 (left-hand side before and right-hand side after improvement) in your prescribed book.*
- (2) *With reference to question 2, when the performance of a process is evaluated, it is important to make sure that the process is appropriate for whatever it is supposed to achieve. This could be done by assessing how well the process has been designed (see the example of the new customised sandwich in table 4.2 of your prescribed book).*
- (3) *One of the most practical and fundamental laws in waiting-line theory is Little’s Law, which relates the number of customers in a waiting-line system to the arrival rate and the waiting time of customers.*

The mathematical equation for Little’s Law can be expressed as follows:



The triangle above represents three different ways of expressing Little's Law. They are presented below:

$Tt = WIP \times Ct$	$WIP = \frac{Tt}{Ct}$	$Ct = \frac{Tt}{WIP}$
----------------------	-----------------------	-----------------------

The definitions of the applicable variables are offered in the table below:

Variable	Abbreviation	Definition
Throughput time	Tt	The time for a unit to move through a process
Work-in-process	WIP	The number of units within a process waiting to be processed further
Cycle time	Ct	The average time between units of output emerging from a process

The Little's Law relationship holds for a wide variety of arrival processes and service-time distributions. The practical advantage of Little's Law is that you only need to know two of the parameters to calculate the third.

To illustrate the value of using Little's Law we begin by working through the example presented under the heading "Little's Law" in your prescribed textbook, in which you are asked to calculate the number of servers required to sell sandwiches.

LITTLE'S LAW PROBLEM 1:

Suppose it is decided when a new process is being introduced, that the average number of customers in the process should be limited to around ten and the maximum time a customer is in the process should be four minutes on average. The time required to assemble and sell a sandwich (from customer request to the customer leaving the process) in the new process has been reduced to 1.2 minutes. How many staff should be serving?

The calculations follow in the block below. The diagram below the calculations should help you to understand the applicable variables in this specific calculation as well as Little's Law in general.

$Tt = 4$ minutes

WIP = 10 customers

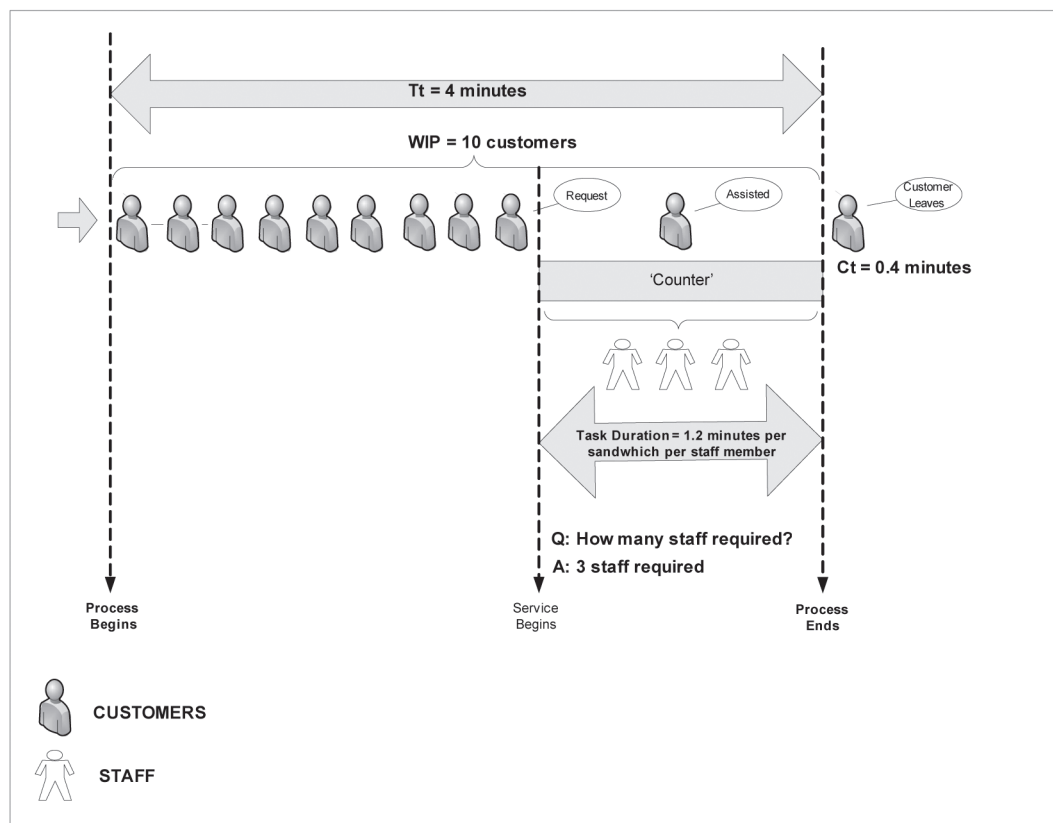
$$\begin{aligned} Ct &= \frac{Tt}{WIP} \\ &= \frac{4 \text{ minutes}}{10 \text{ customers}} \\ &= 0.4 \text{ minutes} \end{aligned}$$

A customer should emerge from the process every 4 minutes, on average.

Given that an individual can be served in 1.2 minutes

$$\begin{aligned} \text{The number of servers required} &= \frac{1.2 \text{ (Task duration)}}{0.4 \text{ (Ct)}} \\ &= 3 \end{aligned}$$

In other words, three servers would serve three customers in 1.2 minutes or one customer in 0.4 minutes



LITTLE'S LAW PROBLEM 2:

This is the second worked example under the heading Little's Law in your prescribed textbook. We encourage you to work through the example with the assistance of the diagrams below to help you understand how we apply Little's Law.

Every year all the workstations in the building had to be renovated (tested, new software installed, etc) in the space of a single week. That week fell in the middle of the August vacation period, when the renovation process would cause minimum disruption to normal work. Last year the company's 500 workstations were all renovated within one working week (40 hours).

Each renovation last year took two hours on average and 25 technicians completed the process within the week. This year there will be 530 workstations to renovate but the company's IT support unit has devised a faster testing and renovation routine that will only take 1.5 hours on average instead of 2 hours. How many technicians will be needed this year to complete the renovation processes within one week?

LAST YEAR

Tt = 40 Hours
WIP = 500 workstations

$$Ct = \frac{Tt}{W.I.P}$$

$$= \frac{40 \text{ Hours}}{500 \text{ workstations}}$$

$$= 0.08 \text{ Hours}$$

A workstation should be completed every 0.08 hours on average
Given that one work stations lead time is 2 hours per technician

$$\text{The number of servers required} = \frac{2 \text{ (Task duration)}}{0.08 \text{ (C.t)}}$$

$$= 25$$

In other words, 25 technicians would upgrade 25 workstations in 2 hours or 1 workstation in 0.08 hours (4.8 minutes)

THIS YEAR

Tt = 40 Hours
WIP = 530 workstations

$$Ct = \frac{Tt}{W.I.P}$$

$$= \frac{40 \text{ Hours}}{530 \text{ workstations}}$$

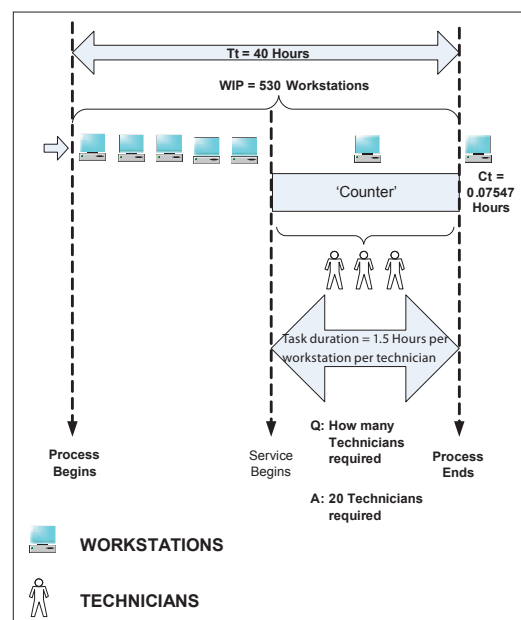
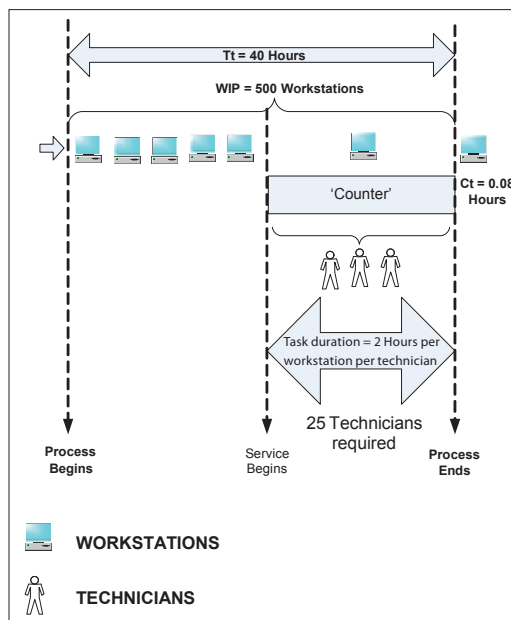
$$= 0.07547 \text{ Hours}$$

A W/S should be completed every 0.07547 hours on average
Given that one work stations lead time is 1.5 hours per technician

$$\text{The number of servers required} = \frac{1.5 \text{ (Task duration)}}{0.07547 \text{ (C.t)}}$$

$$= 19.8754 \text{ workers}$$

In other words, 20 technicians would upgrade 20 workstations in 1.5 hours or 1 workstation in 0.07547 hours (4.53 minutes)



The question above (Problem 2) has been answered and illustrated in a similar fashion to Problem 1 to show you that although examples may differ, the principles and methodologies remain similar.

Also work through the worked example dealing with Little's Law in chapter 4 of your prescribed book (the example in which Mike and Dick watch people having coffee and biscuits).

Make sure you can solve these and similar types of problems using Little's Law.

(4) With reference to question 4, the implication is that for a percentage of the time that represents the difference between the two (eg assume throughput time is 4 minutes but work content requires only 1.2 minutes) no actual work is being done on the unit being processed (eg $1.2/4 = 30\%$). This is a measure of throughput efficiency. Please study the worked example of the vehicle licensing centre dealing with throughput efficiency in chapter 4 of your prescribed book, and indicate whether you agree or disagree with the analysis. Also note that a further refinement can be made in the calculation, namely value-added

throughput efficiency, where the work content is limited to that portion of the tasks that actually did add value to whatever was being processed. Usually such percentages are even lower than the normal throughput efficiency percentages, as the example (again the vehicle licensing centre) in chapter 4 of your prescribed book shows.

5.6 THE EFFECTS OF PROCESS VARIABILITY



Activity 5.5

Study the fourth section of chapter 4 of your prescribed book, entitled “The effects of process variability” and then answer the questions and/or complete the following:

- (1) What is process variability, why does it occur and which two fundamental types can be identified?
- (2) Illustrate the relationship between process utilisation and the number of units waiting to be processed for constant and variable arrival and process times by means of a diagram. What are the implications in terms of process design or managing process capacity (utilisation), waiting time and/or variability?
- (3) How may simulation be used in design and what are its advantages?



Feedback on Activity 5.5

- (1) *With reference to question 1, process variability occurs either in the demand to which the process is expected to respond (more, or less demand) or in the time it takes to perform the various activities that make up the process (longer or shorter process). There are a number of reasons for this. These could include the late or early arrival of materials, information or customers/clients; a temporary malfunction or breakdown in process technology during a stage of the process; the recycling of badly processed material; or a significant change in the requirements for the items being processed. These sources of variation interact with each other and result in the following two fundamental kinds of variation: (i) variability in demand for a stage within the process (earlier or later arrival times) (ii) variation in the time it takes to process or perform the activities required (longer or shorter).*
 - (2) *With reference to question 2, carefully study the discussion in chapter 4 of your prescribed book and note figure 4.17, which illustrates the relationship between process utilisation and the number of units waiting to be processed for constant and variable arrival and process times. This relationship implies three options (see fig 4.18 [b]): accept long average waiting times but achieve high utilisation (point X); accept short waiting times but achieve low utilisation (point Y); reduce variability in arrival times, activity times or both and achieve higher utilisation and shorter waiting times (point Z).*
 - (3) *With reference to question 3, simulation in design, in the words of the authors of your prescribed book, “involves making decisions in advance of the final process being created”. The purpose is therefore to simulate how the product/service would function in practice under normal working conditions. Simulation models, which may take many forms, such as computer-based models, are used to gain insights and explore possibilities though formalised “pretending”.*
-

CHECKLIST



Now that you have completed study unit 5 on process design in operations, you should be able to do the following:

- explain what the design activity in production/operations management entails
- distinguish between the various process types and explain the volume-variety effect on process design
- explain what the detailed process design in production/operations management entails
- explain the effects of process variability on process design
- use the knowledge, insight and skills you have gained regarding
 - the design activity in production/operations management
 - the various process types and the volume-variety effect on process design
 - the detailed process design
 - the effects of process variability on process design
- justify the importance of process design and demonstrate this by identifying both correct and incorrect statements in multiple-choice format or in essay-type questions
- provide evidence of your appreciation for the general nature of design in operations by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills

SUMMARY



Now that you are familiar with the purpose and objectives of process design in production and operations management, you will probably agree that products cannot be manufactured successfully and services cannot be provided efficiently if sequences are wrong. It is necessary to start with design, that is, to take the real needs of customers/clients into consideration. This study unit examined process design; the next one, study unit 6, focuses in detail on the stages of design.

ASSESSMENT



- (1) Consider the case study entitled “The Central Evaluation Unit” at the end of chapter 4 and then answer the three questions posed at the end.
- (2) Do problems 1 to 8 at the end of chapter 4 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 5 at the end of chapter 4 of your prescribed book.

Note: Some study activities may merely require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experience of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Study unit 6

Design of products and services



Key concepts

good design enhances profitability	bill of materials (BOM)
design concept	design complexity
design package	standardisation
design process	commonality
core products and services	modularisation
supporting products and services	design evaluation and improvement
stages of design	quality function deployment (QFD)
concept generation	"house of quality"
ideas from customers/clients	"voice of the customer"
listening to customers	QFD matrix
ideas from competitor activity	value engineering (VE)
reverse engineering	Taguchi methods
ideas from staff	prototyping and final design
ideas from research and development	virtual prototype
R&D research and development	computer-aided design (CAD)
concept screening	interactive design
design criteria	simultaneous development
feasibility	simultaneous (or concurrent) engineering
acceptability	time-to-market (TTM)
vulnerability	early conflict resolution
design funnel	project-based organisation structures
design creativity	task force
preliminary design	matrix organisation
component (or product) structure	

GETTING AN OVERVIEW

In this study unit we address the fifth, sixth and seventh learning outcomes of topic 4. We explain why good design is important, distinguish between the sequence of stages in the design of products/services, namely concept generation, concept screening, preliminary design, design evaluation and improvement, prototyping and final design, and justify the use of interactive design.

6.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 5 of your prescribed book. Then study figure 5.1 of your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is (are) discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 5:

- “Why is good product and service design important?” (3 points)
- “What are the stages in product and service design?” (5 points)
- “Why should product and service design and process design be considered interactively?” (1 point)
- “How should interactive design be managed?” (3 points)

6.2 OPERATIONS IN PRACTICE



Activity 6.1

Read the case study entitled “Adcock Ingram’s product strategy” in chapter 5 of your prescribed book and then answer the following questions:

- (1) Differentiate clearly between the two types of product approach followed by Adcock Ingram.
- (2) From a design point of view, what are the main points of consideration in both approaches?



Feedback on Activity 6.1

- (1) *The first approach is applied to proprietary drugs where the patent is owned by Adcock Ingram. The second approach involves generic drugs where the patents on the formula have expired and regional players mass-produce what has become a commodity, and brand this to secure a differentiated position in the market.*
 - (2) *From a design point of view, special consideration is required by the proprietary drug approach (patented drugs). As the case study illustrates, new product and service design at Adcock Ingram is particularly important because it takes such a long time and billions of rand to develop a drug, test it and have it approved by the regulatory authorities. For this reason it is vital for any pharmaceutical company pursuing this approach to have plenty of potential marketable drugs in the pipeline. Design for this type of drug would probably begin with an attempt to solve a new problem. The second approach, which relates to the manufacture of generic drugs, would be entirely different. In this instance, a drug has already been developed to solve a particular problem. Adcock then manufactures a generic of the drug. Of importance here is that Adcock Ingram manufactures drugs that have proved to be effective in the market and then brands these drugs to ensure that there is a differentiating factor in the local market.*
-

6.3 THE COMPETITIVE ADVANTAGE OF GOOD DESIGN



Activity 6.2

Study the first section of chapter 5 entitled “Why is good design so important” in your prescribed book and then answer the following questions:

- (1) How can a good design influence the competitive position of a business/organisation?
- (2) Why does a good design enhance profitability?
- (3) All products and services comprise or encompass the following three aspects:
 - (i) a concept, which means ...
 - (ii) a package, which means ...
 - (iii) a process, which means ...
- (4) Identify and describe the product/service (a) concept (b) package and (c) process for
 - (i) a stove
 - (ii) a consultation with your lawyer
 - (iii) a motor car
 - (iv) postal delivery to your house or postbox
- (5) The design activity itself is a transformation process. This means Also illustrate this idea by means of a diagram.



Feedback on Activity 6.2

- (1) *With reference to question 1, it should be noted that a business/organisation generally finds itself in a competitive position when customers/clients consistently purchase its products/services rather than those of its competitors. Such a position may be the result of a factor or combination of factors that makes the particular business more successful than others. A good design implies that the needs of the customers/clients will be satisfied by the products/services offered. If a certain design satisfies the needs of customers/clients more effectively than other designs, or if the design exceeds the expectations of customers/clients, it may be said to influence the competitive position of the business positively. Thus good design satisfies customer needs, communicates the purpose of the product or service to them and, by selling well, brings financial rewards to the business. Study the short case study entitled “Ocean Observations” in chapter 5 of your prescribed book and answer the two questions that follow in the prescribed book.*
- (2) *With reference to question 2, the authors note the findings of the Design Council in the UK, which confirms that “using design throughout the business ultimately boosts the bottom line by helping to create better products that compete on value rather than price”. It “helps businesses connect strongly with their customers by anticipating their real needs”. It “gives them the ability to set themselves apart in increasingly tough markets”, and “using design both to generate new ideas and turn them into reality allows businesses to set the pace in their markets and create new ones rather than simply responding to the competition”. Note the critical commentary by the authors of your prescribed book concerning the fact that new products and services are not always created in response to a clear and articulated customer need. Sometimes customers are not even aware of certain needs until a new product or service arrives on the market!*
- (3) *With reference to question 3, it must again be emphasised that the final design of a product or service is usually the result of the investment of considerable time and effort in its development. At the outset there is an initial, somewhat vague product/service concept. Eventually, after many hours of screening, evaluation and testing, there is a final design for a product/service that will hopefully be successful and make money. This does*

not happen coincidentally – it is the result of a formal development process. Ignoring or neglecting any of the stages of design may result in a suboptimal product/service design with no potential to gain a competitive advantage for the business when the product is manufactured or the service rendered.

- (4) *With reference to question 4, note that the design of a product/service includes more than the design of the tangible product (the stove or car) or the intangible service (the consultation with your lawyer or the delivery of post to your house). Each of the examples firstly comprises a product/service concept, which encompasses all the expected benefits that the product/service will provide. Secondly, they all represent a combination of a product and a service. Thirdly, more attention should be given to processes (process design) when services are designed. Finally, study the short case study entitled “Spangler, Hoover and Dyson” in chapter 5 of your prescribed book (a case where a radically new design captured a large piece of the market) and answer the four questions that follow in the prescribed book.*
- (5) *With reference to question 5, take note of figure 5.2 in chapter 5 of your prescribed book. This figure basically depicts the input-transformation-output model previously encountered in chapter 1 (see fig 1.2). The inputs will again consist of transformed resources (technical, market information) and transforming resources (design staff and design equipment). The design process will be guided by the objective’s quality (designs that are error-free and fulfil their purpose), speed (designs that have moved from concept to detailed specification in a short period of time), dependability (designs that are delivered when promised), flexibility (designs that incorporate the latest ideas) and cost (designs that do not consume excessive resources). Finally, the output will be a fully detailed specification, which should include details of the overall concept (the form, function, overall purpose and benefits it will provide), its package (the collection of the individual component products and services required to provide and support the concept) and the process by which the design will fulfil its concept.*

6.4 THE STAGES OF DESIGN FROM CONCEPT GENERATION TO PROTOTYPING AND FINAL DESIGN



Activity 6.3

Study the second, third, fourth, fifth, and sixth sections of chapter 5 of your prescribed book, entitled “Concept generation”, “Concept screening”, “Preliminary design”, “Design evaluation and improvement”, and “Prototyping and final design” and then answer the following questions and/or complete the statements:

- (1) The approximate sequence of stages in the design process is as follows:
(a) (b) (c) (d) (e)
- (2) From which sources inside and outside the business do new ideas for products or services originate? How are these ideas translated into product/service concepts?
- (3) Why would reverse engineering be used? Do you think it is an ethical business practice?
- (4) Which design criteria are used in the concept screening stage to assess the worth of the number of options available?
- (5) What does the “design funnel” entail? Which two important considerations does the design funnel clarify?
- (6) What role does creativity play in the design of products and services?
- (7) Illustrate a component (or product/service) structure for the following examples:

(a) a wooden dining-room suite (table and 8 chairs)

(b) a visit to the nearest automatic drive-in car wash

Further, what does design complexity entail and which three approaches may be used to reduce it?

(8) How can QFD (quality function deployment), VE (value engineering) and Taguchi methods be used to evaluate and improve the preliminary design of a cell phone?

(9) What do prototyping and final design stage entail? What benefits may be derived from CAD (computer-aided design)?



Feedback on Activity 6.3

- (1) *With reference to question 1, note the five stages of the design activity as described and illustrated in figure 5.3 of your prescribed book.*
- (2) *With reference to question 2, new product or service ideas may originate from sources outside the business/organisation (customers/clients or competitor activity) or from staff (research and development). Study the short case study entitled “Mxit’s customer driven design philosophy” and answer the two questions at the end. Also read “mpesa – Designing for low income banking clients”. Take note of how the users influenced the way in which the product evolved*
- (3) *With reference to question 3, reverse engineering means taking a product apart to find out how a competing organisation has made it. Interestingly enough, a recent programme on the History Channel of DSTv (South African satellite channel) contained an account of the exact copying of the B-52 American bomber by the Soviets during the Second World War. The ethics of re-engineering remains debatable, but we would venture to say that as long as patents and trademarks are respected, it seems to be acceptable.*
- (4) *With reference to question 4, note that the design criteria used will depend on the nature and circumstance of the particular exercise, but may be considered in terms of the following broad categories: feasibility (can the business do it?), acceptability (does the business want to do it?), vulnerability (does the business want to take the risk?). Study figure 5.4 in your prescribed book and note the broad categories of design criteria and the focus of their evaluation.*
- (5) *With reference to question 5, the design funnel is used to narrow down a large number of design options, as illustrated in figure 5.5 in your prescribed book. This helps to cut down the number of options that will be available further along in the design activity, and consequently reduces uncertainty about the product, service or process. In addition, it shows what it will cost to change one’s mind about details of the design – during the early stages of the design activity the costs of change are relatively low, but later on changes become increasingly expensive. For a more balanced view, read the critical comments by the authors of your prescribed book on the design funnel. The authors say (a) that managers do not necessarily start out with an infinite number of options; (b) that the number of options often increases as the design activity progresses; and (c) that the real process of design often involves back-tracking, “as potential design solutions raise fresh questions or become dead ends”.*
- (6) *With reference to question 6, creativity per se is a very difficult aspect to quantify in terms of its contribution or effect. Creativity is closely linked to innovation, which is an important prerequisite for the long-term survival of many if not most industries.*
- (7) *With reference to question 7, a component structure similar to figure 5.6 in your prescribed book should be drawn up for the dining room set and the automatic drive-in car wash. The component structure indicates the order in which the component parts of the package have to be put together, while the bill of materials (BOM) specifies the quantities of each*

component part required to make up the total package. Note that the purpose of the preliminary design is “a first attempt at both specifying the component products and services in the package, and defining the processes to create the package”. Because the best design is usually the simplest, design complexity may be reduced by following these approaches: standardisation (restrict variety to that which has real value for customers/clients); commonality (use the same components across a range of products or services); and modularisation (design standardised subcomponents that can be put together in different ways). Study the worked example that deals with “Activo Health Club” in chapter 5 of your prescribed book. Note the emphasis on designing products and services while at the same time defining the process that will be used for their production and the linkage between the component structure (fig 5.7 in your prescribed book) and process map (fig 5.8 in your prescribed book).

- (8) With reference to question 8, quality function deployment, value engineering and Taguchi methods are three techniques that may be used in the design evaluation and improvement stage of the design of products and services. They are used to investigate the preliminary design to see if it can be improved before it is tested in the market. Note figure 5.9 in your prescribed book as an example of a QFD matrix. This matrix is referred to as the “house of quality” (because of its shape) and the “voice of the customer” (because of its purpose). Figure 5.10 in your prescribed book gives a further example of linked QFD matrices, where the “hows” of one matrix form the “whats” of the next. The basic purpose of VE is to try to reduce costs and prevent any unnecessary costs that do not contribute to the value and performance of the product and service. Taguchi methods test the “robustness” of a design. The product/service thus needs to perform even in extreme conditions and must be able to cope with all probable eventualities.
- (9) With reference to question 9, prototyping and final design are the last stage in the design of products and services. The purpose is to take the improved design at this stage of the design activity and turn it into a prototype that can be tested in the market. Nowadays computers make it possible to create a virtual prototype, which can be tested with virtual reality-based simulations. The advantages of CAD is that the design data can be stored electronically and retrieved easily, making it possible to manipulate or change designs quickly and inexpensively. It can thus serve as a prototyping device as well as a drafting device, and when it is combined with the virtual prototyping approach, the designer can model the design to assess its suitability prior to full production.

At this point you might be somewhat surprised at the amount of work that goes into the design of products/services we use every day to satisfy our needs. Not only should you now have some understanding of the importance of each stage in the sequence of design from concept to final design and specification, but you should also be able to apply the selected tools and techniques for design evaluation and improvement to practical examples.

6.5 INTERACTIVE DESIGN



Activity 6.4

Study the seventh section of chapter 5 in your prescribed book, entitled “The benefits of interactive design” and then answer the questions and/or complete the following:

- (1) Interactive design refers to the merging of
- (2) What are the benefits of reducing the time-to-market (TTM) for a business with many competitors? Are these benefits the same for a business with no or only a few competitors?

- (3) If the following concepts were applied, how could they significantly reduce the TTM for a product or service?
- (a) simultaneous development (also referred to as simultaneous engineering in manufacturing)
 - (b) early resolution of conflict and uncertainty in design
 - (c) adopting more project-based organisational structures for design projects



Feedback on Activity 6.4

- (1) *With reference to question 1, it should be noted that interactive design, which means that the design of the product/service and the design of the processes for its manufacture/provision are both done at virtually the same time or are more closely integrated with one another, offers greater potential to reduce the time-to-market (TTM) of a product or service.*
- (2) *With reference to question 2, figure 5.11 in your prescribed book illustrates the possible effects of a delay in the time-to-market of a product or services. Note that not only are the development costs higher, but there are also delays on sales revenue and cash flow, and therefore a delay in reaching the financial break-even point.*
- (3) *With reference to question 3, to reduce the TTM, many previously sequential activities could be done concurrently or simultaneously (see fig 5.12 in your prescribed book). Quicker decisions impacting on other parties involved in the design process could be taken if uncertainty could be reduced (see fig 5.13 in your prescribed book), and more suitable organisational structures could be selected for project-type or task-team work (see fig 5.14 in your prescribed book).*
-

CHECKLIST



Now that you have completed study unit 6 on the design of products and services, you should be able to do the following:

- explain why good design is important
- distinguish between the sequence of stages in the design of products/services, namely concept generation, concept screening, preliminary design, design evaluation and improvement, prototyping and final design
- justify the use of interactive design
- use the knowledge, insight and skills you have acquired regarding
 - good design
 - the sequence of design stages, namely concept generation, concept screening, preliminary design, design evaluation and improvement, prototyping and final design
 - interactive design
- to justify the importance of design and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions

- show evidence of your appreciation for the design of products and services by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

SUMMARY



Now that you are more knowledgeable about what occurs in each of the stages of the design of products/services, you ought to have a greater appreciation of what design in these areas requires. The previous study unit, study unit 5, along with earlier study units grouped under topic 4, considered process design in production and operations management. This study unit focused exclusively on the design of the products and services. The next topic continues to focus on design activities within production and operations management, but also deals more specifically with the design of the supply network.

ASSESSMENT



- (1) Consider the case study entitled “Chatsworth House – the adventure playground decision” at the end of chapter 5 and then answer the three questions posed at the end.
- (2) Do problems 1 to 5 at the end of chapter 4 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 5 at the end of chapter 5 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 5

Design and location of production/operations capacity

AIM



To provide learners with an understanding of what the design of the supply network entails and why it is important for production/operations management.



Learning outcomes

When you have worked through this study unit, you should be able to do the following:

- explain what the supply network perspective entails
- explain how decisions relating to the configuration of the supply network could impact on the production/operations capacity
- explain what the location of the production/operations capacity entails
- demonstrate how the production/operations capacity should be managed in the long term
- use the knowledge, insight and skills you have attained regarding
 - the supply network perspective
 - decisions relating to the configuration of the supply network and its impact on production/operations capacity
 - the location of production/operations capacity and
 - the management of production/operations capacity in the long term
- to justify the importance of the management of production/operations capacity and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation of the design and location of the production/operations capacity by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 7

Supply network design



Key concepts

supply network perspective	wide process span
supply network	in-house supply
supply side	outsourced supply
demand side	operations location decision
"first tier"/"second tier" suppliers and customers	spatially variable costs
customers/clients of customers and clients	supply-side influences
suppliers of suppliers	demand-side influences
immediate supply network	weighted-score method
total supply network	centre-of-gravity method
customer/client-supplier relationship	optimum capacity level
downstream (forward)	variable/fixed cost
upstream (backward)	long-term capacity management decision
outsourcing	fixed cost breaks
vertical integration	economies/diseconomies of scale
location	average cost/unit
long-term capacity management	demand-capacity balance
disintermediation	scale of capacity
co-opetition	capacity leading strategies
business process outsourcing (BPO)	over-capacity (low capacity utilisation)
direction of vertical integration	under-capacity (high capacity utilisation)
extent of vertical integration	capacity lagging strategies
balance among stages	capacity leads/lags
narrow process span	"smoothing" with inventory strategies
	break-even analysis of capacity extension

GETTING AN OVERVIEW

In this study unit we address all four learning outcomes of topic 5. We explain what the supply network perspective entails, and how decisions relating to the configuration of the supply network could impact on the production/operations capacity. We also explain what the location of the production/operations capacity entails, and demonstrate how the production/operations capacity should be managed in the long term.

7.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 6 in your prescribed book. Then study figure 6.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 6 in your prescribed book:

- (1) "Why should an organisation take a total supply network perspective?" (1 mark)
- (2) "What is involved in configuring a supply network?" (3 marks)
- (3) "Where should an operation be located?" (1 mark)
- (4) "How much capacity should an operation plan to have?" (3 marks)

7.2 OPERATIONS IN PRACTICE



Activity 7.1

Read the case example entitled "Botswana diamond extraction and recovery" and then answer the following questions:

- (1) The whole range of operations processes are mentioned in this case study, beginning with the feeding of diamond-rich ore into the crushes, up to the stage where the diamonds are sent to jewellers for mounting. In your opinion, at what stage does the "integrated supply chain" end in this instance?



Feedback on Activity 7.1

- (1) *This case focuses largely on the processes carried out within a system called CARP (Completely Automated Recovery Process) and FISH (Fully Integrated Sort House). In the context of vertical integration, it would seem that these systems fall under a network of processes which are owned by the Botswana Diamond Industry. The supply network only extends beyond these systems when the diamonds "in sealed containers" are transported to the airfield for final shipment out of the country. From this point onwards, value is provided through various external parties who sell the diamonds at "sights", then polish, finish and ultimately mount the diamonds in finished jewellery. The "integrated supply chain" in this instance therefore ends where we differentiate between the internal processes owned by the Botswana Diamond Industry and the external processes, where other parties (also part of the supply network) add further value to the diamonds.*
-

7.3 SUPPLY NETWORK PERSPECTIVE



Activity 7.2

Study the first section of chapter 6 entitled “The supply network perspective” and then answer the following questions:

- (1) What are the main features of a supply network?
- (2) Why should the whole supply network be considered in design decisions relating to production and operations management (POM)?
- (3) What design decisions need to be taken with regard to the supply network?



Feedback on Activity 7.2

- (1) *With reference to question 1, study figure 6.2 in your prescribed book. This figure illustrates the supply networks for a plastic homeware manufacturer and a shopping mall. Note the meaning of the key words/concepts: supply network, supply side, demand side, first-tier, second-tier, immediate supply network and total supply network.*
 - (2) *With reference to question 2, the network perspective offers a better opportunity to consider the implications of design decisions in POM at the strategic level. Understanding and improving the various customer-supplier links or relationships within a particular supply network helps the operation to understand how it can compete effectively. It helps the business/organisation to identify significant links in the network and to focus on its long-term position in this network.*
 - (3) *With reference to question 3, the design decisions in supply networks that need to be taken are (i) how the supply network is to be configured (shape and extent of ownership or vertical integration decision); (ii) where each part of the network is to be located (location decision); and (iii) what the physical capacity of each part of the (owned) network has to be at any point in time (long-term capacity management decision).*
-

7.4 CONFIGURING THE SUPPLY NETWORK



Activity 7.3

Study the second section of chapter 6, entitled “Configuring the supply network”, and then answer the questions below and/or complete the statements:

- (1) How can the shape of the supply network be changed? What does “disintermediation” mean and what are the advantages for customers/clients? What does “co-opetition” mean and what is the implication of customers and suppliers having “symmetric” roles?
- (2) What does the “do or buy” decision (also referred to as the vertical integration decision) involve? How does it relate to outsourcing and business process outsourcing (BPO)?
- (3) An organisation’s vertical integration strategy may be defined in terms of
 - (a) the ... , which can be expanding strategically on the supply side of the network (which is called ...) and/or expanding on the demand side of the network (which is called ...)

- (b) the ... , which determines how far the business chooses to integrate, if at all, from its original part of the network
- (c) the ... among stages, which can either be total (meaning ...) or less than fully balanced (meaning ...)
- (4) How would the choices relating to do or buy, vertical integration or not, in-house or outsource, affect the performance objectives of an operation?



Feedback on Activity 7.3

- (1) *With reference to question 1, note that even when an operation does not directly own or even control other parts of the supply network, it may use its influence to change the shape of the network itself. This move is referred to as network reconfiguration. It means that a business/organisation can attempt to merge parts of the network with the operation, or can reduce the number of suppliers it has direct contact with. Also study the short case study entitled "Automotive components in South Africa" in chapter 6 and answer the questions that follow in the prescribed book. Disintermediation means that a business/organisation may bypass customers or suppliers in the supply network and make direct contact with these customers' customers or suppliers' suppliers. It is therefore a way of "cutting out the middleman". Co-opetition is the term used to capture the idea that all parties or "players" in a particular supply network (whether they are customers, suppliers, competitors or complementors) can be either "friends" or "enemies" at different times. In this regard, a complementor makes a particular business's products or services more valued by customers because of the way the complementor's products or services complement the business's own. Competitors are the opposite in that their products or services detract from the value that customers attribute to a particular business's own products and services. However, competitors can also be complementors, and vice versa. The term used to capture this idea is called "co-opetition". It is thus considered important to distinguish between the way businesses cooperate in increasing the total size of the market, and the way they compete for a share of that market. Note that because customers and suppliers have "symmetric" roles, harnessing the value added by a supplier is just as important as listening to the needs of a customer. Therefore, destroying the value of a supplier to create value in a customer does not increase the total value of the network. Thus, in the long term, it is better to find ways to increase the total value of the network for both customers and suppliers alike, rather than increasing the value of one at the expense of the other.*
- (2) *With reference to question 2, it should be noted that most (if not all) businesses/organisations do not "do everything and all things alone" to produce their products and/or services. They not only buy-in their transformed input resources (materials, information, customers/clients), but also some of their transforming input resources (facilities, staff). This action is referred to as outsourcing, which means that a business/organisation buys-in (or outsources) parts, components or subassemblies from specialist suppliers. Study the short case example entitled "Behind the brand names" in chapter 6 of your prescribed book and answer the questions at the end. It is estimated that around 60 percent of all notebook computers (including those bearing well-known brand names like Dell, Compaq, Apple, etc) are manufactured (therefore outsourced) by Taiwanese firms, which achieve the economies of scale that come with high volume. Taking this concept even further, some businesses/organisations may outsource some of their indirect processes. This is called business process outsourcing (BPO). Note the critical comments made by the authors of your prescribed book about BPO and the negative reaction of trade unions to BPO, because it threatens employees' job security – sometimes staff find that they themselves have been outsourced.*
- (3) *With reference to question 3, note that decisions relating to the vertical integration of the capacity of the production/operations network may impact on its direction (upstream*

or downstream), its extent (narrow or wide process span), and its balance among stages (total or less than full balance). This is illustrated by figure 6.3 in your prescribed book.

- (4) With reference to question 4, note the influence that the “do or buy, vertical integration or not, in-house or outsource” decisions may have on the operation’s performance objectives, including quality, speed, dependability, flexibility and costs, as illustrated in table 6.1 of your prescribed book. Clearly, the decision whether to utilise in-house or outsourced supply must be considered in view of the possible effect on the operation’s performance objectives as well as the future. A number of questions need to be answered before the decision is taken: Is the activity of long-term strategic importance? Does the business/organisation have the necessary specialised knowledge to perform the task? Would the business/organisation perform better with an in-house or an outsourced supply? Is significant improvement in operations performance likely? These questions must be answered in a logical sequence (see fig 6.4 in your prescribed book) to arrive at the “best” possible decision.
-

7.5 THE LOCATION OF CAPACITY



Activity 7.4

Study the third section of your prescribed book, entitled “The location of capacity”, and then answer the questions and/or complete the following:

- (1) Why is the location of the production/operations capacity so important?
- (2) Location decisions are prompted by changes in (which by the increase in its aggregated volume means capacity) and changes in (which by either becoming too expensive means and/or more scarce because).
- (3) What are the objectives of location decisions in profit-seeking organisations and nonprofit organisations? Why do they differ?
- (4) How can the supply-side factors and demand-side factors influence the location of an operation?
- (5) The location decision must be made at three levels:
 - (a) for, such as South Africa, Europe
 - (b) for, such as Gauteng, Western Province, Kwazulu-Natal
 - (c) for, such as plot 233 Rosslyn, plot 564 Pretoria North
- (6) How can the weighted-score and centre-of-gravity methods be used to make better location decisions?



Feedback on Activity 7.4

- (1) With reference to question 1, it should be noted that decisions relating to the location of the production/operations capacity are important. They may impact on the cost base of the operation itself and the business’s ability to serve its customers/clients (and thus its revenue-earning potential). Also, once the decisions have been made, it is often very difficult and expensive to reverse them or to relocate. Study the short case example entitled “Mr Delivery” in chapter 6 and answer the question that follows in the prescribed book. Note that the location decision has two aspects, namely (i) whether the company should open a theme park in Europe and (ii) where in Europe the theme park should be located.
- (2) With reference to question 2, note that location decisions are prompted by changes in the demand for goods and services, and changes in the supply of inputs. When goods and

services change, the operation may need more space. An existing site could be expanded, or a larger site could be found somewhere else. Alternatively, a second site could be used in addition to the original one. All these options involve location decisions. Changes in the supply of inputs may mean that inputs become too expensive. The operation may then relocate to a part of the world where the labour costs are lower and/or inputs are less expensive or more readily available.

- (3) With reference to question 3, the objectives of the location decision are to achieve an appropriate balance between (i) the spatially variable costs of the operation or the costs that change with the geographical location; (ii) the service the operation is able to offer its customers/clients; and (iii) the revenue potential of the operation. Note that in for-profit organisations the last two points are of prime importance. In not-for-profit organisations the first two points are of prime concern because the revenue potential is not relevant.
 - (4) With reference to question 4, it should be noted that the location decision of any operation is determined by the relative strength of supply side influences (labour, land costs, energy, transportation costs and other community factors) and demand side influences (labour skills, suitability of site, image of the location, convenience for customers/clients), as illustrated in figure 6.5 of your prescribed book. Community factors form a supply side influence and include the costs derived from the social, political and economic environment of the proposed site. This may often turn out to be the single most important factor of all. Study the short case example entitled "Aluminium smelters in Africa" and answer the questions that follow in the prescribed book.
 - (5) With reference to question 5, the location decision must be made at three levels. It should be determined (i) in which country the operation should be located, but it will also have to be decided (ii) in which region of the country and (iii) on which specific site in that region the operation needs to be located.
 - (6) With reference to question 6, study the worked example of the Irish company that had to decide where to build a new factory in one of the Benelux countries and used the weighted-score method. Also read the worked example of the company that operated four out-of-town garden centres, but kept all its stock in one single warehouse. How could the company use the centre-of-gravity method in chapter 6 of your prescribed book to select the best site?
-

7.6 THE LONG-TERM MANAGEMENT OF CAPACITY



Activity 7.5

Study the fourth section in your prescribed book entitled "Long-term capacity management" and then answer the following questions:

- (1) What does the long-term management of capacity entail?
- (2) How would the following decisions relate to
 - (a) determining the optimum capacity level for an operation (including the issues of the scale of capacity and the demand-capacity balance)?
 - (b) balancing various capacity levels of operations in the supply network?
 - (c) timing capacity changes in each part of the network (including the use of capacity-leading, and capacity-lagging strategies and the practice of "smoothing" with inventory), which impact on the long-term management of production/operations capacity?
- (3) How could break-even analysis be used in examining capacity expansion?



Feedback on Activity 7.5

- (1) With reference to question 1, having determined the configuration of the supply network and direction, the span and balance of vertical integration (see section 7.3 in this study guide) and the location of production/operations capacity (see section 7.4 in this study guide), the next step is to determine the size or capacity of each part of the network. At this point a long-term general view of capacity is taken and the forecasts of demand (see topic 8, study unit 13, section 13.4) is one of the main inputs used.
- (2) With reference to question 2 (i), the optimum capacity level is where the average cost of producing a certain level of output is the lowest. The average cost = total cost at the specific output level (both fixed and variable costs) divided by the output level. Read the discussion that precedes and follows figure 6.8 in your prescribed book and note the explanation of the terms “fixed cost breaks”, economies and diseconomies of scale. Note that the scale of capacity must be in balance with the projected or forecast demand (referred to as demand-capacity balance). That means that if the units of capacity used are too large, the business may have substantial amounts of over-capacity for periods when the demand is on the increase – see figure 6.9 in your prescribed book.

With reference to question 2 (ii), it should be noted that all parts or stages of the production/operations network must have the same capacity, otherwise bottlenecks will occur in the supply network. If not, the capacity of the whole network will be limited to the capacity of the slowest link.

With reference to question 2 (iii), the timing of capacity change is not just about deciding on the best size of a capacity increment. It also means deciding when new “on-stream” capacity must be brought in. Study figure 6.10 in your prescribed book and note how important the timing of capacity change is. Note the arguments for and against pure leading and pure lagging strategies of capacity timing as summarised in table 6.4 of your prescribed book. Also note that it is possible to implement a strategy that does not result in the accumulation of inventories. This requires that the capacity is introduced in such a way that demand is met by a combination of production and inventory. Look at figure 6.11 and table 6.5 in your prescribed book, which highlight the advantages and disadvantages of a “smoothing with inventory” strategy.

- (3) With reference to question 3, an alternative view of capacity extension is presented. This view considers the cost implication of adding increments of capacity on a break-even basis. Figure 6.12 in your prescribed book illustrates how a capacity increase may move an operation from profitability to loss as the result of the additional lump of expenditure of fixed costs (fixed-cost break), which has to be incurred before additional volumes result and revenues increase. Study the worked example of the business process outsourcing company considering building a processing centre in India, and note that only by quarter six and thereafter is the sales revenue (cash inflow) sufficient to cover the operating and capital costs (cash outflows). Also study the worked example of the specialist graphics company in chapter 6 of your prescribed book and pay careful attention to the calculation of the specialist printing company’s profits for years one and two.

CHECKLIST



Now that you have completed study unit 7, you should be able to do the following:

- explain what the supply network perspective entails
- explain how decisions relating to the configuration of the supply network could impact on the production/operations capacity

- explain what the location of the production/operations capacity entails
- demonstrate how the production/operations capacity should be managed in the long term
- use the knowledge, insight and skills you have acquired regarding
 - the supply network perspective
 - the decisions relating to the configuration of the supply network and its impact on the production/operations capacity
 - the location of the production/operations capacity, and
 - the management of the production/operations capacity in the long term
- to justify the importance of the management of production/operation capacity and to demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation of the supply network design by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

NOTE: The supplement to chapter 6 entitled “Forecasting” in your prescribed book (2010:166–173) is discussed in topic 8, study unit 11.5, Activity 11.4, entitled “Measuring demand and capacity in operations”.

SUMMARY



The supply network consists of all the various first-tier, second-tier, third-tier etc customer-supplier relationships involved in the manufacture/provision of a particular product/service. This study unit examined certain design considerations relating to the basic form of the supply network, the configuration of this network and its impact on production/operations, and the location and long-term management of the production/operations capacity.

Having first considered the design of processes (topic 4, study unit 5), the design of products and services (topic 4, study unit 6) and the design of the supply network (topic 5, study unit 7), we now move on to the design of the layout and workflow (topic 6, study unit 8).

ASSESSMENT



- (1) Consider the case study entitled “Delta Synthetic Fibres” at the end of chapter 6 and then answer the four questions posed at the end.
- (2) Do problems 1 to 8 at the end of chapter 6 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 6 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 6

Design of layout and workflow

AIM



To provide learners with an understanding of what is entailed in the design of the layout and workflow of an operation and why it is important for production/operations management.



Learning outcomes

When you have worked through this study unit, you should be able to do the following:

- explain what the layout of an operation entails
- explain what the basic layout types are and how one would be selected
- use the knowledge, insight and skills you have acquired regarding
 - the layout procedure of the operation, and
 - the basic layout types and their selection
- to justify their importance and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation of the design of layout and workflow by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 8

Layout and flow of an operation



Key concepts

layout of an operation or process
positioning of transforming resources
allocation of tasks to transforming resources
workflow through the operation
layout decision
inherent safety
length of flow
clarity of flow
staff conditions
management coordination
accessibility
use of space
long-term flexibility

process types
physical manifestation
volume-variety characteristics process type spectrum
basic layout types
fixed-position layout
process layout (lately called functional layout)
cell layout
product layout
mixed layouts
hybrid layout
intermittent workflow
continuous workflow

GETTING AN OVERVIEW

In this study unit we address both learning outcomes of topic 6. We explain what the layout of an operation entails, and explain the basic layout types and how to select one.

8.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 7 in your prescribed book. Then study figure 7.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally read the points listed under the following questions at the end of chapter 7 in your prescribed book:

- (1) "What are the basic layout types used in operations?" (1 mark)
- (2) "What type of layout should an operation choose?" (2 marks)
- (3) "What is layout design trying to achieve?" (1 mark)
- (4) "How should each basic layout type be designed in detail?" (4 marks)

8.2 OPERATIONS IN PRACTICE



Activity 8.1

Read the case example of “Layout impacts supermarket profits” in chapter 7 of your prescribed book and then answer the following questions:

- (1) What aspects need to be considered in the layout of a supermarket?
- (2) Which “general rules” regarding the actual location or placement of products can be followed to maximise sales?



Feedback on Activity 8.1

- (1) *With regard to question 1, we may initially think that the layout of a supermarket is pretty complicated, but in reality it has much in common with layout decisions faced by other operations. Of prime importance is that the supermarket “must maximise the revenue and contribution per square metre as well as minimising the costs of operating the store, in terms of material handling and check-out productivity, etc. Customers’ comfort and convenience is an important factor, as is the ability to evacuate the store quickly in the case of an emergency ... At a basic level, supermarkets have to get the amount of space allocated to the different areas right.” The space allocated for checkouts, circulation, the width of aisles, etc must all be carefully considered.*
 - (2) *With reference to question 2, the actual location of products is very important, because it affects customers’ convenience, the level of spontaneous purchasing and the cost of filling the shelves. Fruit and vegetables are often displayed close to the main entrance to convey the idea that they are fresh and wholesome. Basic products such as flour, sugar and bread are placed at the back of the store and away from each other to encourage customers to pass many higher margin items. High margin items themselves are displayed at eye level on shelves, while lower margin items are placed lower down or higher up. Impulse-bought goods should not be displayed in the first metre or so of an aisle, because this space is considered “dead space”. The “gondolas” or shelves at the end of the aisle have to be fully utilised, because products displayed there may sell 200 to 300 percent better.*
-

8.3 THE LAYOUT OF AN OPERATION



Activity 8.2

Study the first section of chapter 7 in your prescribed book entitled “What is layout” and then answer the questions and/or complete the following:

- (1) What does the layout of an operation or process entail and why is it an important operations design decision?
- (2) The general objectives of a good layout that are relevant to all operations are
 - (a) inherent safety, which refers to
 - (b) length of flow, which refers to
 - (c) clarity of flow, which refers to
 - (d) staff conditions, which refers to
 - (e) management coordination, which refers to

- (f) accessibility, which refers to
- (g) use of space, which refers to
- (h) long-term flexibility, which refers to
- (3) What is the relationship between the process types (for both manufacturing and services) and the basic layout types?



Feedback on Activity 8.2

- (1) *With reference to question 1, the layout of an operation or process refers to the positioning of the transforming resources relative to each other, including the allocation of the various tasks to these transforming resources. Layout therefore dictates the pattern of flow for transformed resources as they progress through the operation (see fig 7.2 in your prescribed book). As noted by the authors in the previous edition of your prescribed book (2004:205), there is "double pressure" on the layout decision because of the importance of adopting the correct layout from the beginning. If the layout is later found to be faulty and has to be changed, this causes disruption and customer dissatisfaction – in some cases it might even be impossible to change the layout later. A wrong or unsuitable layout might have serious consequences, including over-long or confused flow patterns, excessive inventory, customer queues becoming long, customers being inconvenienced, long process times, inflexible operations, unpredictable flow and high costs.*
 - (2) *With reference to question 2, note the discussion in your prescribed book of the general objectives of the layout decision relevant to all operations. After studying this section you should appreciate that the positioning of transforming resources and allocation of tasks of an operation are NOT (or should not be viewed as) merely "an incidental happening, which can be rushed and be done with as soon as possible!"*
 - (3) *With reference to question 3, study table 7.1, which illustrates the relationship between process types and basic layout types. The layout of an operation is related to the selection of an appropriate process type as described earlier, but in a narrower sense, as it represents the physical manifestation of a particular process type. Note that the product layout is more suited to a continuous process (eg an electricity utility), while the fixed-position layout is better suited to a project process (eg the construction of a dam wall). Similarly, the product or cell layout is preferred for a mass service (eg an international airport), while the fixed-position layout should be used for a professional service (eg a dentist). Where more than one type of process is possible for a particular operation, the relative importance of the operation's performance objectives will influence the decision.*
-

8.4 THE BASIC LAYOUT TYPES



Activity 8.3

Study the second section of chapter 7 of your prescribed book entitled "The basic layout types" and then answer the following questions:

- (1) What are the unique characteristics of each of the four basic layout types?
- (2) What is a "mixed layout" and in which circumstances would such a layout be used?
- (3) How would high/low volume and/or high/low variety affect the importance and feasibility of flow in operations?

- (4) What are the advantages and disadvantages of each of the basic layout types that could influence the final selection of a layout type?



Feedback on Activity 8.3

- (1) With reference to question 1, note the discussion and practical examples of types of operation with regard to each layout type in chapter 7 of your prescribed book. Also study the short case examples entitled “Factory flow helps surgery productivity” for the fixed-position layout, and “Grootgeluk Coal Mine” and “Maropeng: cradle of humankind” for the product layout. Then answer the questions that follow in the prescribed book. For the functional (process) layout, study figure 7.4; for a cell layout, figure 7.5; and for a product layout, figures 7.6 and 7.7 in your prescribed book.
 - (2) With reference to question 2, a mixed layout comprises the use of the pure layout types for different parts of an operation. Study figure 7.8 in your prescribed book and note the appearance of a fixed-position layout (in the service restaurant), a process layout (in the kitchen), a cell layout (for the buffet) and a product layout (in the self-help cafeteria). Hybrid layouts can also be chosen – this means that the operation itself designs the layout, using a combination or elements of some or all the pure basic layout types.
 - (3) With reference to question 3, study figure 7.9 in your prescribed book. Note that the selection of a basic layout type depends on the volume-variety characteristics of a particular operation. In the top left-hand corner of the table the flow is intermittent (low volume with high variety), but in the bottom right-hand corner of the table the flow becomes continuous (high volume with low variety). “Stepping” down diagonally from top left to bottom right, the fixed-position, process layout, cell layout and product layout (with or without overlaps between the preceding and/or following layout-type) would be the most appropriate.
 - (4) With reference to question 4, note that the final choice of a basic layout type may appear somewhat difficult, because it is possible that one or more of the layout types may be suitable for two seemingly similar operations. (Note that by considering the volume-variety characteristics of the operation, the number of options is already narrowed down to one or two.) It is therefore also important to give careful consideration to the advantages and disadvantages of each layout type as it would apply to each particular operation in practice. Table 7.2 in your prescribed book highlights some of the advantages and disadvantages of the basic layout types. According to the authors of your prescribed book, another, perhaps even more significant factor when deciding on the final layout type is the unit cost implications of each layout choice. Figure 7.10(a) illustrates the total costs associated with each layout type. (Fixed and variable costs element must be considered separately before a particular layout type is adopted.) Note that total costs depend on the volume in units of output of the operation. In practice it is extremely difficult to draw an exact distinction between fixed and variable costs, therefore broad bands for lowest-cost basic layouts may be used to suggest one or more appropriate basic layout types. See figure 7.10(b) in your prescribed book.
-

CHECKLIST



Now that you have completed study unit 8 on the layout and workflow of an operation, you should be able to do the following:

- explain what the layout of an operation entails
- explain what the basic layout types are and how one would be selected
- use the knowledge, insight and skills you have acquired regarding
 - the layout of the operation, and
 - the the basic layout types and their selection
- to justify the importance of the layout and to demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- provide evidence of your understanding of the layout and flow of an operation by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

NOTE: The whole of the third section of chapter 7 entitled “Detailed design of the layout” in your prescribed book (2010:189–205) may be left out and should NOT be studied for examination purposes for MNO2601. Also ignore the case study entitled “Weldon Hand Tools” and you do NOT have to be able to do problems 1 to 4 at the end of chapter 7 of your prescribed book.

SUMMARY



In studying the design of processes (topic 4, study unit 5), we have considered the design of products and services (topic 4, study unit 6) and related process design, the design of the supply network (topic 5, study unit 7), and the design of the layout and workflow (topic 6, study unit 8). We now move on to the third aspect of the design of processes, namely the design of jobs and the organisation of work.

ASSESSMENT



(1) Complete study activities 1 to 4 at the end of chapter 7 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 7

Job design and work organisation

AIM



To provide understanding of what the design of jobs and work organisation entails and why it is important for production/operations management.



Learning outcomes

When you have worked through this study unit you should be able to do the following:

- explain the nature and elements of job design and work organisation
- explain the role of ergonomics in designing for environmental conditions and designing the human interface
- explain what it means to design task allocation through the division of labour
- explain how the design of job methods and job time is influenced by the principles of the scientific management approach
- explain the influence of behavioural approaches on designing for job commitment
- use the knowledge, insight and skills you have acquired regarding
 - the nature and objectives of job design and work organisation
 - the role of ergonomics in designing for environmental conditions and the technological-human interface
 - designing task allocation through the division of labour
 - designing job methods and job time through the principles of the “scientific management” approach and
 - designing for job commitment through the influence of the “behavioural approaches” to job design and work organisation
- to justify the importance of job design and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for job design and work organisation by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 9

Elements, objectives and approaches to job design



Key concepts

job design	activity sampling
work organisation	designing for job commitment
key elements of job design	behavioural approaches to job design
ergonomics	quality of working life (QWL)
human factors engineering	techniques of job design
ergonomic environmental design	combining tasks
occupational health and safety	forming natural work units
working temperature	establishing client relationships
illumination levels	vertical loading
noise levels	opening feedback channels
ergonomics in the office	core job characteristics
ergonomic workplace design	skills variety
repetitive strain injury (RSI)	task identity
anthropometric aspects	task significance
neurological aspects	autonomy
designing task allocation	feedback
division of labour	mental states
designing job methods	performance and personal outcomes
scientific management	job rotation
work study	job enlargement
method study	job enrichment
work measurement	empowerment
principles of motion economy	suggestion involvement
qualified worker	job involvement
defined level of performance	high involvement
specified job	team working
standard performance	team-based work organisation
basic time	flexible working multitasking
time study	time flexibility
rating	flexi time work
standard times	location flexibility
allowances	teleworking
synthesis from elemental data	control of staff versus job commitment
predetermined motion-time	
synanalytical estimating times	
(PMTS)	

GETTING AN OVERVIEW

In this study unit we address all the learning outcomes of topic 7. We explain the nature and elements of job design and work organisation, explain the role of ergonomics in designing for environmental conditions and designing the human interface, explain what it means to design task allocation through division of labour, explain how the design of job methods and job time is influenced by the principles of the scientific management approach, and explain the influence of behavioural approaches on designing for job commitment.

9.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 9 in your prescribed book. Then study figure 9.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 9 in your prescribed book:

- (1) "What is job design?" (1 mark)
- (2) "What are the key elements of job design?" (1 mark)
- (3) "How do we go about designing jobs and organizing work?" (4 marks)

9.2 OPERATIONS IN PRACTICE



Activity 9.1

Read the case example of "The Giza Quarry Company" in chapter 9 of your prescribed book and then answer the following questions:

- (1) Why is job design such an important aspect of managing the staff at Giza Quarry Company?
- (2) What are the main aspects of job design for quarry operations at the company?



Feedback on Activity 9.1

- (1) *With regard to question 1, designing jobs for operations that are physically demanding and have to be performed under harsh environmental conditions is particularly important in managing staff, as "badly designed jobs will quickly have an impact on the motivation and physical safety of all operations staff".*
- (2) *With reference to question 2, note that the jobs designed must firstly be appropriate for the harsh environmental conditions (desert). Secondly, the company needs to work with the technology available to them (the quarry site lies in a designated environmentally*

sensitive area, which rules out the use of large machinery and generally excludes all but manual operations). Thirdly, specific tasks have to be allocated to specific members of staff based on skills levels (eg, blocks are cut from the quarry face and then roughly cut to size – a semi-skilled task – after which the blocks are dressed – a highly skilled task). Methods of performing each task need to be devised, time estimates must be made to determine the number of people needed, etc. It is very important to motivate and build commitment among the employees, because the work and working conditions are extremely demanding!

9.3 NATURE AND KEY ELEMENTS OF JOB DESIGN



Activity 9.2

Study the first section of chapter 9 of your prescribed book, entitled “What is job design?” and then answer the following questions:

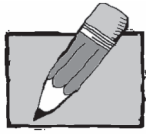
- (1) What is the purpose of job design and why is it particularly important in production/operations management?
- (2) What elements of job design need to be considered when defining people’s jobs/tasks and how would these elements apply to practical examples of jobs/tasks?



Feedback on Activity 9.2

- (1) *With reference to question 1, according to the authors of your prescribed book, the essence of job design (how each individual’s job is structured, the workplace or environment in which they work and the interface with technology) and work organisation (the organisation of the operation as a whole, material, technology and people) is to define “the way in which people go about their working lives”. It includes “their expectations of what is required of them, ... influences their perceptions of how they contribute to the organization, ... defines their activities in relation to their colleagues and channels the flows of communication between different parts of the operation ... [and] develops the culture of the organization – its shared values, beliefs and assumptions”.*
 - (2) *With reference to question 2, note that the descriptions of the job/task and the stipulations as to who is to perform it (or the questions of what the tasks are and who should be responsible for them) are merely one of the key elements of job design. It is just as important to consider the environmental factors (What are the environmental conditions at the workplace?), the technology interface (What technology is available and how will it be used?), the appropriate methods (What is the best method of performing each job?), the duration of the job and the number of people working together (How long will it take and how many people will be needed?) and their commitment to the job (How do we maintain their commitment?). Also study figure 9.2 in your prescribed book. Note how the six key elements of job design are illustrated in the form of a puzzle – all the pieces must fit together precisely to make up the full picture.*
-

9.4 THE ROLE OF ERGONOMICS IN DESIGNING FOR ENVIRONMENTAL CONDITIONS AND DESIGNING THE HUMAN INTERFACE



Activity 9.3

Study the second and third sections of chapter 9 of your prescribed book entitled “Designing environmental conditions – ergonomics” and “Designing the human interface – ergonomic workplace design” and then answer the questions and/or complete the following:

- (1) Ergonomics is concerned primarily with ... and involves two aspects, namely ... and
- (2) What does ergonomic environmental design entail? How would occupational health and safety issues affect the design of the workplace? How could ergonomic design principles affect the office environment?
- (3) What does ergonomic workplace design entail? How should anthropometric and neurological aspects be taken into consideration?



Feedback on Activity 9.3

- (1) *With reference to question 1, note that ergonomics is primarily concerned with the physiological aspects of job design, that is, how the human body fits into its surroundings. It is concerned with how a person interfaces with both environmental conditions in the immediate working area (temperature, lighting, noise, etc) and the physical aspect of the workplace (tables, chairs, desks, machines, computers, etc). Ergonomics is sometimes referred to as human factors engineering. It is linked to the idea that (1) there must be a fit between people and the jobs they do and (2) it is important to adopt a scientific approach to job design.*
 - (2) *With reference to question 2, note that the ergonomic environmental design of the workplace must focus on aspects provided for in occupational health and safety legislation or standards (OSHAS 18 000). These aspects include working temperatures (comfortable temperature range, the effectiveness of people, likelihood of accidents when temperatures are above or below the comfortable range), illumination levels (intensity of lighting), and noise levels (damaging effect of excessive noise). Study figure 9.3 in your prescribed book for an illustration of the various design principles and aspects that need to be considered when an office environment is designed. Also study the short case example entitled “Mponeng’s safety strategy focuses on behavioural change” in chapter 9, and answer the questions that follow in the prescribed book.*
 - (3) *With reference to question 3, note that ergonomic workplace design investigates how people interface with the physical aspects of their jobs to avoid repetitive strain injuries (RSI), impaired vision, and so on. It is therefore important to understand how the workplace affects performance, fatigue, physical strain and injury. The anthropometric aspects of job design are those related to people’s size, shape and physical abilities. Neurological aspects focus on how people’s sensory capabilities (ie sight, touch, sound, smell) are engaged in their jobs.*
-

9.5 DESIGNING TASK ALLOCATION – THE DIVISION OF LABOUR



Activity 9.4

Study the fourth section of chapter 9 entitled “Designing task allocation – the division of labour” and then answer the questions and/or complete the following:

- (1) The division of labour means
- (2) What are the advantages of applying division of labour principles in job design?
- (3) The potential drawbacks of highly specialised or highly divided jobs are
 - (a) greater monotony because
 - (b) physical injury because
 - (c) low flexibility because
 - (d) poor robustness because



Feedback on Activity 9.4

- (1) *With reference to question 1, note that the division of labour refers to dividing the total task into a number of smaller manageable parts. Each of these smaller tasks is the responsibility of a single person or team.*
 - (2) *With reference to question 2, the division of a particular job into a number of smaller, more manageable tasks that can each be performed by a different person is a basic principle we know well and practise every day. It has real advantages because it promotes faster learning, facilitates automation and reduces the amount of nonproductive work.*
 - (3) *With reference to question 3, note that despite the above potential advantages of the division of labour, there are also possible drawbacks in terms of monotony (repeating the same task over and over again can become boring), physical injury (over-use of some parts of the body can result in pain, a reduction in physical capability and increasing RSIs), low flexibility (doing only one thing over and over again leads to rigidity that makes change difficult) and poor robustness (having highly divided jobs implies many different stages that could all be affected by the failure of one stage).*
-

9.6 DESIGNING JOB METHODS AND JOB TIME – SCIENTIFIC MANAGEMENT



Activity 9.5

Study the fifth and sixth sections of chapter 9, entitled “Designing job methods – scientific management” and “Work measurement in job design”, and then answer the questions and/or complete the following:

- (1) The basic principles of scientific management are
- (2) Which of the criticisms of scientific management do you believe are valid and probably the most difficult to overcome?
- (3) Work study is a generic term for
 - (a) method study, which is
 - (b) work measurement, which is

- (4) What do each of the basic steps of method study entail and how can the principles of motion economy be used in the development of a new work/job method?
- (5) How can time study as a work measurement technique be used to establish the time needed for a qualified worker to carry out a specified job at a defined level of performance? Which other work measurement techniques can be used?



Feedback on Activity 9.5

- (1) *With reference to question 1, note the basic principles of scientific management as summarised in chapter 9 of your prescribed book. Based on this approach, a discipline called “work study” (consisting of two separate but related fields of study, namely method study and work measurement) eventually emerged.*
- (2) *With reference to question 2, scientific management may be criticised because it is not particularly scientific and because the method concentrates on relatively limited and sometimes trivial objectives. Also note the first critical comment on scientific management made by the authors of your prescribed book. They say (1) it “inevitably results in standardization of highly divided jobs and thus reinforces the negative effects of excessive division of labour” and (2) it “formalizes the separation of the judgmental, planning and skilled tasks, which are done by ‘management’, from routine, standardized and low-skill tasks, which are left for ‘operators’”. Both these points lead to the same conclusion, namely that “jobs designed under strict scientific management principles lead to low motivation among staff, frustration at the lack of control over their work, and alienation from the job”. Pretty harsh criticism, or perhaps not?*
- (3) *With reference to question 3, note the composition of work study and detailed descriptions of its two constituent parts, namely method study and work measurement, in figure 9.5 of your prescribed book.*
- (4) *With reference to question 4, method study concentrates on determining the methods and activities that should be included in jobs. It is a systematic approach to finding the best method and it follows six steps: (a) selecting the work to be studied, (b) recording the present method, (c) examining the facts, (d) developing a new method, (e) installing the new method and (f) maintaining the method by periodic checking. In step (c), examining the facts, the “questioning technique” is used to detect weaknesses in the rationale for existing methods – see table 9.4 in your prescribed book, which indicates the focus of broad questions and their accompanying detailed questions. In step (d), developing a new method, the principles of motion economy can assist with this process and act as a checklist. The principles of motion economy are described in table 9.5 of the prescribed book. Note the three main focus areas, namely, “using the human body the way it works best”, “arranging the workplace to assist performance” and “using mechanical devices to reduce the human effort”.*
- (5) *With reference to question 5, work measurement is defined by the authors of your prescribed book as “the process of establishing the time for a qualified worker, at a defined level of performance, to carry out a specified job”. A qualified worker is “one who is accepted as having the necessary physical attributes, intelligence, skill, education and knowledge to perform the task to satisfactory standards of safety, quality and quantity”. Standard performance is “the rate of output which qualified workers will achieve without over-exertion as an average over the working day provided they are motivated to apply themselves to the work”. Basic time is “when a qualified worker is working on a specified job at standard performance”. To establish the basic time for a job, several techniques may be used, but the best known technique is time study. Time study involves three steps that are taken to derive the basic times for the elements of a job, namely (a) observing, measuring and rating, (b) adjusting observed times and (c) averaging the basic times.*

Study the worked example in your prescribed book, where the basic time is adjusted to normalise the observed time based on a performance rating of 90, where 100 represents standard performance. Standard time is an extension of the basic time, which is “the time allowed for the job under specific circumstances” – it includes allowances for rest and relaxation because of the conditions under which the job is performed. Allowances are also made for contingencies, interfaces or synchronisation, introduction, unusual conditions, or unoccupied time. Other work measurement techniques mentioned in the prescribed book are (a) synthesis from elemental data, (b) predetermined motion-time systems (PMTS), (c) analytical estimating, and (d) activity sampling. Also note the second critical comment made by the authors of your prescribed book about work measurement. Finally, study the worked example of the two work teams in the Monrovia Embassy in chapter 9 of your prescribed book and indicate whether you agree with the calculations and initial conclusion that Team A was better organised.

9.7 THE BEHAVIOURAL APPROACHES TO JOB DESIGN



Activity 9.6

Study the seventh section of chapter 9 of your prescribed book, entitled “Designing for job commitment – behavioural approaches to job design” and then answer the following questions:

- (1) What was the reasoning behind the movement towards the behavioural approaches to job design?
- (2) Why did the behavioural approaches represent a major shift in the historical approaches to job design at the time?
- (3) How could the “techniques of job design” as termed in the typical “behaviour” job design model by Hackman and Oldham influence the core job characteristics, mental states as they relate to work and finally, the expected performance and personal outcomes?
- (4) What are the differences and/or similarities between job rotation, job enlargement, job enrichment and empowerment, and how could these affect job design?
- (5) What do “team-working” and “flexible working” entail? How could these and the other more traditional approaches to job design affect the degree of control versus commitment?



Feedback on Activity 9.6

- (1) *With reference to question 1, note that the ideas and concepts concerning motivation theory contributed to the behavioural approach to job design. Where jobs were previously designed purely on division of labour, scientific management or even purely ergonomic principles, they alienated the people performing the jobs. Job design should take into account individuals’ need for self-esteem and personal development.*
- (2) *With reference to question 2, the major shift in taking individuals’ desires into account when designing jobs met two important objectives. (i) It provided jobs that have an intrinsically higher quality of working life (QWL), which is an ethically desirable end in itself, and (ii) it brought about higher levels of motivation, which were instrumental in achieving better performance in terms of quality and quantity of output. Thus the new approach to job*

design involves two stages, namely, exploring how various characteristics of the job affect people's motivation, and exploring how individuals' motivation affects their work performance.

- (3) With reference to question 3, study figure 9.7 in your prescribed book. This figure shows a number of recommended job design techniques that affect particular core characteristics of the job itself. They influence various positive mental states, and these in turn lead to certain performance outcomes. Note the further explanations of job design techniques in the first left-hand column of figure 9.7.*
 - (4) With reference to question 4, study the sections in your prescribed book that deal with job rotation (moving periodically between different sets of tasks), job enlargement (increasing the number of tasks or doing a more complete set of tasks), job enrichment (adding extra tasks, which involves more decision making, greater autonomy and therefore greater control over the job), and empowerment (giving staff greater autonomy and the authority to make changes to the job itself). Note the basic difference between job enrichment and job enlargement as illustrated in figure 9.8 of your prescribed book. Also note how empowerment through "suggestion involvement", "job involvement" or "high involvement" can be designed into jobs. Finally, study the short case example entitled "McDonald's lets families share job", answer the questions at the end and note the difference between the declared purpose of the scheme and the other unstated objective. Note also how the building was designed to promote "flexible working".*
 - (5) With reference to question 5, "team-working" or "team-based work organisation" is a development in job design closely linked to the empowerment concept. It means that staff with overlapping skills "collectively perform a defined task and have a high degree of discretion over how they actually perform the task". Teams are a useful organisational device, because they often lead to an improvement in productivity and quality; they encourage innovation; they increase individual job satisfaction and they make it easier to implement technological changes, because such challenges can be shared. For a more balanced view, read the critical comments about teams made by the authors of your prescribed book. Would you agree that these comments seem to be pretty sound? Flexible working goes even further – it allows for multiskilling (individuals can then migrate from one skill or job to another as the demand for their skills changes), time flexibility (apart from certain core working days and hours, and a total for the required compensatable hours that individuals may work when they want to, taking the demand for their services into account), and location flexibility (individuals are not necessarily location-specific and the necessary communication links enable them to work away from the organisation itself). Certain degrees of teleworking are already recognised and described in the prescribed book. They are occasional telecommuting (fixed offices, but occasional work at home), hotelling (no fixed offices provided, but workers occupy office cubicles when they visit the organisation), home working (no office at the organisation, but a fully set-up office at home) and fully mobile (no office at work or at home). Note the critical comments about teleworking made by the authors of your prescribed book. Would you agree that these comments seem pretty sound? Also note how the different approaches to job design imply a difference in balance between control and commitment (see fig 9.9 in your prescribed book). Study the short case example entitled "The Great Pyramid of Giza". The pyramid is seen as a tribute not only to the people who built it, but also to the managers who designed the jobs and organised the building of this phenomenal structure.*
-

CHECKLIST



Now that you have completed study unit 9, you should be able to do the following:

- explain the nature and elements of job design and work organisation
- explain the role of ergonomics in designing for environmental conditions and designing the human interface
- explain what designing the task allocation through division of labour entails
- explain how the design of job methods and job time is influenced by the principles of the scientific management approach
- explain the influence of the behavioural approaches on designing for job commitment
- use the knowledge, insight and skills you have acquired regarding
 - the nature and objectives of job design and work organisation
 - the role of ergonomics in designing for environmental conditions and the technological-human interface
 - designing the task allocation through the division of labour
 - designing job methods and job time through the principles of the scientific management approach and
 - designing for job commitment through the influence of the “behavioural approaches” to job design and work organisation
- to justify the importance of job design and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for the elements, objectives and approaches to job design by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

SUMMARY



The design of jobs and the organisation of work can therefore be seen as the starting point for completing an undertaking successfully. However, job design involves more than merely deciding on the content of the job/task. At least eight interrelated elements should be considered when defining the jobs/tasks. Various approaches, which have been used over the last hundred years or so, still influence job design and work organisation today.

This study unit completes the four topics that focus on the design aspects of the production/operations management system. By now you should be quite familiar with some of the issues pertaining to the design of processes for products and services, the design and location of the production/operations capacity, the design of layout and workflow, and finally, job design and work organisation.

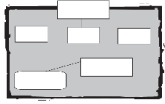
In the next part of this study guide, part III, we turn our attention to the aspects involved in the planning and control of the production/operations management system.

ASSESSMENT



- (1) Consider the case study entitled “ZZ2, the largest tomato in South Africa” at the end of chapter 9 and then answer the four questions posed at the end.
- (2) Do problems 1 to 5 at the end of chapter 9 of your prescribed book. Note that these will give you practice in analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 9 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.



PART III

Planning and controlling the production/operations management system

TOPIC 8

CAPACITY PLANNING AND CONTROL

Topic 8 introduces the general nature of planning and control in operations and explains how capacity management aims to reconcile demand with supply.

- Study unit 10: General nature of planning and control in operations (chapter 10)
- Study unit 11: Capacity management (chapter 11)

TOPIC 9

AGGREGATE PLANNING AND MASTER SCHEDULING

Topic 9 explains aggregate planning and master scheduling in operations and introduces the concept of MRP (material requirements planning).

- Study unit 12: Enterprise resource planning (chapter 14)

TOPIC 10

INVENTORY AND SUPPLY CHAIN PLANNING AND CONTROL

Topic 10 examines inventory management in order to facilitate the smoothing of supply and demand for stocks and introduces the concept of supply chain management.

- Study unit 13: Inventory planning and control (chapter 12)
- Study unit 14: Supply chain planning and control (chapter 13)

Topic 8

Capacity planning and control

AIM



To provide learners with an understanding of the general nature of planning and control activities in operations, and why capacity planning and control is important for production/operations management.



Learning outcomes

When you have worked through this study unit you should be able to do the following:

- explain what the planning and control of operations entails
- demonstrate how the nature of demand and supply could affect planning and control in operations
- describe the tasks of planning and control in operations and give practical examples to illustrate how each could be applied
- explain the volume-variety effect on the planning and control task of operations
- explain the meaning of capacity in the context of operations
- explain what the task of capacity planning and control entails
- demonstrate how demand and capacity are measured in operations
- explain which alternative capacity plans or methods of responding to demand fluctuations are available in operations
- demonstrate how a capacity planning and control approach would be selected for a particular operation
- use the knowledge, insight and skills you have acquired regarding
 - the general nature of planning and control of the operation
 - the nature of demand and supply and their effect on planning and controlling the operation
 - the tasks of planning and control in operations
 - the volume-variety effect on the planning and control task of operations
 - the meaning of capacity in the context of operations
 - the task of capacity planning and control
 - measuring demand and capacity in operations
 - alternative capacity plans or methods of responding to demand fluctuations in operations
 - selecting a capacity planning and control approach for a particular operation
- to justify the importance of capacity planning and control and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for capacity planning and control by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

Study unit 10

General nature of planning and control in operations



Key concepts

planning and controlling the operation

planning task in operations

balance of planning and control over time

uncertainty in demand

dependent demand

resource-to-order

make-to-stock

P:D ratios

D – demand time

finite/infinite loading

customer/client priority

LIFO – last in first out

longest operation/longest total job time first (LOT)

Johnson's Rule

complexity of scheduling

$(n!)^m$ number of possible schedules for m numbers of machines

Gantt Charts

rostering

push/pull control

volume-variety effect on planning and control

reconciling supply with demand

control task in operations

long-term, medium-term and short term planning and control

uncertainty in supply

independent demand

make-to-order

make-to-stock and assemble-to order hybrid

P – total throughput time

loading

sequencing

due date (DD)

FIFO – first in first out

shortest operation/shortest total job time first (SOT)

scheduling

$n!$ number of possible schedules for a single process (fractional n)

forward/backward scheduling

scheduling work patterns

monitoring and controlling the operation

drum, buffer, rope

GETTING AN OVERVIEW

In this study unit we address the first four learning outcomes of topic 8. We explain what the planning and control of operations entail, demonstrate how the nature of demand and supply could affect planning and control in operations, describe the tasks of planning and control in operations, give practical examples to illustrate how each could be applied, and explain the volume-variety effect on the planning and control task in operations.

10.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 10 in your prescribed book. Then study figure 10.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 10 of your prescribed book:

- (1) "What is planning and control?" (1 mark)
- (2) "What is the difference between planning and control?" (2 marks)
- (3) "How does the nature of demand affect planning and control?" (3 marks)
- (4) "What is involved in planning and control?" (2 marks)

10.2 OPERATIONS IN PRACTICE



Activity 10.1

Read the case example entitled "Joanne manages the schedule" in chapter 10 of your prescribed book and then answer the following questions:

- (1) What are the three types of work at the BMW dealership for which workshop time needs to be scheduled?
- (2) How does the computer-based scheduling system at the BMW dealership help Joanne and her colleagues to handle 50 to 80 jobs per day?



Feedback on Activity 10.1

- (1) *With reference to question 1, there are three types of work that need to be organised at the BMW dealership, namely (i) repair work on customers' vehicles, which must normally be done as soon as possible; (ii) routine servicing, which is usually not so urgent and is done by appointment; and (iii) the predelivery service and extensive checks on pre-owned cars before they are sold (this job is also flexible, but good service is still required).*
- (2) *With reference to question 2, note that all jobs are entered into the computer-based scheduling system, which shows the total daily capacity, all the jobs that have been booked and the customers' details. It is possible to see at any point in time whether any jobs can be added and whether there is still "surplus" capacity left. BMW provides standard times for all major repair jobs, but these jobs often need to be modified for local circumstances. Nevertheless, the jobs and their expected duration can be determined daily. The planning system can provide a print-out of all the jobs that are to be undertaken per day, the parts that will probably be needed, a stock list of parts, etc. However, note that they also have to cope with the unexpected on a daily basis – extra unexpected work may be required, customers may want something extra done, technicians may become ill, parts may not be available, or customers may forget and just not turn up. Joanne says: "We can cope*

with most of these uncertainties because our technicians are flexible in terms of the skills they have and are willing to work overtime when needed."

10.3 PLANNING AND CONTROL IN OPERATIONS



Activity 10.2

Study the first section of chapter 10 of your prescribed book, entitled "What is planning and control?" and then answer the questions and/or complete the following:

- (1) The purpose of planning and control in operations is
- (2) Which two "entities" in an operation do the planning and control activities reconcile?
- (3) Why are planning and control in operations linked in your prescribed book, while it may be argued that, theoretically, they involve separate concepts?
- (4) How would the decreasing flexibility in the operation's ability to change plans as time shortens influence the nature of planning and control over the long, medium and short terms?
- (5) How could high/low volume and high/low variety operations affect customer/client responsiveness, the planning horizon, major planning decisions, control decisions and robustness?



Feedback on Activity 10.2

- (1) *With regard to question 1, note that "planning and control is concerned with the reconciliation between what the market requires and what the operation's resources can deliver". While the "planning and control activities provide the systems, procedures and decisions which bring together different aspects of supply and demand", the purpose of planning and control is "to ensure that the operation's processes run effectively and efficiently and produce products and services as required by customers".*
- (2) *With reference to question 2, planning and control activities provide the systems, procedures and decisions to bring together the resources of the operation (with a general ability to supply customers/clients with products/services) and a set of general and specific demands from actual and potential customers/clients for products/services. Planning and control therefore reconcile supply and demand.*
- (3) *With reference to question 3, planning is "a formalization of what is intended to happen at some time in the future", although a plan does not guarantee that an event will actually happen. It is, therefore, rather "a statement of intention". Control is "the process of coping with changes" in variables such as what the customers want and when they want it, suppliers who are not able to deliver on time, machine failure and absent staff members. Control "makes the adjustment which allows the operation to achieve the objectives that the plan has set, even when assumptions on which the plan was based do not hold true".*
- (4) *With reference to question 4, note that the nature of planning and control activities change over time. In long-term planning and control, operations managers plan their actions, they estimate what resources they will need and they formulate their objectives. The emphasis is, therefore, on planning and not on control. Medium-term planning and control is "concerned with planning in more detail (and replanning if necessary)" and "looks ahead to assess the overall demand which the operation must meet in a partially disaggregated manner". In short-term planning and control, "many of the resources will*

have been set and it will be difficult to make large-scale changes" although "short-term interventions are possible if things are not going to plan" and "by this time, demand will be assessed on a totally disaggregated basis". Study figure 10.2 in your prescribed book and note the balance between planning and control activities as the time horizon changes from hours/days to months/years.

- (5) With reference to question 5, note that operations that produce a relatively low volume but a high variety of products and services have different customers who require a different set of factors. They use processes that have a different set of needs from those that create standardised products and services in high volume. Study table 10.1 and note the differences in the volume-variety effect on the planning horizon, major planning decisions, control decisions and robustness.

10.4 NATURE OF DEMAND AND SUPPLY IN OPERATIONS



Activity 10.3

Study the second section of chapter 10 in your prescribed book entitled "The nature of demand and supply" and then answer the following questions:

- (1) Why could uncertainty in demand and supply affect the planning and control activities in operations?
- (2) What is the difference between dependent and independent demand and how would each affect the planning and control activities of such operations?
- (3) What are the differences in the planning and control activities for "resource-to-order", "create-to-order/make-to-order" and "make-to-stock", and how would their total throughput time (P) and demand time (D) differ?
- (4) How could the P and D ratios over the life cycle (from obtaining resources-to-producing products/services-to-delivering to customers) differ for the following different types of planning and control: "produce-to-stock", "part produce-to-order", "produce-to-order" and "resource-to-order"?



Feedback on Activity 10.3

- (1) With reference to question 1, some operations are reasonably predictable and run to plan, so the need for control is minimal. In others, supply and/or demand may be unpredictable and it may, therefore, become necessary to rethink plans quite often and to step up control activities.
- (2) With reference to question 2, dependent demand is "relatively predictable because it is dependent upon some factor which is known", so dependent demand planning and control concentrate on the consequences of the demand within the operation. (In this case the materials requirements planning approach – MRP – may be used.) On the other hand, independent demand operations will supply demand without having any firm idea of customers' orders. The business/organisation therefore runs the risk of stockouts on certain items because demand may exceed the available stock. This is normal for independent demand planning and control where the operation attempts to draw on inventory to respond quickly by putting resources in place to satisfy demand. Study the short case example entitled "Operations control at Air France" in chapter 10 and answer

the questions that follow in the prescribed book. Also study figure 10.3 in your prescribed book. This figure illustrates dependent/independent demand operations.

- (3) Note that the dependent and independent demand concepts (discussed in question 2) are closely related to the three different ways of responding to demand (resource-to-order, make-to-order and make-to-stock).

In conditions of dependent demand, an operation will only begin the process of producing goods or services when a customer order is received. The order will trigger the planning and control activities.

In conditions of independent demand, operations will supply demand without having any firm forward visibility on customer orders.

Classifying the relationship between dependent demand, independent demand and the three ways of responding to demand does not result in a strictly ordered classification. However, it can be said that there appears to be a trend.

For example, make-to-stock scenarios, where goods are made and ready for purchase ahead of firm demand, like movies in cinema houses and stock in supermarket retail outlets, are typically examples of independent demand in which orders are based on experience and understanding of the market, independent of what may actually happen. They make a "best guess" concerning future demand and attempt to put the resources in place which can satisfy this demand.

Resource-to-order scenarios, where parts are ordered and goods manufactured only after a customer order has been received, like construction of a house or organisation of a conference, are typically examples of dependent demand. The firm customer order results in exact specifications for parts or materials, and a production schedule.

When the operation is more confident of the nature of demand but still does not know its volume or timing, it may still only produce the actual product or service after receiving a firm customer order, but it will keep the resources in stock to satisfy demand quickly (make-to-order). Determining the P:D ratios (P stands for "total throughput time" and D for "demand time") for various operations is another way of characterising the relationship between the time when a customer/client "asks" for a product/service and the time it takes the operation to obtain the resources and produce and/or deliver the product/service. High P:D ratios generally mean that the operations need a long time before they can respond to customers/clients needs. These operations normally also have a higher degree of uncertainty in their planning and control activities as a result of the longer time span involved in "order-by-customer/client-to-production/delivery-by-operation". Nevertheless, it should be noted that because demand (D) per se is usually only a prediction, subject to considerable uncertainty, no matter how much greater P is than D, the risk of inaccurate operations planning and control decisions is greater. Reducing the P:D ratios, therefore, becomes a way of removing some of the risk involved in operations planning and control.

- (4) With reference to question 4, study figure 10.4 in your prescribed book, which illustrates P and D ratios for the following different types of planning and control: "produce-to-stock", "part produce-to-order", "produce-to-order" and "resource-to-order".

10.5 PLANNING AND CONTROL TASKS



Activity 10.4

Study the third section of chapter 10 of your prescribed book entitled “Planning and control activities” and then answer the questions and/or complete the following:

- (1) Planning and control require the reconciliation of supply and demand in terms of
 - (a) loading, which means determining the
 - (b) sequencing, which means determining the
 - (c) scheduling, which means deciding on
- (2) When loading a work centre, why is there normally a difference between the maximum time available and the actual running time?
- (3) What is the difference between finite and infinite loading, and in which types of operation would each be relevant?
- (4) How would the priorities given to work be affected by using each of the following sequencing rules: customer/client priority, due dates, LIFO, FIFO, longest operation/longest total job time first, and shortest operation/shortest total job time first?
- (5) How can Johnson’s Rule be used to optimise the scheduling of many jobs through two work centres?
- (6) What is a Gantt Chart and how can it be used to track progress in jobs and work centres?
- (7) Why is scheduling said to be one of the most complex tasks in production/operations management and how many possible schedules could there be if we had three different jobs and two machines that operated independently of each other?
- (8) What is the difference between forward and backward scheduling, and which approach complements just-in-time (JIT) operations? Where is rostering used and why is scheduling staff times considered even more complex?
- (9) What does the monitoring and control activity entail with regard to the volume and timing reconciliation between supply and demand? How could push and pull control affect inventory build-up in operations?



Feedback on Activity 10.4

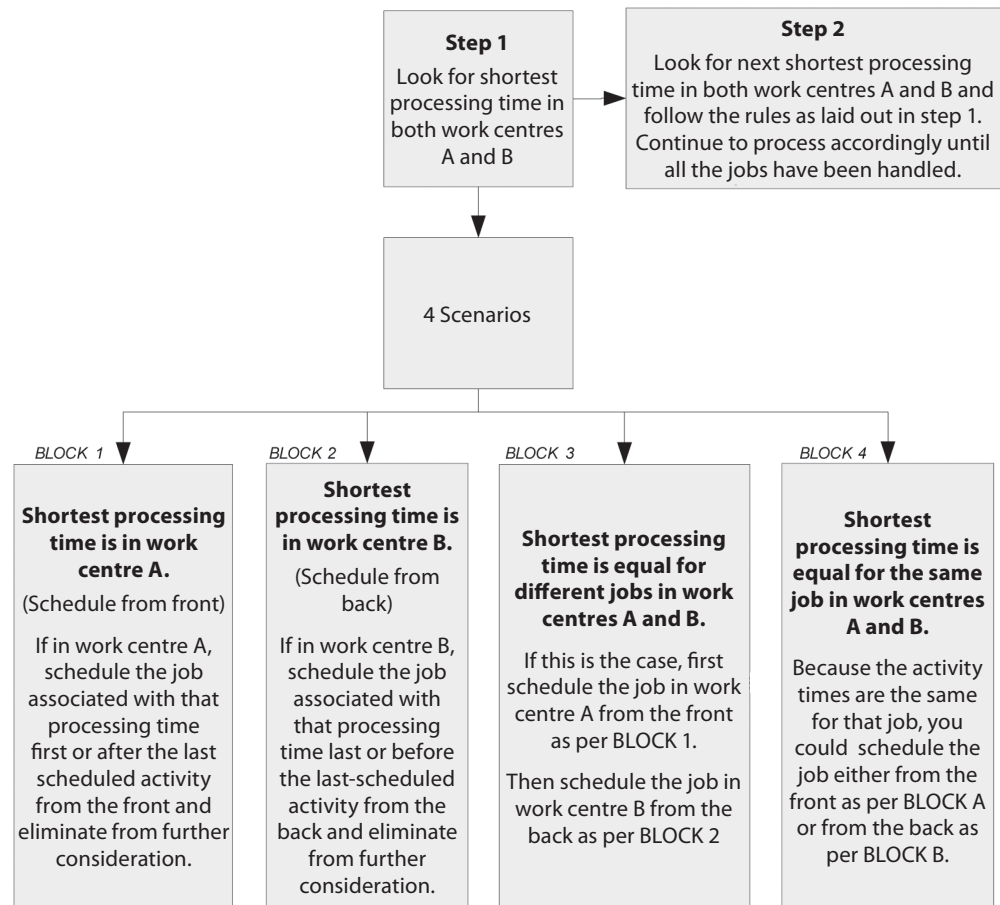
- (1) *With reference to question 1, note that “planning and control require reconciliation of supply and demand in terms of volumes, timing and quality”. To reconcile volume and timing (quality is discussed in our third-level module MNO3701), four overlapping activities are performed: loading, sequencing, scheduling, and monitoring and control. Loading is the amount of work that is allocated to a work centre. Sequencing is the order in which the work will be performed. Scheduling involves a detailed timetable showing dates and times for starting and completing jobs. And finally, given the plan of the operation through loading, sequencing and scheduling, each part of the operation needs to be monitored to ensure that planned activities are indeed undertaken (ie to exercise control). Study figure 10.5 in your prescribed book for an illustration of planning and control activities for operations, and what each activity attempts to achieve.*
- (2) *With reference to question 2, the difference between maximum available time and valuable operating time (eg of a machine) takes into consideration certain unavoidable time losses (public holidays, weekends, equipment idling, set-up and changeover – planned periods of unproductivity, etc) and avoidable losses in time (quality losses, slow running equipment, breakdown, failure – unplanned periods of unproductivity, etc) in well-run operations. Study figure 10.6 in your prescribed book.*

- (3) With reference to question 3, finite loading is “an approach which only allocates work to a work centre up to a set limit”, which is the estimate of the capacity of the work centre. Infinite loading is “an approach to loading work which does not limit accepting work, but tries to cope with it”. Study figure 10.7 for illustrations of both finite and infinite loading, and note where each approach is particularly relevant to an operation.
- (4) With reference to question 4, note that when priorities are given to work in an operation, a predefined set of rules may apply, or physical constraints (physical nature of materials being processed) may determine the priority. This predefined set of sequencing rules includes customer/client priority sequencing, due date (DD), last-in-first-out (LIFO), first-in-first-out (FIFO), longest operation time first (LOT) and shortest operation time first (SOT). Customer/client priority sequencing is when important or aggrieved customers/clients are processed prior to others, irrespective of the order of arrival – see figure 10.8 in your prescribed book for such examples. DD sequencing is when work is sequenced according to the due date for delivery, irrespective of the size of each job or importance of the customer/client. LIFO sequencing is when work is selected for practical reasons, meaning those last in move out first. With FIFO sequencing customers/clients are served as they arrive – this is also known as first-come-first-served. LOT sequencing is when jobs that take the longest are done first, and SOT is usually when cash-constrained operations do the quickest jobs first to invoice, receive payment quicker and improve cash flow. Study the short case example “Triage at Chris Hani Baragwanath Hospital” in chapter 10 and then answer the questions that follow in the prescribed book. Also study the worked example dealing with “Steve Smith, the website designer” in chapter 10 of your prescribed book. Do you agree with his calculations of FIFO, DD and SOT? Compare your calculations with those in table 10.2 of your prescribed book. Finally, note that the five performance objectives may also be used to judge the effectiveness of the sequencing rules. These performance objectives include dependability (meeting due dates as promised), speed (minimising the time a job spends in process) and cost (minimising work-in-process inventory and minimising idle time at work centres).
- (5) Johnson’s Rule is a technique that managers can use to minimise the makespan for a group of jobs to be processed on two machines or at two successive work centres (sometimes referred to as a two-machine flow shop). It also minimises the total idle time at the work centres. For the technique to work, several conditions must be satisfied:
1. Job time (including setup and processing) must be known and constant for each job at each work centre.
 2. Job times must be independent of the job sequence.
 3. All jobs must follow the same two-step work sequence.
 4. Job priorities cannot be used.
 5. Units in a job must be completed at the first work centre before the job moves on to the second work centre.

To guide you through the procedure of using Johnson’s Rule, the diagram below outlines the steps involved as well as the rules associated with the different scenarios when using Johnson’s Rule.

Once you have browsed through the diagram, we will work through an example, using the steps as outlined in the diagram. To ensure that you understand how and why each step has been applied, explanations will be provided next to each step.

The figure below offers a step-by-step guide on Johnson’s Rule for scheduling n jobs through two work centres. Assume “A” represents work centre A and “B” represents work centre B/ Jobs must be processed through work centre A and then through work centre B.



An example which strikes close to home is the process of washing and drying with a washing machine and a tumble dryer respectively. Assume five items (jobs) need to be processed separately. Firstly they need to be washed (work centre A) and then tumble-dried (work centre B). The processing times (in minutes) for each item in the respective work centres are reflected below. Using Johnson's Rule, determine the optimal job processing sequence.

Follow **THE ARROWS** in their numbered sequence in the table below to gain an understanding of how one should apply the steps reflected on the previous page. Start at **ARROW 1** and progress towards **ARROW 4**, reading the accompanying notes on the right hand side.

When you have completed this example, work through figure 10.9 in your prescribed textbook, applying the same principles, to make sure you have grasped the concept.

JOB	PROCESSING TIME WORK CENTRE A (WASHING MACHINE) (MINUTES)	PROCESSING TIME WORK CENTRE B (TUMBLE DRYER) (MINUTES)	SEQUENCE	REASONING ACCORDING TO RULES
A: BLANKET	80	30	4	Arrow 4 in work centre B indicates where our eyes should glance after arrow 3 in work centre A. See notes next to arrow 3. This job is processed as per BLOCK 2.
B: DUVET	40	50	3	This job is the last job on the list and therefore takes the last remaining place in the sequence, which happens to be in the middle at number 3
C: SHEET	20	30	1	Arrow 2 indicates where our eyes should glance next. They glance at the "20" because "20" is the next lowest of all the processing times for the jobs that have NOT been eliminated from the list. In which work centre is it? Work centre A. Therefore, process according to BLOCK1 and eliminate from the list.
D: COAT	30	50	2	Arrow 3 indicates where our eyes should glance next. We glance at the "30" because "30" is the next lowest processing time for any of the jobs not yet eliminated. BUT, there is also a "30" for another job not eliminated in work centre B. In this instance, first process the job in A as per BLOCK 1. Then process the job in work centre B as per BLOCK 2.
E: TABLE CLOTH	20	10	5	Arrow 1 indicates where our eyes should glance first. They first glance at the "10" because "10" is the lowest of all the processing times. In which work centre is it? Work centre B. Therefore, process according to BLOCK 2 and eliminate from the list.

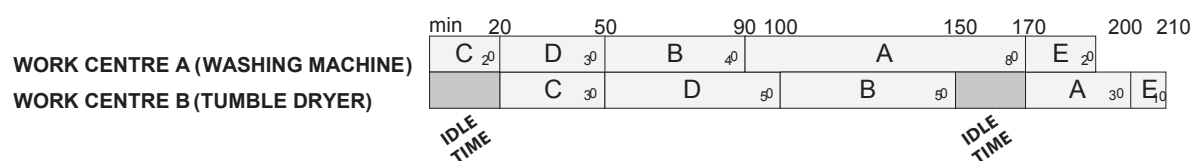
INFORMATION PROVIDED WITH QUESTION
ANSWER
EXPLANATIONS TO AID YOUR UNDERSTANDING

The correct sequence is therefore C D B A E.

This sequence can be reflected in a Gantt Chart.

The Gantt Chart below reflects the sequence.

The grey areas indicate idle time. Johnson's Rule helps us to minimise idle time.



(6) With reference to question 6, Gantt Charts are used as a method of scheduling. It is a simple device, which represents time as a bar or channel on a chart. Figure 10.10 is an illustration of a Gantt Chart.

(7) With reference to question 7, note that schedules are statements of the volume of work to be done and the timing of operations. Their aim is to ensure that customers' demands are met. In rapid-response service operations where customers arrive in an unplanned manner, schedules do not work – the operation can merely respond once demands are made. Note that the scheduling activity is considered to be one of the most complex tasks in production/operations management, because it deals with several different types of resources simultaneously (eg machines with different capabilities and capacities) and the number of possible schedules increase rapidly as the number of activities and processes increase. For example, with one machine and five different jobs, there are 120 different possible schedules. For [n] jobs there are [n!] (factorial n) different ways of scheduling

the jobs through a single process. For more than one machine, the possible number of schedules are $[(n!)^m]$ – for three different jobs and two machines we could have $36261 = 666 = 36$ possible schedules. Study the short case example entitled “The life and times of a chicken salad sandwich – part one” in chapter 10 and then answer the questions that follow in your prescribed book.

- (8) With reference to question 8, forward scheduling involves starting work as soon as it arrives, while backward scheduling involves starting the jobs at the last possible moment. Study table 10.4 in your prescribed book and note the advantages of forward and backward scheduling. In theory both MRP (material requirements planning) and JIT (just-in-time) use backward scheduling, which means that the work is only started when it is required. Scheduling work patterns refers to a situation where the dominant resource in the operation is its staff, and the schedule of work times effectively determines the capacity of the operation itself. The main task is therefore to make sure that enough people are working at any time to provide the capacity appropriate to the level of demand at that time. Note some of the issues involved in scheduling staff times as described by the authors of your prescribed book. These issues include the following: capacity must match demand; the length of each shift must not be too long or short; working unsociable hours must be minimised; days off must match staff's conditions of employment, vacation and other times off; and sufficient flexibility must be maintained to cover unexpected changes in staff due to sickness and demand surges.
- (9) With reference to question 9, note that after a plan has been created for the operation through loading, sequencing and scheduling, each part of the operation has to be monitored to ensure that the planned activities do take place. Any deviation from the plans must be rectified through some kind of intervention and may involve replanning. Study figure 10.14 in your prescribed book for an illustration of a simple model of control. An important distinction is made between intervention signals that push work through the process within the operation (work is pushed out without considering whether the succeeding work centre can use it and idle time occurs – inventory build-up and queues may result) and intervention signals that pull work only when it is required (the customer works as the trigger to pull the work from the preceding work station). Note that in pull systems an inventory build-up is less likely and, therefore, they are favoured in JIT operations. Study figure 10.15, which illustrates push and pull control systems. Note the inventory build-up in the systems. It is not always easy to decide exactly where there should be control in the process. Goldratt (accredited author of the “theory of constraints” [TOC] and “optimised production technology” [OPT] concepts) argued that the point of control should be the bottleneck. It is called the “drum”, as it sets the beat for the rest of the process to follow and it should be working all the time, because by definition it does not have enough capacity. It is therefore sensible to set up a buffer of inventory in front of the bottleneck to make sure it always has something to work on. It is not worthwhile for the parts in the process before the bottleneck to work to their full capacity as this will cause inventory build-up. Some form of communication (called the rope) is needed between the bottleneck and the input to the process to make sure that activities before the bottleneck do not overproduce. Study figure 10.16 in your prescribed book where the concepts of “drum/bottleneck”, “buffer” and “rope” are illustrated. Also study the critical comments by the authors of your prescribed book about the degree of difficulty with which an operation is controlled. We agree with the authors. Note the set of useful questions that can be asked to assess the difficulty associated with the control of any operation. This is illustrated in figure 10.17 in your prescribed book (ie consensus over the operation's objectives, the measurability of the operation's outputs, knowledge of the operation's behaviour and the degree of repetition of activities). Finally, study the short case example entitled “Routing and scheduling helps milk processor gain an extra collection trip a day” in chapter 10 of your prescribed book, and then answer the questions that follow at the end.

CHECKLIST



Now that you have completed study unit 10 on the general nature of planning and control in operations you should be able to do the following:

- explain what the planning and control of operations entail
- demonstrate how the nature of demand and supply could affect planning and control in operations
- describe the tasks of planning and control in operations and give practical examples to illustrate how each could be applied
- explain the volume-variety effect on the planning and control task of operations
- use the knowledge, insight and skills you have acquired regarding
 - the general nature of planning and control of the operation
 - the nature of demand and supply and their effect on planning and control of the operation
 - the tasks of planning and control in operations, and
 - the volume-variety effect on the planning and control task of operations
- to justify the importance of planning and control in operations and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation of the general nature of planning and control in operations by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

SUMMARY



This study unit examined the general aspects of planning and control in the production/operations management system. The next study unit, study unit 11, focuses specifically on capacity planning and control.

ASSESSMENT



- (1) Consider the case study entitled "Air traffic control: a world-class juggling act" at the end of chapter 10 of your prescribed book and then answer the three questions posed at the end.
- (2) Do problems 1 to 11 at the end of chapter 10 of your prescribed book. Note that these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 10 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experience of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Study unit 11

Capacity management



Key concepts

capacity planning and control	level capacity plan
aggregate planning and control	chase demand plan
capacity of an operation	overtime/idle time
maximum processing capacity	annualised hours
capacity constraints	"hire and fire"
capacity ceiling	part-time staff
"bottleneck" constraint	subcontracting
long-term capacity planning and control	managed demand plan
medium-term and short-term capacity	mixed capacity plans
aggregate demand and capacity planning and control	yield management
measuring demand and capacity	cumulative demand/production
forecasting demand fluctuations	cumulative representation
seasonality of demand	capacity change costs
output capacity measure	waiting/queuing theory
input capacity measure	parallel servers
activity mix	source of customers/calling population
design capacity	finite/infinite
effective capacity	source of customers/clients
utilisation as measure of operation's performance	arrival rate
efficiency as measure of operation's performance	queue
overall equipment effectiveness (OEE)	rejecting
availability rate $[a]$	balking
performance rate $[p]$	reneging
quality rate $[q]$	queue discipline
alternative capacity plans	servers
	balancing capacity and demand
	variability in demand and supply
	customers' perceptions of queuing

GETTING AN OVERVIEW

In this study unit we address the last five learning outcomes of topic 8. We explain the meaning of capacity in the context of operations, explain what the task of capacity planning and control entails, demonstrate how demand and capacity are measured in operations, explain which alternative capacity plans or methods of responding to demand fluctuations are available in operations, and demonstrate how a capacity planning and control approach for a particular operation would be selected.

11.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 11 of your prescribed book. Then study figure 11.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 11 of your prescribed book:

- (1) "What is capacity planning and control?" (3 marks)
- (2) "How is capacity measured?" (2 marks)
- (3) "What are the ways of coping with demand fluctuation?" (4 marks)
- (4) "How can operations plan their capacity level?" (2 marks)
- (5) "How can operations control their capacity level?" (1 mark)

11.2 OPERATIONS IN PRACTICE



Activity 11.1

Read the case example entitled "South African Breweries" in chapter 11 of your prescribed book and then answer the following questions:

- (1) What are the capacity planning and control considerations that SAB must take into account?
- (2) Describe SAB's challenges related to decision making and capacity. Can SAB deal with these challenges? Justify your answer.



Feedback on Activity 11.1

- (1) *Capacity planning and control is the task of setting the effective capacity of the operation so that it can respond to the demands made on it. In the case of SAB and the Alrode plant, production runs 24 hours a day every day of the week, so the question in this instance is not whether to produce, but rather, what brands to produce and when to produce them (ie volume and timing calculations for the respective brands). They forecast potential sales 21 days (three weeks) into the future. The logistics department issues the brewing department with a brewing plan, specifying each week how many brews of each brand they are expected to produce. Different brands have specific implications for productions runs and the timing and volume of the runs. For example, Castle Lager is not a big production run; as few as two brews can be brewed in this brewhouse. The balance of the volume is made up from the two major brands produced by the brewhouse – Hansa Pilsner and Carling Black Label.*
- (2) *SAB's brewing plan assumes that there are no breakdowns or unforeseen circumstances. It also assumes that all necessary raw materials will be available. If there are breakdowns or raw materials shortages, the team leader of the shift needs to make the decisions on what brands to brew and how to adjust priorities. The team leaders have no model to enable*

them to make an informed decision on how to allocate the available resources. SAB can deal with these challenges by using medium-term planning models (now common practice for most plants such as Alrode) with batch sequencing capabilities to make optimum use of capacity. Short-term adjustments on these models allow a more dynamic response to market demand and make it possible to cope with other unforeseen events.

11.3 THE MEANING OF CAPACITY IN OPERATIONS



Activity 11.2

Study the first section of chapter 11 in your prescribed book entitled “What is capacity?” and then answer the questions and/or complete the following:

- (1) The capacity of an operation is (definition) and takes the scale of the operation into account, which means and with which a time dimension must be incorporated, which means
- (2) Why do many operations operate at levels below their maximum processing capacity?
- (3) What is a “bottleneck” constraint and how could it impact on the capacity of an operation?



Feedback on Activity 11.2

- (1) *With reference to question 1, the “common use of the word capacity is in the static, physical sense of the fixed volume of”, for instance a container. While this measure of capacity describes the scale of the operation, it does not reflect the processing capacity, which must include a time dimension. The definition of the capacity of an operation as put forward by the authors of your prescribed book reads: “[Capacity] is the maximum level of value-added activity over a period of time that the process can achieve under normal operating conditions.”*
 - (2) *With reference to question 2, note that some organisations operate below their maximum processing capacity because (a) there may be insufficient demand for the products or services to use up all available capacity, or (b) as a deliberate policy that enables the operation to respond quickly to new “unplanned or unexpected” orders.*
 - (3) *With reference to question 3, note that a capacity constraint will occur when some part(s) of the operation is/are operating at its (their) “capacity ceiling” (at maximum processing capacity level) even while other parts may be operating below the capacity ceiling. Various parts of the operation may be pushed to their capacity ceiling and act as a constraint on the operation as a whole.*
-

11.4 THE TASK OF CAPACITY PLANNING AND CONTROL



Activity 11.3

Study the second section of chapter 11 of your prescribed book entitled “Planning and controlling capacity” and then answer the questions and/or complete the following:

- (1) The task of capacity planning and controlling involves
- (2) How do medium-term and short-term capacity decisions tie in with long-term capacity strategies?
- (3) Aggregated demand and capacity are used to set capacity levels over the medium and short terms. In the case of a motor vehicle manufacturer, the aggregate measure of capacity and demand would comprise and in the case of a dentist, it would comprise
- (4) How would decisions relating to capacity planning and control affect the following aspects of an operation’s performance: costs, revenues, working capital, quality, speed, dependability and flexibility?
- (5) The sequence of capacity planning and control involves the three steps of
 - (a) measuring
 - (b) identifying
 - (c) choosing



Feedback on Activity 11.3

- (1) *With reference to question 1, the capacity planning and control task is to set the effective capacity of the operation so that it can respond to the demands placed upon it. This usually determines how the operation will react to fluctuations in demand.*
- (2) *With reference to question 2, long-term capacity was previously examined (see topic 5, study unit 7 in this study guide). The introduction (or deletion) of major increments of physical capacity, which depends on long-term changes in demand and alternative capacity strategies, was discussed. When long-term capacity has been established, it must be decided how capacity should be adjusted in the medium term (based on the assessment of demand forecasts for a period of 2 to 18 months) and in the short term (flexibility owing to demand changes on a month-to-month, week-to-week, day-to-day or even hour-to-hour basis, as well as on short notice).*
- (3) *With reference to question 3, aggregated demand and capacity deal with “broad measures of output” (they are not concerned with the details of the individual products and services offered) and involve some degree of approximation (especially if the mix of products and services varies significantly). In the case of the motor vehicle manufacturer, it could be the number of units for a given model of car (eg 40 000 BMW 3 series vehicles per year) and in the case of the dentist it could be number of patients (eg 10 patients per day or 50 per week).*
- (4) *With reference to question 4, note that decisions taken while capacity plans are devised may affect several different aspects of performance as described by the authors of your prescribed book under the subheading “The objectives of capacity planning and control”. Apart from the aspects of performance discussed earlier, namely quality, speed, dependability, flexibility and cost, other areas influenced by capacity planning and control decisions include revenues (fluctuations in demand may cause less or more sales) and working capital. Working capital involves “tying up” money in finished goods through inventory prior to the probable realisation of demand, and/or maintaining unnecessary surplus capacity in case unrealistic sales opportunities should arise.*

- (5) *With reference to question 5, the sequence and steps of capacity planning and control are illustrated in figure 11.2 of your prescribed book. Note: The forecast in actual demand is likely to be uncertain and fluctuating, therefore the operation is likely to work on a measure of aggregate demand that it believes it would be able to meet with its current capacity.*
-

11.5 MEASURING DEMAND AND CAPACITY IN OPERATIONS



Activity 11.4

Study the third section of chapter 11 of your prescribed book entitled “Measuring demand and capacity” as well as the supplement to chapter 6 entitled: “Forecasting”. Then answer the questions and/or complete the following:

- (1) From a capacity planning and control point of view, what are the three requirements for a demand forecast for an operation?
- (2) What is the difference between seasonality of demand and weekly and/or daily demand fluctuations?
- (3) What are the two main approaches to forecasting and how can the following techniques be used in practice to forecast the demand for products and/or services: panel approach; Delphi methods; scenario planning; time series analysis (including moving-average forecasting and exponential smoothed forecasting); and causal models (regression modelling)? Which of the demand forecasting methods and/or techniques would we expect to be more accurate?
- (4) How is capacity measured? Why is it sometimes more appropriate and better to use an input measure of capacity than an output measure?
- (5) Why does capacity depend on the activity mix? In the worked example of the air conditioner factory in chapter 11, where output capacity varies according to the activity mix, the number of units produced per week if the ratio of deluxe, economy and standard units was 1:3:3 would be
- (6) What is the difference between design and effective capacity and how could the measures of utilisation and efficiency be used to judge the performance of the operation?
- (7) What does the overall equipment effectiveness (OEE) measure entail and how can it be used to judge the performance of operations equipment?



Feedback on Activity 11.4

- (1) *With reference to question 1, note that while the demand forecast is usually the primary responsibility of the sales or marketing functions, it has a significant impact on capacity planning and control. (Without an estimate of future demand, it is not possible to plan effectively for future events.) It is therefore important that operations managers should understand the basis and rationale of the demand forecast. Three specific requirements of the demand forecast for capacity planning and control purposes are:*
- (a) *It should be expressed in terms that are useful for capacity planning and control (eg machine hours per year and not just a monetary value like R2 million per year).*
 - (b) *It should be as accurate as possible. (Demand may fluctuate, but there is a time-lag between the decision to change capacity and the implementation of the decision, when the change actually takes effect.)*

- (c) *It must give an indication of its level of uncertainty (also referred to as “accuracy”) through demand statistics, possible deviation of actual demand from average, probabilistic estimates, etc.*
- (2) *With reference to question 2, note that almost all products and services have some demand seasonality. Some also have supply seasonality. While the seasonality usually occurs over a year, similar predictable variations can also occur in a shorter cycle (eg daily or weekly). Study figure 11.3 in your prescribed book and note some of the typical causes of seasonality, including climate, the festive season, behavioural factors, and political, financial and social influences. Also study the short case example entitled “Producing while the sun shines” and answer the question that follows in the prescribed book. Finally, study figure 11.4, which illustrates the demand patterns of four different types of operation (a woollen knitwear factory, a city hotel, a retailer and an aluminium producer). Note that the first three have seasonal demand patterns (the result of climatic patterns, demand from business people who take vacations from work at Christmas, pre-vacation peaks and reduced sales during vacation periods) while the last, the aluminium producer’s demand pattern, shows virtually no seasonality and a steady growth in sales over the forecast period.*
- (3) *With reference to question 3, note that the information needed to answer this question is contained in the supplement to chapter 6 in your prescribed book, entitled “Forecasting”. The two basic approaches to forecasting are:*
 - (a) *The qualitative approach, which involves collecting and appraising judgements, considering various options, looking at past performances, and inviting experts to make a prediction (the methods available are the panel approach, Delphi methods and scenario planning).*
 - (b) *The quantitative approach, which includes the time series analysis (where the pattern of past behaviour of a single phenomenon over time is examined to forecast the phenomenon’s future behaviour – the methods available are moving average forecast and exponential smoothing) and causal modelling (where complex cause and effect relationships are described and evaluated – simple regression is used to determine the “best fit” between two variables).*

Note figure S6.3 in the supplement to chapter 6 of your prescribed book. This figure illustrates a time series with a trend, a seasonality pattern and random variation. Also study the examples of the moving average forecast (table S6.2), exponential smoothing (table S6.3 and figure S6.4) and regression modelling (figure S6.5) in the supplement to chapter 6 of your prescribed book. Finally, note the discussion by the authors of your prescribed book on the performance of forecasting models. It appears that certain forecasting techniques perform better under certain circumstances and that “econometric methods offer more accurate long-range forecasts than do expert opinion or time series analysis, and that the superiority of objective causal methods improves as the time horizon increases”.

- (4) *With reference to question 4, capacity can be measured by using an output capacity measure (the most appropriate measure for an operation where the nature of its output does not vary much, eg 2 000 Model A television sets per week) or input capacity measures (when a much wider range of outputs occurs, which places varying demands on the operation, eg number of customers per night at a restaurant). Also study table 11.1 in your prescribed book and note the input and output capacity measures for various types of operation.*
- (5) *With reference to question 5, capacity depends on the activity mix of the operation. Study the worked example in chapter 11 of your prescribed book on the air-conditioner factory that produces three different models of air-conditioner units (deluxe, standard and economy). With reference to the air-conditioner factory, if the activity mix is 1:3:3, it would mean 6.75 hours are needed to assemble 7 units ($1 + 3 + 3 = 7$) – the total output would be 829.6 ($800 \div 6.75 \times 7 = 829.6$) units per week. Note that in this case an output measure of capacity (ie number of units per week) is used.*

- (6) With reference to question 6, the design capacity of an operation is the theoretical capacity that the designers of the operation had in mind when commissioning the operation. As the operation will not run continuously at maximum rate (owing to losses in the operational time because of stoppages for maintenance, set-up, etc), the effective capacity is equal to the design capacity minus the capacity lost due to the technical and market demands placed on the operation. Note that the actual capacity will probably be even lower than the effective capacity as further losses could occur as a result of quality problems, machine breakdowns, absenteeism, etc. The ratio of the operation's actual output to design capacity is a measure of utilisation, while the ratio of actual output to effective capacity is a measure of the efficiency of the operation. Study the worked example of the photographic paper manufacturer, in chapter 11 of your prescribed book. Do you agree with the way the utilisation and efficiency ratios have been calculated? Also study the critical comments by the authors of your prescribed book on the importance of capacity measures as diagnostic measures that can highlight the root causes of the inefficient use of capacity.
- (7) With reference to question 7, the use of the overall equipment effectiveness (OEE) measure is based on three aspects, namely (i) the time that equipment is available to operate – denoted by symbol $[a]$ = total operating time divided by the loading time; (ii) the quality of the product or service it produces – denoted by symbol $[q]$ = valuable operating time divided by net operating time; (iii) the speed or throughput rate of the equipment – denoted by symbol $[p]$ = net operating time divided by total operating time and calculated as $OEE = [a] \times [p] \times [q]$, which represents the valuable operating time as a percentage of the design capacity. Figure 11.5 illustrates the calculation of OEE. Study the worked example of OEE for a machine. Finally, study the short case exercise entitled "British Airways London eye" in chapter 11 in your prescribed book and then answer the question posed there.
-

11.6 ALTERNATIVE CAPACITY PLANS IN OPERATIONS



Activity 11.5

Study the fourth section of chapter 11 in your prescribed book entitled "The alternative capacity plans" and then answer the questions and/or complete the following:

- (1) There are three different methods (or "pure" options) for responding to demand fluctuations, namely
 - (i) a level capacity plan, which means
 - (ii) a chase demand plan, which means
 - (iii) a managed demand plan, which means
- (2) What are the main differences and advantages or disadvantages of a level capacity, a chase demand and a managed demand capacity plan? How would the level capacity and chase demand plans affect the operations of the aluminium producer, the woollen knitwear factory, the hotel and the retail store?
- (3) How can capacity be adjusted in the chase demand approach and how can demand be changed in the managed demand approach?
- (4) Why do many operations choose to follow a mixture of the three "pure" approaches to capacity planning and control?
- (5) Another approach that is used by operations with relatively fixed capacities to maximise their potential to generate profit is called yield management. It is a collection of methods that are especially useful when,, and

- (6) In the case of an airline, yield management methods could include over-booking capacity, price discounting and varying service types. How could these methods be used in the case of a movie theatre to maximise the yield from its fixed capacity?



Feedback on Activity 11.5

- (1) With reference to question 1, the alternative options available for coping with variations due to demand fluctuations are (i) to ignore the fluctuations and keep activity levels constant – the level capacity plan, (ii) to adjust capacity to reflect the fluctuations in demand – chase demand plan and (iii) to attempt to change demand to fit the capacity available – demand management.
- (2) With reference to question 2, note that each of the three “pure” options described above deals differently with demand fluctuations (and therefore with production/operations capacity). When using a level capacity plan, demand fluctuations are ignored. The production/operations capacity is held constant over time, resulting in periods of undersupply and oversupply (see figs 11.7 and 11.8 for each of the four examples indicated). Should a chase demand plan be used, the production/operations capacity is increased or decreased according to the actual patterns and/or fluctuations in demand (see figs 11.9 and 11.10 for each of the four examples indicated). In an attempt to create a stable and uniform demand on one hand and to gain the benefits of reduced costs and improve service through optimum capacity utilisation on the other, a managed demand plan could be considered. Also study the short case example entitled “Seasonal salads” in chapter 11 of your prescribed book and then answer the questions that follow.
- (3) With reference to question 3, in the case of the chase demand approach, a number of different methods can be used to adjust capacity by some means. Firstly, capacity can be adjusted by varying the number of productive hours worked by staff (ie when demand is higher than nominal capacity, staff work overtime, or when demand is lower, staff use the idle time for engaging in other activities like maintenance, cleaning, etc). Note that there is a limit to the amount of extra working time that a workforce can deliver before productivity levels start to decrease. Study the short case example entitled “Working by the year” in chapter 11 of your prescribed book and then answer the questions that follow. Secondly, the size of the workforce can also be varied by hiring extra staff during periods of high demand or laying them off (“firing” them) as demand falls. Thirdly, part-time staff may be recruited for part of a normal working day. Fourthly, capacity might be bought from another organisation by means of subcontracting. Study the critical comments made by the authors of your prescribed book about the controversial and perhaps “unethical” practice of fluctuating the workforce to match demand by either using part-time staff or hiring and firing. While the argument against this practice is quite convincing in as much as “it is any business’s responsibility ... to engage in a set of activities which are capable of sustaining employment at a steady level”, we are of the opinion that the workings of the “free-market” economy cannot be “regulated” with legislation preventing seasonal fluctuations in the demand for employment. What is your opinion?

In the case of the managed demand plan, some organisations attempt to transfer customer/client demand from peak periods to quiet periods by changing demand (through pricing, like offering discounts in off-peak periods) and/or creating alternative products/services to fill the capacity (by developing new outputs which can be produced using the existing processes). The latter, “more radical approach” of creating new products/services to fill the periods of low demand is discussed in the case example entitled “Getting the message” in chapter 11 of your prescribed book. Study the example and then answer the questions that follow.

- (4) *With reference to question 4, note that each of the three “pure” plans described above will only be applied when its advantages clearly outweigh its disadvantages. Many organisations only adopt one of these approaches and their choice does not always match their combination of competitive and operational objectives. Therefore, these organisations should follow a mixture of the three “pure” approaches to “simultaneously ... reduce costs and inventory, ... minimise capital investment, and yet to provide a responsive and customer-oriented approach at all times”. Study figure 11.11, which illustrates a mixed capacity plan for the woollen knitwear factory. According to this plan, some of the peak demand is moved forward by offering discounts (managed demand plan) and adjusting its capacity twice a year (chase demand plan) to reflect the broad changes in demand.*
 - (5) *With reference to question 5, yield management is used where the operation has a relatively fixed capacity, and it is important for generating revenues at full potential. It is a collection of methods used when (a) capacity is relatively fixed; (b) the market is fairly segmented; (c) the service cannot be stored; (d) the services are sold in advance; and (e) the marginal cost of making a sale is relatively low.*
 - (6) *With reference to question 6, the yield management approaches of over-booking capacity (selling more tickets for a movie than there are seats available on a particular day, but allowing cinemagoers to attend either morning, afternoon, evening or late evening sessions), price discounting (by offering discounts to pensioners, housewives, etc for quiet sessions) and varying service types (opening or closing cafeteria facilities, including or excluding new movie trailers, etc) could be used.*
-

11.7 SELECTING A CAPACITY PLANNING AND CONTROL APPROACH



Activity 11.6

Study the fifth section of chapter 11 of your prescribed book entitled “Choosing a capacity planning and control approach” and then answer the following questions:

- (1) Why is it important to be aware of the consequences of adopting a particular capacity planning and control approach for an operation, and how can this assessment be aided?
- (2) How can cumulative representations be used to assess various capacity plans?
- (3) How can the queuing theory be used to reduce the average time a customer/client spends waiting in a queue to an acceptable level?
- (4) What are the dynamics of capacity planning and control as governed partly by the combination of long-term and short-term outlooks, and how can the outlook matrix be used to manage capacity when the long-term and short-term outlooks are in various combinations of poor, normal and good?



Feedback on Activity 11.6

- (1) *With reference to question 1, the importance of being aware of the consequences of adopting a particular capacity planning and control approach lie in the objective as expressed by the authors of your prescribed book in the subsection on mixed plans. The objective is to “reduce costs and inventory, ... [and] minimise capital investment, and yet to provide*

a responsive and customer-oriented approach at all times". Deciding which capacity planning and control approach to select for a particular operation, can be greatly aided by considering the cumulative representation of demand and capacity and the queuing theory.

- (2) *With reference to question 2, first study figure 11.12 in your prescribed book. The forecast aggregated demand and capacity levels are plotted in this figure. Note the periods of "overcapacity" and "undercapacity". Also consider the problems that flow from this approach, because (i) each month may not have the same amount of productive time owing to vacation periods, which reduce the available capacity and (ii) the capacity level that seems adequate may only be able to supply the products after the demand has arisen for them and no inventory has been accumulated (if there was a period of undercapacity at the beginning of the year, no inventory would yet have been accumulated). Now study figure 11.13. This figure illustrates the capacity plan, with demand plotted on a cumulative basis. The cumulative representation reveals more information than figure 11.12, for the following reasons:*

- (a) The total demand peaks in September, but the peak demand per productive day peaks a month earlier, in August.*
- (b) The fluctuation in demand over the full year is greater ($50:5 = 10:1$) than initially thought (ie $650:100 = 6.5:1$).*
- (c) The feasibility and consequences of various capacity plans can be plotted and assessed (see the inventory build-up and shortages effect of a level cumulative production rate of 14.03 tonnes per day in fig 11.13, and with inventory build-up but without shortages at a level cumulative production rate of 15.28 tonnes per day in fig 11.14).*
- (d) The inventory implications in terms of the cost of carrying inventory (ie total cost of R1 437 810 for the year) and inventory levels (average monthly peak of 806 tonnes in July).*

Chase demand plans can also be illustrated on a cumulative representation. With a level capacity plan the line would have a constant gradient, but for the chase demand plan the line would have a varying gradient exactly matched by the cumulative production line and equal to the cumulative demand line, hence there is no space between the lines and zero inventory. While inventory is eliminated in this case, costs are associated with adjustment in capacity levels – these costs are determined by the degree of change (ie 5% or 15% increase), the direction of change (ie it is usually more expensive to increase than to decrease capacity) and the capacity level from which the change is being made (it is usually less expensive to change capacity to what is regarded as "normal" than vice versa). Study the worked example of the chocolate manufacturer who is unhappy about the inventory costs of its level capacity plan and wishes to explore two alternative plans. Do you agree with the preference for plan 2?

- (3) *With reference to question 3, note that cumulative representation of capacity plans is useful where the operation can store its finished goods as inventory. In operations where goods and services cannot be produced before demand has occurred (like most service operations), capacity planning and control are best considered by using the waiting or queuing theory. Seen from this perspective, the queuing theory accepts that, while demand will sometimes be satisfied immediately, there are other times when some customers may have to wait because it is difficult to predict the arrival of individuals and/or because the time required to produce a product or service is uncertain. Study figure 11.16 in your prescribed book. This figure depicts the general form of the capacity decision in queuing systems. Note the various descriptions and explanations of the following terms: source of customers or calling population, finite or infinite source of customers, arrival rate, the queue, rejecting, balking, reneging, queue discipline and servers. Also study table 11.2, which provides some examples of operations that have parallel processors. The main dilemma in managing the capacity of queuing systems lies in determining how many servers to have available at*

any time to avoid unacceptably long queuing times or unacceptably low server utilisation. Capacity planning and control for this type of problem are often presented as a trade-off between customer waiting time and system utilisation. If the operation has too few servers, queues will build up and customers/clients will get dissatisfied with the time they have to wait before being served, but staff utilisation will be high. If the operation has too many servers, the waiting time will be short but the utilisation of staff low. Study the short case example entitled “Queuing, stress and airport capacity” in chapter 11 of your prescribed book and then answer the questions that follow. Note the discussion in your prescribed book concerning the effect of variability in demand and supply on the effective capacity of an operation. It reduces the operation’s ability to process inputs and, similarly, its effective capacity. Figure 11.17 in your prescribed book illustrates this effect. The implication according to the authors of your prescribed book “is that the greater the variability, the more extra capacity will need to be provided to compensate for the reduced utilization of available capacity”. Also study the principles that influence how customers perceive waiting times, as described by the authors of your prescribed book. The time spent idle is perceived as longer than the time spent occupied. The waiting time before the service starts is perceived as more tedious than waiting during the service process. Anxiety heightens the perception that time spent waiting is long. A wait of unknown duration is perceived as more tedious than a wait whose duration is known. Unexplained waiting is perceived as more tedious than explained waiting. The higher the value of the service to the customer/client, the longer the wait will be tolerated; and waiting on one’s own is more tedious than waiting in a group.

- (4) With reference to question 4, note that in practice, the management of capacity is a much more dynamic process than the planning of capacity. It involves controlling and reacting to actual demand and actual capacity, as shown in figure 11.18 of your prescribed book. In period $[t]$, production/operations management must consider its updated forecasts, its current capacity estimates and, where applicable, the inventory or shortages or queues being carried forward from the previous period $[t - 1]$, to plan the capacity level to meet actual demand for the following period $[t + 1]$. One of the main influences when making period-by-period capacity decisions is the confidence the production/operations managers have in future demand matching future capacity.
-

CHECKLIST



Now that you have completed study unit 11 on capacity management in operations you should be able to do the following:

- explain the meaning of capacity in the context of operations
- explain what capacity planning and control entails
- demonstrate how demand and capacity are measured in operations
- explain which alternative capacity plans or methods of responding to demand fluctuations are available in operations
- demonstrate how a capacity planning and control approach for a particular operation would be selected
- use the knowledge, insight and skills you have acquired regarding
 - the meaning of capacity in the context of operations
 - the task of capacity planning and control
 - measuring demand and capacity in operations

- alternative capacity plans or methods of responding to demand fluctuations in operations
- selecting a capacity planning and control approach for a particular operation
- to justify the importance of capacity planning and control and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for capacity management by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

NOTE: The whole of the supplement to chapter 11 entitled “Analytical queuing models” in your prescribed book (2010: 336–341) may be left out and is NOT to be studied for examination purposes for MNO2601.

SUMMARY



This study unit examined capacity planning and control in operations. You ought to have a good understanding of the importance of and steps in capacity planning and control, and be able to apply the concepts in practice. Together with the previous study unit, study unit 10, this study unit on the general aspects of planning and control in the production/operations management system completes the discussion of topic 8. In the next topic, topic 9, we examine issues in aggregate planning and master scheduling.

ASSESSMENT



- (1) Consider the case study entitled “Beyerskloof Farm” at the end of chapter 11 of your prescribed book and then answer the two questions posed at the end.
- (2) Do problems 1 to 6 at the end of chapter 11 of your prescribed book. Note that these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 11 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 9

Aggregate planning and master scheduling

AIM



To provide learners with an understanding of what the task of aggregate planning and master scheduling entails and why it is important for production/operations management.



Learning outcomes

When you have worked through this study unit you should be able to do the following:

- describe what enterprise resource planning (ERP) entails and how it has developed
- explain what the information input requirements of MRP are and what the outputs derived from this will be
- demonstrate how MRP calculations are done
- describe what manufacturing resource planning (MRP II) entails and explain how ERP, including web-integrated ERP, supply chain ERP and optimised production technology (OPT), are developing
 - use the knowledge, insight and skills you have acquired regarding
 - enterprise resource planning (ERP)
 - the information input requirements of MRP and the outputs derived from these
 - MRP calculations, and
 - manufacturing resource planning (MRP II) and ERP, including web-integrated ERP, supply chain ERP and optimised production technology (OPT)
- to justify the importance of aggregate planning and master scheduling and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for aggregate planning and master scheduling by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 12

Enterprise resource planning



Key concepts

enterprise resource planning (ERP)	"V"-shape product structure
material requirements planning (MRP)	"X"-shape product structure
MRP/MRP one (MRP I)	single-level BOM
manufacturing resource planning (MRP II)	indented BOM
MRP II/MRP two	inventory records
dependent demand system	item master file
demand management	transaction file
customers/clients orders	location file
forecast demand	MRP calculations
make-to-order	MRP netting process
make-to-stock	back-scheduling
purchase-to-order	closed-loop MRP
resource-to-order	resource requirements plan (RRP)
master production schedule (MPS)	rough-cut capacity plans (RCCP)
chase/level master production schedule	capacity requirements plan (CRP)
available to promise (ATP)	web-integrated ERP
bills of materials (BOM)	e-commerce
levels of assembly	supply chain ERP
product structure	collaborative commerce
product structure shapes	c-commerce (collaborative commerce)
"A"-shape product structure	optimised production technology (OPT)
"T"-shape product structure	theory of constraints (ToC)
	OPT systems
	OPT terminology (drum, buffer, rope)

GETTING AN OVERVIEW

In this study unit we address all five learning outcomes of topic 9. We describe what enterprise resource planning (ERP) entails and how it has developed, explain what the information input requirements of MRP are and what the outputs derived from it will be, demonstrate how MRP calculations are done, describe what manufacturing resource planning (MRP II) entails and explain how ERP, including web-integrated ERP, supply chain ERP and optimised production technology (OPT), is developing

12.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 14 of your prescribed book. Then study figure 14.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally read the points listed under the following questions at the end of chapter 14 of your prescribed book:

- | | |
|------------------------------|-----------|
| (1) "What is ERP?" | (2 marks) |
| (2) "How did ERP develop?" | (2 marks) |
| (3) "What is MRP?" | (3 marks) |
| (4) "What is MRP II?" | (2 marks) |
| (5) "How is ERP developing?" | (2 marks) |

12.2 OPERATIONS IN PRACTICE



Activity 12.1

Read the case example entitled "SAP in Financial Services" in chapter 14 of your prescribed book and then answer the following questions:

- (1) What would be the advantage of using the SAP ERP Banking solution set, as opposed to developing a new customised ERP banking solution for a company like Standard Bank?



Feedback on Activity 12.1

- (1) *In this specific question, we are asked why a company like Standard Bank could benefit from an SAP ERP solution set as opposed to developing their own ERP solutions. Take note that Standard Bank in South Africa became lead users in the banking industry as the SAP ERP solution set was developed. Four advantages are mentioned below. By using SAP's ERP solution set, Standard Bank: 1) Benefits from a broader base of intellectual capital. SAP's approach was to engage thought leaders from multiple banks to scope and architect the entire solution environment. 2) Benefits from shared processes among banks that have collaborated in non-competitive areas. These areas range from "open account" to "fetch balance" to "find business partner". 3) Benefits from SAP's Enterprise Services Architecture (ESA), engines, service repositories etc. Standard Bank benefits from "tried and trusted" credible expertise which ultimately offers stability in a market where stability is imperative. 4) Benefits from being able to focus its attention on its core competency, leaving the development of software and computerised ERP systems to recognised experts in the field.*
-

12.3 THE CONCEPT OF ENTERPRISE RESOURCE PLANNING



Activity 12.2

Study the first section of chapter 14 entitled “What is ERP?” and then answer the questions and/or complete the following:

- (1) ERP stands for and is the most recent development from the original, which as a system helped calculate and the they are required.
- (2) What is the essence of ERP and how did it develop?



Feedback on Activity 12.2

- (1) *With reference to question 1, ERP stands for Enterprise Resource Planning and is the most recent development to have evolved from the original material requirements planning (MRP), which as a system helped calculate the quantity and types of materials required (or volume calculations) and the times when they are required (or the timing calculations). See figure 14.2 in your prescribed book, which illustrates the development path to ERP and beyond.*
 - (2) *With reference to question 2, note that to “make sure that the right materials in the right quantities arrive at the right place at the right time” requires “planning and control, not only of materials but also finance, purchasing, people, equipment and many other activities”. ERP helps organisations to plan ahead for decisions of this kind and to understand the implications of any changes to the plan. It integrates information from all parts of the organisation, as illustrated in figure 14.3 of your prescribed book. It developed out of the original material requirements planning (MRP) philosophy, which could drive the basic planning and control mathematics of the volume and timing calculations for manufacturing in a fast, efficient and flexible way through computer processing. Consequently, manufacturing resource planning (MRP II) expanded the capabilities of MRP by allowing for LAN (Local Area Network) communication between different parts of the organisation and forward modelling of “what-if” scenarios. The strength of both MRP and MRP II lay in their ability to assess the consequences of required changes quickly and accurately (eg, if demand changes, the MRP systems could calculate all the “knock-on” effects and issue change instructions). In ERP systems, the same principle applies on a much wider scale, because all the parts of the organisation are integrated on the same databases. As a result, the consequences of any decisions affecting the planning and control in one part of the organisation will be reflected throughout the rest or other parts of the organisation. A further ERP development was made possible through web-based communication – parts of the business, suppliers and customers/clients that are geographically located at a distance, but that have similar ERP capabilities, may tie in and communicate with one another. This degree of integration, which could include technical, organisational and strategic concerns, could become possible with web-integrated ERP systems (also known as “collaborative commerce”).*
-

12.4 INFORMATION INPUTS AND OUTPUTS OF MATERIAL REQUIREMENTS PLANNING



Activity 12.3

Study the second section of chapter 14 of your prescribed book entitled “Material requirements planning (MRP)” and then answer the questions and/or complete the following:

- (1) The inputs to MRP I are the
 - (a) master production schedule (MPS), which provides information on, which has as its inputs customer/client orders, which provide information on and the demand forecast, which provides information on
 - (b) bills of materials, which provide information on
 - (c) inventory records, which provide information on
- (2) The outputs of MRP I are
 - (a) purchase orders, which show
 - (b) materials plans, which show
 - (c) work orders, which show
- (3) What does the term “demand management” encompass?
- (4) How are orders and forecasts combined to represent demand, and what is the significance of forecasting for different types of operation?
- (5) What are the inputs to the master production schedule and how would a chase and level MPS impact on the number of finished items of simple practical examples that need to be completed (or produced), that are available (inventory) and that are available to promise (ATP) in the weeks ahead?
- (6) How is the product structure broken down into various levels of assembly, and what are the distinguishing features of A-shape, T-shape, V-shape and X-shape product structures?
- (7) What is the difference between single-level and indented bills of materials, and how would they differ practically in appearance?
- (8) How is inventory managed in MRP systems, and why is the accuracy of the inventory files so important?



Feedback on Activity 12.3

- (1) *With reference to question 1, the inputs to MRP I are the following: (Also see fig 14.4).*
 - (a) *master production schedule (MPS), which provides information on demand, has as its inputs customer/client orders, which provide information on firm orders scheduled and the demand forecast, which provides information on realistic estimates of the quantity and timing of future orders*
 - (b) *bills of materials, which provide information on the product structure (level, part number, description and quantity)*
 - (c) *inventory records, which provide information on the parts or items that are already in stock in the form of finished goods, work in progress or raw materials, components, etc*
- (2) *With reference to question 2, the outputs of MRP I are the following: (See fig 14.4.)*
 - (a) *purchase orders, which show the quantity and time required for the net requirements of items, raw materials, components, parts, etc, that are bought from suppliers*

- (b) *materials plans, which show the material requirements at each level and part or component of the bill of materials for each product*
- (c) *work orders, which show the net requirements for items, components, parts, etc, that are made in-house*
- (3) *With reference to question 3, demand management combines the management of customer orders and sales forecasts, which feeds into the MPS and encompasses the set of processes that interface with customer and market.*
- (4) *With reference to question 4, confirmed customer orders provide a record of exactly what each customer has ordered, how many the customers have ordered and when they require delivery (known orders). (Note that the details of these sales orders may change several times after they have been placed.) The forecast of demand tries to predict what the likely orders (forecast orders) will be. The organisation uses the combination of confirmed customer orders and forecast of demand to represent demand. Note that the further ahead in the future, the less certainty there will be about demand. The mix of known orders and forecast orders will also be different for different types of operation. In a make-to-order business, the visibility of known orders over time will be greater than in a make-for-stock business. For the purchase-to-order business, most raw materials will only be ordered once confirmed customer orders are received. In a resource-to-order business, raw materials will not be ordered, and the business will not enter into contracts for labour and equipment.*
- (5) *The master production schedule (MPS) indicates the **quantity** and **timing of planned production**. This planned production will take into account the **quantity** and **timing** of desired delivery. It will also take on-hand inventory into consideration.*

*Against this backdrop, we acquire an understanding of the **three inputs** to the master production schedule, namely:*

- *Input 1: The beginning inventory*
- *Input 2: Forecasts for each period*
- *Input 3: Customer orders (Quantities already committed to customers)*

*As far as **outputs** are concerned, the master scheduling process uses this information on a period-by-period basis to determine:*

- *Output 1: Projected inventory*
- *Output 2: Production requirements*
- *Output 3: Resulting uncommitted inventory. This is referred to as available-to-promise (ATP) inventory.*

*Production requirements (Output 2) for a master production schedule (MPS) will differ according to the **MPS strategy** adopted by the relevant company/ Master production scheduler.*

MPS Strategies:

Typically, an MPS schedule will follow either a chase production schedule or a level production schedule

- **Chase production schedule**

Chase production scheduling involves the MPS increasing as demand increases and aims to keep available inventory low.

- **Level production schedule**

Level production scheduling involves averaging the amount required to be completed to smooth out peaks and troughs; it generates more inventory than chase production scheduling.

The best means of explaining how to complete an MPS schedule is to work out an example and explain the actions step by step.

Details will be provided of a master production scheduling scenario. The details will include information relating to the three inputs (referred to above). You will be required to derive information relating to the three outputs (referred to above).

We will be working out the same example three times. The reason for this is that you will be expected to adopt a different MPS strategy each time. The first **two** attempts fall into the **chase production schedule** category. The **third** attempt falls into the **level production schedule** category.

MPS PROBLEM 1:

Assume you are required to complete the MPS time-phase record for the production of tables as an end item. The schedule below outlines the **three inputs**, including the relevant data, as well as the **three outputs**. The three outputs do not yet contain any data. This data will be calculated on the basis of the inputs as we go through the example:

		June				July			
		1	2	3	4	5	6	7	8
Input 2	Forecast	30	30	30	30	40	40	40	40
Input 3	Customer orders	33	4	10	4	2			
Output 1	Projected on-hand inventory								
Output 2	MPS (planned production)								
Output 3	Available-to-promise Inventory (uncommitted)								
Input 1	On hand inventory: 64								

Further to these three inputs, you will need information relating to the following:

- The **safety stock size**: Stocks in excess of average demand to cater for variability in demand and lead time.
- **The MPS strategy**

The **quantity** and **timing** of production (Output 2) will differ according to the MPS strategy adopted. The example above will be worked out three times according to three different MPS strategies:

- Chase demand – with fixed batch quantities (example 1.1)
- Chase demand – without fixed batch quantities (example 1.2)
- Level demand – with fixed batch quantities (example 1.3)

The **fixed batch quantity** is the number of items produced in a batch each time production is called for.

NB: Another term for “fixed batch quantities” is “production lot sizes”. These terms are used interchangeably, so make sure you are familiar with both!

Safety stock size and production (quantities and timing) are interconnected: Safety stock must never fall below a minimum level. This minimum level will differ from one organisation to the next. As stock approaches the minimum level, production is called for (in line with the MPS strategy), so as to avoid safety stock running below the minimum level.

The safety stock for all three scenarios will be 5 units.

EXAMPLE 1.1: CHASE DEMAND - WITH FIXED BATCH QUANTITIES

You are required to compile an MPS schedule for “MPS Problem 1” using a Chase Demand Strategy with Fixed Batch Quantities of 70.

MPS schedules should be carried out in two separate steps:

- Step 1: Completion of Output 1 (Projected on-hand inventory) and Output 2 (Planned production)
- Step 2: Completion of Output 3 (Available-to-promise inventory)

STEP 1: COMPLETION OF OUTPUT 1 (PROJECTED ON-HAND INVENTORY) AND OUTPUT 2 (PLANNED PRODUCTION)

First, we need to calculate the projected on-hand inventory, one week at a time, until it falls below the safety stock level. The projected on-hand inventory is calculated as follows:

$$\text{Projected on-hand inventory} = \text{Inventory from previous week} - \text{Current weeks requirements}$$



Current week's requirements = the greater of the forecast (Input 2) and customers orders (Input 3)

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Output 1 → Projected on-hand inventory	31	71	41	11	41	71	31	61
Output 2 → MPS (planned production)		70			70	70		70
Available-to-promise (uncommitted)								
On-hand inventory: 64								
Production lot size = 70								
Safety stock = 5								

64 (starting on-hand inventory) – 33 (customer orders in period 1) = 31 (on-hand inventory at end of period 1). We always begin with the starting on-hand inventory. In period 1 there is a forecast of 30 and customer orders of 33. We need to subtract the larger of the two, namely the 33 customer orders, from the starting on-hand inventory of 64 to get to the projected-on-hand inventory of 31 at the end of period 1.

The projected on-hand inventory at the end of period 2 is 71. The method of arriving at this figure is as follows: The on-hand inventory at the end of period 1 is 31. We then need to subtract the greater of the forecast and customer orders for period 2 to get to the projected on-hand inventory at the end of period 2. This sum would be as follows: $31 - 30 = 1$.

The problem with this projected on-hand inventory of 1 is that it is below the safety stock level of 5. To avoid this, we require a production, according to the lot size, of 70 units. The projected on-hand inventory for the end of period 2 now changes as follows: 31 (projected inventory on-hand at the end of period 1) – 30 (forecast for period 2) plus 70 (production lot size required in period 2) = 71 (projected inventory on-hand at end of period 2).

71 (on-hand inventory at end of period 2) – 30 (forecast in period 3) = 41 (on-hand inventory at end of period 3). We do not require production for this period, as projected on-hand inventory of 41 at the end of period 3 is greater than the safety stock level of 5.

41 (on-hand inventory at end of period 3) – 30 (forecast in period 4) = 11 (on-hand inventory at end of period 4). We do not require production for this period, as projected on-hand inventory of 11 at the end of period 4 is greater than the safety stock level of 5.

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Output 1 → Projected on-hand inventory	31	71	41	11	41	71	31	61
Output 2 → MPS (planned production)		70			70	70		70
Available-to-promise (uncommitted)								
On-hand inventory: 64								
Production lot size = 70								
Safety stock = 5								

11 (on-hand inventory at end of period 4) – 40 (forecast in period 5) = -29 (on-hand inventory at end of period 5). We therefore require production of 70 units for this period.

The projected on-hand inventory for the end of period 5 now changes as follows: 11 (on-hand inventory at end of period 4) – 40 (forecast for period 5) plus 70 (production lot size required in period 5) = 41 (projected inventory on hand at end of period 5).

41 (on-hand inventory at end of period 5) – 40 (forecast in period 6) = 1 (on-hand inventory at end of period 6). Since this is below the safety stock level of 5, we require production of 70 units for this period.

The projected on-hand inventory for the end of period 6 now changes as follows: 41 (on-hand inventory at end of period 5) – 40 (forecast for period 6) plus 70 (production lot size required in period 6) = 71 (projected inventory on hand at end of period 6).

71 (on-hand inventory at end of period 6) – 40 (forecast in period 7) = 31 (on-hand inventory at end of period 7). We do not require production for this period, as projected on-hand inventory of 31 at the end of period 7 is greater than the safety stock level of 5.

31 (on-hand inventory at end of period 7) – 40 (forecast in period 8) = -9 (on-hand inventory at end of period 8). We therefore require production of 70 units for this period.

The projected on-hand inventory for the end of period 8 now changes as follows: 31 (on-hand inventory at end of period 7) – 40 (forecast for period 8) plus 70 (production lot size required in period 8) = 61 (projected inventory on hand at end of period 8).

STEP 2: COMPLETION OF OUTPUT 3 (AVAILABLE-TO-PROMISE INVENTORY)

Now that we have completed Output 1 (Projected on-hand inventory) and Output 2 (Planned production), it is possible to calculate Output 3 (Inventory that is available-to-promise, and hence uncommitted).

Available-to-promise (ATP) inventory is that portion of a firm's inventory and planned production that is not already committed and is available to the customer. This allows delivery promises to be made and customer orders and deliveries to be scheduled accurately. Several methods are used in practice. The one we shall employ involves a "look-ahead" procedure. This method assumes that the ATP will be sold before the next scheduled receipt arrives. In the "look-ahead" procedure, the ATP quantity is only calculated for the **first week** and for **other weeks in which there is an MPS quantity**.

- **First week calculations**

In the first week, you are required to total customer orders for all the weeks **up to but excluding** the customer orders in the week in which there is an MPS quantity. This quantity is then subtracted from the on-hand inventory at the beginning to give you your ATP amount in week 1.

- **Other weeks in which there is an MPS quantity**

In the weeks in which there is an MPS quantity, you are required to total the customer orders for **that particular week** and all the weeks thereafter **up to but excluding** the customer orders in the week in which there is an MPS quantity. This quantity is then subtracted from the MPS (planned production quantity in that particular week).

Against the backdrop of this methodology, let us work through the calculations of the ATP in example 1.1:

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Projected on-hand inventory	31	71	41	11	41	71	31	61
MPS (planned production)		70			70	70		70
Output 3 Available-to-promise Inventory (uncommitted)	31	52			68	70		70
On hand inventory: 64								
Production lot size = 70								
Safety stock = 5								

The customer orders in week 1 are 33. We will NOT total the customer orders in week 2 because there is an MPS quantity in week 2.

We then subtract 33 from the on-hand inventory of 64 to get to the ATP quantity of 31.

The customer orders in week 2 are 4. In week 3, they are 10, and in week 4, they are 4. We therefore total the customer orders for weeks 2, 3 and 4 to get to the total customer orders of 18. We do not take the customer orders in week 5 into account, because there is an MPS quantity in week 5.

We then subtract 18 from the MPS quantity of 70 in week 2 to get the ATP of 52.

The customer orders in week 5 are 2. We do not take the customer orders in week 6 into account, because there is an MPS quantity in week 6.

We then subtract the 2 customer orders from the MPS quantity of 70 in week 5 to get the ATP in week 5 of 68.

The customer orders in week 6 are 0 and in week 7 they are also 0. Total customer orders in week 6 and 7 are therefore 0. We do not take the customer orders in week 8 into account, because there is an MPS quantity in week 8.

We then subtract 0 from the MPS quantity of 70 in week 6 to get the ATP in week 6 of 70.

The customer orders in week 8 are 0 and there is no further info with regard to customer orders.

We therefore subtract 0 from the MPS quantity of 70 in week 8 to get the ATP in week 8 of 70.

EXAMPLE 1.2: CHASE DEMAND - WITHOUT FIXED BATCH QUANTITIES

You are required to compile an MPS schedule for “MPS Problem 1” using a chase demand strategy whereby production should match demand, with a safety stock level of 5 units.

The difference in example 1.2 is that production will not be carried out in fixed quantities. We will produce only what we are required to produce, according to demand. We are also required to ensure that we never allow our stock to drop below the safety stock level of 5 units.

Once again, the MPS schedule should be done in two separate steps:

- Step 1: Completion of Output 1 (Projected on-hand inventory) and Output 2 (Planned production)
- Step 2: Completion of Output 3 (Available-to-promise inventory)

STEP 1: COMPLETION OF OUTPUT 1 (PROJECTED ON-HAND INVENTORY) AND OUTPUT 2 (PLANNED PRODUCTION)

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Output 1 → Projected on-hand inventory	31	5	5	5	5	5	5	5
Output 2 → MPS (planned production)		4	30	30	40	40	40	40
Available-to-promise (uncommitted)								
On hand inventory: 64								
Production lot size : n/a								
Safety stock = 5								

64 (starting on-hand inventory) – 33 (customer orders in period 1) = 31 (on-hand inventory at end of period 1). We always begin with the starting on-hand inventory. In period 1 there is a forecast of 30 and customer orders of 33. We need to subtract the larger of the two, namely the 33 customer orders, from the starting on-hand inventory of 64 to get to the projected on-hand inventory of 31 at the end of period 1.

The projected on-hand inventory at the end of period 2 is 5. The method of arriving at this figure is as follows: The on-hand inventory at the end of period 1 is 31. We then need to subtract the greater of the forecast and customer orders for period 2, to get to the projected on-hand inventory at the end of period 2. This sum would be as follows: $31 - 30 = 1$.

The problem with this projected on-hand inventory of 1 is that it is below the safety stock level of 5. To avoid this, we require sufficient production, to match demand **KEEPING A SAFETY STOCK LEVEL OF 5 UNITS**. With our 1 unit of inventory falling below the safety stock level of 5 units, we would therefore require production of 4 units in period 2 to end up with a safety stock level of 5 units.

5 (on-hand inventory at the end of period 2) – 30 (forecast in period 3) = -25 (inventory short at end of period 3). We therefore require production of 30 units to satisfy demand and keep the required safety stock level of 5 units.

		June				July			
		1	2	3	4	5	6	7	8
Forecast		30	30	30	30	40	40	40	40
Customer orders		33	4	10	4	2			
Output 1	Projected on-hand inventory	31	5	5	5	5	5	5	5
Output 2	MPS (planned production)		4	30	30	40	40	40	40
Available-to-promise (uncommitted)									
On hand inventory: 64									
Production lot size : n/a									
Safety stock = 5									

5 (on-hand inventory at end of period 3) – 30 (forecast in period 4) = -25 (inventory short at end of period 4).
We therefore require production of 30 units to satisfy demand and keep the required safety stock level of 5

5 (on-hand inventory at end of period 4) – 40 (forecast in period 5) = -35 (inventory short at end of period 5).
We therefore require production of 40 units to satisfy demand and keep the required safety stock level of 5

5 (on-hand inventory at end of period 5) – 40 (forecast in period 6) = -35 (inventory short at end of period 6).
We therefore require production of 40 units to satisfy demand and keep the required safety stock level of 5

5 (on-hand inventory at end of period 6) – 40 (forecast in period 7) = -35 (inventory short at end of period 7).
We therefore require production of 40 units to satisfy demand and keep the required safety stock level of 5

5 (on-hand inventory at end of period 7) – 40 (forecast in period 8) = -35 (inventory short at end of period 8).
We therefore require production of 40 units to satisfy demand and keep the required safety stock level of 5

STEP 2: COMPLETION OF OUTPUT 3 (AVAILABLE TO PROMISE INVENTORY)

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Projected on-hand inventory	31	5	5	5	5	5	5	5
MPS (planned production)		4	30	30	40	40	40	40
Available-to-promise Inventory (uncommitted)	31	0	20	26	38	40	40	40
On hand inventory: 64								
Production lot size : n/a								
Safety stock = 5								

The customer orders in week 1 are 33. We will NOT total the customer orders in week 2 because there is an MPS quantity in week 2.

We then subtract 33 from the on-hand inventory of 64 to get to the ATP quantity of 31.

The customer orders in week 2 are 4. We have not taken the customer orders in week 3 into account, because there is an MPS quantity in week 3.

We then subtract 4 (the customer orders in week 2) from the MPS quantity of 4 in week 2 to get the ATP of 0.

The customer orders in week 3 are 10. We have not taken the customer orders in week 4 into account, because there is an MPS quantity in week 4.

We then subtract 10 (the customer orders in week 3) from the MPS quantity of 30 in week 3 to get the ATP of 20.

The customer orders in week 4 are 4. We have not taken the customer orders in week 5 into account, because there is an MPS quantity in week 5.

We then subtract 4 (the customer orders in week 4) from the MPS quantity of 30 in week 4 to get the ATP of 26.

The customer orders in week 5 are 2. We have not taken the customer orders in week 6 into account (even if there had been orders), because there is an MPS quantity in week 6.

We then subtract 2 (the customer orders in week 5) from the MPS quantity of 40 in week 5 to get to the ATP of 38.

There are no customer orders in weeks 6, 7 and 8, but since there are MPS quantities in each of these weeks, we need to calculate the ATP. The ATP equals the MPS quantities because of the fact that there were no customer orders for these weeks.

EXAMPLE 1.3: LEVEL DEMAND - WITH FIXED BATCH QUANTITIES

You are required to compile an MPS schedule for “MPS Problem 1” using a level demand strategy with fixed batch quantities of 30, with a safety stock level of five units.

In example 1.3, production will be carried in fixed quantities of 30 units. We will produce 30 units within each period, regardless of demand. The lot size of 30 is sufficient to ensure that the safety stock does not drop below the minimum level of five units.

Once again, the MPS schedule should be carried out in two separate steps:

- Step 1: Completion of Output 1 (projected on-hand inventory) and Output 2 (planned production)
- Step 2: Completion of Output 3 (available-to-promise Inventory)

STEP 1: COMPLETION OF OUTPUT 1 (PROJECTED ON-HAND INVENTORY) AND OUTPUT 2 (PLANNED PRODUCTION)

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Output 1 → Projected on-hand inventory	61	61	61	61	51	41	31	21
Output 2 → MPS (planned production)	30	30	30	30	30	30	30	30
Available-to-promise (uncommitted)								
On hand inventory: 64								
Production lot size: 30								
Safety stock = 5								

Owing to the fact that we are dealing with level demand strategy and have been told that the fixed batch quantity is 30, the MPS figures will be 30 units for every period regardless of the forecast.

Calculation of inventory on-hand for period 1 is as follows: 64 (on-hand inventory) – 33 (customer orders in period 1) + the MPS quantity of 30 in week 1 = 61 units

The projected on-hand inventory at the end of period 2 is 61. Calculation of inventory on-hand for period 2 is as follows: 61 (on-hand inventory at end of period 1) – 30 (forecast in period 2) + the MPS quantity of 30 in week 2 = 61 units

The projected on-hand inventory at the end of period 3 is 61. Calculation of inventory on-hand for period 3 is as follows: 61 (on-hand inventory at end of period 2) – 30 (forecast in period 3) + the MPS quantity of 30 in week 3 = 61 units

		June				July			
		1	2	3	4	5	6	7	8
Forecast		30	30	30	30	40	40	40	40
Customer orders		33	4	10	4	2			
Output 1	Projected on-hand inventory	61	61	61	61	51	41	31	21
Output 2	MPS (planned production)	30	30	30	30	30	30	30	30
Available-to-promise (uncommitted)									
On hand inventory: 64									
Production lot size = 30									
Safety stock = 5									

The projected on-hand inventory at the end of period 4 is 61. Calculation of inventory on-hand for period 4 is as follows: 61 (on-hand inventory end of period 3) – 30 (forecast in period 4) + the MPS quantity of 30 in week 4 = 61 units.

The projected on-hand inventory at the end of period 5 is 51. Calculation of inventory on hand for period 5 is as follows: 61 (on-hand inventory end of period 4) – 40 (forecast in period 5) + the MPS quantity of 30 in week 5 = 51 units.

The projected on-hand inventory at the end of period 6 is 41. Calculation of inventory on hand for period 6 is as follows: 51 (on-hand inventory at end of period 5) – 40 (forecast in period 6) + the MPS quantity of 30 in week 6 = 41 units.

The projected on-hand inventory at the end of period 7 is 31. Calculation of inventory on hand for period 7 is as follows: 41 (on-hand inventory at end of period 6) – 40 (forecast in period 7) + the MPS quantity of 30 in week 7 = 31 units.

The projected on-hand inventory at the end of period 8 is 21. Calculation of inventory on hand for period 8 is as follows: 31 (on-hand inventory at end of period 7) – 40 (forecast in period 8) + the MPS quantity of 30 in week 8 = 21 units.

STEP 2: COMPLETION OF OUTPUT 3 (AVAILABLE-TO-PROMISE INVENTORY)

	June				July			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Customer orders	33	4	10	4	2			
Projected on-hand inventory	61	61	61	61	51	41	31	21
MPS (planned production)	30	30	30	30	30	30	30	30
Output 3 Available-to-promise Inventory (uncommitted)	61	26	20	26	28	30	30	30
On hand inventory: 64								
Production lot size = 30								
Safety stock = 5								

The customer orders in week 1 are 33. We have not totalled the customer orders in week 2 because there is an MPS quantity in week 2.

The ATP for week 1 is 61 units, calculated as follows: 64 (starting on-hand inventory) + 30 (week 1 MPS quantity) – 33 (week 1 customer orders) = 61.

The customer orders in week 2 are 4. We have not totalled the customer orders in week 3, because there is an MPS quantity in week 3.

The ATP for week 2 is 26 units and is calculated as follows: 30 (week 2 MPS) – 4 (week 2 customer orders) = 26.

The customer orders in week 3 are 10. We have not totalled the customer orders in week 4, because there is an MPS quantity in week 4.

The ATP for week 3 is 20 units, calculated as follows: 30 (week 3 MPS) – 10 (week 3 customer orders) = 20.

The customer orders in week 4 are 4. We have not totalled the customer orders in week 5, because there is an MPS quantity in week 5.

The ATP for week 4 is 26 units, calculated as follows: 30 (week 4 MPS) – 4 (week 4 customer orders) = 26.

The customer orders in week 5 are 2. We will not total the customer orders in week 6 (even if there are orders), because there is an MPS quantity in week 6.

The ATP for week 5 is 28 units, calculated as follows: 30 (Week 4 MPS) – 2 (Week 5 Customer orders) = 28.

There are no customer orders in weeks 6, 7 and 8, but since there are MPS quantities in each of these weeks, we need to calculate the ATP. The ATP equals the MPS quantities because of the fact that there are no customer orders for these weeks.

Also study table 14.1 for an example of a “chase” MPS, and note the calculations of the available stock and MPS requirements. Also study table 14.2 for an example of a “level” MPS, and note the calculations of the available stock and MPS requirements.

- (6) With reference to question 6, note that the MPS drives the rest of the MRP process. It works out the volume and timing of assemblies, subassemblies and materials required to meet the master production schedule. The product structure will show the various parts that go into making the final or end product and, in MRP terms, will break down the finished product into different levels of assembly with level 0 being the finished product, level 1 the parts and subassemblies that go into making the finished product, level 2 the parts and subassemblies that go into level 1, etc. Study figure 14.6 in your prescribed book regarding

a board game known as “Treasure Hunt”. Note the product structure and its breakdown into the various levels of assembly. The product structure is closely related to the design of the product itself and is reflected in the product structure shape. The product structure shape is determined by the number of components and parts used at each level (the more components and parts used the wider the structure would be) as well as the number of items made in-house versus those bought in complete (only requiring assembling). More bought-in assemblies make the product structure “shallow”, but if many parts or items are manufactured under the same manufacturer’s roof, the product structure is “deep”. Note that the product structure stops when it gets down to parts that are not made by the particular organisation but are bought from another operation and merely assembled in-house. Study figure 14.7 in your prescribed book and note the four different product structures and the typical operations associated with them (“A” shape – only one finished product that goes into a larger number of components; “T” shape – small number of raw materials and a standard process with a very wide range of highly customised end products; “V” shape – like the “T” shape, but with less standardisation; and “X” shape – a wide range of finished products with the cost saving and stability of large-volume production of modularised manufactured components).

- (7) With reference to question 7, the single-level bills of materials provide the details of the relationships between parts and subassemblies at one level at a time – see table 14.4 in your prescribed book and note that the format is level 0, level 1 (all items under each other) and level 2 (all items under each other). The relationships for several levels of the indented bills of materials are shown at the same time – see table 14.5 in your prescribed book and note the format is level 0 and level 1 (individual items with their level 2 parts, if applicable). Also study the worked example concerning the “Novelty Pencil Company” in chapter 14 of your prescribed book. Do you agree with the suggested solution, namely that the BOM information should be translated into a product structure?
- (8) With reference to question 8, first note that the BOM provides the MRP with the base data on the “ingredients” or structure of the products. Rather than simply taking the “ingredients” and multiplying them by the number of items demanded, thereby determining the total material requirements, MRP recognises that some of the required items may be available in stock in the form of finished goods, work in progress or raw materials. MRP therefore requires that inventory records be kept of these items in stock so that only the extra requirements (net requirements) needed to supplement the inventory for total demand are determined. Three files kept in the MRP system help manage inventory, namely the item master file (contains the unique standard identification code for each part or component), the transaction file (which tracks the quantity of inventory of each part kept) and the location file (identifies where the parts of inventory are physically kept or stored).

12.5 MRP CALCULATIONS



Activity 12.4

Study the third and fourth sections of chapter 14 of your prescribed book entitled “MRP calculations” and “Manufacturing resource planning (MRP II)”, then answer the questions and/or complete the following:

- (1) The core of the MRP procedure involves calculating and requirements.
- (2) What is the purpose of the MRP netting process and how is the process performed (or carried out) in simple practical examples?

- (3) Why is it necessary to back-schedule the quantities required in MRP and how would this be done in simple practical examples?
- (4) What do the MRP capacity checks entail (also referred to as “closed-loop MRP”), and how do the planning routines of resource requirements planning (RRP), rough-cut capacity plans (RCCP) and capacity requirements planning (CRP) enable an operation to check its production plans against sufficient resources?
- (5) What is the difference between materials requirements planning (MRP I), manufacturing resource planning (MRP II) and enterprise resource planning (ERP)?



Feedback on Activity 12.4

- (1) With reference to question 1, the core of the MRP procedure involves calculating the volume (how much) and timing (when or at which time) of requirements of materials that will satisfy the demand for the finished or end products.
- (2) With reference to question 2, the MRP netting process takes the master production schedule and “explodes” the schedule through a single-level bill of materials to determine how many subassemblies or parts are required. Before moving down to the next level of the product structure, it checks to see how many of the required parts are already in stock. Then it generates work orders or requests for the net requirements of items made in-house and/or purchase orders for the net requirements for items that are bought from suppliers. Study figure 14.8 in your prescribed book for an illustration of the MRP netting process for the “Treasure hunt” board game example.
- (3) Under feedback on activity 12.3, question 5 (master production scheduling), we described the role of the master production schedule (MPS) in showing the **end items** that manufacturing intends to produce/assemble.

The **end items** are produced/assembled from **components** that must be available in the right quantities and at the right time to meet the MPS requirements.

Materials requirements planning (MRP) helps us to establish an MRP schedule showing the components required at each level of the assembly and, based on lead times, calculates when these components will be needed.

This MRP schedule has the following three inputs:

Input 1: The MPS schedule

The MPS schedule indicates the **quantity** and **timing** of the end items required. To see how these relate to the previous section (master production scheduling under feedback on activity 12.3), look at the MPS schedule for the first example discussed in this section. The MPS indicates (on the basis of a chase strategy with fixed batch quantities of 70) that production is required in weeks 2, 5, 6, and 8

	June				July			
	1	2	3	4	5	6	7	8
MPS (planned production)		70			70	70		70

These end items could have been anything, but for the sake of clarity, let’s assume they were tables. We are therefore required to produce/assemble 70 tables in weeks 2, 5, 6 and 8.

To do that, we need to know **the parts (and the number of parts)** required to make one table. We therefore require Input 2 for an MRP schedule.

Input 2: The bill of materials

Simplified, the bill of materials (BOM) is a listing of the **parts required** to produce/assemble one end item. It also includes the **quantities of each part required** to produce/assemble one end item.

In the case of a table, perhaps we would require one wooden table top, four wooden legs and sixteen screws (four for each wooden leg)

To assemble these tables, we need to order the respective parts (listed above), but before we order, it is imperative that we find out:

1. How much **stock we currently have** of the respective items.
2. The **lead times** of the various items (time between when we order the parts and when we receive the parts).

Therefore, for an MRP schedule, we require Input 3.

Input 3: Inventory records

Inventory records supply two kinds of information. The first is called **inventory status**. The second is called **planning factors**.

1. **Input 3.1: Inventory status** – How much material is available of each part for future demand
2. **Input 3.2: Planning factors** – Includes information such as order quantities, lead times, and safety stock

To help us compile MRP schedules we make use of **two tools**:

Tool 1: Product structure tree

The product structure tree is a visual description of the requirements in a bill of materials, where all components are listed by levels.

Tool 2: Indented bill of materials

A table reflecting the following details for each part used in the production/assembly of the end item

- Part level (determined by Tool 1: product structure tree)
- Part number (provided)
- Part description (provided)
- Lead time (provided)
- Quantity required (determined with assistance from Tool 1: product structure tree)

As can be seen from Tool 2, the **indented bill of materials**, some items reflected in this tool are calculated with the assistance of Tool 1, the **product structure tree**. It is therefore necessary to complete the product structure tree first before you complete the **indented bill of materials**.

Against the backdrop of the underlying theoretical components to MRP schedules, the best means of explaining how to complete an MRP schedule in practice is to work an example and explain the actions step by step.

MRP PROBLEM 1:

A company manufactures product A, which is made up of one unit of B, and half a unit of C. Each unit of B is made up of one unit of D, two units of E, and one unit of F. Each unit of C requires half a unit of G and three units of H. The lead times to manufacture all these components are as follows:

- A 2 weeks
- B 1 week
- C 2 weeks
- D 2 weeks
- E 3 weeks
- F 1 week

G 2 weeks

H 1 week

All these parts have 20 units in stock. The company is required to make 100 units of A for delivery in seven weeks' time.

You are required to:

- a. Draw up a product structure tree.
- b. Draw up an indented bill of materials for the product.
- c. Draw up a gross materials requirements plan for the manufacturer of the product.
- d. Construct a net materials requirements plan for the manufacturer of the product.

Before we begin the example, let's take a brief look to see whether we have all the necessary information relating to the three inputs.

Input 1: The MPS schedule

"The company is required to make 100 units of A for delivery in seven weeks' time."

Input 2: The bill of materials

"product A, which is made up of one unit of B, and half a unit of C. Each unit of B is made up of one unit of D, two units of E, and one unit of F. Each unit of C requires half a unit of G and three units of H."

Input 3: Inventory records

Input 3.1 Inventory status:

"All these parts have 20 units in stock."

Input 3.2 Planning factors:

"The lead times to manufacture all these components are as follows:"

A 2 weeks,

B 1 week

C 2 weeks

D 2 weeks

E 3 weeks

F 1 week

G 2 weeks

H 1 week

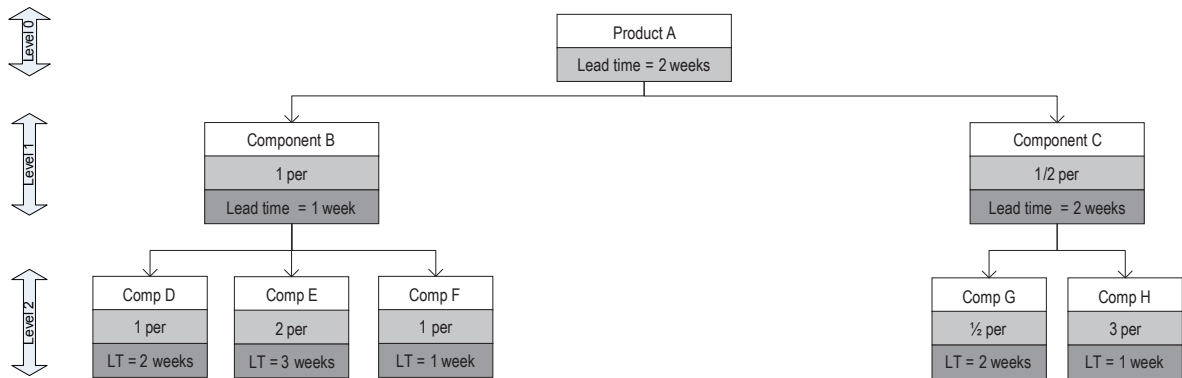
We see from the above exercise that, according to the three types of input, we have all the information we require in order to complete an MRP schedule for product A.

Let's approach the question as follows:

a. Draw up a product structure tree

To draw up a product tree, we need to source information from Input 2: the bill of materials and Input 3.2: the planning factors. We then break down the end item (product A in this instance), level by level, to reflect the parts (and number of parts) required to produce/assemble the end item. The product structure for product A is revealed below:

Product structure tree for item A



b. Draw up an indented bill of materials for the product.

Indented bill of materials for:

Part:	Product A
Description:	None
Level:	0

Level	Part no	Description	Lead time	Quantity
0	A	None	2 weeks	1
.1	B	None	1 weeks	1
.1	C	None	2 weeks	1/2
..2	D	None	2 weeks	1
..2	E	None	3 weeks	2
..2	F	None	1 weeks	1
..2	G	None	2 weeks	1/2
..2	H	None	1 weeks	3

c. Draw up a gross materials requirements plan for the manufacturer of the product.

d. Construct a net materials requirements plan for the manufacturer of the product.

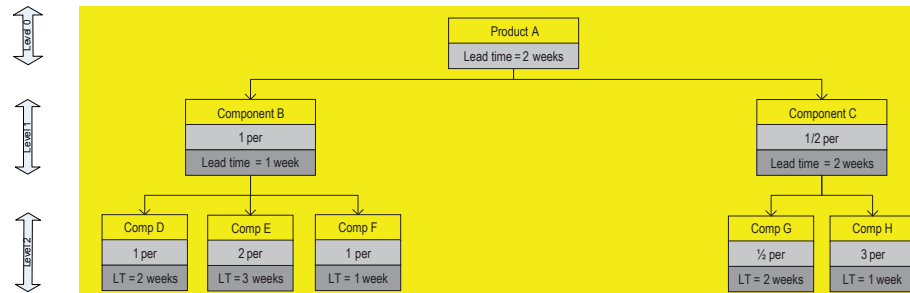
Part (c) of the question required you to draw up a gross materials requirements plan (without taking available stock into account) and part (d) a net materials requirements plan (taking available stock into account). The combined MRP schedules that follow include the information relating to both (c) and (d). The information relating to (c) is included in the line item "GROSS REQUIREMENTS" within the respective MRP schedules. The information relating to (d) is included in the line item "PLANNED ORDER RELEASE" within the respective MRP schedules.

The product structure for A as well as the MRP SCHEDULES OF ALL PARTS (A to H) are all included on the next page.

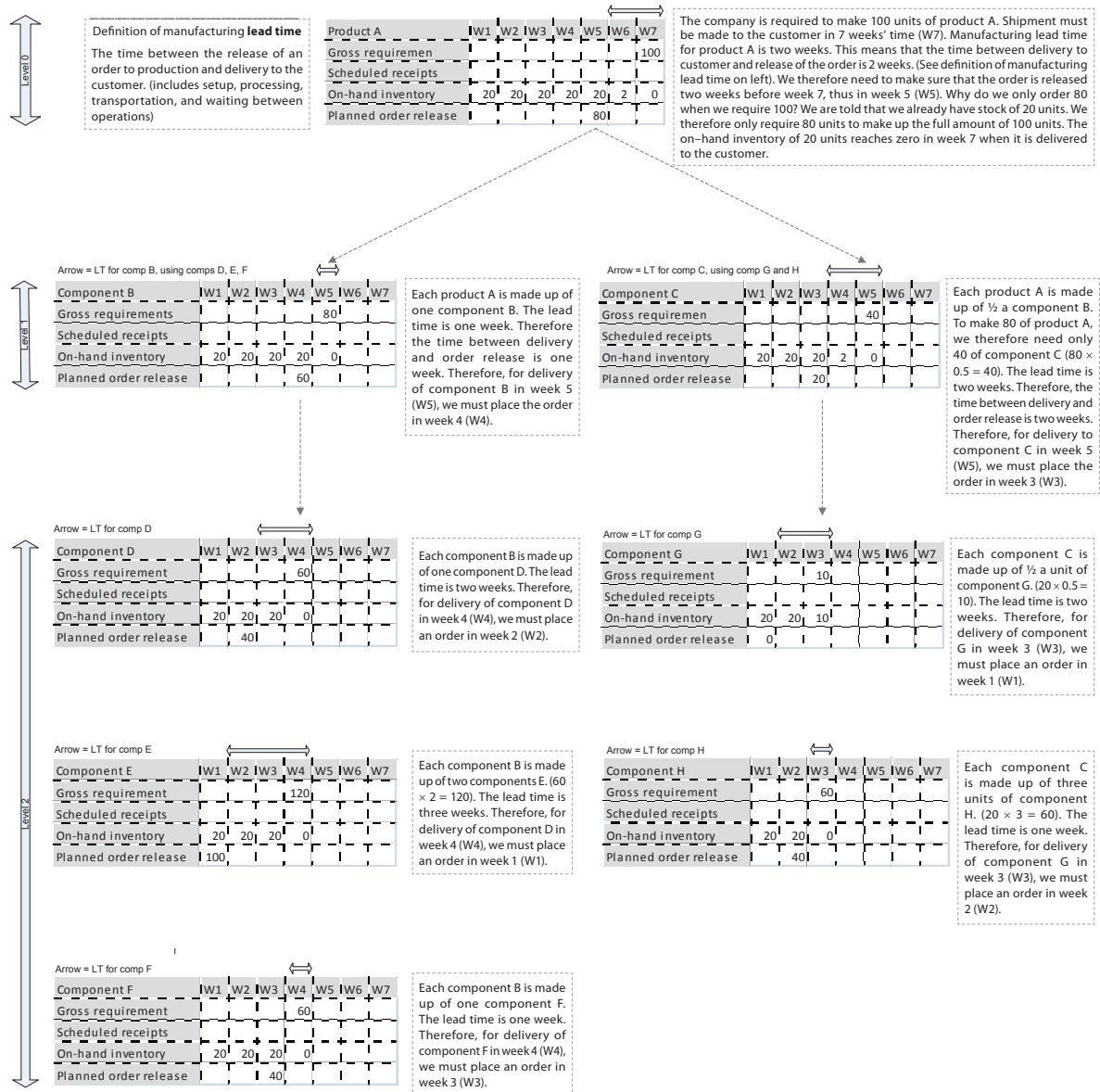
By placing all the information on the same page, it is easier to acquire an understanding of the logic and flow of information through the various levels. Please read the answer thoroughly, level by level, and read the EXPLANATORY NOTES that accompany each MRP schedule on the right-hand side.

Extra information has been included for the sake of EXPLAINING the logic.

PRODUCT STRUCTURE FOR PRODUCT A



MRP SCHEDULE CALCULATIONS (WITH EXPLANATIONS)



Also study table 14.7 ("Back-scheduling of requirements in MRP") and note the calculations in figure 14.9 ("Extract of the MRP records for the board game").

- (4) With reference to question 4, note that the MRP process needs checking to determine whether a plan is achievable. The "closed-loop" MRP therefore includes a feedback loop that facilitates the checking of production plans against available resources. If the plans

are not achievable at any level, they are revised through three planning routines. These are resource requirements plans (RRP), rough-cut capacity plans (RCCP) and capacity requirements plans (CRP). RRP focus on the long term to predict the requirements for large structural parts of the operation and are referred to as infinite capacity plans as they assume an almost infinite ability to set up production capacity if demand warrants it. RCCP are referred to as finite capacity plans because they have to operate within certain constraints. In the medium to short term, the MPS must use the capacity that is available and RCCP check the levels of capacity against known bottlenecks. CRP are infinite capacity plans in that they do not take the capacity constraints of each machine or work area into account. With CRP, the work orders may have a variable effect on the loading of particular machines and individual workers on a day-to-day basis. Study figure 14.10 in your prescribed book for an illustration of a closed loop MRP with its three planning routines of RRP, RCCP and CRP.

- (5) With reference to question 5, note that MRP was essentially aimed at the planning and control of production and inventory in manufacturing organisations. It was extended to MRP II, which is one integrated system containing a database that could be accessed by the whole business, including marketing, finance and engineering. ERP is a further development of MRP II. Its aim is to “integrate the management of the different functions within the business as a whole in order to improve the performance of all the interrelated processes in a business”.

12.6 ENTERPRISE RESOURCE PLANNING (ERP)



Activity 12.5

Study the fourth and fifth sections of chapter 14 in your prescribed book entitled “Enterprise resource planning (ERP)” and “Web-integrated ERP” and then answer the questions and/or complete the following:

- (1) What does ERP entail? What are the benefits that could be expected from its implementation? Why is it considered a “powerful” planning and control tool?
- (2) What benefits can be expected from the further development of ERP in the form of web-integrated ERP and supply chain ERP?
- (3) The OPT approach recognises the importance of planning to known capacity constraints and is based on the theory of constraints (TOC), which entails
- (4) What are the principles of OPT? How are these principles applied in practice?
- (5) To what extent are OPT and MRP/ERP compatible?



Feedback on Activity 12.5

- (1) With reference to question 1, ERP is a complete system of software support modules that integrates marketing, sales, product design, production and inventory control, procurement, distribution, process design and development, manufacturing, quality, human resources, finance and accounting etc and enables the sharing of information among these functions. The ERP discipline which offers greatly enhanced visibility that results from information integration is seen as a “double-edged” sword. On the one hand it keeps the management of every process within the organisation “on their toes” and promotes best practices, but on the other the rigidity of this discipline is both difficult to achieve and possibly not appropriate

for all parts of the business. Generally, accepted benefits include: absolute visibility of what is happening in all parts of the business; business process-based changes that are used to make all parts of the organisation more efficient; a better “sense of control” of operations as the basis for continuous improvement; more sophisticated and accurate communication with customers, suppliers and other business partners; and the integration of whole supply chains, including suppliers’ suppliers and customers’ customers. ERP is further considered to be a powerful planning and control tool because it is based on client/server architecture (the information systems are open to all stakeholders whose computers are linked to the central computer); it includes decision support facilities (decision makers have access to the latest company information); it can be linked to external extranet systems (the company’s supply chain partners through electronic data interchange [EDI]); it can interface with other standard application programmes (programmes like spreadsheets etc which are widely used by managers); and it is able to operate on most common platforms (operating systems like Windows, etc). Study the short case example entitled “The life and times of a chicken salad sandwich – Part two” in chapter 14 of your prescribed book and then answer the questions that follow. Finally, study the critical comments made by the authors of your prescribed book in chapter 14 about the implementation of ERP. Note that the implementation of ERP resulted in a negative or even zero return on investment in some companies because of the high expense of the software and associated expenses of consulting, training, etc. In addition the implementation of ERP may have a very disruptive impact on the existing organisation. We have heard very similar negative comments from organisations (public sector departments and private sector companies alike), but would venture to say that most “negatives” could be countered or avoided with sound prior planning, training, etc. ERP is an organisation-wide, integrated planning and control system that could result in a multitude of advantages if implemented correctly.

- (2) With reference to question 2, note that an ERP system gives the organisation the potential to link up with the “outside world” (its customers/clients and suppliers). It would therefore be much easier for an organisation to move to internet-based trading (e-commerce) if it could integrate its external internet systems with its internal ERP systems. While problems resulting from different information requirements (internal users, external customers/clients, suppliers, etc) may lead to increased ERP complexity, the next step is to integrate all ERP systems along the whole supply chain (“supply chain ERP”).
- (3) With reference to question 3, the OPT (optimised production technology) approach recognises the importance of planning to known capacity constraints. It is based on the theory of constraints (TOC), which entails focusing on the capacity constraints or bottlenecks in an operation, working to remove them, and then looking for the next constraint to improve the pace of output or throughput. The contribution of Eliyahu Goldratt to the development of the theory of constraints (TOC) is widely acknowledged. Additional material on this topic that is well worth reading may be found in the following books by Goldratt: *The goal: a process of ongoing improvement*, *It’s not luck* and *The critical chain*. These works are available through most of the major bookstores.
- (4) With reference to question 4, study the basic 10 principles of OPT as described by the authors of your prescribed book. Note that they primarily deal with production flow or throughput, bottlenecks, process batches and schedules.
- (5) With reference to question 5, OPT should not be viewed as a replacement for MRP/ERP, and they can run together, although conflict may arise in practice. MRP/ERP does not prescribe fixed lead times and batch sizes, although the systems are usually run that way. The focus on bottlenecks (which may change their location and severity because of the dynamic nature of unplanned variations in demand, supply and the process of manufacture) necessitates that lead times and batch sizes change throughout the operation (depending on whether a particular work centre is a bottleneck or not).

CHECKLIST



Now that you have completed study unit 12 on materials requirements planning in operations, you should be able to do the following:

- describe what enterprise resource planning (ERP) entails and how it has developed
- explain what the information input requirements of MRP are and what the outputs derived from these will be
- demonstrate how MRP calculations are done
- describe what manufacturing resource planning (MRP II) entails and explain how ERP, including web-integrated ERP, supply chain ERP and optimised production technology (OPT), is developing.
- use the knowledge, insight and skills you have attained regarding
 - enterprise resource planning (ERP)
 - the information input requirements of MRP and the outputs derived from these
 - MRP calculations, and
 - manufacturing resource planning (MRP II) and ERP, including web-integrated ERP, supply chain ERP and optimised production technology (OPT)
- to justify the importance of manufacturing resource planning and to demonstrate this by identifying both correct and incorrect statements with regard to, specifically the study material but also practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for materials requirements planning by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



This study unit examined enterprise resource planning in operations. It formed part of topic 9, which is entitled “Aggregate planning and master scheduling”. You may have noticed in the prescribed book (beginning of ch 11), which is also the study material for topic 8, study unit 11, that capacity planning and control are sometimes also referred to as “aggregate planning and control” or sometimes just “aggregate planning”. Aggregate planning as part of this topic was covered in study unit 11, while the main focus for the remainder of topic 9, namely master scheduling, was dealt with in this study unit (study unit 12).

ASSESSMENT



- (1) Consider the case study entitled “Psycho Sports Ltd” at the end of chapter 14 of your prescribed book and then answer the two questions posed at the end.
- (2) Do problems 1 to 5 at the end of chapter 14 of your prescribed book. Note: these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 14 of your prescribed book.

Note: Some study activities only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experiences of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Topic 10

Inventory and supply chain planning and control

AIM



To provide learners with an understanding of the task of inventory and supply chain planning and control and explain why they are important for production/operations management.



Learning outcomes

When you have worked through this study unit you should be able to do the following:

- explain the role of inventory in operations
- explain what the volume decision of inventory entails and demonstrate how this decision should be made by solving practical problems
- explain what the timing decision of inventory entails and demonstrate how this decision should be made by solving practical examples
- explain how inventory analysis and control systems can be used in operations and demonstrate their practical application
- explain what supply chain management entails
- explain the activities of supply chain management (including purchasing and supplier management on the supply side and physical distribution management on the demand side of the operation, and logistics and materials management)
- explain the types of relationship in supply chains
- explain supply chain behaviour
- use the knowledge, insight and skills you have acquired regarding
 - the role of inventory in operations
 - the volume decision of inventory
 - the timing decision of inventory
 - inventory analysis and control systems used in operations
 - supply chain management
 - the activities of supply chain management (including purchasing and supplier management on the supply side and physical distribution management on the demand side of the operation, logistics and materials management)
 - the types of relationship in supply chains, and
 - supply chain behaviour
- to justify the importance of supply chain planning and control and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for inventory and supply chain planning and control by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

Study unit 13

Inventory planning and control



Key concepts

inventory planning and control

rate of supply

supporting the performance

objectives

buffer inventory

cycle inventory

anticipation inventory

position of inventory

two-stage inventory systems

raw materials/component inventory

finished goods inventory

volume decision

inventory holding costs

consignment stock

$[Q]$ = order quantity

$[D]$ = demand over a period

$[Q] \div [D]$ the time interval between deliveries

economic order quantity (EOQ)

$[Ch]$ = cost of holding inventory

EOQ formula = $\frac{\sqrt{2[CoD]}}{[Ch]}$

order frequency = $\frac{[D]}{EOQ}$ per period

economic batch quantity (EBQ)

EBQ formula = $\frac{\sqrt{2[CoD]}}{[Ch](1 - [D] \div [P])}$

the inventory timing decision

re-order point (ROP)

buffer or safety stock

optimum time interval between

orders $[t_r]$

$[t_r] = \frac{EOQ}{[D]}$

inventory analysis and control

ABC system

Pareto law

inventory or "stock"

rate of demand

types of inventory

safety inventory/stock

de-coupling inventory

pipeline inventory

single-stage inventory systems

multistage inventory systems

work-in-progress inventory

multi-echelon inventory systems

inventory costs

inventory ordering costs

inventory profiles

$[Q] \div 2$ = the average inventory

$[P]$ = production over a period

$[D] \div [Q]$ the frequency of deliveries

$[Co]$ = cost of ordering inventory

$[D]$ = demand

time between orders = $\frac{EOQ}{[D]}$

economic manufacturing quantity (EMQ)

lead time

re-order level (ROL)

continuous/periodic review approach

inventory priorities systems

usage value = usage rate $\times$ individual value

80/20 rule (80% of sales account for 20% of stock)

stock turn

GETTING AN OVERVIEW

In this study unit we address the first four learning outcomes of topic 10. We explain the role of inventory in operations, explain what the volume decision of inventory entails and demonstrate how it should be made by solving practical problems, explain what the timing decision of inventory entails and demonstrate how it should be made by solving

practical examples, explain how inventory analysis and control systems can be used in operations and demonstrate their practical application.

13.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 12 in your prescribed book. Then study figure 12.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 12 in your prescribed book:

- "What is inventory?" (2 marks)
- "Why is inventory necessary?" (2 marks)
- "What are the disadvantages of holding inventory?" (3 marks)
- "How much inventory should an operation hold?" (3 marks)
- "When should an operation replenish its inventory?" (2 marks)
- "How can inventory be controlled?" (2 marks)

13.2 OPERATIONS IN PRACTICE



Activity 13.1

Read the case example entitled "South Africa's National Blood Transfusion Service (SANBS)" in chapter 12 of your prescribed book and then answer the following questions:

- (1) What are the challenges with respect to inventory management at the South African National Blood Transfusion Service (SANBS)?



Feedback on Activity 13.1

- (1) *With reference to question 1, a number of issues related to inventory management are of particular importance to the SANBS. One of these is shelf life. Blood stock has to be replaced regularly to keep a current supply of fresh blood and blood products. The case highlights the different periods of shelf life for red blood cells, platelets, fresh frozen plasma, cryoprecipitate and frozen plasma, which vary between five days and four years. Another challenge they face is "counter cyclical patterns" around supply and demand. Winter and holidays are peak periods in terms of demand owing to illness and road accidents, but low periods in terms of supply owing to illness and travel. This problem reinforces the importance of shelf-life and careful planning from an inventory management point of view.*
-

13.3 THE ROLE OF INVENTORY IN OPERATIONS



Activity 13.2

Study the first section of chapter 12 of your prescribed book entitled “What is inventory?” and then answer the questions and/or complete the following:

- (1) Inventory or stock is defined as and it is also referred to as transformed input resources because
- (2) Why is it necessary for all operations to keep inventories? What does the strategic role of inventories entail?
- (3) What are the disadvantages of holding inventory? In which circumstances would the following types of inventory be held: buffer inventory, cycle inventory, de-coupling inventory, anticipation inventory and pipeline inventory?
- (4) Inventory can be held at several points between different stages in the operation where there is an imbalance between supply and demand. Where is/are inventory/inventories held in single-stage, two-stage, multistage and multi-echelon inventory systems?
- (5) Which three day-to-day inventory decisions need to be taken concerning the management of the inventory system? Why are they important for all operations?



Feedback on Activity 13.2

- (1) *With reference to question 1, note that inventory or stock is defined by the authors of your prescribed book as “the stored accumulation of material resources in a transformation system”. It is also referred to as transformed input resources, because it primarily comprises materials that are treated, transformed or converted in the transformation process to produce products.*
- (2) *With reference to question 2, keeping inventory for the operation in the form of raw materials, components, parts, finished products etc has a direct bearing on all its performance objectives, including speed, flexibility, quality, cost and dependability. Study table 12.1 in your prescribed book for a list of examples of inventory held in different types of operations. Note that the value of the inventories may differ considerably – some may be relatively small in comparison with the total input cost (for example the hotel) while others may be high (for example the retail store). Inventory is sometimes seen as a “necessary evil”. Without it or with too little of it, the operation would probably not be able to run smoothly, but with too much inventory, the operation becomes too expensive to run. Between these extremes it must find and maintain a position where the operation can function optimally. Study figure 12.2 in your prescribed book and note the common analogy of the water tank. If the rate of supply exceeds the rate of demand, the tank will fill up (inventory), but if the rate of supply is less than rate of demand, the tank will run dry.*
- (3) *With reference to question 3, note that while inventory plays an important role in helping the operation to achieve its performance objectives, there are several good reasons why too much inventory should not be held. Inventory is an asset that ties up money (in the form of working capital). The money tied up in inventory cannot be used for other purposes like reducing debt or new investments in productive assets, and is therefore an “opportunity” cost associated with the money. Also, inventory does not directly add value to the operational process or the product. It can become obsolete or damaged, or its condition can deteriorate. It can get lost or be difficult to retrieve among other inventory. It can be expensive to store inventory because of its hazardous nature, or it can take up excessive storage space, incur high administrative and insurance costs, etc. For various reasons, an*

imbalance between the rates of supply and demand might arise at different points in the operation, and therefore different types of inventory are required. These include buffer inventory, cycle inventory, de-coupling inventory, anticipation inventory and pipeline inventory. Buffer inventory is also known as "safety inventory or stock", and is needed to compensate for any unexpected fluctuations in supply and demand. Cycle inventory is needed when products are produced in batches, and extra quantities of the products must be made and kept in stock to satisfy the demand until the next time a batch is produced (see fig 12.3 in your prescribed book). Decoupling inventory is where the operation uses a process layout (eg a kitchen), and work can be done relatively independently at each work centre. Work-in-progress inventory or decoupling inventory is required before each work centre can ensure its maximum local utilisation and efficiency. Anticipation inventory is needed where fluctuations in demand are high but predictable and extra products are made and held in stock for future demand. Finally, pipeline inventory is needed where products or materials cannot be transported instantaneously between the point of supply and the point of demand, and are held in stock until they can be transported.

- (4) *With reference to question 4, the position of inventory can also differ, depending on the location of the imbalance between supply and demand in the operation. Study figure 12.4 in your prescribed book. Note the four types of inventory relationships and the different points where inventory or stock is held. In the multistage inventory system, inventory is held for raw materials and components, work-in-progress and finished goods. In the multi-echelon inventory system, inventory is held between the various operations within the supply network as a whole.*
 - (5) *With reference to question 5, production/operations managers need to take three types of inventory decisions on a day-to-day basis, namely, how much to order (the volume decision), when to order (the timing decision) and how to control the inventory (inventory analysis and control systems).*
-

13.4 DETERMINING THE QUANTITY AND TIMING OF INVENTORY TO ORDER



Activity 13.3

Study the second and third sections of chapter 12 in your prescribed book entitled "The volume decision – how much to order" and "The timing decision – when to place an order" and then answer the following questions:

- (1) Which types of inventory costs need to be considered before management can decide how much to order?
- (2) What is an inventory profile? How is it used to determine the order quantity and order timing?
- (3) How could the standard economic order quantity (EOQ) formula be used to determine the quantity of a particular item to be ordered, and in which circumstances could it be used?
- (4) How could the standard economic batch quantity (EBQ) formula be used to determine the quantity of a particular item to be ordered, and in which circumstances could it be used?
- (5) What are the criticisms levelled at the EOQ approach and how valid are they?

- (6) How could calculations of the re-order point (ROP) and the re-order level (ROL) be used to determine when a particular item should it be ordered, and in which circumstances could it be used?
- (7) What is the difference between the continuous and periodic review approaches to replenishment timing decisions?
- (8) How could the EOQ formula be used to determine the time interval between placing orders, and what do two-bin and three-bin systems entail?



Feedback on Activity 13.3

- (1) *With reference to question 1, the decision about order quantities balances two sets of costs, namely the cost of ordering and obtaining the supplies on the one hand and the cost associated with keeping the supplies in stock on the other. Note the seven types of costs associated with the order-quantity decision as described by the authors of your prescribed book. The first three (the cost of placing the order, price discount costs and stockout costs) decrease as the order size increases, while the remainder (working capital costs, storage costs, obsolescence costs and production inefficiency costs) generally increase as the order size increases. Some of these costs are not necessarily incurred by the same organisation – in the case of consignment stock, for example, a supplier may agree to deliver large quantities of inventory to a customer, but only actually charge for the goods as and when they are used.*
- (2) *With reference to question 2, inventory profiles are visual representations of inventory levels over time. Study figure 12.5 in your prescribed book and note the terminology and calculation of $[Q]$, the order quantity; $[Q] \div 2$, the average inventory; $[Q] \div [D]$, the time interval between deliveries; and $[D] \div [Q]$, the frequency between deliveries.*
- (3) *With reference to question 3, note the standard EOQ formula as shown in your prescribed book. This formula is used in an attempt to find the best balance between the cost of holding stock over a period ($[Ch]$) and the cost of placing an order ($[Co]$). Study table 12.2 in your prescribed book, which illustrates the cost of adopting plans with different order quantities. Note that the minimum total cost for the example is found where the order quantity $[Q]$ equals 200. Using the EOQ formula, and given a demand $[D]$ of 1000 units per year, order costs $[Co]$ of R20.00 per order, and holding costs $[Ch]$ of R1.00 per item per year:*

$$\begin{aligned}
 \text{EOQ} &= \frac{\sqrt{2[CoD]}}{[Ch]} \\
 &= \frac{\sqrt{2 \times 20 \times 1\,000}}{1} \\
 &= \frac{\sqrt{40\,000}}{1} \\
 &= 200 \text{ units}
 \end{aligned}$$

Study figure 12.7, which depicts a graphical representation of the economic order quantity for the example above, where the total costs (holding costs + ordering costs) are minimised at $\text{EOQ} = 200$ units. However, note that any small deviation from the EOQ quantity will not increase the total costs significantly and therefore a near optimum, provided the value of $[Q]$ is close to the EOQ, would still be acceptable, especially should discounts for certain quantity intervals be on offer. Also study the worked example of the building materials supplier in chapter 12 of your prescribed book and note the small total cost difference for the more convenient placement of the order for 100 tonnes of cement rather than 91,287 tonnes.

- (4) With reference to question 4, note that in some situations replenishment may occur on a gradual basis over a period of time rather than in a single batch according to the standard EOQ formula. In such cases the economic batch quantity (EBQ), also known as the economic manufacturing quantity (EMQ), is more appropriate. Study figure 12.8, which depicts the inventory profile for gradual replacement of inventory and note the EBQ formula:

$$\text{EBQ formula} = \frac{\sqrt{2[CoD]}}{[Ch](1-[D] \div [P])}$$

Also study the worked example of the bottle-filling plant in chapter 12 of your prescribed book and note that $[D]$, demand (80 000 bottles per month) must be expressed in the same unit of measurement as $[P]$, production rate (3 000 bottles per hour). Thus $[D]$, demand is 80 000 divided by the number of production hours per month (160 hours) = 500 bottles per hour.

- (5) With reference to question 5, note the critical comments made by the authors of your prescribed book about the criticism levelled at the EOQ and EBQ models. Their comments cover the following points:
- (a) The simplistic nature of the assumptions of EOQ models. (Assumptions were made that would rarely prove to be true in practice, although most can approximate reality. Where they are found to be completely incorrect, the use of EOQ models should be considered inappropriate.)
 - (b) The argument that the “real” cost of stock does not correspond to the assumed cost in the EOQ models. (Should the slope of the holding costs be increased to reflect a more accurate picture, the total costs of inventory would be higher and the minimum cost point would fall to the left of the previous point. This favours lower economic order quantities and implies that operations less willing to hold stock on the grounds of cost should order smaller quantities but do so more frequently. (See fig 12.9.)
 - (c) The argument that the models are descriptive and should not be used prescriptively. (This is primarily a philosophical criticism from the Japanese-inspired just-in-time [JIT] approach, which focuses on reducing the overall level of inventory rather than on determining the optimum order quantity. It also tries to reduce the cost of placing orders by reducing changeover times or cost of changeover.)

Also note the argument that cost minimisation is not an appropriate objective for inventory management. The EOQ model does not address objectives like maximising the investment in inventory. (Organisations whose main investment is inventory focus on making a good return on their inventory investment and the highest possible inventory turnover.) Study the short case example entitled “Sappi Paper” in chapter 12 and then answer the questions that follow in the prescribed book.

- (6) With reference to question 6, if demand were steady and orders arrived instantaneously, the decision of when to place an order would be self-evident – namely, the order would be placed as soon as the stock level reached zero. However, if replenishment orders do not arrive instantaneously and a lag occurs between when the order is placed and when it arrives in inventory, the timing of the replacement order needs to be determined by taking into account the order lead time. Study figure 12.11, which is a representation of the timing of a replacement order by determining the re-order point (ROP) (the point in time when stock falls to zero minus the lead time [week 4 minus 2 weeks = week 2]) or re-order level (ROL) (the level that the inventory will have reached when the replacement order must be placed [demand per week multiplied by lead time = $100 \times 2 = 200$ items]). The assumption here is that demand and the order lead times are perfectly predictable, which they will not be – see figure 12.12 in your prescribed book. Safety stock is held because of the unpredictability and variability of both lead times $[t]$ and demand rate $[d]$. The key statistic in calculating how much safety stock to allow is the probability statistic, which shows the lead-time usage. Its distribution is a combination of the distributions, which describe lead-time variation and the demand rate during lead time. When the safety stock is set below the lower limit

of the distribution, shortages will occur in every replenishment cycle, but if the level is set above the lower limit, there will be no chance of stockouts. Study the worked example of the company that imports running shoes in chapter 12 of your prescribed book. Note the following:

- (a) The order lead-time histogram at the top left-hand side of figure 12.13 reflects the data given on previous orders. Out of 10 orders, one order took one week ($1 \div 10 =$ probability thus 0,1), two took two weeks ($2 \div 10 =$ probability thus 0,2), four took three weeks ($4 \div 10 =$ probability thus 0,4), two took four weeks ($2 \div 10 =$ probability thus 0,2) and one took five weeks ($1 \div 10 =$ probability thus 0,1). These lead-time probabilities are transferred to the horizontal part of the matrix of table 12.3 in your prescribed book.
- (b) The demand rate histogram on the top right-hand side of figure 12.13 in your prescribed book reflects the data given on shoes. They range from 110 to 140 pairs per week (0,2 probability of demand being either 110 or 140 pairs per week, 0,3 probability of demand being either 120 or 130 pairs per week). These demand probabilities are transferred to the vertical part of the matrix of table 12.3 in your prescribed book.
- (c) Table 12.3 in your prescribed book shows a matrix of the combined probabilities of the demand rate multiplied by the lead-time (ie first column to its last row – $110 \times 1 = 110$; $0,2 \times 0,1 = 0,02$; $120 \times 1 = 120$; $0,3 \times 0,1 = 0,03$; $130 \times 1 = 130$; $0,3 \times 0,1 = 0,03$; $140 \times 1 = 140$; $0,2 \times 0,1 = 0,02$, second column to its last row – $110 \times 2 = 220$; $0,2 \times 0,2 = 0,04$; $120 \times 2 = 240$; $0,3 \times 0,2 = 0,06$; $130 \times 2 = 260$; $0,3 \times 0,2 = 0,06$; $140 \times 2 = 280$; $0,2 \times 0,2 = 0,04$, third column to its last row – $110 \times 3 = 330$; $0,2 \times 0,4 = 0,08$; $120 \times 3 = 360$; $0,3 \times 0,4 = 0,12$; $130 \times 3 = 390$; $0,3 \times 0,4 = 0,12$; $140 \times 3 = 420$; $0,2 \times 0,4 = 0,08$, fourth column to its last row – $110 \times 4 = 440$; $0,2 \times 0,2 = 0,04$; $120 \times 4 = 480$; $0,3 \times 0,2 = 0,06$; $130 \times 4 = 520$; $0,3 \times 0,2 = 0,06$; $140 \times 4 = 560$; $0,2 \times 0,2 = 0,04$, fifth column to its last row – $110 \times 5 = 550$; $0,2 \times 0,1 = 0,02$; $120 \times 5 = 600$; $0,3 \times 0,1 = 0,03$; $130 \times 5 = 650$; $0,3 \times 0,1 = 0,03$; $140 \times 5 = 700$; $0,2 \times 0,1 = 0,02$).
- (d) Table 12.4 in your prescribed book shows the combined probabilities for lead-time usage ranges of 100–199 (probability = $0,1 = 0,02 + 0,03 + 0,03 + 0,02$), 200–299 (probability = $0,2 = 0,04 + 0,06 + 0,06 + 0,04$), 300–399 (probability = $0,32 = 0,08 + 0,12 + 0,12$), 400–499 (probability = $0,18 = 0,08 + 0,04 + 0,06$), 500–599 (probability = $0,12 = 0,06 + 0,04 + 0,02$), 600–699 (probability = $0,06 = 0,03 + 0,03$), 700–799 (probability = $0,02 = 0,02$) with the probabilities being added as indicated here from table 12.3 in your prescribed book.
- (e) Table 12.5 in your prescribed book is then derived from table 12.4 and shows the lead-time usage and probability of stockout for the various re-order levels. Re-order level 100 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,1 + 0,2 + 0,32 + 0,18 + 0,12 + 0,06 + 0,02 = 1,0 = 100\%$). Re-order level 200 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,2 + 0,32 + 0,18 + 0,12 + 0,06 + 0,02 = 0,9 = 90\%$). Re-order level 300 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,32 + 0,18 + 0,12 + 0,06 + 0,02 = 0,7 = 70\%$). Re-order level 400 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,18 + 0,12 + 0,06 + 0,02 = 0,38 = 38\%$). Re-order level 500 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,12 + 0,06 + 0,02 = 0,20 = 20\%$). Re-order level 600 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,06 + 0,02 = 0,08 = 8\%$). Re-order level 700 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,02 = 2\%$). Re-order level 800 = probability that lead-time usage will exceed available inventory and stockout will occur ($0,00 = 0\%$).
- (f) The question is when the company should place a replenishment order if the probability of stockout is to be less than 10%. The answer is, therefore, at re-order level 600, where

the probability that lead-time usage will exceed available inventory and stockout will occur is 8%.

- (7) With reference to question 7, the continuous review approach makes the decision concerning the timing of stock replenishment by continuously reviewing the current level of stock and re-ordering as soon as it reaches its re-order level. This means the timing of orders is irregular, but there is a constant re-order size [Q], which can be set at the optimum economic order quantity. With a periodic review approach, the fixed order quantity, which could be the optimum economic order quantity, is sacrificed, but the decision to replenish stock is made at a fixed, regular time interval. Study figure 12.14, which is a representation of the periodic review approach with probabilistic demand and lead-time elements. Note that the re-order quantities [Q1, Q2, Q3] vary in size each time a review is done at fixed intervals [T1, T2, T3].
- (8) With reference to question 8, the optimum time interval between placing orders, $[t_r]$ is usually calculated on a deterministic basis and derived from the EOQ using the formula:

$$[t_r] = \frac{EOQ}{[D]}$$

Therefore if the EOQ equals 447 units and the demand over the year is 2 000 units, the time interval between orders would be $447 \div 2\,000 = 0,2235 \text{ years} \times 12 = 2,68 \text{ months}$.

Two-bin and three-bin systems may be used as part of a continuous review approach. They are easy to use – see figure 12.15 in your prescribed book. In a two-bin system, the items used are placed in the first bin and the re-order-point quantity and safety stock are placed in the second bin. As soon as the first bin empties, it signals that the next re-order quantity should be ordered. In a three-bin system, the second bin (as described above) is split, and the re-order quantity and safety stock are held separately.

13.5 INVENTORY ANALYSIS AND CONTROL SYSTEMS



Activity 13.4

Study the fourth section of chapter 12 in your prescribed book, entitled “Inventory analysis and control systems” and then answer the questions and/or complete the following:

- (1) To cope with a complex situation whereby a large number of stocked items are supplied by many different suppliers for a great number of individual customers/clients, the inventory analysis and control system must be able to discriminate between the different stocked items because and have an information processing system to
- (2) How could inventory priorities be determined by the ABC system and how could this system be extended to measure the absolute level of inventory at any point in time?
- (3) What other criteria, apart from the annual usage and value as used by the ABC system, could also be used to classify or prioritise inventory?
- (4) What are the common functions of an inventory information system?



Feedback on Activity 13.4

- (1) With reference to question 1, to cope with a complex situation whereby a large number of stocked items are supplied by many different suppliers for a great number of individual customers/clients, the inventory analysis and control system must be able to discriminate between the different stocked items. This is because there is a need to apply a degree of

control to each item that is appropriate to its importance, and to have an information processing system to cope with their particular inventory control circumstances.

- (2) With reference to question 2, in any inventory that contains more than one stocked item, some items may be more important to the organisation than others because (i) they may have a high usage rate and stockout would be very disappointing to customers; and (ii) they may be of high value, so it would be very costly to hold high inventory levels. Different stock items can be identified on the basis of their usage value (the usage rate multiplied by the individual item value), where high usage value items require more careful control than lower usage value items. The ABC system for prioritising inventory enables managers to focus control efforts on a few items that represent the major proportion of inventory value (class A – 20% high usage value items account for 80% of total usage value; class B – next 30% of items account for 10% of the total usage value; and class C – remaining 50% of items account for 10% of the total usage value). Study the worked example of the electrical wholesaler in chapter 12 of your prescribed book and note figure 12.16, which illustrates a Pareto curve for the items in the warehouse – 20% of the items represent 80% of the value. In ABC the monetary value of the annual usage of each item was used as a measure of inventory usage. It can also be used to measure the absolute level of inventory at any point in time by calculating the “cover of stock” (amount of time that the inventory would last subject to normal demand if items were not replenished) and “stock turn” (how often stock is used up in a period). Also study the worked example of the small specialist wine importer and note the calculations for the average stock cover and average stock turn. Finally, note the critical comments made by the authors of your prescribed book about the fact that the Pareto law is often misquoted as “80 per cent of the SKUs (stock keeping units) account for only 20 per cent inventory value” and that class B and C items actually need more “control”, as they are slow moving items that remain in stock for a long time and require between half and two-thirds of the total investment in inventory.
 - (3) With reference to question 3, note that although annual usage and value are the two criteria most often used to determine the priority of inventory items, other criteria may also be used. They are (i) the consequences of stockout (items that may seriously delay or disrupt other operations or the high priority given to customers/clients); (ii) uncertainty of supply (items whose supply may be erratic or uncertain); and (iii) a high obsolescence or deterioration risk (items lose their value very quickly).
 - (4) With reference to question 4, computerised inventory information systems are used to manage most inventories that are of any significant size and have the common functions of updating stock records, generating orders and reports and forecasting. Most computer-based inventory management systems are based on the perpetual inventory principle. This means that inventory records are automatically updated when items are received and taken out of inventory using the formula: $\text{opening stock level} + \text{receipts in} - \text{dispatches out} = \text{new stock level}$. Common problems with inventory systems lie in data inaccuracy as a result of keying errors, quantity errors, damaged inventory not being recorded, wrong items being taken out of stock, delays between transactions and the updating of records, and items stolen from inventory. Study the short case example entitled African Oxygen in chapter 12 of your prescribed book and answer the questions that follow.
-

CHECKLIST



Now that you have completed study unit 13 on inventory planning and control in operations you should be able to do the following:

- explain the role of inventory in operations
- explain what the volume decision of inventory entails and demonstrate how it should be made by solving practical examples
- explain what the timing decision of inventory entails and demonstrate how it should be made by solving practical examples
- explain how inventory analysis and control systems can be used in operations and demonstrate their practical application
- use the knowledge, insight and skills you have acquired regarding
 - the role of inventory in operations
 - the volume decision of inventory
 - the timing decision of inventory, and
 - inventory analysis and control systems used in operations
- to justify the importance of inventory planning and control and demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for inventory planning and control by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner

SUMMARY



This study unit examined inventory planning and control. Determining the quantity to order and the timing of placing the order is underpinned by a system of inventory analysis and control. The next study unit (study unit 14) focuses specifically on supply chain planning and control.

ASSESSMENT



- (1) Consider the case study entitled “Trans-European Plastics” at the end of chapter 12 in your prescribed book and then answer the six questions posed at the end.
- (2) Do problems 1 to 6 at the end of chapter 12 of your prescribed book. Note that these will help you practise analysing operations.
- (3) Complete study activities 1 to 4 at the end of chapter 12 of your prescribed book.

Note: Some study activities may only require you to re-read a portion of the chapter you have just studied, while others may draw on your general knowledge and experience of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

Study unit 14

Supply chain planning and control



Key concepts

supply chain planning and control	responsive supply chain policies
supply chain(s)	customer interface process
purchasing (procurement) and supply management	supply chain management
logistics	supply chain pipeline
business-to-consumer (B2C) relationship	end or final customer
customer-to-consumer (C2C) relationship	supply network
virtual operations	physical distribution management
relationships as “exchange”	materials management
efficient supply chain policies	business-to-business (B2B) relationship
“partnership” supply relationships	consumer-to-business (C2B) relationship
	traditional market supply relationships
	supply chain behaviour
	customer relationship management (CRM)

GETTING AN OVERVIEW

In this study unit we address the last four learning outcomes of topic 10. We explain what supply chain management entails, explain the activities of supply chain management (including purchasing and supplier management on the supply side and physical distribution management on the demand side of the operation, and logistics and materials management), explain the types of relationship in supply chains, and explain supply chain behaviour.

14.1 INTRODUCTION IN THE PRESCRIBED BOOK



Read

First read the introduction to chapter 13 of your prescribed book. Then study figure 13.1 in your prescribed book and specifically note which part(s) of the model of the design activities in production and operations management is/are discussed in this particular topic. Finally, read the points listed under the following questions at the end of chapter 13 of your prescribed book:

- “What is supply chain management and its related activities?” (2 marks)
- “How can the relationship between operations in a supply chain affect the way it works?” (4 marks)

- “Are different supply chain objectives needed in different circumstances?” (1 mark)
- “How do supply chains behave in practice?” (3 points)

14.2 OPERATIONS IN PRACTICE



Activity 14.1

Read the case example entitled “Suppliers are vital to Lucent’s success” in chapter 13 of your prescribed book and then answer the following questions:

- (1) Why is it important for Lucent Technologies to develop good supplier relationships?
- (2) What does Lucent’s Supplier Relationship Programme entail? What are the advantages of the programme?



Feedback on Activity 14.1

- (1) *With reference to question 1, even though Lucent had the support of the world leader in technology research, Bell Labs, the company realised that it needed close working relationships with a group of carefully chosen suppliers. Lucent needed suppliers that could be relied upon to provide the materials and services it needed, when it needed them and at a price it could afford. Lucent understands that it depends on its suppliers for new ideas, knowledge, products and services. These (the ideas, knowledge, products and services) must also benefit both Lucent’s customers and Lucent’s customers’ customers.*
 - (2) *With reference to question 2, Lucent’s Supplier Relationship Programme (SRP) is its mechanism for dealing with all supply chain issues and managing its supply base. To be a Lucent supplier means more than merely being able to supply high-quality materials and services at competitive prices. Lucent expects its suppliers to “work beside” the company during all its operations, from design to the point of delivery and even beyond. The objectives of the SRP are to evaluate suppliers to determine whether they qualify to be suppliers; to provide suppliers with a single contact point at Lucent; to provide a web-based supply chain portal for members of the supply chain network so that everyone has a clear view of demand, availability and state of delivery; and to organise supplier forums to find ways of cutting costs, reducing lead times and improving delivery. The advantages of the SRP include increased profit margins, a 66 percent cut in inventory, and a 35 to 55 percent cut in component costs.*
-

14.3 TERMINOLOGY USED IN SUPPLY CHAINS



Activity 14.2

Study the first section of chapter 13 of your prescribed book entitled “What is supply chain management?” and then answer the questions and/or complete the following:

- (1) Previously the term “supply network” was used to refer to all the operations that were linked together to provide goods and services to the end customers (see topic 5, study unit 7). A supply chain is ...

- (2) What are the objectives of supply chain management?
- (3) What are the supply chain objectives?



Feedback on Activity 14.2

- (1) *With reference to question 1, a supply chain refers to the individual channels or strands that represent the flow of goods and services through the supply network of linked operations. Note the analogy the authors of your prescribed book use to compare supply chains with liquids or fluids flowing through a pipeline. In long pipelines it will take longer for the fluids (fluids used as the metaphor for goods or services) to flow than if the pipeline was shorter. Similarly, bigger (larger diameter) pipelines can accommodate more fluids than smaller (narrower) ones. The stocks of inventory held at various points in the supply chain could be represented by storage tanks.*
 - (2) *With reference to question 2, supply chain management refers to “the management of the interconnection of organizations that relate to each other through upstream and downstream linkages between the different processes that produce value in the form of products and services to the ultimate consumer”. Supply chain management embodies a holistic approach to the management of the flow of goods and services across company boundaries. An analogy often used to describe supply chains is that of a pipeline. Goods and services must flow through the supply chain pipeline and are processed and stored by different operations in the chain. Also study figure 13.2 in your prescribed book, which illustrates a customer perspective on supply chain performance.*
 - (3) *With reference to question 3, the supply chain objectives are expressed in terms of the five performance objectives of all operations, namely, quality, speed, dependability, flexibility and cost. The final product/service quality is a function of the quality performance of each and every operation in the whole chain. The speed at which customers can be served, from requesting the product/service until receiving it, and/or the time it takes products/services to move through the entire supply chain is the second performance objective. Dependability refers to the “on-time delivery” guarantee and/or “throughput time” dependability, which reduces uncertainty. Flexibility is the ability to cope with variability, changes and deviations, and cost is the additional cost apart from the cost of each operation transforming its inputs to outputs in doing “business” with other operations in the supply chain as a whole.*
-

14.4 ACTIVITIES OF SUPPLY CHAIN MANAGEMENT



Activity 14.3

Study the second section of chapter 13 entitled “The activities of supply chain management” and then answer the questions and/or complete the following:

- (1) To which part or parts of the supply chain do the following terms refer?
 - (a) purchasing and supply management, which encompasses ... and refers to ... parts
 - (b) physical distribution management, which encompasses ... and refers to ... parts
 - (c) logistics, which encompasses ... and refers to ... parts
 - (d) materials management, which encompasses ...
- (2) The purchasing function enters into contracts with suppliers to buy in materials and services, some of which are ... (termed transformed resources) and others, which are ... but are still essential purchases for the smooth running of the operation.

- (3) The physical distribution management, limited to manufacturing operations, involves ...
- (4) When several supplier-purchaser linkages are considered as an integrated whole, concepts such as a materials management (which involves ...) and merchandising (which involves ...) become the subjects of focus.



Feedback on Activity 14.3

- (1) With reference to question 1, (a) purchasing and supply management, which encompasses the operation's interface with its supply markets and refers to the supply part of the chain; (b) physical distribution management, which encompasses the supplying of immediate customers and refers to the demand part of the chain; (c) logistics (an extension of physical distribution management), which encompasses the management of materials and information flow of the business through a distribution channel and refers to the demand part of the chain; (d) materials management, which encompasses the management of the flow of materials and information through the immediate supply chain, including the activities of purchasing, inventory management, stores management, operations planning and control, and physical distribution management. Study figure 13.3 in your prescribed book and note what part of the supply chain as a whole each of the terms used covers.
 - (2) With reference to question 2, the purchasing function enters into contracts with suppliers to buy in materials and services. Some of these materials and services are used directly in the production of goods and services (they are called transformed resources) and others are used in the running of the business itself. These may not be directly involved in the production of goods and services, but they are nevertheless essential for the smooth running of the operation.
 - (3) With reference to question 3, the physical distribution management limited to manufacturing operations involves the physical transportation of goods from the manufacturing operation to the customer.
 - (4) With reference to question 4, when several supplier-purchaser linkages are considered as an integrated whole, concepts such as a materials management (which involves the traditional function of purchasing, expediting, inventory management, stores management, production planning and control, and physical distribution management), and merchandising (which involves the purchasing task combined with sales and the physical distribution task in retail operations) become the subjects of focus.
-

14.5 TYPES OF RELATIONSHIPS IN SUPPLY CHAINS



Activity 14.4

Study the third section of chapter 13 entitled "Types of relationships in supply chains" and then answer the questions:

- (1) How should operations manage their relationships with their immediate suppliers? What do each of the four general categories of business/consumer supply chain relationships involve? What types of business-to-business relationship can develop, in terms of what is outsourced and who is chosen to supply products and services?
- (2) How do traditional market supply relationships work? What are the advantages and disadvantages of wholly free-market practices like these?

- (3) What does the extreme form of outsourcing operational activities known as “virtual operations” entail?
- (4) What do “partnership” supply relationships entail? Which different types of relationship and main elements of exchange can be identified?
- (5) What does customer relationship management (CRM) entail and how can it be used to gain a better understanding of the customer interface process?



Feedback on Activity 14.4

- (1) *With reference to question 1, it should be noted that the behaviour of the supply chain as a whole is the result of the relationships that develop between the individual parts of the operation within the chain itself. It is necessary to develop a framework to help the business understand the different ways in which the various business/consumer relationships can develop. Study figure 13.6 in your prescribed book. Note that the growth in e-commerce has made a broad range of relationships possible, for example, business-to-business (B2B) relationships (between two commercial businesses), business-to-consumer (B2C) relationships (between traditional retailers, online retailers and consumers), consumer-to-business (C2B) relationships (consumers post their needs on the web to find companies that are willing to supply) and customer-to-consumer (C2C) relationships (online exchange and auction of services between consumers). With regard to business-to-business supply chain relationships, these are characterised by the extent and nature of what is bought from suppliers (What is outsourced?), and who is chosen to supply the products and services (To whom is the supply outsourced?). Study figure 13.7 in your prescribed book and note the identified business-to-business supply chain relations (traditional market supply, virtual operations and “partnership” supply relationships). Also study the short case example entitled “Barloworld Logistics” in chapter 13 and then answer the questions that follow in the prescribed book.*
- (2) *With reference to question 2, traditional market supply relationships involve purchasing goods and services from outside the organisation in a “pure” market fashion, where the best supplier is sought for every purchase. Such a relationship is usually very brief, and once goods or services are delivered and payment is made, all contact may come to an end. The advantages of this traditional market supplier relationship are the following: competition between alternative suppliers is maintained, a supplier specialises in a small number of products and services, the inherent inflexibility of outsourced suppliers is overcome by changing to a number of suppliers, innovations can be exploited no matter where they originate, and it helps operations concentrate on core activities. However, there are also disadvantages, including the following: there are supply uncertainties; choosing the best supplier may involve considerable effort; there is a strategic risk in subcontracting too many activities; and over-reliance on outsourcing may “hollow out” the business, leaving it with no internal competencies to exploit in the market. Study the short case example entitled “Extracts from Levi Strauss’s global sourcing policy” in chapter 13 in your prescribed book and note the declared policy to do business only with partners who adhere to certain guidelines on child labour, prison/forced labour, disciplinary practices, working hours, wages and benefits, freedom of association, and discrimination. Finally, also study the short case example entitled “Purchasing from China” in chapter 13 and then answer the questions that follow in the prescribed book.*
- (3) *With reference to question 3, virtual operations do very little themselves – they rely on a network of suppliers who provide the products and services on demand. Advantages centre around the flexibility and speed of the virtual operations, and the risks of investing in production facilities are far lower. Disadvantages are the “hollowing out” effect virtual operations may have, because without a solid base of resources it is difficult to hold onto*

and develop unique core technical expertise. In addition, the resources of the virtual operation will almost certainly also be available to competitors and, therefore, the virtual operation's only core competence may lie in actually managing its supply network.

- (4) With reference to question 4, partnership supply relationships are seen as a compromise between vertical integration (owning the resources that supply the organisation) and a pure market supply relationship (only transactional involvement with the organisation's suppliers). Study the formal definition given by the authors of your prescribed book in section 13.3. Note that they focus on cooperation between suppliers and customers, even to the extent of sharing skills and resources to achieve a joint benefit beyond those that could be achieved if the partners had acted alone. Note the factors that may affect the degree of "closeness" of the partnership, including the sharing of success, long-term expectations, multiple points of contact, joint learning, fewer relationships, joint coordination of activities, information transparency, joint problem solving and increased trust.
 - (5) With reference to question 5, customer relationship management (CRM) is a "method of learning more about customers' needs and behaviours in order to develop stronger relationships with them" and further ... "it is a process that helps to understand customers' needs and develop ways of meeting those needs while maximizing profitability". CRM builds a number of steps into the customer interface process. (a) The business must determine the exact needs of its customers and how best to meet those needs. (b) The business must examine all the different ways and places where customer-related information is collected, stored and used (c) All customer-related data must be analysed to obtain a holistic view of each customer and where service can be improved. However, note the critical commentary by the authors of your prescribed book concerning CRM. Is it just the latest fad?
-

14.6 SUPPLY CHAIN BEHAVIOUR



Activity 14.5

Study section 4 of chapter 13 in your prescribed book entitled "Supply chain behaviour" and then answer the following questions:

- (1) How should supply chains be managed when operations compete in different ways in different markets?
- (2) How do the natural dynamics of supply chain behaviour affect different parts of the supply chain?
- (3) How can the organisation's supply chain performance be improved?
- (4) What does supply chain vulnerability entail?



Feedback on Activity 14.5

- (1) With reference to question 1, supply chains serving individual markets should be organised in different ways. For example, supply chain policies for functional products should have what is termed "efficient supply chain policies", which means that inventories are kept low, especially downstream, to maintain fast throughput and to reduce the amount of working capital tied up in inventory. It also means that utilisation should remain high and manufacturing costs low. Supply chain policies for innovative products should have what is termed "responsive supply chain policies", which means that high service levels are emphasised, especially close to the end customer, and inventory is kept as close to

the customer as possible. In addition, fast throughput from upstream parts of the chain is still needed to replenish downstream stocks to ensure high levels of availability even if changes occur in customer demand. Study figure 13.8 and note the matches/mismatches between the resources in the supply chain and market requirements.

- (2) With reference to question 2, certain dynamics that exist between organisations in the supply chain can cause errors, inaccuracies and volatility, which are more pronounced further upstream in the chain. One is the effect known as the Forrester Effect or Bullwhip Effect, because a small disturbance at one end of the chain causes increasingly large disturbances as it works its way to the end of the chain. Study table 13.3 in your prescribed book concerning the original equipment manufacturer (OEM), served by three tiers of suppliers. Note that the further upstream (or further back in the supply chain) an operation is placed, the more drastic the fluctuations caused by a relatively small change in demand from the final customer will be. The effect of miscommunication in the supply chain (see figure 13.9) can be illustrated by the children's game "Chinese whispers". When one child whispers a message to the next child, who repeats it to the next, etc, until the last child repeats the message out loud, everyone is amazed at how the original message has been distorted. In the same way, a relatively small fluctuation in the market can cause great volatility further back in the chain.
- (3) With reference to question 3, the attempt to improve the supply chain performance is important, given the nature of the previously described supply chain dynamics. Most improvement efforts focus on coordination in the supply chain activity and include the development of e-business, information sharing, channel alignment and operational efficiency. New information technology applications combined with internet-based e-business allow for accurate, "near real-time" information by which the disparate elements of the supply chain can integrate their efforts for the benefit of the whole chain and, ultimately, the final customer – see table 13.5 in your prescribed book. It is better to share or transmit information so that all operations can monitor true demand without the interference of distortions. Channel alignment involves the adjustment of scheduling, material movements, stock levels and pricing to bring all operations in line with each other. Finally, operational efficiency amounts to the efforts that each operation in the supply chain can make to reduce its own complexity, reduce the cost of doing business in the chain and increase throughput. Time compression (ie speeding up the flow of material down the chain and the flow of information back) is one way of improving operational efficiency and has been noted previously in the discussion about the supply chain pipeline (see sect 14.1 in this study guide). Figure 13.11 in your prescribed book illustrates the way time compression can both reduce costs and increase revenues.
- (4) With reference to question 4 and supply chain vulnerability, note that an agile supply chain can bring about serious supply chain risks and disruption. This may be the result of major unplanned events such as a key supplier becoming insolvent, global outsourcing (parts are shipped around the world, which involves increased risk), increased demand volatility in certain areas, natural disasters, terrorist attacks, industrial action, port blockades, accidents, etc.

CHECKLIST



Now that you have completed study unit 14 on supply chain planning and control in operations you should be able to do the following:

- explain what supply chain management entails

- explain the activities of supply chain management (including purchasing and supplier management on the supply side and physical distribution management on the demand side of the operation, logistics and materials management)
- explain the types of relationship in supply chains
- explain supply chain behaviour
- use the knowledge, insight and skills you have acquired regarding
 - supply chain management
 - the activities of supply chain management (including purchasing and supplier management on the supply side and physical distribution management on the demand side of the operation, logistics and materials management)
 - the types of relationship in supply chains and
 - supply chain behaviour
- to justify the importance of supply chain planning and control and to demonstrate this by identifying both correct and incorrect statements with regard to the study material and practical applications in general, in either multiple-choice format or essay-type questions
- show evidence of your appreciation for supply chain planning and control by endorsing the theoretical concepts and demonstrating the application of the relevant production/operations skills in a practical manner.

SUMMARY



This study unit examined supply chain planning and control. Together with the previous study unit (study unit 13 – inventory planning and control) it formed topic 10. Supply chain planning and control examines different parts and types of relationship within the total supply network.

ASSESSMENT



- (1) Consider the case study entitled “Supplying fast fashion” at the end of chapter 13 of your prescribed book and then answer the question posed at the end.
- (2) Do problems 4 to 6 at the end of chapter 13 of your prescribed book. Note that these will help you practise analysing operations.
- (3) Complete study activities 1 to 5 at the end of chapter 13 of your prescribed book.

Note: Some study activities may only require you to reread a portion of the chapter you have just studied, while others may draw on your general knowledge and experience of a particular business/organisation or other institution. Finally, others may even send you on a trip to your local equivalent business/organisation or other institution to do your own investigation.

REFLECTION AND A LOOK AHEAD

TOPIC 10: Inventory and supply chain planning and control

This module examined the specialist discipline of production and operations management within the context of a business. It consisted of three major parts, where we focused our attention on (1) the foundations of production/operations and productivity management, (2) some of the design aspects of the production/operations management system, and (3) some of the planning and control aspects of the production/operations management system.

We hope you enjoyed learning about production and operations management and that you will be able to use the knowledge, skills and attitudes you have acquired regarding this very important and exciting field of business expertise. What you have learnt should equip you as a specialist production/operations manager in whatever business management field you decide to enter.

Should you see the value of gaining further knowledge of and skills in production and operations management which, as we said at the beginning, could be considered the core function of any business, you might like to continue your studies by registering for our third-year module, MNO3701. In that module we examine the following:

PART I	DEVELOPING PRODUCTION/OPERATIONS MANAGEMENT STRATEGIES FOR COMPETITIVE ADVANTAGE
Topic 1	The strategic role and objectives of production/operations management
Topic 2	Production/operations management strategies
PART II	DESIGNING, PLANNING AND CONTROLLING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM FOR WORLD-CLASS PERFORMANCE
Topic 3	Process technology
Topic 4	Enterprise resource planning (material requirements planning)
Topic 5	Lean operations and just-in-time planning and control
Topic 6	Quality planning and control
Topic 7	Project management
PART III	IMPROVING THE PRODUCTION/OPERATIONS MANAGEMENT SYSTEM FOR THE EMERGING CHALLENGES OF THE 21ST CENTURY
Topic 8	Measuring performance and improving the production/operations management system
Topic 9	Maintaining the production/operations management system
Topic 10	New challenges and contemporary issues of production and operations management

Best of luck for your preparation for the examination on this module and we hope that you will register for our third-year module MNO3701!

ANNEXURE A: 327 VARIABLES CONSIDERED BY THE INSTITUTE FOR MANAGEMENT DEVELOPMENT (IMD) IN DETERMINING THE RANKINGS OF ECONOMIES IN THE ANNUAL WORLD COMPETITIVENESS REPORT (WCS)

TABLE 1: 76 Economic Performance variables consider by the IMD in determining rankings in the WCS

Economic Performance

DOMESTIC ECONOMY

Size	
I.1.01 Gross domestic product (GDP)	US\$ billions
I.1.02 GDP (PPP)	Estimates; US\$ billions at purchasing power parity
I.1.03 World GDP contribution	Percentage share of world GDP in market prices
I.1.04 Household consumption expenditure	US\$ billions
I.1.05 Household consumption expenditure	Percentage of GDP
I.1.06 Government consumption expenditure	US\$ billions
I.1.07 Government consumption expenditure	Percentage of GDP
I.1.08 Gross fixed capital formation	US\$ billions
I.1.09 Gross fixed capital formation	Percentage of GDP
I.1.10 Gross domestic savings	US\$ billions
I.1.11 Gross domestic savings	Percentage of GDP
I.1.12 Economic sectors	Breakdown as a percentage of GDP
I.1.13 Diversification of the economy	Diversification of the economy is sufficient
Growth	
I.1.14 Real GDP Growth	Percentage change, based on national currency in constant prices
I.1.15 Real GDP Growth per capita	Percentage change, based on national currency in constant prices
I.1.16 Household consumption expenditure - real growth	Percentage change, based on constant prices
I.1.17 Government consumption expenditure - real growth	Percentage change, based on constant prices
I.1.18 Gross fixed capital formation - real growth	Percentage change, based on constant prices
I.1.19 Resilience of the economy	Resilience of the economy to economic cycles is strong
Wealth	
I.1.20 GDP per capita	US\$ per capita
I.1.21 GDP (PPP) per capita	Estimates; US\$ per capita at purchasing power parity
Forecasts	
I.1.22 Forecast: Real GDP Growth	Percentage change, based on national currency in constant prices
I.1.23 Forecast: Inflation	Percentage change
I.1.24 Forecast: Unemployment	Percentage of total labor force
I.1.25 Forecast: Current account balance	Percentage of GDP/GNP
INTERNATIONAL TRADE	
I.2.01 Current account balance	US\$ billions (minus sign = deficit)
I.2.02 Current account balance	Percentage of GDP
I.2.03 Balance of trade	US\$ billions (minus sign = deficit)
I.2.04 Balance of trade	Percentage of GDP
I.2.05 Balance of commercial services	US\$ billions (minus sign = deficit)
I.2.06 Balance of commercial services	Percentage of GDP
I.2.07 World exports contribution	Percentage share of world exports (goods and commercial services)
I.2.08 Exports of goods	US\$ billions
I.2.09 Exports of goods	Percentage of GDP
I.2.10 Exports of goods - growth	Percentage change based on US\$ values
I.2.11 Exports of commercial services	US\$ billions
I.2.12 Exports of commercial services	Percentage of GDP
I.2.13 Exports of commercial services - growth	Percentage change based on US\$ values
I.2.14 Exports breakdown by economic sector	Percentage of total exports
I.2.15 Imports of goods & commercial services	US\$ billions
I.2.16 Imports of goods & commercial services	Percentage of GDP
I.2.17 Imports of goods & commercial services - growth	Percentage change based on US\$ values
I.2.18 Imports breakdown by economic sector	Percentage of total imports
I.2.19 Trade to GDP ratio	(Exports + Imports) / (2 x GDP)
I.2.20 Terms of trade index	Unit value of exports over unit value of imports (2005 = 100)
I.2.21 Tourism receipts	International tourism receipts as a percentage of GDP
I.2.22 Exchange rates	Exchange rates support the competitiveness of enterprises

INTERNATIONAL INVESTMENT**Investment**

I.3.01 Direct investment flows abroad	US\$ billions
I.3.02 Direct investment flows abroad	Percentage of GDP
I.3.03 Direct investment stocks abroad	US\$ billions
I.3.04 Direct investment stocks abroad	Percentage of GDP
I.3.05 Direct investment flows inward	US\$ billions
I.3.06 Direct investment flows inward	Percentage of GDP
I.3.07 Direct investment stocks inward	US\$ billions
I.3.08 Direct investment stocks inward	Percentage of GDP
I.3.09 Balance of direct investment flows	US\$ billions (flows abroad minus flows inward)
I.3.10 Balance of direct investment flows	Percentage of GDP (flows abroad minus flows inward)
I.3.11 Net position in direct investment stocks	US\$ billions (stocks abroad minus stocks inward)
I.3.12 Net position in direct investment stocks	Percentage of GDP (stocks abroad minus stocks inward)
I.3.13 Relocation threats of production	Relocation of production is not a threat to the future of your economy
I.3.14 Relocation threats of R&D facilities	Relocation of R&D facilities is not a threat to the future of your economy
I.3.15 Relocation threats of services	Relocation of services is not a threat to the future of your economy

Finance

I.3.16 Portfolio investment assets	US\$ billions
I.3.17 Portfolio investment liabilities	US\$ billions

EMPLOYMENT

I.4.01 Employment	Total employment in millions
I.4.02 Employment	Percentage of population
I.4.03 Employment - growth	Estimates: percentage change
I.4.04 Employment by sector	Percentage of total employment
I.4.05 Employment in the public sector	Percentage of total employment
I.4.06 Unemployment rate	Percentage of labor force
I.4.07 Long-term unemployment	Percentage of labor force
I.4.08 Youth unemployment	Percentage of youth labor force (under the age of 25)

PRICES

I.5.01 Consumer price inflation	Average annual rate
I.5.02 Cost-of-living index	Index of a basket of goods & services in major cities, including housing (New York City = 100)
I.5.03 Apartment rent	3-room apartment monthly rent in major cities, US\$
I.5.04 Office rent	Total occupation cost (US\$/Sq.M per year)

Source: IMD Online (2011)

TABLE 2: 71 Government Efficiency variables considered by the IMD in determining rankings in the WCS

Government Efficiency

PUBLIC FINANCE

2.1.01 Government budget surplus/deficit	US\$ billions
2.1.02 Government budget surplus/deficit	Percentage of GDP
2.1.03 Total general government debt	US\$ billions
2.1.04 Total general government debt	Percentage of GDP
2.1.05 Total general government debt - real growth	Percentage change, based on national currency in constant prices
2.1.06 Central government domestic debt	Percentage of GDP
2.1.07 Central government foreign debt	Percentage of GDP
2.1.08 Interest payment	Percentage of current revenue
2.1.09 Management of public finances	Management of public finances over the next two years is likely to improve
2.1.10 Tax evasion	Tax evasion does not damage public finances
2.1.11 Pension funding	Pension funding is adequately addressed for the future
2.1.12 General government expenditure	Percentage of GDP

FISCAL POLICY

2.2.01 Collected total tax revenues	Percentage of GDP
2.2.02 Collected personal income tax	On profits, income and capital gains, as a percentage of GDP
2.2.03 Collected corporate taxes	On profits, income and capital gains, as a percentage of GDP
2.2.04 Collected indirect tax revenues	Taxes on goods and services as a percentage of GDP
2.2.05 Collected capital and property taxes	Percentage of GDP
2.2.06 Collected social security contribution	Compulsory contribution of employees and employers as a percentage of GDP
2.2.07 Effective personal income tax rate	Percentage of an income equal to GDP per capita
2.2.08 Corporate tax rate on profit	Maximum tax rate, calculated on profit before tax
2.2.09 Consumption tax rate	Standard rate of VAT / GST
2.2.10 Employee's social security contribution rate	Compulsory contribution as a percentage of an income equal to GDP per capita
2.2.11 Employer's social security contribution rate	Compulsory contribution as a percentage of an income equal to GDP per capita
2.2.12 Real personal taxes	Real personal taxes do not discourage people from working or seeking advancement
2.2.13 Real corporate taxes	Real corporate taxes do not discourage entrepreneurial activity

INSTITUTIONAL FRAMEWORK

Central Bank

2.3.01 Real short-term interest rate	Real discount / bank rate
2.3.02 Cost of capital	Cost of capital encourages business development
2.3.03 Interest rate spread	Lending rate minus deposit rate
2.3.04 Country credit rating	Rating on a scale of 0-100 assessed by the Institutional Investor Magazine ranking
2.3.05 Central bank policy	Central bank policy has a positive impact on economic development
2.3.06 Foreign currency reserves	US\$ billions
2.3.07 Exchange rate stability	Parity change from national currency to SDR, 2009 / 2007

State Efficiency

2.3.08 Legal and regulatory framework	The legal and regulatory framework encourages the competitiveness of enterprises
2.3.09 Adaptability of government policy	Adaptability of government policy to changes in the economy is high
2.3.10 Government decisions	Government decisions are effectively implemented
2.3.11 Transparency	Transparency of government policy is satisfactory
2.3.12 Bureaucracy	Bureaucracy does not hinder business activity
2.3.13 Bribing and corruption	Bribing and corruption do not exist

BUSINESS LEGISLATION**Openness**

2.4.01 Tariff barriers	Tariffs on imports: Most favored nation simple average rate
2.4.02 Customs' authorities	Customs' authorities do facilitate the efficient transit of goods
2.4.03 Protectionism	Protectionism does not impair the conduct of your business
2.4.04 Public sector contracts	Public sector contracts are sufficiently open to foreign bidders
2.4.05 Foreign investors	Foreign investors are free to acquire control in domestic companies
2.4.06 Capital markets	Capital markets (foreign and domestic) are easily accessible
2.4.07 Investment incentives	Investment incentives are attractive to foreign investors

Competition and Regulations

2.4.08 Government subsidies	To private and public companies as a percentage of GDP
2.4.09 Subsidies	Subsidies do not distort fair competition and economic development
2.4.10 State ownership of enterprises	State ownership of enterprises is not a threat to business activities
2.4.11 Competition legislation	Competition legislation is efficient in preventing unfair competition
2.4.12 Parallel economy	Parallel (black-market, unrecorded) economy does not impair economic development
2.4.13 Ease of doing business	Ease of doing business is supported by regulations
2.4.14 Creation of firms	Creation of firms is supported by legislation
2.4.15 Start-up days	Number of days to start a business
2.4.16 Start-up procedures	Number of procedures to start a business

Labor Regulations

2.4.17 Labor regulations	Labor regulations (hiring/firing practices, minimum wages, etc.) do not hinder business activities
2.4.18 Unemployment legislation	Unemployment legislation provides an incentive to look for work
2.4.19 Immigration laws	Immigration laws do not prevent your company from employing foreign labor
2.4.20 Redundancy costs	Number of weeks of salary
2.4.21 Labor market flexibility	Index on rigidity of employment (index 0-100)

SOCIETAL FRAMEWORK

2.5.01 Justice	Justice is fairly administered
2.5.02 Personal security and private property	Personal security and private property are adequately protected
2.5.03 Ageing of society	Ageing of society is not a burden for economic development
2.5.04 Risk of political instability	The risk of political instability is very low
2.5.05 Social cohesion	Social cohesion is a priority for the government
2.5.06 Gini index	Equal distribution of income scale: 0 (absolute equality) 100 (absolute inequality)
2.5.07 Income distribution - lowest 10%	Percentage of household incomes going to lowest 10% of households
2.5.08 Income distribution - highest 10%	Percentage of household incomes going to highest 10% of households
2.5.09 Equal opportunity	Equal opportunity legislation in your country encourages economic development
2.5.10 Females in parliament	Percentage of total seats in Parliament
2.5.11 Female positions	Percentage of total legislators, senior officials and managers
2.5.12 Gender income ratio	Ratio of estimated female to male earned income globally

Source: IMD Online (2011)

TABLE 3: 67 Business Efficiency variables considered by the IMD in determining rankings in the WCS

Business Efficiency

PRODUCTIVITY AND EFFICIENCY

3.1.01 Overall productivity (PPP)	Estimates: GDP (PPP) per person employed, US\$
3.1.02 Overall productivity	GDP per person employed, US\$
3.1.03 Overall productivity - real growth	Estimates: Percentage change of real GDP per person employed
3.1.04 Labor productivity (PPP)	Estimates: GDP (PPP) per person employed per hour, US\$
3.1.05 Labor productivity (PPP) growth	Percentage change of GDP (PPP) per person employed per hour
3.1.06 Agricultural productivity (PPP)	Estimates: Related GDP (PPP) per person employed in agriculture, US\$
3.1.07 Productivity in industry (PPP)	Estimates: Related GDP (PPP) per person employed in industry, US\$
3.1.08 Productivity in services (PPP)	Estimates: Related GDP (PPP) per person employed in services, US\$
3.1.09 Large corporations	Large corporations are efficient by international standards
3.1.10 Small and medium-size enterprises	Small and medium-size enterprises are efficient by international standards
3.1.11 Productivity of companies	Productivity of companies is supported by global strategies (supplies, offshoring, outsourcing)

LABOR MARKET

Costs

3.2.01 Compensation levels	Total hourly compensation for manufacturing workers (wages + supplementary benefits), US\$
3.2.02 Unit labor costs in the manufacturing sector	Percentage change
3.2.03 Remuneration in services professions	Gross annual income including supplements such as bonuses, in US\$
3.2.04 Remuneration of management	Total base salary plus bonuses and long-term incentives, US\$

Relations

3.2.05 Working hours	Average number of working hours per year
3.2.06 Labor relations	Labor relations are generally productive
3.2.07 Worker motivation	Worker motivation in companies is high
3.2.08 Industrial disputes	Working days lost per 1,000 inhabitants per year (average 2006-2008)
3.2.09 Employee training	Employee training is a high priority in companies

Availability of Skills

3.2.10 Labor force	Employed and registered unemployed (millions)
3.2.11 Labor force	Percentage of population
3.2.12 Labor force growth	Percentage change
3.2.13 Part-time employment	Percentage of total employment
3.2.14 Female labor force	Percentage of total labor force
3.2.15 Foreign labor force	Percentage of total labor force
3.2.16 Skilled labor	Skilled labor is readily available
3.2.17 Finance skills	Finance skills are readily available
3.2.18 Attracting and retaining talents	Attracting and retaining talents is a priority in companies
3.2.19 Brain drain	Brain drain (well-educated and skilled people) does not hinder competitiveness in your economy
3.2.20 Foreign high-skilled people	Foreign high-skilled people are attracted to your country's business environment
3.2.21 International experience	International experience of senior managers is generally significant
3.2.22 Competent senior managers	Competent senior managers are readily available

FINANCE**Bank Efficiency**

3.3.01 Banking sector assets	Percentage of GDP
3.3.02 Financial cards in circulation	Number of cards per capita
3.3.03 Financial card transactions	US\$ per capita
3.3.04 Investment risk	Euromoney country credit-worthiness scale from 0-100
3.3.05 Banking and financial services	Banking and financial services do support business activities efficiently.
3.3.06 Financial institutions' transparency	Financial institutions' transparency is sufficiently implemented.
3.3.07 Finance and banking regulation	Finance and banking regulation is sufficiently effective.
3.3.08 Financial risk factor	The risk factor in the financial system is adequately addressed.

Stock Market Efficiency

3.3.09 Stock markets	Stock markets provide adequate financing to companies.
3.3.10 Stock market capitalization	US\$ billions
3.3.11 Stock market capitalization	Percentage of GDP
3.3.12 Value traded on stock markets	US\$ per capita
3.3.13 Listed domestic companies	Number of listed domestic companies
3.3.14 Stock market index	Percentage change on index in national currency
3.3.15 Shareholders' rights	Shareholders' rights are sufficiently implemented.

Finance Management

3.3.16 Credit	Credit is easily available for businesses.
3.3.17 Venture capital	Venture capital is easily available for businesses.
3.3.18 Corporate debt	Corporate debt does not restrain the ability of enterprises to compete.

MANAGEMENT PRACTICES

3.4.01 Adaptability of companies	Adaptability of companies to market changes is high
3.4.02 Ethical practices	Ethical practices are implemented in companies.
3.4.03 Credibility of managers	Credibility of managers in society is strong
3.4.04 Corporate boards	Corporate boards do supervise the management of companies effectively.
3.4.05 Auditing and accounting practices	Auditing and accounting practices are adequately implemented in business.
3.4.06 Customer satisfaction	Customer satisfaction is emphasized in companies.
3.4.07 Entrepreneurship	Entrepreneurship of managers is widespread in business.
3.4.08 Social responsibility	Social responsibility of business leaders is high
3.4.09 Health, safety & environmental concerns	Health, safety & environmental concerns are adequately addressed by management

ATTITUDES AND VALUES

3.5.01 Attitudes toward globalization	Attitudes toward globalization are generally positive in your society.
3.5.02 Image abroad	The image abroad of your country encourages business development.
3.5.03 National culture	The national culture is open to foreign ideas.
3.5.04 Flexibility and adaptability	Flexibility and adaptability of people are high when faced with new challenges.
3.5.05 Need for economic and social reforms	The need for economic and social reforms is generally well understood.
3.5.06 Value system	The value system in your society supports competitiveness.
3.5.07 Corporate values	Corporate values take into account the values of employees.

Source: IMD Online (2011)

TABLE 4: 113 Infrastructure Factors variables considered by the IMD in determining rankings in the WCS

Infrastructure

BASIC INFRASTRUCTURE

4.1.01 Land area	Square kilometers ('000)
4.1.02 Arable area	Square meters per capita
4.1.03 Water resources	Total internal renewable per capita in cubic meters
4.1.04 Access to water	Access to water is adequately ensured and managed.
4.1.05 Access to commodities	Access to commodities (basic resources, food, etc.) is adequately addressed.
4.1.06 Urbanization	Urbanization of cities supports business development.
4.1.07 Population - market size	Estimates in millions
4.1.08 Population under 15 years	Percentage of total population
4.1.09 Population over 65 years	Percentage of total population
4.1.10 Dependency ratio	Population under 15 and over 64 years old, divided by active population (15 to 64 years)
4.1.11 Roads	Density of the network, km per square km
4.1.12 Railroads	Density of the network, km per square km
4.1.13 Air transportation	Number of passengers carried by main companies, thousands
4.1.14 Quality of air transportation	Quality of air transportation encourages business development.
4.1.15 Distribution infrastructure	The distribution infrastructure of goods and services is generally efficient
4.1.16 Water transportation	Water transportation (harbors, canals, etc.) meets business requirements.
4.1.17 Maintenance and development	Maintenance and development of infrastructure are adequately planned and financed.
4.1.18 Energy infrastructure	Energy infrastructure is adequate and efficient
4.1.19 Future energy supply	Future energy supply is adequately ensured.
4.1.20 Total indigenous energy production	Millions MTOE
4.1.21 Total indigenous energy production	Percentage of total requirements in tons of oil equivalent
4.1.22 Total final energy consumption	Millions MTOE
4.1.23 Total final energy consumption per capita	MTOE per capita
4.1.24 Electricity costs for industrial clients	US\$ per kwh
4.1.25 Gasoline prices	Premium unleaded gasoline (95 Ron) US \$ per litre

TECHNOLOGICAL INFRASTRUCTURE

4.2.01 Investment in telecommunications	Percentage of GDP
4.2.02 Fixed telephone lines	Number of main lines per 1000 inhabitants
4.2.03 Fixed telephone tariffs	US\$ per 3 minutes local call (peak)
4.2.04 Mobile telephone subscribers	Number of subscribers per 1000 inhabitants
4.2.05 Mobile telephone costs	Mobile cellular tariffs - US\$ per minute local call, off-net (peak)
4.2.06 Communications technology	Communications technology (voice and data) meets business requirements.
4.2.07 Connectivity	Connectivity of people and firms (telecom, IT, etc.) is highly extensive
4.2.08 Computers in use	Worldwide share / Source: Computer Industry Almanac
4.2.09 Computers per capita	Number of computers per 1000 people / Source: Computer Industry Almanac
4.2.10 Internet users	Number of internet users per 1000 people / Source: Computer Industry Almanac
4.2.11 Fixed broadband tariffs	Monthly fee (residential), US\$
4.2.12 Broadband subscribers	Number of subscribers per 1000 inhabitants
4.2.13 Internet bandwidth speed	Per internet user (kbps)
4.2.14 Information technology skills	Information technology skills are readily available.
4.2.15 Qualified engineers	Qualified engineers are available in your labor market.
4.2.16 Technological cooperation	Technological cooperation between companies is developed
4.2.17 Public and private sector ventures	Public and private sector ventures are supporting technological development.
4.2.18 Development and application of technology	Development and application of technology are supported by the legal environment.
4.2.19 Funding for technological development	Funding for technological development is readily available.
4.2.20 Technological regulation	Technological regulation supports business development and innovation.
4.2.21 High-tech exports	US\$ millions
4.2.22 High-tech exports	Percentage of manufactured exports

SCIENTIFIC INFRASTRUCTURE

4.3.01 Total expenditure on R&D	US\$ millions
4.3.02 Total expenditure on R&D	Percentage of GDP
4.3.03 Total expenditure on R&D per capita	US\$ per capita
4.3.04 Business expenditure on R&D	US\$ millions
4.3.05 Business expenditure on R&D	Percentage of GDP
4.3.06 Total R&D personnel nationwide	Full-time work equivalent (FTE thousands)
4.3.07 Total R&D personnel nationwide per capita	Full-time work equivalent (FTE) per 1000 people
4.3.08 Total R&D personnel in business enterprise	Full-time work equivalent (FTE thousands)
4.3.09 Total R&D personnel in business per capita	Full-time work equivalent (FTE) per 1000 people

SCIENTIFIC INFRASTRUCTURE

4.3.10 Science degrees	Percentage of total first university degrees in science and engineering
4.3.11 Scientific articles	Scientific articles published by origin of author
4.3.12 Nobel prizes	Awarded in physics, chemistry, physiology or medicine and economics since 1950
4.3.13 Nobel prizes per capita	Awarded in physics, chemistry, etc and economics since 1950 per million people
4.3.14 Patent applications	Number of applications filed for residents and non-residents
4.3.15 Patents granted to residents	Number of patents granted to residents (average 2006-2008)
4.3.16 Number of patents in force	Per 100,000 inhabitants
4.3.17 Patent productivity	Patents granted to residents / R&D personnel in business ('000s)
4.3.18 Scientific research	Scientific research (public and private) is high by international standards.
4.3.19 Researchers and scientists	Researchers and scientists are attracted to your country.
4.3.20 Scientific research legislation	Laws relating to scientific research do encourage innovation.
4.3.21 Intellectual property rights	Intellectual property rights are adequately enforced.
4.3.22 Knowledge transfer	Knowledge transfer is highly developed between companies and universities.
4.3.23 Innovative capacity	Innovative capacity of firms to generate new products, processes and/or services is high

HEALTH AND ENVIRONMENT

4.4.01 Total health expenditure	Percentage of GDP
4.4.02 Total health expenditure per capita	US\$ per capita
4.4.03 Public expenditure on health	Percentage of total health expenditure
4.4.04 Health infrastructure	Health infrastructure meets the needs of society.
4.4.05 Life expectancy at birth	Average estimate
4.4.06 Healthy life expectancy	Average estimate
4.4.07 Infant mortality	Under five mortality rate per 1000 live births
4.4.08 Medical assistance	Number of inhabitants per physician and per nurse
4.4.09 Urban population	Percentage of total population
4.4.10 Human development index	Combines economic - social - educational indicators / Source: Human Development Report
4.4.11 Health problems	Health problems (sicknesses, AIDS, alcohol, etc.) do not have a significant impact on companies.
4.4.12 Energy intensity	Commercial energy consumed for each dollar of GDP in kilojoules
4.4.13 Paper and cardboard recycling rate	Percentage of apparent consumption
4.4.14 Waste water treatment plants	Percentage of population served
4.4.15 Water consumption intensity	Water withdrawal for each 1000 US\$ of GDP in cubic meters
4.4.16 CO2 emissions	Metric tons of carbon dioxide
4.4.17 CO2 emissions intensity	CO2 emissions in metric tons per one million US\$ of GDP
4.4.18 Renewable energies	Share of renewable in total energy requirements, %
4.4.19 Green technologies	Green technologies are quickly turned into competitive advantages.
4.4.20 Total biocapacity	Global hectares per capita of biologically productive space
4.4.21 Ecological footprint	Global hectares per person
4.4.22 Ecological balance Reserve / Deficit	Total biocapacity minus total footprint in global hectares per capita
4.4.23 Sustainable development	Sustainable development is a priority in companies.
4.4.24 Pollution problems	Pollution problems do not seriously affect your economy.
4.4.25 Environmental laws	Environmental laws and compliance do not hinder the competitiveness of businesses.
4.4.26 Climate change	Climate change is being sufficiently addressed by your government.
4.4.27 Quality of life	Quality of life is high

EDUCATION

4.5.01 Total public expenditure on education	Percentage of GDP
4.5.02 Total public expenditure on education per capita	US\$ per capita
4.5.03 Pupil-teacher ratio (primary education)	Ratio of students to teaching staff
4.5.04 Pupil-teacher ratio (secondary education)	Ratio of students to teaching staff
4.5.05 Secondary school enrollment	Percentage of relevant age group receiving full-time education
4.5.06 Higher education achievement	Percentage of population that has attained at least tertiary education for persons 25-34
4.5.07 Student mobility inbound	Foreign tertiary-level students per 1000 inhabitants
4.5.08 Student mobility outbound	National tertiary-level students studying abroad per 1000 inhabitants
4.5.09 Educational assessment	PISA survey of 15-year olds
4.5.10 English proficiency	TOEFL scores
4.5.11 Educational system	The educational system meets the needs of a competitive economy.
4.5.12 Science in schools	Science in schools is sufficiently emphasized.
4.5.13 University education	University education meets the needs of a competitive economy.
4.5.14 Management education	Management education meets the needs of the business community.
4.5.15 Illiteracy	Adult (over 15 years) illiteracy rate as a percentage of population
4.5.16 Language skills	Language skills are meeting the needs of enterprises.

Source: IMD Online (2011)

